



TEAMCENTER

Structure Management on Active Workspace — Usage

Software Version 2606

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1. About Structure Management

Teamcenter Structure Manager provides a way to create and maintain your product structure comprising assemblies and parts. Specifically, you can import, view, and modify the structure, which is displayed as a multilevel indented list. You can also configure the structure by managing revisions, effectivity, and structure variants.

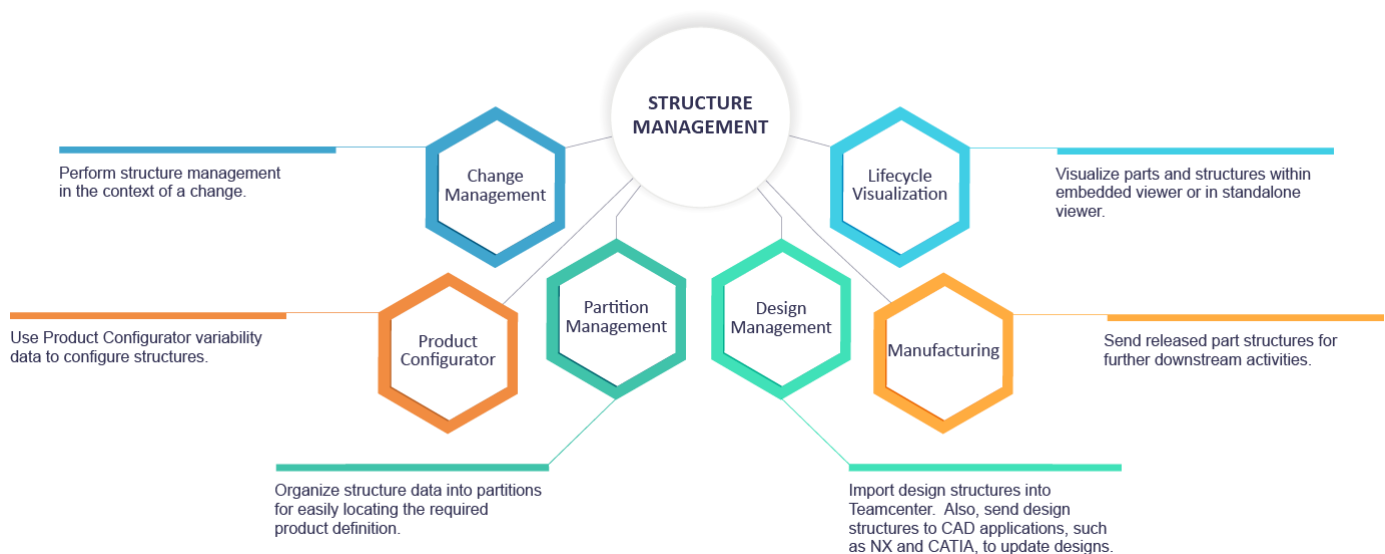
When working with large assemblies comprising thousands of parts, such as cars, Structure Management helps you manage these volumes by performing several tasks. You can view and compare multiple structures, edit structures, configure views, track the changes, and personalize and share views.

As the complexity and variability of the product increases, the amount of data increases even more. Using Smart Discovery for Structures, your site administrator can index structures so that you can find the correct product definition required to do your work. Indexing structures makes the search faster and provides advanced structure filtering and configuration capabilities.

Example

To fix a design issue, an engineer wants to work with the engine block of the car. When the car structure is indexed, the engineer can search for the engine assembly within the structure without having to expand the entire structure. The engineer can further filter the search results to locate all elements that are at a proximity of 10 mm from the engine block.

The following graphic shows how different areas in Teamcenter can play a role in managing a structure:



Where do I go from here?



Administrator

To deploy Structure Management, see [Structure Management — Deployment and Administration](#).

	To index structures by using Smart Discovery, see Structure Indexing Using Smart Discovery .
 Business User	
I want to perform structure management tasks in Rich Client.	See Structure Management on Rich Client .
I want to create a structure.	You can create structures by: • Duplicating a product structure. • Importing a structure from Excel.
I want to make changes to a structure and its properties.	You can add or remove content from a structure or replace a structure component . You can also edit the properties of a part or an assembly .
I want to find differences between my current structure configuration and another structure.	You can do this by comparing product structures .
I want to configure a structure based on the current product or project requirements.	To do this, follow the procedure to configure structures by revision rules, variant rules, and effectivity. You can also configure structures by selection and proximity if your administrator has set up Smart Discovery for Structures.
I want to filter a structure so that I can work with a specific product definition.	See filtering structures. You can filter a structure only if the structure is indexed using Smart Discovery indexing. You can save a filtered and configured structure within a session. Sessions help you to easily locate the product definition that you are currently working with.

2. Structure Manager terms

The terms explained in this topic help you better work with Teamcenter and manage your structures.

Term	Description
Element or occurrence	An item or an item revision within the structure of a parent.
Baseline	A copy of the currently configured structure at the time the baseline was created. Baselining configures a structure and thereby guarantees that the structure is always the same as that when the baseline was created.
Release effectivity	A term used in manufacturing. You configure the structure for a specified date or unit (serial) number by applying a revision rule. Teamcenter shows the revision of each item that is in effect for the specified date, unit number, or range.
Occurrence (structure) effectivity	A term used by manufacturers in the military and aerospace domains. You configure the structure in a way similar to revision effectivity, and Teamcenter shows the actual occurrences that are in effect. Occurrence effectivity applies to the specific element's usage and not globally. It indicates when a child item is effective in the parent assembly.
Working context	A way to set your work aside and then pick up where you left off when you return. The saved working context captures the structure configuration with the context applied, including revision rules, effectivity, and variant rules. It also captures the current focus of navigation and any active selections.
Solution variant	A variant created to derive a buildable (100%) variant of a product that has variability. To create a solution variant, you apply a valid variant configuration to the variable structure. The solution variant has a unique item ID and is linked to the source structure.
Workset	A collection of design components. A workset can contain one or more structures, items, item revisions, BOM views, and BOM view revisions that represent existing parts or assemblies.

3. Find, filter, and update structures using Teamcenter Copilot for Structures

Use Teamcenter Copilot for Structures to find, filter, and change structures and review the updated structure or visual representation in real time.

Prerequisites

Your administrator must install and configure Teamcenter Artificial Intelligence (AI) services.

Procedure

1. Click Copilot .
2. Select **Teamcenter Structures**.
3. Use natural language to type your prompt.

Example

Locate and open the most recent hard drive BOM.

4. Type additional requests to filter the structure, find specific parts, complete weight and balance rollups, or replace parts.
5. Click **Submit**  or press **Enter** to perform the action.
Copilot completes the action and updates the structure and visual representation based on your request.

Copilot also suggests a list of follow-up prompts for the next steps. You can either use the suggested follow-up prompts or type a different prompt.
6. (Optional) You can **Copy**  the Copilot response for future use.
7. (Optional) To clear the current conversation and start a new chat, click **New Chat** .

Examples

- You have learned that one of your BOM components needs a clearance of 5 cm. You need to quickly identify if there are any other pieces within 5 cm of this component. When you select the applicable component and ask Copilot to show you any content within 5 cm of the selected item, the BOM structure and visual representation are filtered according to these parameters.
- You want to work on a chassis and any parts within 30 mm of this chassis. When you ask Copilot to show you any parts within 30mm of the chassis, the structure and visual representation are filtered according to your request. You can then ask Copilot to create and name a workset. Copilot provides a link to the workset after it is created.

- You have learned that one of your structures exceeds the recommended weight limit. You need to find a way to reduce the weight. Therefore, you can ask Copilot to identify the heaviest item in the structure by creating a rollup report, suggest replacements, and even replace the item in the structure.
- You know that the *Aerospace Fastener Torque Standards - Rev C* document, stored in Teamcenter, contains the definitive torque specifications for various materials and fastener types. Specifically, this document mandates that for *Titanium Alloy (Ti-6Al-4V)* fasteners used in critical shear applications, the torque value must be within the range of 150 Nm to 160 Nm.

Instead of manually searching through hundreds of BOM lines, you can use Copilot to efficiently identify non-compliant parts by asking it to find all the parts in your current wing assembly BOM that are made of Titanium Alloy (Ti-6Al-4V), have a specified torque value, and whose torque value falls outside the range of 150 Nm to 160 Nm, as specified in the *Aerospace Fastener Torque Standards - Rev C* document.

- You can ask Copilot to save a series of your favorite actions, and then run them by prompting a keyword. For example, you could ask Copilot to remember the command *Open in context*, which would then perform the following sequence of actions: open the BOM structure, filter the electrical components, and finally, highlight all items that are assigned to you. The next time you prompt *Open in context*, Copilot automatically performs these tasks for you.
- You have identified that certain parts require updates, which must be processed through an Engineering Change Notice (ECN). You can instruct Copilot to manage these structural updates within an engineering change. This includes asking Copilot to create an engineering change, automatically associating affected items to a change, replacing identified parts with recommendations, creating an impact analysis workset for affected items, and finally, visualizing change markups across multiple structures.
- You have learned that a part of an assembly is impacted as a part of the requirement R123. You can prompt Copilot and understand all the impacted items due to this requirement. Copilot analyzes the cross-structure relationships, and displays all affected items, which can include linked designs, associated parts, supplier components, and dependent assemblies.
- You can ask Copilot to configure a structure by applying the *Any Status; Working* revision rule and the *Physical* partition scheme.

4. Work in the context of a change

Set a change context to track structure updates

You can track structure updates by setting an engineering change notice as an active change context. The updates can be viewed by accessing the change notice.

Procedure

To set a change context, click **No Active Change**  in the global header and select the required change notice.

Review active or closed changes for a structure

You can track changes (added, modified, replaced, revised, or deleted) to the assemblies using a change request or a change notice. When the ECN or ECR is an Active change, redlines are shown in the structure. To view the closed changes, use the **Show Redlines** command.

Prerequisites

Your administrator can configure the various methods to effectively track and display relevant changes to an assembly.

Procedure

- 1. Search for and open the structure. If the structure has changes, they are indicated by redlining. Expand the assembly to view the details of the change.

Element 	ID 	Revision 
 Engine	035611	B 
 Piston	035612	A
 <i>Enhanced Valve</i> Valve	<i>035617</i> 035613	A
 Connecting Rod	035614	A
 <i>Connecting Rod Assembly</i>	<i>035621</i>	<i>A</i>

- Deleted parts are indicated by a red strikethrough.
- The added parts display in green and are italicized.
- For replaced or revised parts, the old and the new values are shown side by side.

After the structure is revised, the redlines are shown for all changes including the edits that you make for the part attributes.

2. (Optional) To disable highlighting the changes in the structure, go to **More Commands *** > View**  icon and turn off **Show Redlines** from the **Page level** toolbar.
3. Only active changes are shown by default. To view the closed changes, click **More Commands *** > View**  > **Show Redlines** from the **Page level** toolbar.

These changes are also shown in the packed mode.

You can return a redline change back to its original impacted version by reverting the BOM modifications.

Update multiple structures simultaneously in a change notice

Update multiple impacted structures simultaneously for the problem items that are a part of a change notice by adding siblings, child items, and occurrence substitutes simultaneously as a part of the solution items.

When updating the structure, ensure that the selected item is not in the **Released** state.

When updating multiple structures simultaneously, you can perform the following tasks:

- Add siblings, child items, and occurrence substitutes.
- Revise the structures.
- Replace and remove the items from the structures.

All updates that are made as a part of the change notice are indicated as redline text.

Restrictions and limitations

You can update multiple structures simultaneously if your Teamcenter setup includes BOM Structure Management license.

Procedure

1. In an active change notice, perform the impact analysis for the problem items and then select the impacted items.
The impacted items are the structures or immediate parent assemblies for the problem items. If a problem item is a part of multiple structures, you can select all these structures as impacted items.
2. Click the **Change Content** tab to view all the problem items and their immediate parent assemblies, which are identified as impacted items.

Tip

To navigate to or locate specific items on the **Change Content** tab, use filters and sorting in columns.

After you apply a filter to a column, all items that satisfy the filter criteria are displayed on the **Change Content** tab. For example, suppose that you are searching for an item with ID as 12. In such a case, in the **ID** column, select the **Contains** operator and type **12**. The **Change Content** tab displays all items that have ID as 12. After a filter is applied, the **Change Content** tab considers the redlining and shows the modified content. If an item is removed on the **Change Content** tab as a part of the change, and its content satisfy the filter criteria, then that item is still shown on the **Change Content** tab. For example, suppose that the 8cm nut is removed and you entered the filter specifying the 8cm size. In such a case, the **Change Content** tab shows all 8cm nuts including the one that you removed.

You cannot filter the visual indicator columns like . After applying column filters to locate the required content, if you click another tab in the change and then return to the **Change Content** tab, the column filtering is removed. You must again apply the column filters.

3. Perform the following actions as appropriate to select and show the required items on the **Change Content** tab.

If you want to...	Perform this actions...
Select all items including the parent and child items.	Click Select  > All  .
Select all top parents assemblies.	Click Select  > All top parents .
Select all occurrences of the selected item.	Click Select  > All matching elements .
View all child items for the selected parent node apart from the ones that are a part of the change notice and also view the rest of the siblings and sub-assemblies. After viewing the non-impacted child items, you can update their properties and include them in the change. You might see all child items by default if your administrator has set it for you. Furthermore, if all child items are shown, the parent are always collapsed. You must manually expand them.	Click Show  > All Children  .
View only the child items that are a part of the change notice and do not want to view the siblings and sub-assemblies.	Click Show  > Children in Change  .

4. To revise one or more items, select them and click **Revise** .
- If you select only one item for revision, the **Revise** panel appears. Review the information on the **Revise** panel and click **Revise**.
- If you select multiple items, a confirmation message appears. Click **Revise** on the confirmation message.
- The selected items are revised and are shown with the redlines.

5. If you want to edit the description, quantity, and any other properties of the selected one or more child items, click **Edit** .
After you are finished editing the properties, click **Save Edits** .
6. If you want to add a child item to one or more items:
 - a. Select the items under which you want to add the child item and click **Add**  > **Child**.
 - b. In the **Add Child** panel, search an existing child item or create a new one, and then click **Add**.
The child is added to the selected items.
7. If you want to add a sibling item to one or more items:
 - a. Select the items for which you want to add the sibling item and click **Add**  > **Sibling**.
You cannot add sibling to the parent item in the structure.
 - b. In the **Add Child** panel, search for an existing child item or create a new one, and then click **Add**.
The child item is added to the selected items.
8. If you want to add occurrence substitutes:
 - a. Select one or more items and click **Add**  > **Substitute** .
 - b. In the **Add Substitute** panel, search for an item or create a new item and click **Add**.
The occurrence of the selected item is added as a substitute. The **Substitutes** and **Primary Component** columns show the information of substitutes and primary items. The **Substitutes** section in the **Overview** tab for the primary item also show the list of the substitutes items. You can further set substitutes as primary components.

After the substitutes are added, if you want to hide them from being shown in the structure, right-click a structure and select **Hide Substitute Components** .
9. If you want to remove some of the items from the structure, select the items and click **Edit Structure**  > **Remove** .

The selected items are removed from the structures and redlines appear for them.

10. If you want to replace the selected items by another item:
 - a. Select one or more items and click **Edit Structure**  > **Replace** .
 - b. In the **Replace** panel, search for an existing item or create a new one, and then click **Replace**.
All selected items are replaced with the chosen item.
11. If required, select the required revision rules and apply date and unit effectivity on the **Change Content** tab.

What to do next

After the impacted structures are updated and the solution items are provided, you can view the change summary and then send the change for downstream process.

Track BOM modifications in the Change Summary

Applying an active change context for BOM revisions automatically tracks the modifications in the Change Summary.

See [set a change context to track structure updates](#) for more information.

With the change notice set as an active change, BOM modifications and properties are identified by redlines. For more information on the **Show Redlines** feature, refer to [Review active or closed changes for a structure](#).

The screenshot displays the Siemens CAD software interface. The top bar shows the project name 'Rev1_3-004-006660_ChangeECI_Deloo_TC/B:1-Beta25_Truck_3 ...' and the user 'SIEMENS'. The breadcrumb navigation indicates the current view is 'Shock absorber' within the 'Front Axle (20L steered, freewheel, long leafspring, r5)' assembly.

The main BOM table lists various components and their revisions. The 'Shock absorber' row is highlighted in blue, and its revision 'A' is marked with a redline. The 'Find Number' column shows '530' for the 'Shock absorber' and '530' for its sub-components.

The right-hand panel shows the 'Details' view for the 'Shock absorber' component. It includes tabs for 'Overview', 'Weight and Balance', 'Parameters', 'Finishes', 'Classification', and 'Made From'. The 'Overview' tab is active, showing 'Aligned Parts' and 'Substitutes'.

Object	Type	Find No.
033710_sub1_Deloo_TC/A:1-Shock absorber sub1	Design Revision	530
033711_sub2_Deloo_TC/A:1-Shock absorber sub2	Design Revision	530
033712_sub3_Deloo_TC/A:1-Shock absorber sub3	Design Revision	530
033714_sub5_Deloo_TC/A:1-Shock absorber sub5	Design Revision	530
033870/A:1-Shock absorber sub6	Design Revision	530
033713_sub4_Deloo_TC/A:1-Shock absorber sub4	Design Revision	530

These modifications, not the full assemblies, are tracked in the Change Summary of the change notice.

View the authoring change notice and the changes made to the structure element, and the change dashboard for the product

ECN-00033/A:1-Truck_Change_Charles

OverviewAffected ItemsChange ContentParticipantsReference ItemsWorkflowDependenciesRelationsReportsMass Update

Owner: charles (charles)Date Modified: 21-Mar-2025Release Status:Type: Change Notice RevisionClosure: OpenDisposition: ApprovedECN-00033/A:1-Truck_Change_Charles

Release effectivityGenerate Report

▼ Task to Perform

Workflow: NX_ChangeNoticeRevisionDefaultWorkflowTemplate : ECN-00033/A:1-Truck_Change_Charles

Name:

Comments:

ElaboratingReviewingExecutingComplete

▼ Impacted Items (7)

ID	Object	Vendor Part Name	
Rev1_3-004-006660_ChangeECI_Deloo_TC	Rev1_3-004-006660_ChangeECI_Deloo_TC/A:1-Beta25_Truck_3 Axle	Beta25_Truck_3 Axle	1
Rev1_3-004-006604_ChangeECI_Deloo_TC	Rev1_3-004-006604_ChangeECI_Deloo_TC/A:1-Truck Front Axle (r5)	Truck Front Axle (r5)	2
033669_Deloo_TC	033669_Deloo_TC/A:1-Touchscreen Display	Touchscreen Display	7
Rev1_3-004-006639_ChangeECI_Deloo_TC	Rev1_3-004-006639_ChangeECI_Deloo_TC/A:1-Front (Bare) Axles	Front (Bare) Axles	3
Rev1_3-004-006455_ChangeECI_Deloo_TC	Rev1_3-004-006455_ChangeECI_Deloo_TC/A:1-Front Axle (20L steered, freewh...	Front Axle (20L steered, ...	4
Rev1_3-004-005190_ChangeECI_Deloo_TC	Rev1_3-004-005190_ChangeECI_Deloo_TC/A:1-Leafspring	Leafspring	5
Rev1_3-004-006378_ChangeECI_Deloo_TC	Rev1_3-004-006378_ChangeECI_Deloo_TC/A:1-Torsian Sway Bar (extra bended)	Torsian Sway Bar (extra ...	6

▼ Change Summary

Impacted Items: ALL

Changes for	Merge Status	Action	Element	ID	Revision
		Modified	Beta25_Truck_3 Axle	Rev1_3-004-006660_Char	B A
		Modified	Touchscreen Display	033669_Deloo_TC	B A
		Modified	Truck Front Axle (r5)	Rev1_3-004-006604_Char	B A
		New	Torsian Sway Bar Doc5	033868	A
		New	TouchScreen_Xenarc	033869	A

▼ Details

Synopsis: Truck_Change_Charles

Description: Truck_Change_Charles

Date Released:

Effectivity:

View the authoring change notice and the changes made to the structure element, and the change dashboard for the product

View the authoring change notice for a selected structure element, or access an interactive change dashboard to see how change notices impact a top-level product and its immediate sub-assemblies. You can also review the modification history of a structure element.

Procedure

1. Open the required structure and select the structure element.
2. Go to the **Changes** tab. In the **Changes** section, you can see the authoring change notice for the selected structure element.

If the selection is top level product, then you can see the interactive change dashboard listing all the change notices under which the top product is impacted along with change notices in which its first level part usages are added or updated. Here, click on the PI chart to filter between the change notices as required. This dashboard is only visible for top product but your administrator can enable it for assemblies below the top product as well.

3. Go to the **History** tab. All the modifications or revisions made to the selected structure element with their respective change notices are listed under **Change History**.

Cancel a change notice

Based on a specific product requirement, an engineering change notice is created. The engineering BOM content is authored or updated in the context of the change notice. After authoring or updating the BOM content, the product requirements may get canceled. In such a case, you must also cancel the corresponding change notice before it is released. On canceling the change notice, the updates made to the BOM content also get canceled.

Procedure

1. From **Launcher**, click **Changes** .

If this option is unavailable, search for it. You can pin it from the search results for future access.

2. Select the change under which the structure is created or modified.
3. Click **Cancel Change** in the **Overview** tab.
All the **Solution Items** of the change notice get canceled and their property **Cancelled** is set as `<true>`.

Canceling the engineering change notice will cancel all its associated solution objects. A canceled revision can never configure. However, you can create a new revision from a canceled revision. You cannot reopen a canceled change and release its content.

If there are sequential changes in ECN2 that have been created based on ECN1, and ECN1 is canceled, ECN2 is not canceled automatically. You can choose to cancel it manually.

When you cancel a change notice which contains a workset or a session having a structure, this cancel change behavior is not supported.

Every cancellation of ECN may not inherit this behavior. For example, cancellation of ECN having workset and subset, which contains the structure.

5. Access structure data within partitions

A BOM architect organizes a large product into smaller, accessible sections called *partitions* and arranges structure elements logically and hierarchically within the partitions. Partitions can belong to different partition schemes such as functional, physical, or spatial.

The BOM architect can create partitions only if **Partitions for Structure** is installed in your Teamcenter set up. If it is installed, you can access the structure data and perform the various structure related tasks from within the partitions.

For information on partition schemes and partitions, see *Structure Partitions — Usage*.

6. Find and navigate structures

Finding structures and structure elements

Discover various methods for locating structures and structure elements within Teamcenter, including global, advanced, and in-context searches, to efficiently find the data you need.

You can search for structures or structure components using the following methods:

Global Search	Searches all indexed data in Teamcenter.
Advanced Search	Searches for data by using specific criteria based on predefined queries.
In-context search	Searches for content within an open structure, a working context, or a session that is composed of structures. The structure can be indexed by using either Active Content Structure indexing or Smart Discovery Indexing. It may be non-indexed as well. The non-indexed search requires a Context Management User license. This is used for Quick Find searches.

You can perform a simple search to find an occurrence of an object in an open structure. You can enter any attribute or text associated with the occurrence (for example, a name) and then click **Search** . Search results span the entire context.

You can use the same syntax as for full-text searches, including operators such as OR and AND. You can use the **Filter** tab to narrow the results by selecting specific filters and categories.

Find elements within a structure

Perform keyword searches within an open structure or a selected subassembly to quickly locate specific elements, including those based on content, and narrow down your search results.

To narrow your search results to a specific branch in the structure, you can perform a keyword search within the selected subassembly from the **Find** panel.

Procedure

1. Open a structure from the search results.
2. Click **Find** .

3. In the **Find** panel, enter the search parameters, and click **Search** .

If file content indexing is deployed in your Teamcenter configuration, you can specify dataset filenames as the search parameter. You can also specify the text within a file as the search parameter. For example, consider that the dataset *EngineSpecification.pdf* contains the text **RPM**. If you enter **RPM** as the search parameter, the search result displays elements that have the dataset *EngineSpecification.pdf* as an attachment. The search result also displays elements that have datasets with the text **RPM** in their content.

Additionally, you can specify the name of a .prt file as the search parameter.

When a structure is indexed using Smart Discovery and the administrator has enabled indexing for an element's occurrence properties, you can search for elements using those properties. For example, if the **Assembly Instructions** occurrence property is marked for Smart Discovery Indexing and the structure is already indexed using Smart Discovery, you can search for elements based on their **Assembly Instructions** occurrence property.

4. (Optional) To narrow the results:
 - a. Select the subassembly to which you want to limit your search.
 - b. Select the **Find within** check box, enter the search parameters, and click **Search**.
5. Select an element from the search results to view it in the structure.

If you have **Partitions for Structure** deployed in your Teamcenter configuration *and* if you select an element from the search results on the **Find** panel, the element is highlighted in the work area within the partition that contains it.

Note

Clicking **Select All**  does not select the corresponding elements in the structure.

Searching structured content if you do not have access to objects

If you find an element in a structure that you do not have read access to, Teamcenter does not show that occurrence and excludes it from the find-in-context results. If the occurrence exists in an assembly to which you *do* have access, Teamcenter shows an **Access Denied** indicator in its place.

Note

Access to occurrences is set by your system administrator by using Access Manager. Access controls protect intellectual property and prevent general access to data. To view restricted content that you do not have access to, you can request the owner of the assembly to use project-level security to allow collaboration.

If you do not have access to a revision configured by the revision rule, Teamcenter searches for the latest revision you have access to, and configures the access accordingly.

Automatic searching for the latest accessible revision is not supported in indexed structures. However, it is supported if Smart Discovery Indexing is used.

Navigate a structure

Learn how to open and navigate through a structure in Teamcenter, including expanding to view child parts and using the breadcrumb trail to move to higher-level assemblies.

You can select a structure from the search results and then click **Open**  to view the structure.

Navigate to child parts

The assemblies or subassemblies that have child parts are listed with the **Show Children**  button.

Click **Show Children**  to navigate to child parts.

Navigate to a higher level using the breadcrumb trail

Assembly nodes are visible in the breadcrumb trail. Use the breadcrumb trail to navigate to a higher level (such as parent parts) in a structure.

HDD-0527/A;1-Hard Drive Assembly > HDD-0507/A;1-Baseplate Assembly > HDD-0522/A;1-Motor Electronics Assembly

Note

The back button  does not take you to the parent parts. It takes you to the previous location visited in Active Workspace. It might also not show the latest structure. You may refresh the structure to view the latest one.

If you use the breadcrumb trail to navigate to a different parent, the first leaf in that structure is displayed, the breadcrumb is updated with the path to that leaf node, and the display is switched back to the hierarchical view.

Navigate to a higher level using the breadcrumb trail

7. Import and export structures

About importing structures from Excel

You can import structure data into Teamcenter from a Microsoft Excel spreadsheet and create new structures with new or existing components. This activity is useful for integrating data from external sources or vendors.

In Teamcenter, you can import a structure from a Microsoft Excel spreadsheet. This feature is useful when you need to import data from an external source, for example, from a design contractor who does not have Teamcenter. You can import relatively simple and small structures, such as structure data provided by vendors. Importing from Excel might not be suitable for very large and complicated structures.

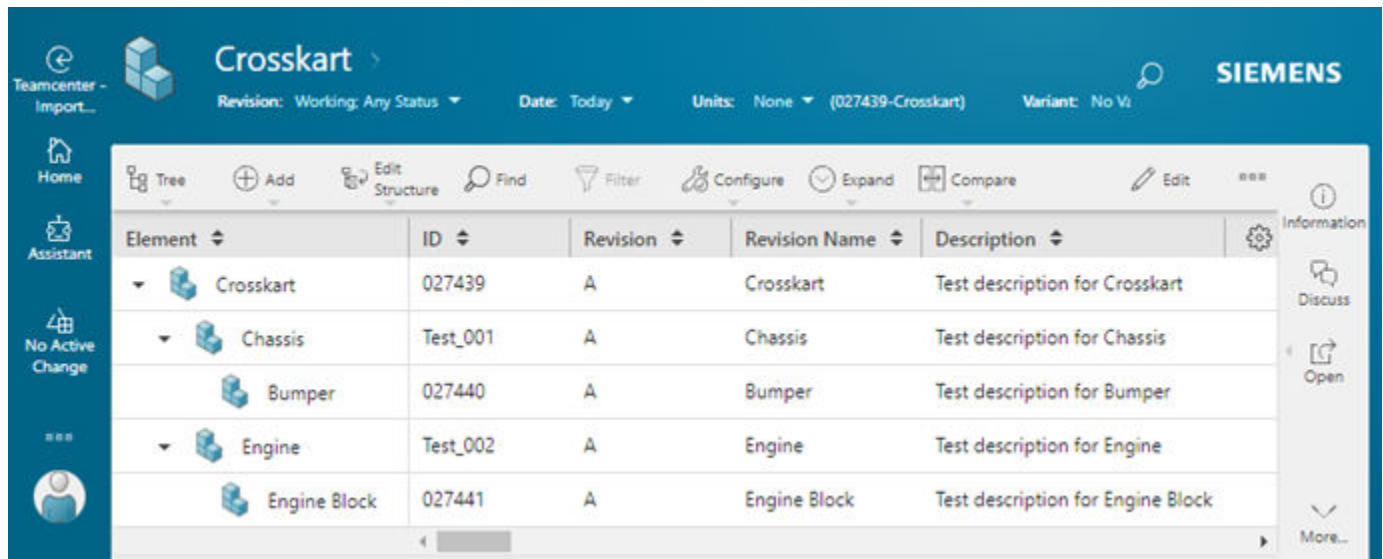
The following scenarios explain how you can **import structures** in Teamcenter.

Scenario 1

You want to create a new *Crosskart* structure with all new components. To do this, you can create an Excel spreadsheet similar to the following and then import it in Teamcenter:

	A	B	C	D	E	F
1	Product Structure					
2	Tc_Level	Tc_ObjectType	Name	ID	Revision	Description
3	0	Item	Crosskart			Test description for Crosskart
4	1	Item	Chassis	Test_001		Test description for Chassis
5	2	Item	Bumper			Test description for Bumper
6	1	Item	Engine	Test_002		Test description for Engine
7	2	Item	Engine Block			Test description for Engine Block
8	<endtag>					

After the import, the following structure is created in Teamcenter:



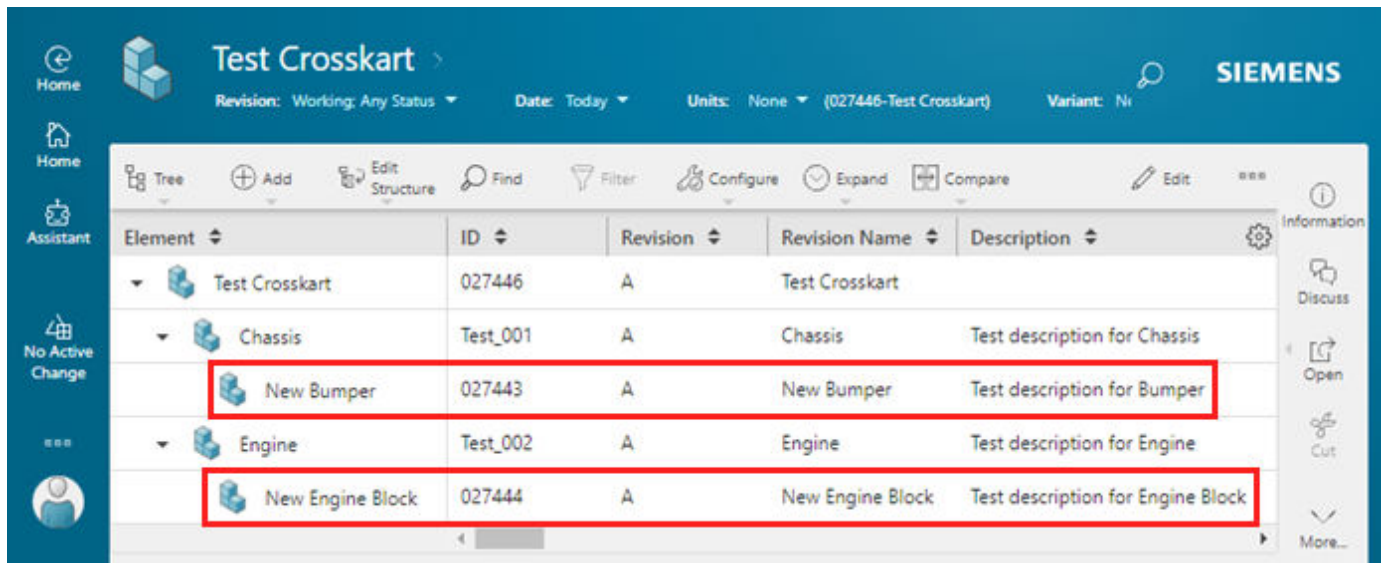
Even if you did not specify an ID for some of the elements in the input spreadsheet, Teamcenter generates the IDs for those elements automatically.

Scenario 2

You want to create a new structure called *Test Crosskart* by reusing some of the existing elements in Teamcenter. Therefore, in the input spreadsheet, you add the name for the new structure and only the names and IDs of the existing subassemblies. Your input spreadsheet appears as follows.

	A	B	C	D	E	F
1	Product Structure					
2	Tc_Level	Tc_ObjectType	Name	ID	Revision	Description
3	0	Item	Test Crosskart			
4	1	Item	Chassis	Test_001		
5	1	Item	Engine	Test_002		
6	<endtag>					

After the import, Teamcenter creates the new *Test Crosskart* by reusing the existing elements from the sub-assemblies. You only need to specify the required properties of the subassemblies and not of the existing elements.



Prepare the Excel file for importing a structure

Prepare an Excel spreadsheet for importing structures into Teamcenter by following specific guidelines for mandatory fields, reference properties, vendor-part data, and handling multiple parts, assemblies, or occurrence-based substitutes.

To import a structure from Excel, you must prepare the Excel file in a specific way. The guidelines for preparing the Excel file are listed here:

- Mandatory points
 - The top row must have a title, for example, **Primary Object**.
 - The cells in the second row must have the headers **Tc_Level** and **Tc_ObjectType**. The header **Tc_Level** must be in column A and **Tc_ObjectType** must be in column B.
 - The properties **Name**, **ID**, and **Revision** are mandatory.
 - While you import a structure, the elements that occur multiple times under a single parent in the structure are uniquely identified by certain properties. It is recommended that you use these properties as columns in the input spreadsheet. To get these properties, you may refer to the values in the **AWC_Occ_Unique_Identifier** preference. When you reimport the same structure, make sure that the values for these properties are correctly set. Otherwise, you might not get the expected results.
 - You must include **<endtag>** in column A at the end.
 - If you do not want to update the part or occurrence properties, leave the corresponding cells in the input spreadsheet blank. Teamcenter ignores blank cells in the input spreadsheet.
- Import a structure along with reference properties
 - You can import structures with certain reference properties, such as **Unit of Measure**, **Owner**, and **Group**. You can also import any other reference property with an LOV.

- After you add the display string of a reference property to the Excel sheet, Teamcenter maps this string to the correct reference object.
- Import a structure along with vendor-part data
 - To import vendor-part data, you must specify the secondary objects along with the primary ones. Thus, you must fill in the **Tc_Secondary_ObjectType**, **Tc_Secondary_Relation**, and **Vendor ID** fields.
 - The **Vendor ID** that you specify must exist in the Teamcenter database.
 - **Tc_Secondary_ObjectType** must be the first column in the **Secondary Object** section.
- Import multiple parts
 - To import multiple parts, you can add them, excluding any assemblies, to the Excel file. You must specify the **Tc_Level** of each part as zero (0).
 - As Teamcenter ignores occurrence properties, you need not specify them.
 - After you import multiple parts, Teamcenter opens only the first part.
- Import multiple assemblies
 - To import multiple assemblies, you can add them to the Excel file. You must specify the **Tc_Level** of each top-level part in an assembly as zero (0) and its children as 1 or 2 and so on as per the hierarchy of items.
 - Teamcenter considers each top-level part as a new structure.
 - After you import multiple assemblies, Teamcenter opens only the first assembly.
- Import multiple types of secondary objects
 - You can import multiple types of secondary objects for both an item and an item revision.
 - While importing related objects, you can specify the object type and the relation type.
 - If a related object already exists, Teamcenter relates the primary object to the related object. If a related object does not exist, Teamcenter creates a new related object.
 - If you do not specify any value in the **Tc_Primary_To_Secondary_Relation** column, Teamcenter considers the following values, by default: For vendor-manufacturing related parts, the value is **item-to-rev** and for any other parts that are not related to vendor manufacturing, the value is **rev-to-rev**.
- Import occurrence-based substitutes
 - To import the occurrence-based substitutes, set the **Is Substitute** column to **Yes** in Excel for every substitute part.
 - The primary part and its substitute must have the same value in the **Tc_Level** column in Excel.
 - The primary part must be defined above the substitute in the Excel.
 - The primary part is a line with no value in the **Is Substitute** column.
 - The substitute part is a line defined below the primary part.

	A	B	C	D	E	F
1	Primary Object					
2	Tc_Level	Tc_ObjectType	Name	ID	Revision	Is Substitute
3	0	Item	Front Door Assembly	FD-100	A	
4	1	Item	Door shell assembly	DS-200	A	
5	2	Item	Steel Door Shell	DS-200-ST	A	
6	2	Item	Aluminium Door Shell	DS-200-AL	A	Yes
7	1	Item	Door handle Assembly	DH-300	A	
8	2	Item	Carbon Fiber Door Shell	DS-200-CF	A	Yes
9	2	Item	Standard Handle Black	DH-300-SH	A	
10	2	Item	Chrome-plated Handle	DH-300-CH	A	Yes
11	2	Item	Sport Handle with Keyless Entry	DH-300-SP	A	Yes
12	1	Item	Window Regulator	WR-400	A	
13	2	Item	Manual Window Regulator	WR-400-R	A	
14	2	Item	Electric Window Regulator	WR-400-E	A	Yes
15	2	Item	Premium Electric Regulator with Anti-pinch	WR-400-EP	A	Yes
16	1	Item	Side Mirror	SM-500	A	
17	2	Item	Basic Side Mirror	SM-500-BM	A	
18	2	Item	Heated Mirror	SM-500-H	A	Yes
19	2	Item	Power-folding Mirror with Turn Signal	SM-500-PF	A	Yes
20	<endtag>					

The above image shows that the structure of the front door assembly is being imported from Excel. In this structure:

- Aluminum Door Shell and Carbon Fiber Door Shell are substitutes for Steel Door Shell
- Chrome-plated Handle and Sport handle with Keyless Entry are substitutes for Standard Handle Black
- Electric Window Regulator and Premium Electric Regulator with Anti-pinch are substitutes for Manual Window Regulator
- Heated Mirror and Power-folding Mirror with Turn Signal are substitutes for Basic Side Mirror

Import and export structures and changes using Excel round-trip

Import structures and structure changes from Excel

Import new structures or apply changes to existing ones in Teamcenter; using data from Excel spreadsheets. This includes adding and removing elements, updating item and element properties, and managing relations, whether from a prepared Excel file for initial import or a round-trip Excel file for modifications.

Use the following guidelines for importing Excel to create structures, elements, and relations in Teamcenter:

- The structure can include all reference properties with an LOV, such as unit of measure, owner, and group.

- You can import structures with the global revision rule and structures using different ID types that use a Multi-Field Key (MFK). You can import vendor-part data, multiple elements and assemblies, and multiple types of secondary objects.
- You can also import the relational properties (like preferred or backup vendor preference status). If an array property contains multiple values in the Excel, you can import them too. For example, if the Excel contains a column for Production Sites and the values are Milford and Troy, both these values are imported in Teamcenter.

When you import changes from a previously exported Excel, you can add or remove structure levels, add or remove elements from the structure. However, do not make changes to the **TCUID** column in Excel. When you import the changes from a previously exported Excel, Teamcenter detects the original structure and imports changes to it.

Restrictions and limitations

You can perform this task if it is configured for you by your administrator.

Procedure

1. Prepare the Excel with the structure data, elements, or relations you intend to import.
 - For an external file, ensure that the columns are organized for mapping to Teamcenter properties. These files can represent simple structures or structures with relations.
To import a structure from Excel, you must prepare the Excel file in a specific way. See *Prepare the Excel file for importing a structure*.
 - For a round-trip file that was previously exported from Teamcenter, you can modify:
 - The item and element properties.
 - Modify structure including adding and removing elements and replacing items.
 - Modify the relation properties, including adding new relations and modifying existing relation properties.
2. In Teamcenter, navigate to the location where you want to import or update the structures.
The location can be a folder in Teamcenter, the **Content** tab of an engineering change, or an existing open structure in Teamcenter.
3. Perform one of the following actions:
 - Click **Excel Round-trip**  > **Import from Excel** .
 - Click **More commands ***** > **Import/Export**  > **Import from Excel** .
4. In the Import from Excel panel, click **Choose File** and select the Microsoft Excel file from which you want to import the structure.
You must review and configure the import options.
If you are importing an external file, map the Excel columns to Teamcenter columns and properties. Furthermore, the structures, its relations, and elements are imported by default. If you are importing

a round-trip file, Teamcenter recognizes the column mapping, and you are not required to map the columns manually.

5. If you are importing an external Excel file, map the properties between Teamcenter and Excel.

- Select the required mapping from the **Saved Mappings** list. The list shows the mappings created and saved by the administrator or previous users.

Select a saved mapping that matches your Excel file's column structure. Mapping names typically indicate the structure type or source. For example, BOM_Import_Standard or Vendor_Parts_Mapping. If unsure, create a new mapping or consult your administrator about which mapping to use for your specific import scenario.

- Alternatively, create a new mapping by entering a name in the **Saved Mappings** field and selecting the **Mapped Attributes** for the **Excel Headers**. Even if an error occurs while importing a structure, the created mapping is saved. You need not select the mapped attributes again for the Excel headers.
- If a mapped attribute is not available for an Excel header:
 - a. In the **Mapped Attributes** list for the corresponding Excel header, click **Add New** to create a new attribute.
 - b. In **Add Properties**, select appropriate **Subtypes**, if you want to change the default one.
The reference properties **Unit of Measure**, **Owner**, and **Group**, along with other properties, are listed in the **Add Properties** section.
 - c. Filter and select the attribute that you want to map.
 - d. Click **Add**.
 - e. In the **Import Structure** panel, choose the newly added attribute.

Import from Excel

File:

Choose File

MobileStructure.xlsx (0.010MB) X

▼ Import Options

Scope:

✓

 Structure

✓

 Relations

✓

 Parts

Structure Options: ⓘ

●

 Merge

○

 Overwrite

▼ Map Properties

Saved Mappings:

Excel Header:

Mapped Attributes

Name:

Name (Required) ▼

ID:

ID (Required) ▼

Revision:

Revision (Required) ▼

☐ Run in Background

Preview

Import

7-8
Software Version 2606

Structure Management on Active Workspace — Usage
Unpublished work. © 2026 Siemens

6. If you are importing a round-trip file that was previously **exported from Teamcenter**, select the following import options:
- Select the **Structure** check box to import the structural changes. The examples of structural change can be a new element is added to the structure, a subassembly is removed from the structure, or the quantity of elements is updated.
 - Select the **Relations** check box to import the relational changes between the primary and secondary objects that are made in the Excel.
 - Select the **Parts** check box to import the item-specific changes made to the structure. The examples of item-specific changes can be new items or updates to an item's description.
 - If you want to merge the changes made to the Excel into the structure and retain all the structure-related data in Teamcenter, click the **Merge** option.

If you want to replace data in Teamcenter with the data from the Excel file, click the **Overwrite** option.

Caution

Always ensure that you understand the implications of overwrite operation and have appropriate backups or version control in place before overriding the structure data.

Note

The **Overwrite** option requires specific settings configured by your administrator. If this option is not available, contact your administrator.

Import from Excel

File:

Choose File **Mobile Phone.xlsxm** (0.015MB) X

▼ **Import Options**

Scope:

☒ Structure

☐ Relations

☐ Parts

Structure Options: ⓘ

☒ Merge

☐ Overwrite

☐ Run in Background

Preview Import

7. (Optional) If you do not want to run the import process in the background, clear the **Run in Background** check box.

Teamcenter notifies you about the result of the import process in the **Alerts** panel.

8. (Optional) Click **Preview** to check the structure before import.

Perform the following tasks in the Preview panel:

- a. Review the structure to be imported along with the **Action** that is performed by default and the **Teamcenter Information** column that shows additional information related to an action.

Mobile Phone

Revision Name	Action	Teamcenter Information
▼  Mobile Phone	Overwrite	
▼  CPU	Revise	
 Power Supply	New	⚠ The object type required to create object at row 4 was not specified.
 Motherboard	Overwrite	
▼  Screen	Reference	
 Gorilla	New	
▼  Camera	Overwrite	
 48MP	New	
 13MP	New	
 SIM tray	Overwrite	

- b. (Optional) Change the action for **Revise** and **Overwrite** considering the business requirement. By default, Teamcenter selects actions based on whether items exist and whether they are writable. Change actions when the default action does not match your requirements. To do so, right click an action and select:
- **Revise** when you want to preserve the existing revision and create a new one with updates.
 - **Reference** when you want to use existing elements without modifications.
 - **Overwrite** when you need to update the existing revisions with new data without creating new revisions.

9. Click **Import** to import the structure in Teamcenter.

Results**If there were no errors while importing the structure:**

The structure is imported in Teamcenter. When you import multiple elements or assemblies simultaneously, all the top-level elements and assemblies are added to the folder you specified previously. You receive a notification in the **Alerts** panel when the import is complete. Click **Alerts**  to open the **Alerts** panel.

The structure is imported using the global revision rule. The same revision rule is used while opening the structure after import.

You can also import a structure with different ID types that use a multi-field key, that is, when you have two different elements with the same ID. You can see the elements on the **Import Preview** screen and

import them successfully. The import structure functionality searches for and creates elements based on the multi-field key.

If there were errors while importing the structure:

You receive an error message stating that the import failed because there were errors while importing the structure. You also receive a notification in the **Alerts** panel, where you can view the error details. Click **Alerts**  to open the **Alerts** panel.

In the **Teamcenter Information** section, you can view a list of all potential errors simultaneously. When you hover over an error, you can view detailed information about that error.

Excel Roundtrip Selection Mode Select All

Name	Action	Teamcenter Information	Release Status	ID	Revision	Secondary Obj...
Computer	New	Structure was not imported because of err...		032391	A	
CPU	New	The "<empty>" value cannot be assigned ...		032392		3 Objects
Power supply	Overwrite	The Vendor ID with value "Company_1" do...		powersupply_1	A	1 Objects
Power supply	New	The object "powersupply_1-P... The Vendor ID with value "Company_1" does not exist at row 9.				
Motherboard	New	Structure was not imported because of err...		032393		
	New	The "<empty>" value cannot be assigned ...				

Tip

At this stage, you need not restart the import process from the beginning. You can make the required corrections in the Excel file and then re-upload it by clicking **Choose File** in the **Import Structure** panel. Once this is done, the saved mappings from the previous import are automatically selected and shown in the **Saved Mappings** list. Then, you can click **Import Structure** to proceed with the import.

When you click the link in the **Alerts** panel, you can view the information about the structures that could not be imported and the related objects.

▼ Properties

Message Subject: Mobile Phone.xlsm failed to import

Priority:

Event Type:

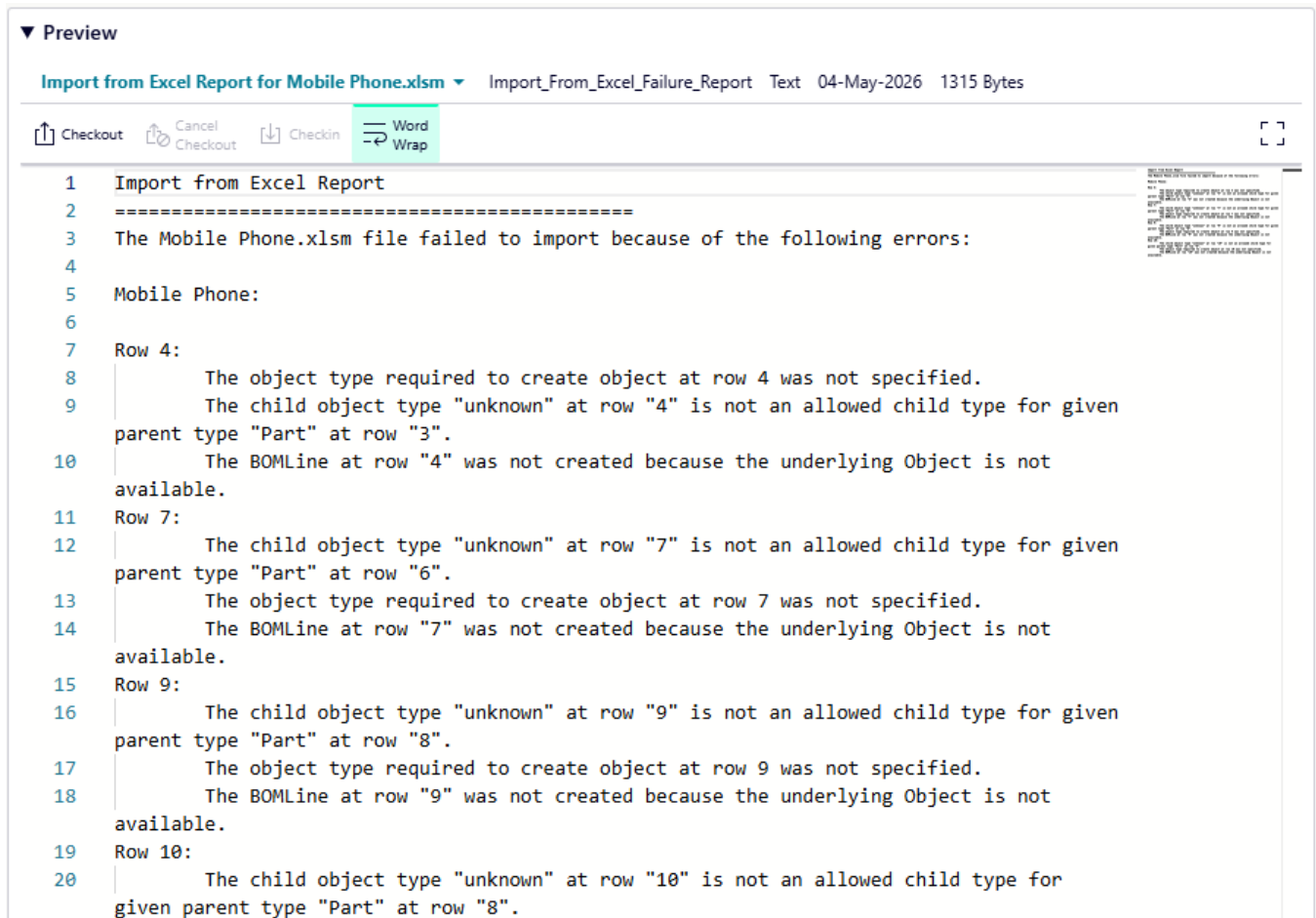
Sent Date: 04-May-2026 14:15

Receiver: ed (ed)

Message Body: Mobile Phone:
"Mobile Phone" failed to import

Related Objects: [Import from Excel Report for Mobile Phone.xlsm](#)

When you click the **Related Objects** link, you receive a detailed report of the errors.



Export a structure to Excel for round-trip editing

You can export a structure to Excel and edit its data. You can then import this Excel back to Teamcenter so that these changes are reflected there. This process is called round-trip in Teamcenter.

Procedure

1. Select a structure element to be exported and click **Excel Round-trip** > **Export to Excel** . The **Export to Excel** panel is displayed.
2. (Optional) In the **Settings** section, select one of the following options.
 - Select **All Levels** to export all levels of the structure to Excel.
 - Select **Levels**, and then enter the number of structure levels to be exported to Excel.

- Click the **Include ID as Hyperlink** check box to include a hyperlink in the **Object ID** column in the exported Excel file. You can click the hyperlink to open the object in your active Teamcenter session. You must be logged into Teamcenter for the hyperlink to function.

3. From the **Template** list in the **Properties** section, select a template or select **Custom**.

- If you select a template from the list, you can export the structure elements into a specified template format. Selecting a template ensures that all required columns are included in Excel.

Templates define the structure and required columns to be exported. Available templates are configured by your administrator based on your organization's business requirement. If a template is not available, contact your administrator.

- If you select **Custom** from the list, you can select the columns that are exported to the Excel. When using the **Custom** option, ensure you include the required properties for successful round-trip import.

Do one of the following:

- Click  to select a saved column arrangement. Saved column arrangements are custom column configurations that you previously saved for reuse.
- Select one or more columns in the **Available Columns** list on the left. Then click  to add the selected column(s) to the **Displayed Columns** list. Select one or more columns in the **Displayed Columns** list on the right. Then, click  to remove the selected columns from the list. You can also choose to move a column up or down the list.

The dynamic compound properties cannot be exported.

4. (Optional) Select the **Run in Background** check box.

If you select this option, the export runs in the background, allowing you to continue working in Teamcenter. You receive an alert  notification when the export process is complete.

5. Click **Export**.

Note

If no objects match the criteria in the Excel template, a blank Excel sheet appears.

Results

After the export is complete, the Excel file is saved to your browser's default download location.

What to do next

You can add and remove rows for the elements, modify property values, and update relationships in the exported Excel. Do not add or delete the columns, as this may cause import errors. After you finish the modifications, the Excel can be **imported back** to Teamcenter.

Guidelines for editing a round-trip Excel file

Use the following guidelines for editing the round-trip Excel file so that it can be imported back to Teamcenter.

Structural modifications

- You can add new and existing elements to the structure and make structural changes.
- You can remove the existing elements from the structure.
- You can update the element properties.
- You can update the item revision properties.
- You can replace an existing item with another item.
- Suppose that an Excel file does not have any parts in an assembly, and then that Excel is imported back into Teamcenter. In such a case, no parts are removed from that assembly in Teamcenter.

In other words, do not remove all child elements from the structure or a subassembly. Retain at least one child element.

Column management

- Do not add or remove columns from a round-trip Excel file.
- The **TCUID** column is hidden. If the **TCUID** column is not hidden for some reasons in the exported Excel, do not edit its values. Teamcenter exports this column as an internal identifier for the exported objects.

Adding objects

- To add a new object to the structure, add a new row to the exported file and ensure that the **TCUID** column is blank if it is visible.
- If you add a new item, ensure that you add its type in the **Type** column and all other required properties.
- If you want to add an existing item to the structure, enter its key fields (for example, **ID**).

Macro requirements

Do not turn off the macro while opening the Excel file for editing. This action ensures that any copied lines do not carry forward the **TCUID** value from the original lines.

Note

The paste operation cannot be undone using Ctrl+Z when the macro is present. The newly pasted row can be deleted, or its contents can be deleted.

Copy and paste operations

If you copy or cut and paste lines in a structure, ensure that the **TCUID** column is blank in the pasted lines. These operations are treated as separate remove and add operations, rather than move operations.

Export and import structures along with partitions

You can export and import structures along with partitions and partition schemes to and from other Teamcenter sites.

For the import and export, you can use:

- Briefcase files
- Multi-Site Collaboration
- PLM XML

Export and import structures along with worksets

You can export and import structures along with worksets to and from other Teamcenter sites.

A workset is a context collector for other structures. If your organization has very large structures with a multitude of occurrences, these occurrences can be collated within a workset to do a *what-if* analysis. But when you send the workset to another site by using a Multi-Site environment or a Briefcase, you might not want to export such a large number of occurrences along with the workset. Therefore, while exporting, the **Include entire BOM** option is not enabled by default. Consequently, the TCXML-based Multi-Site functionality works to share workset and related objects but not the product item or item revisions or their content.

Worksets also support site-consolidation activities through the TCXML-based Multi-Site functionality.

To export and import structures along with worksets, you can use:

- Briefcase files
- Multi-Site Collaboration
- PLM XML

Export a structure to NX

Export a filtered Teamcenter structure to NX, including non-master objects and associated files, to view and work with the assembly within the NX environment.

You can export a filtered structure and open that in NX. That structure does not contain the parts that are filtered out.

Prerequisites

- **NX for Active Workspace** must be installed.
- Briefcase Browser must be installed.

Procedure

1. Search for and open the structure that you want to export.
2. Choose **More Commands *** > Import/Export**  **> Export NX Assembly** .
The **Export NX Assembly** command is not available if Briefcase Browser is not installed.
3. In the **Export NX Assembly** panel, choose:
 - a. **Export Non-Masters** – To export the object types specified as **Non-Master** in the access control list.
 - b. **Export Associated Files** – To export files associated with the selected structure. However, based on the exclusion list defined by your administrator, some file types are not exported.
4. Click **Export**.
The export operation runs in the background. When the export is complete, a notification is displayed in the **Alerts** panel.
5. Click the **Alerts**  icon to see the notification and download the Briefcase (**.BCZ**) file if required.

Note

For additional information about exporting a structure to NX, see the *Teamcenter Integration for NX* guide.

What to do next

After the structure is exported, ensure that any revision rules applied to the structure are reflected in the exported structure.

You can also ensure that

- The default variant condition is reflected in the exported structure. Any user-applied variant rules are ignored.
- Any effectivity applied as a part of a revision rule is reflected in the exported structure.
- Any user-applied effectivity other than those applied as a part of the revision rules is ignored.

Import and export structures and changes individually

Import a structure from Excel

Import a structure into Teamcenter from a prepared Excel file, including mapping properties, previewing changes, and handling potential errors, to efficiently integrate external structure data.

You can import a structure by using an Excel file. The structure can include all reference properties with an LOV, such as unit of measure, owner, and group. You can import structures with the global revision rule and structures using different ID types that use a Multi-Field Key (MFK). You can import vendor-part data, multiple parts and assemblies, and multiple types of secondary objects. You can also import the relational properties (like preferred or backup vendor preference status). If an array property contains multiple values in the Excel, you can import them too. For example, if the Excel contains a column for Production Sites and the values are Milford and Troy, both these values are imported in Teamcenter.

To import a structure from Excel, you must prepare the Excel file in a specific way. See *Prepare the Excel file for importing a structure*.

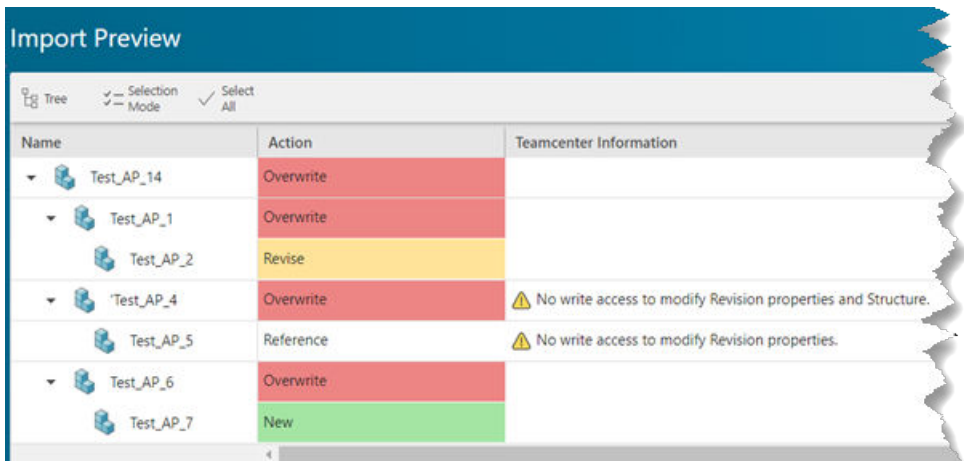
Restrictions and limitations

You can perform this task if it is configured for you by your administrator.

Procedure

1. Navigate to the folder where you want to import the structure, and choose **More Commands** *** > **Import/Export**  > **Import Structure** .
2. In the **Import Structure** panel, select the Microsoft Excel file from which you want to import the structure by clicking **Choose File**.
3. Map the structure properties between Teamcenter and Excel.
 - Select the required mapping from the **Saved Mappings** list. The list shows the mappings created and saved by the administrator or previous users.
 - Alternatively, create a new mapping by entering a name in the **Saved Mappings** field and selecting the **Mapped Attributes** for the **Excel Headers**. Even if an error occurs while importing a structure, the created mapping is saved. You need not select the mapped attributes again for the Excel headers.

- If a mapped attribute is not available for an Excel header:
 - a. In the **Mapped Attributes** list for the corresponding Excel header, click **Add New** to create a new attribute.
 - b. In **Add Properties**, select appropriate **Subtypes**, if you want to change the default one.
The reference properties **Unit of Measure**, **Owner**, and **Group**, along with other properties, are listed in the **Add Properties** section.
 - c. Filter and select the attribute that you want to map.
 - d. Click **Add**.
 - e. In the **Import Structure** panel, choose the newly added attribute.
- 4. (Optional) If you do not want to run the import process in the background, clear the **Run in Background** check box. The system notifies you about the result of the import process in the **Alerts** panel.
- 5. (Optional) Click **Preview** to check the structure before import.
 - a. The preview shows the structure to be imported along with the **Action** that is performed by default. **Teamcenter Information** shows additional information related to an action.



Action	Description
New	A new item or occurrence is created.
Revise	A new revision of the item or occurrence is created.
Overwrite	The existing revision is overwritten with the updated information.
Reference	The existing revision is used as is. For example, if the revision is released, and some changes are made to the revision in the Excel file without revising the item, the action is set as Reference . This indicates that the changes will not be applied.

- b. You can change the action for **Revise** and **Overwrite**. To do so, right click an action and select:
- **Revise**
To create a new revision of the occurrence.
 - **Reference**
To reuse the existing revision in Teamcenter as is. The existing revision will not be updated with the latest information in the Excel file.
 - **Overwrite**
To update the existing revision in Teamcenter with the latest information from the source Excel file.

6. Click **Import Structure** to import the structure in Teamcenter.

If there were no errors while importing the structure:

The structure is imported in Teamcenter. When you import multiple elements or assemblies simultaneously, all the top-level elements and assemblies are added to the folder you specified previously. You receive a notification in the **Alerts** panel when the import is complete. Click **Alerts**  to open the **Alerts** panel.

The structure is imported using the global revision rule. The same revision rule is used while opening the structure after import.

You can also import a structure with different ID types that use a multi-field key, that is, when you have two different items with the same ID. You can see the items on the **Import Preview** screen and import them successfully. The import structure functionality searches for and creates items based on the multi-field key.

If there were errors while importing the structure:

You receive an error message stating that the import failed because there were errors while importing the structure. You also receive a notification in the **Alerts** panel, where you can view the error details. Click **Alerts**  to open the **Alerts** panel.

In the **Teamcenter Information** section, you can view a list of all potential errors simultaneously. When you hover over an error, you can view detailed information about that error.

Excel Roundtrip Selection Mode Select All						
Name	Action	Teamcenter Information	Release Status	ID	Revision	Secondary Obj...
Computer	New	Structure was not imported because of err...		032391	A	
CPU	New	The "<empty>" value cannot be assigned ...		032392		3 Objects
Power supply	Overwrite	The Vendor ID with value "Company_1" do...		powersupply_1	A	1 Objects
Power supply	New	The object "powersupply_1-P... The Vendor ID with value "Company_1" does not exist at row 9.				
Motherboard	New	Structure was not imported because of err...		032393		
	New	The "<empty>" value cannot be assigned ...				

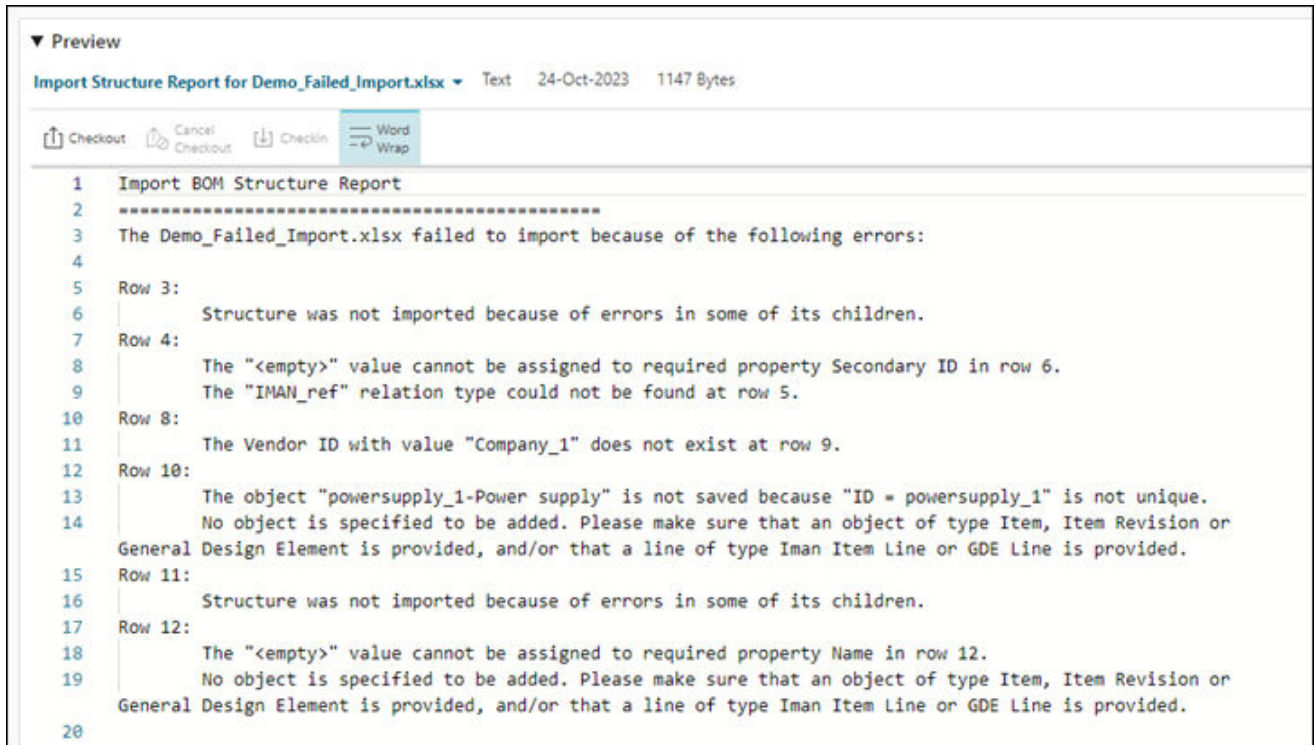
Tip

At this stage, you need not restart the import process from the beginning. You can make the required corrections in the Excel file and then reupload it by clicking **Choose File** in the **Import Structure** panel. Once this is done, the saved mappings from the previous import are automatically selected and shown in the **Saved Mappings** list. Then, you can click **Import Structure** to proceed with the import.

When you click the link in the **Alerts** panel, you can view the information about the structures that could not be imported and the related objects.

▼ Properties	
Message Subject:	Demo_Failed_Import.xlsx failed to import
Priority:	
Event Type:	
Sent Date:	24-Oct-2023 17:46
Receiver:	ed (ed)
Message Body:	"Computer" failed to import "Motherboard" failed to import
Related Objects:	Import Structure Report for Demo_Failed_Import.xlsx

When you click the **Related Objects** link, you receive a detailed report of the errors.



Export a structure to Excel

You can export a structure to Excel. Another user who does not have access to Teamcenter might perform some changes in the Excel. You can then import this Excel back to Teamcenter so that these changes are reflected there.

Restrictions and limitations

You can perform this task if it is configured for you by your administrator.

Procedure

1. Select the structure elements to be exported and click **Excel Round-trip** > **Export to Excel**. The **Export to Excel** panel is displayed.
2. (Optional) In the **Settings** section, select any of the following check boxes:
 - **Allow Structure Changes in Excel:** Selecting this option allows external users to modify the structure in Excel.
 - **Include ID as Hyperlink:** The **Object ID** column and the link to the object are exported to Excel. You can click the link and open the object in Teamcenter.
 - **Include Outline Numbers:** The numbers denoting the hierarchy are exported to Excel.
3. From the **Template** list in the **Properties** section, select a template or select **Custom**.

- If you select a template from the list, you can export the structure elements into a specified template format.

If a template is not available, contact your administrator.

- If you select **Custom** from the list, you can select the columns that are exported to the Excel. Do one of the following:

Select a saved column arrangement. Select one or more columns in the **Available Columns** list on the left. Then click **>** to add the selected column(s) to the **Displayed Columns** list. Select one or more columns in the **Displayed Columns** list on the right. Then click **<** to remove the selected column(s) from the list. You can also choose to move a column up or down the list.

4. (Optional) Select the **Run in Background** check box. If you select this option, you receive an alert  when the export process is complete.
5. Click **Export**.
The export process begins, and a progress bar indicates that the process is still running.
6. Open the exported Excel and make the required changes to the properties listed, and save the Excel.

Note

If no objects match the criteria in the Excel template, a blank Excel sheet appears.

Caution

Do not alter these columns in the Excel sheet: **TC_ObjectType**, **TC_ObjectID**, and **ID**. One exception to altering **TC_ObjectType** is creating a new object in the structure.

7. (Optional) To create a new object in the structure:
 - a. Open the Excel sheet and insert a new row using the Excel insert feature.
 - b. Indicate the object type to be created in the **TC_ObjectType** column, ensuring that the **TC_ObjectID** column is left blank.
 - c. (Optional) To adjust the parent or outline level, use the Excel grouping feature.
 - d. Save the changes to the Excel file.

Import changes from a previously exported Excel

You can import the changes from a previously exported Excel back to Teamcenter.

Restrictions and limitations

- You can perform this task if it is configured for you by your administrator.
- Using the Excel Round-trip functionality, you cannot add or remove structure elements. Additionally, you cannot edit some properties, such as element effectivity and variant formula.

Procedure

- 1. Click **Excel Round-trip** > **Import Changes**.
The **Import Changes** panel is displayed.
- 2. Click **Choose File**, navigate to the Excel file location, and then click **Open**.
- 3. To ensure that your changes are retained in the event of a conflict, select **Overwrite Conflicts**.
- 4. Click **Import**.

View where an element is used across assemblies simultaneously

Export the **Where Used** information for a structure element to an Excel file, allowing you to simultaneously view its usage across all assemblies at all levels or at the product level, for comprehensive analysis.

You can view where a structure element is used across all assemblies up to the top level, starting with the immediate parent and navigating one level up at a time. To do this, you go to the **Used in Structures** section and expand one level at a time. However, if you want to view all this data simultaneously, export it to an Excel file.

Similarly, if you want to see where an element is used at the product level across all assemblies simultaneously, you can export this information to an Excel file as well. However, you must have Smart Discovery Indexing installed to generate this data.

Procedure

- 1. Search for and select the structure element.
- 2. Click the **Where Used** tab.
- 3. Click **Export to Excel** from the following sections depending on the scenario.

Scenario	Action
View where the element is used across all assemblies at all levels.	In the Used in Structures section, click Export to Excel. Note If the number of results in the Used in Structures section goes beyond the threshold value, the Export to Excel button is unavailable. The threshold value and the number of levels to be exported to Excel are set by your system administrator.
View where the element is used across assemblies only at the product level	In the Top Level section, click Export to Excel.

Scenario	Action
(works only if you have Smart Discovery Indexing installed).	

The **Export To Excel** panel is displayed with a list of properties available for export.

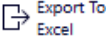
- 4. To change the order of the properties, select a property, and in the **Selected Properties** section, click **Move Up**  or **Move Down** .
- 5. To select the properties to be exported, in the **Selected Properties** section, click **Add Properties** .
- 6. In the **Add Properties** section, clear the check boxes for the properties that you do not want to export. Select the check boxes for the properties that you want to export.
- 7. Click **Add**.
- 8. It is recommended to use the **Run in Background** option while exporting the data from the **Used in Structures** section. However, if you do not want to run the export process in the background, clear the **Run in Background** check box.

While exporting the data from the **Top Level** section, you do not get an option to run the export process in the background.

- 9. Click **Export**.
If the export process ran in the background, view the notification in the **Alerts** panel and click the link to download the Excel file.
If the export process did not run in the background, a dialog box is displayed to download the Excel file to your machine.

Note
You cannot reimport the exported data back into Teamcenter.

The following graphic shows the sample data in the **Used in Structures** section.

▼ Used in Structures		
Revision Rule: Global (Latest Working) ▼		
Object	Type	Release Status 
▼  FUEL TANK ASM/A;1	Item Revision	 TCM Released
▼  FUEL SYS CK/A;1	Item Revision	 TCM Released
▼  CHASSIS CK/A;1-CHASSIS_CK	Item Revision	 TCM Released
 CROSSKART_DC/B;1	Item Revision	

The following graphic shows the sample data in the exported Excel file.

	A	B	C	D	E
1	Level	Object	Type	Release Status	Owner
2	1	FUEL TANK ASM/A;1	Item Revision	TCM Released	Doe, John
3	2	FUEL SYS CK/A;1	Item Revision	TCM Released	Doe, John
4	3	CHASSIS CK/A;1-CHASSIS_CK	Item Revision	TCM Released	Doe, John
5	4	CROSSKART_DC/B;1	Item Revision		Doe, John
6					

8. Create and maintain structures

About creating structures

Create a structure in Teamcenter by manually adding items, duplicating or saving existing structures, or importing structures from Excel or other Teamcenter sites.

You can create a structure by:

- **Manually** adding child items to a parent item.
- **Duplicating** an existing structure.
- **Saving** an existing structure as a new structure.
- Importing a structure.

You can import a structure from **Excel**. You can also **import structures along with their partitions** from other Teamcenter sites.

Create a structure manually

You can create a structure manually by adding child items to it.

Procedure

1. Open the folder in which you want to create a new structure.
2. In the **Overview** tab, click **Add to** ⊕ in the **Contents** section.
3. In the **Add** panel, select the **Type** as **Item**, enter the required information, and click **Add**.

If you want to add further information for an item or assembly, add a secondary item. The **Occurrence Type** of the secondary item is **Reference Geometry**. The secondary components are displayed as a part of the structure. You can hide the secondary components from being displayed into the structure.

4. Open the item that you added, go to the **Content** tab, and click **Add** ⊕ > **Child**.
5. In the **Add Child** panel, enter the required details, and click **Add**.

Add more children to build the structure. To add children that are at the same level, click **Add** ⊕ > **Sibling**.

You can drag an item from one place to another to reorder the structure. To remove an item, right-click it and choose **Remove** ⊖.

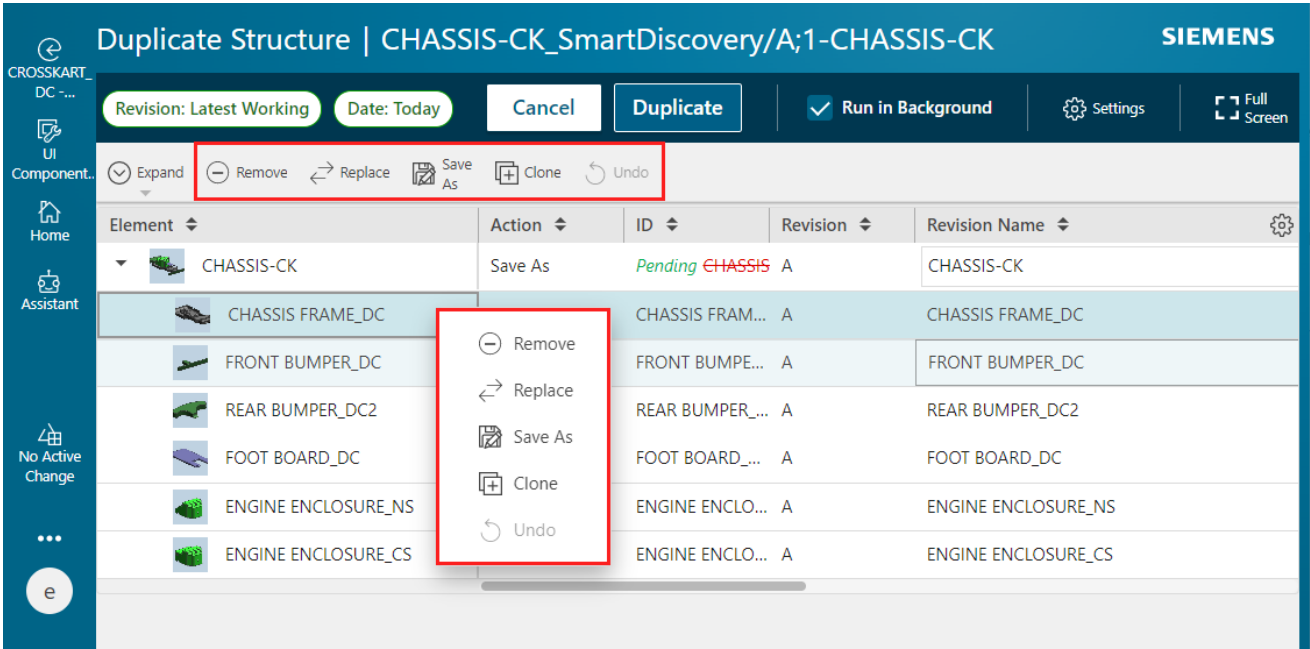
Duplicate a structure

You can create a new structure by duplicating an existing structure. You can also duplicate structures that you do not own. When you duplicate a structure that you do not own, you become the owner of the new structure.

Your site administrator specifies whether the duplicate operation is run at the occurrence level or at the item revision level.

Procedure

- 1. Select the top line of an assembly in the structure that you want to duplicate.
- 2. Click **Duplicate** .
- 3. Right-click each element and select one of the following actions. You can also select these actions from the work area toolbar.



Action	Description
Remove	The element is not included in the new structure.
Replace	The element is replaced with the replacement element that you specified in the Replace panel.
Save As	A copy of the element is added to the new structure. You own this new element. If the element is a structure, all its child elements are only referenced in the new structure. These are still owned by the user who created the source structure.

Action	Description
	You can edit the Revision Name and Description of the element. You can edit the custom properties if your administrator has configured those custom properties accordingly.
Clone	A copy of the element is added to the new structure. You own this new element. If the element is a structure, copies of its child elements are used in the new structure. You can edit the Revision Name and Description of each element. You can edit the custom properties if your administrator has configured those custom properties accordingly.

Note

If any child element is a structure, all its children are also marked with the same action. However, if you change the action of one of its child elements later, the action of the parent element changes accordingly.

For packed lines, the specified duplicate action is applied only on the selected occurrence and not on all occurrences in the line. To apply the selected duplicate option to all *packed* occurrences, you must *unpack* and apply the duplicate action to each occurrence. You can select multiple occurrences and apply the action simultaneously.

If you do not select an action, and the **Action** column is blank, the element is only referenced in the new structure. The element continues to be owned by the user who created the source structure.

4. To undo an action set on an element, select the element, and click **Undo**.
5. (Optional) Click **Settings** , and then perform these additional actions on the duplicated structure:
 - a. Choose an option for the ID naming in the Settings panel:
 - Click the **Default IDs** option to retain the original element IDs for the cloned elements and for any elements that are saved as new elements.
 - Alternatively, click the **ID Naming Rule** option and enter the following details.

ID naming rule	Description
Prefix	The text is prefixed at the start of the existing IDs.
Suffix	The text is appended at the end of the existing IDs.
Replace/With	The text specified in Replace is replaced with the text specified in With . The text specified in Replace must be text that is part of the existing ID, and it is case-sensitive.

- b. To retain the original ownership for cloned elements and the elements that are saved as new elements, click the **Retain Original Ownership** check box.

The elements on which no action is taken are referenced in the duplicated structure and their original ownership is retained.

- c. Ensure that the **Owning Project** section displays the project that is a part of your active Teamcenter session.

You can change your project so that the duplicated structure becomes a part of a different project. If no project is selected in the active Teamcenter session, you do not see the **Owning Project** section.

- d. To include the Designcenter NX dependencies, if any, that are associated with the existing elements in the newly duplicated structure, click the **Include all Designcenter Dependencies** check box.

- e. Click **Close**.

- 6. To run the duplicate process in the background or foreground, select or clear the **Run in Background** check box. If you choose to run the duplicate process in the background, you receive a notification in **Alerts**  once the structure is duplicated.

Note

If your administrator has enabled the parallel duplication of large structures, the duplication process always runs in the background. Contact your administrator to adjust this setting.

- 7. Click **Duplicate**.

Results

By default, your active Teamcenter session project is assigned to the duplicated structure and its elements. If the administrator has enabled the project assignments in the Teamcenter environment, the following actions are performed on the duplicate structure:

- If an item in the original structure is associated to a project, that project is replaced with the project selected in the active Teamcenter session for the corresponding duplicated item.
- If no project is associated to an item in the original structure, no project is assigned to the corresponding duplicated item, even though a project is selected in the Teamcenter session.
- For referenced items that are not cloned but are reused in the new structure, if the original item is associated to a project, the project that is selected in the active Teamcenter session is appended to its existing project list.

Create a new structure from an existing structure

You can create a new structure from an existing structure by saving the existing structure as a new structure.

Procedure

1. Search for a structure.
2. Click **More Commands**  > **New**  > **Save As** .
3. In the **Save As** panel, enter the required information.
4. Click **Save**.

Results

A new structure is created from the existing structure. If the existing structure contains any partitions, the partitions are also carried over to the new structure.

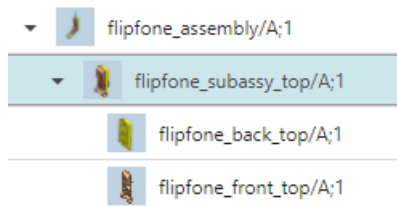
Insert or remove levels in a structure

You must have write access to the structure to insert or remove a level. The insert action adds a level between the selected object and its parent object. The remove action removes the selected element's parent object one level above and attaches the element and all its siblings to its current grandparent object two levels above.

Search for the structure you want to edit and click **Open**.

To	Do this	Result
Insert a level	1. Click the element above which you want to insert a level and click Edit Structure  > Insert Level . 2. Fill in the details and click Insert.	A new level is added above, as the new parent of the selected element.
Remove a level	1. (Optional) If any child of the level you want to remove has effectivity or variant condition set on it or is suppressed, enable the following options: • Show Excluded by Effectivity • Show Excluded by Variants • Show Suppressed 2. Click the element that you want to remove from the assembly and select Edit Structure  > Remove Level .	The selected element is removed from the structure, and its child elements are attached to what was previously their grandparent elements.

The following example shows how the remove-action works in the three-level assembly in the graphic:



Add or remove structure elements

You can add new or existing elements to a Teamcenter structure as child elements or siblings, including multiple occurrences. You can also remove elements from a structure, ensuring proper management of the assembly hierarchy.

Whenever you add an element to a structure, it is always added to the open structure node, which is the right-most element in the breadcrumb.

You can add multiple occurrences of an element to a structure with a single add operation. To add multiple occurrences, the element being added must exist in the database. Multiple occurrences of a new element cannot be added.

If you are working in the markup mode, you must **apply or cancel the markups** to complete the updates.

Add an element to a structure

Procedure

1. Search for the structure where you want to add an element and click **Open** .
2. If you want to add elements in the markup mode, choose **More Commands** *** > **Markup**  to switch to the markup mode.
3. Select the target node where you want to add the element in the structure, click the **Add**  icon, and select one of the following *add* options.
 - Select **Child** to add the element as a child of the target node.
 - Select **Sibling** to add the element at the same level in the structure as the target node. If the top-level node is selected as the target, the **Sibling** option is not available.
4. In the **Add Child** panel (displayed when you are adding a child) or the **Add Sibling** panel (displayed when you are adding a sibling), click one of the following tabs:
 - **New**

Allows you to add a new element from your computer. You can browse to the required file or select it from your **Recent** list. The **Other** list allows you to filter for the relevant file type. You cannot add multiple occurrences from the **New** tab.

- **Palette**

Allows you to paste an element from the clipboard or to select one from your **Favorites** or **Recent** list. You can add multiple occurrences of an element from the **Palette** tab.

- **Search**

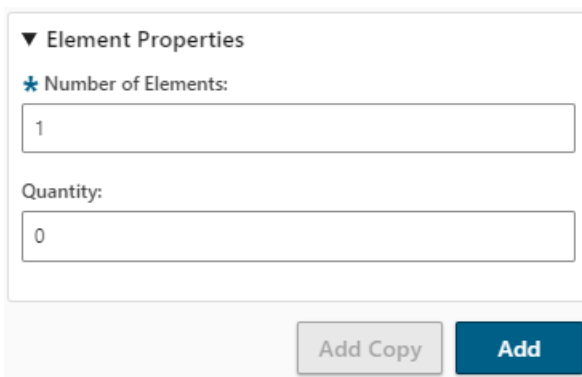
Allows you to search for an element to add. You can add multiple occurrences of an element from the **Search** tab.

5. Enter the ID, name, description, or unit of measure as required.
6. If you are adding a new element from the **New** tab, in **Element Properties**, specify **Number of Elements** and **Quantity**.

The **Quantity** field is available by default. If your administrator has configured to include additional element properties, those properties are also displayed in **Element Properties**.

By default, the **Element Properties** section is available for **Item** and **Part** in the **New** tab. This section appears for other business objects if your administrator has configured the create style sheets for those objects.

7. If you are adding an element from either the **Palette** or the **Search** tab, locate and select the element, and specify the **Number of Elements** and **Quantity**.



▼ Element Properties

* Number of Elements:

1

Quantity:

0

Add Copy Add

8. Click **Add**.

Teamcenter creates the occurrence and refreshes the display.

Remove an element from a structure

Click **Edit Structure**  > **Remove** , and accept the confirmation message.

You can also remove multiple occurrences simultaneously by selecting multiple occurrences using either **Ctrl+click** or **Right-click**.

Note

You cannot remove the top node of the structure, that is, the leftmost element in the breadcrumb.

Add a copy of the existing objects to a structure

In cases where a new occurrence that is very similar to an existing occurrence is added to a structure, you can add a copy of the existing occurrence to the structure instead of creating a new occurrence. The copied occurrence is added with a new ID, and it retains the properties of the occurrence from which it is copied.

Procedure

1. Click the **Add**  icon and select the required option.
2. In the **Add Child** or **Add Sibling** pane (depending on the option that you chose), in the **Palette** or **Search** dialog box, select the object that you want to copy.
3. Click **Add Copy** to add a copy of the existing element with a new ID.

You can also copy an occurrence from one structure to a different structure using the drag-and-drop action. This action copies the occurrence and does not remove it from the source structure.

You can use drag-and-drop between similar web browser windows to copy occurrences from a location in one window to another accepting object or location in another window.

- a. Click and hold the occurrence that you want to copy.
- b. Drag the occurrence to the desired location in the target structure.
- c. Drop the occurrence on to a valid accepting object in the target structure.

Replace a structure component

You can replace a part with another part. You can also replace an existing part or a sub-assembly in an assembly with its copy using the **Save As And Replace** command. If the **Save As And Replace** option is not visible, you may not have the required privileges.

Procedure

1. Search for the element that you want to replace.
2. If you want to edit the structure in the markup mode, choose **More Commands**  > **Markup**  to switch to the markup mode.
3. Select the occurrence and click **Edit Structure**  > **Replace** .
4. In the **Replace** pane, select the replacement and click **Replace**.

Teamcenter replaces the element and refreshes the display.

5. If you are working in the markup mode, you must either **apply or cancel the markups** to complete the updates.

Replace a structure component with its copy

A copy is a new item with identical properties but a new ID. You can copy a part or a subassembly, with or without children, save it with a new name, and replace the existing subassembly in the structure with its copy. If the subassembly is copied with children, the child parts are renamed based on the naming convention specified using the naming rule.

Procedure

1. Select the structure from which you want to copy the element, and click **Open** .
2. Choose **More Commands** *** > **New** ✱ > **Save As and Replace** ⇄.
- The **Save As And Replace** pane is displayed.
3. (Optional) Select the **Clone Children** check box.
- If you choose the **Clone Children** option, the child elements in the assembly are replaced along with the top element.
4. Select **ID Naming Rules** to automatically assign new IDs to the copied elements. The **ID Naming Rules** option is visible only when the **Clone Children** check box is selected.
 - **Prefix**
The text string provided is added to the start of the existing IDs, and new IDs are assigned.
 - **Suffix**
The text string provided is added to the end of the existing IDs, and new IDs are assigned.
 - **Replace/With**
The text string specified in the **Replace** field is replaced with the text string specified in the **With** field in the new IDs.
5. (Optional) Select the **Run in Background** check box.

Note

For the save and replace operation to run in background, your system administrator must configure Dispatcher services.

6. Click **Save As And Replace** at the bottom of the **Save As And Replace** pane.

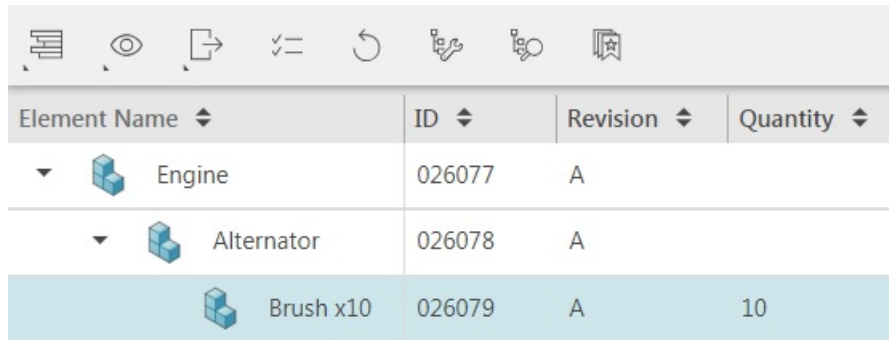
Teamcenter replaces the element and refreshes the display.

If you choose to run the replace operation in the background, a notification is generated once it is complete. Click the notification to view the details of the replacement.

Split an occurrence with multiple quantity values into its individual occurrences

Many structures have multiple occurrences of the same element. For such structures, you can either add all similar occurrences individually or add only one occurrence and specify its quantity.

For example, you can either add 10 brushes individually to an alternator assembly, or add one brush and specify its quantity as 10.



Element Name	ID	Revision	Quantity
Engine	026077	A	
Alternator	026078	A	
Brush x10	026079	A	10

You can split an occurrence that has a quantity of more than one into its individual occurrences. The total number of occurrences after the separation equals the original quantity. All occurrences have the same properties but may be modified independently.

The split command does not work if:

- A unit of measure (UOM) is specified for an occurrence.
- Multiple occurrences already exist and are represented as a packed line.

Procedure

1. Search for the occurrence. The value for **Quantity** must be more than 1.
2. Select the occurrence that you want to separate, choose **Edit Structure** > **Split Quantity**.

The single occurrence is split into several individual occurrences, where the number of occurrences equal the single occurrence's quantity.

 Engine	026077	A
▼  Alternator	026078	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A
 Brush	026079	A

Split the occurrences of an item based on quantity

Considering your business requirement, you can split the occurrences of an element based on quantity.

Suppose a *Light Duty conveyor* for small packages needs only four *M8 Hex Bolts* per *Support Leg Assembly*, whereas a *Heavy Duty conveyor* for bulk materials requires 8 *M8 Hex Bolts* for increased structural integrity. In this situation, you can split the *M8 Hex Bolt* occurrence to define different quantities, with each tied to a specific conveyor load capacity.

You can split an occurrence based on quantity if the quantity of the element is at least two. You cannot split the occurrence of an element if the quantity is one.

Procedure

1. Search for the occurrence and click **Edit Structure**  > **Split by Quantity**.
The Split panel appears.
2. In the **Number of Splits** box, type a number to divide the element by that many splits. Teamcenter suggests the default quantity for each split.
3. Modify the default quantity for the suggested splits in the **Quantity** box as needed.
4. Click **Split**.

Results

The specified number of occurrences for the element are created with the defined quantity. If you split the occurrences of the element as a part of an active change, the changes, including the creation of new occurrences and the modification of existing occurrences, are tracked.

Split the occurrences of an item based on variant criteria

Considering your business requirement, you can split the occurrences of an element based on variant criteria.

Suppose the *Front Exterior Assembly* contains a single, generic *Headlight Assembly* occurrence. However, the bus model is sold in different regions, including North America and Europe, with distinct regulatory requirements for headlights. For example, the headlight requirements are amber turn signals in North America, clear in Europe, and different beam patterns. In this situation, you split the *Headlight Assembly* occurrence into multiple variant-specific occurrences. Each new occurrence contains a region-specific headlight part.

Prerequisites

You can split an occurrence based on variant criteria if the variability scope is defined for the element.

Procedure

1. Search for the occurrence and click **Edit Structure**  > **Split by Variant**.
The Split panel appears and displays that the element is split into two occurrences and it also displays the variant formula.
2. Click **Split**.

Results

The two occurrences of the element are created with the same variant condition. You can adjust the variant formula. If you split the occurrences of the element as a part of an active change, the changes, including the creation of the new occurrences and the modifications of the existing occurrences, are tracked.

What to do next

You must edit the [variant condition](#) for the element.

Move an occurrence within a structure

You can move one or more occurrences of an element within a structure by dragging and dropping them to a new location and reorganize the assembly hierarchy.

You can drag-and-drop one or more occurrences to move it within a structure. You can drop an occurrence on to a valid accepting object only. This action moves the occurrence from the source location to the target location within the structure. It does not copy the occurrence.

You can use drag-and-drop between similar web browser windows to move or copy occurrences from one location to another within the same structure. The two windows must use the same structure with the same configuration in order to move the occurrence. Otherwise the occurrence is copied. For example, if the same structure is open in two windows, you can drag an occurrence from the first window and move it to a different location within the same structure in the second window.

Procedure

1. Click and hold the occurrence that you want to move.
2. Move the occurrence to the target location in the structure.
3. Drop the occurrence on to the target object.

Control edit access to an assembly

As the owner of an assembly, you can control edit access to the assembly in your structure. For example, you can control which users can add or remove a child or modify the occurrence properties of a child in the assembly.

You can do this through a workspace object in Teamcenter called BOM View Revision (BVR). A BVR stores the single-level assembly structure of an item revision. In a structure, whenever an item is an assembly (it has children), the item revision created is a BVR. When you add a child to an item, the system creates a BVR automatically. To edit the assembly, other users must have write access. If you do not want other users to update the assembly, you can check out the BVR. Similarly, you can grant them access to edit the assembly.

Procedure

1. Select the relevant assembly in the structure.
2. On the **Overview** tab, expand the **Assembly Information** section.

This section displays one or more BVR objects associated with your assembly. It also indicates if the assembly is released, or checked out, its last-modified date, and the owner information.

3. Select the relevant BVR.
4. To open the BVR, click **Open**  and to view and edit its properties, click **Information** .

To check out the BVR, choose **More Commands** *** > **Edit**  > **Check Out**  and to check in the checked-out BVR, choose **Edit**  > **Check In** .

Edit the properties of a part or an assembly

To update the properties of a part or an assembly, you must have write access to the parent assembly of the occurrence you want to edit. Additionally, It must not be checked out by any other user.

Procedure

1. If you want to edit the properties in the markup mode, choose **More Commands** *** > **Markup**  to switch to the markup mode.
2. Select the part or assembly in the structure.

3. Click **Edit** .

The editable properties are in boxes. Properties that are not in boxes are *read-only* and cannot be edited.

4. Change the properties you want to edit. Edited values are highlighted until you save the edits.

To edit occurrence properties, you must have write access to the object containing the property. For example, if a property in the **Overview** tab is pulled from a form that is related to the object, you must have write access to the form to edit the property.

To edit the absolute or relative transform properties of a structure element:

- a. Click a value in the **Absolute Transform** or **Relative Transform** column.
 - b. In the **Edit Transform** panel, modify the values as required.
 - c. Click **Modify**.
5. To save changes, click **Save Edits** . To discard them, click **Cancel Edits** .
 6. If you are working in the markup mode, you must **apply or cancel the markups** to complete the updates.

Edit the occurrence properties when a new occurrence is added as a markup

Procedure

1. Search for the occurrence or assembly where you want to add an element and click **Open** .
2. If you want to add elements in the markup mode, choose **More Commands** *** > **Markup**  to switch to the markup mode.

3. Select the target node, where you want to add the element in the structure, click the **Add**  icon, and select **Child** or **Sibling** as required.
4. In the **Add Child** or **Add Sibling** panel (depending on the option that you chose), create a **New** element or use the **Palette** or **Search** function to find an existing element.
5. Enter the ID, name, description, or unit of measure as required. The following properties can be marked up for edits:
 - Sequence
 - Quantity
 - Occurrence name
 - All Notes
 - Any other custom occurrence properties
6. Click **Add**.

Open and update two structures side by side in the split view

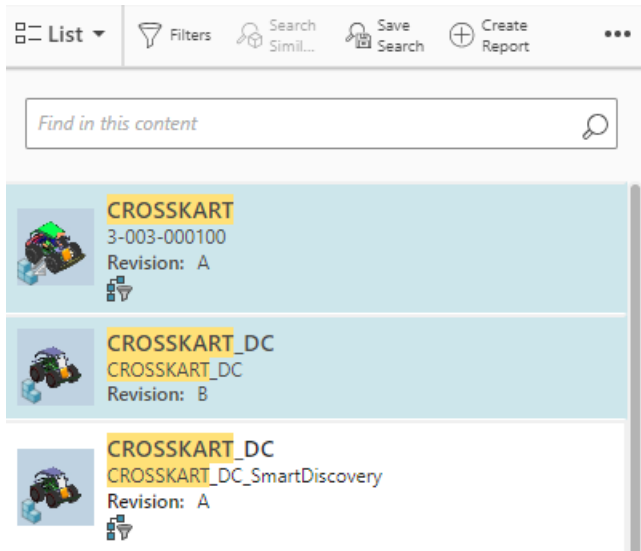
You can open and compare two different structures or two different configurations of the same structure for efficient side-by-side analysis and comparison.

You can use the split feature to view two structures, or two configurations of a structure simultaneously. However, to select two structures for opening, both structures must be listed on the same page.

Open two different structures

Procedure

1. To open two structures simultaneously, select the first structure using **Ctrl+click**.
This activates the multiselect mode.
2. Select the second structure.



3. Choose **Open** > **Open**.

The selected structures are displayed side-by-side in **Split View**. You can navigate to the child or parent parts of each structure as needed.

4. You can drag-and-drop occurrences between the two views to move or copy them from one structure to another. If the two views have the same structure with the same applied configuration, then the occurrences are moved. Otherwise the occurrences are copied.
 - a. Click and hold the occurrence that you want to move.
 - b. Move the occurrence to the target location in the structure.
 - c. Drop the occurrence on to the target object.
5. To close the split view, click **Exit Split** in the header.

The view that you selected last remains open, and the other view is closed.

Open the same structure in both views and update structure configurations

Procedure

1. Search for the structure.
2. Select the structure from the search results, and then choose **Open** > **Open**.
3. Click **Split Context**.

The selected structure is displayed side-by-side in two views. You can update the structure configuration in either view.

4. Change the structure configuration for one or both views.
 - a. Click **More Commands** > **Configure**.
 - b. Change the **Revision Rule**, **Effectivity**, or **Variant Rule** as required.

The configuration is updated for the selected structure. You can work with two different configurations of the same structure simultaneously in the split view.

- c. You can independently control how structures are loaded in either view using the **Show Suppressed** command.

You can also use the **Show Excluded by Effectivity** and **Show Excluded by Variants** commands.

5. To close the split view, click **Exit Split** in the header.

The view that you selected last remains open, and the other view is closed.

Open a structure in a split view

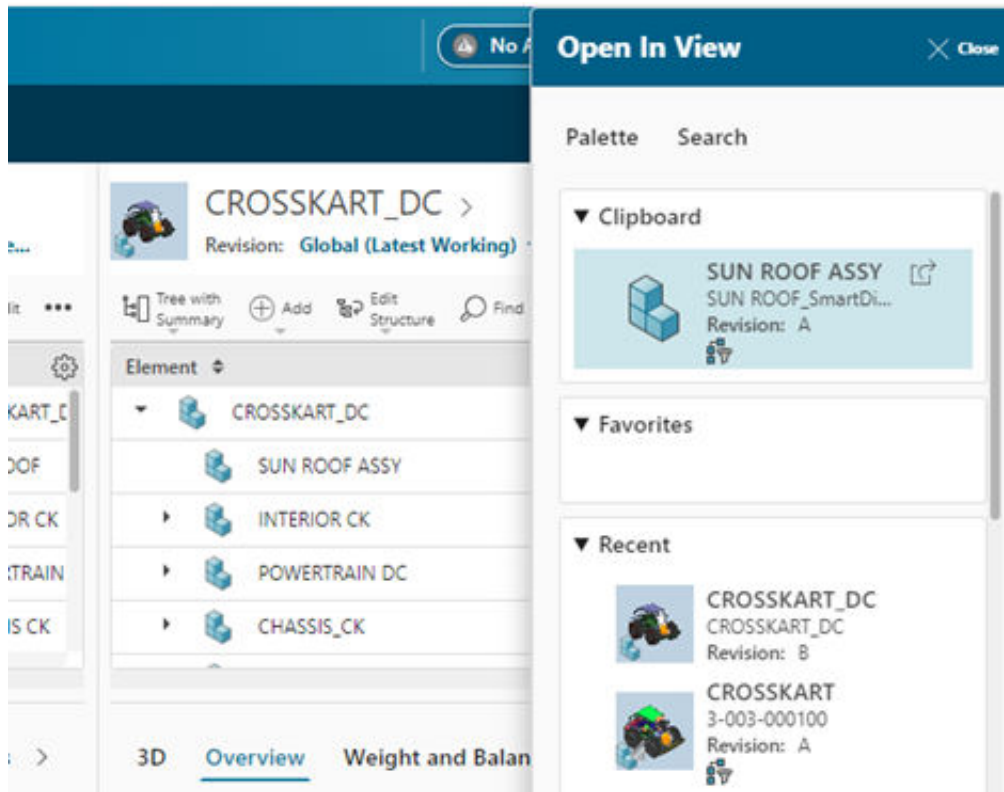
Procedure

1. Search for and open the structure you want to work with.
2. Click **Split Context** .

The selected structure is displayed side-by-side in two views. You can update the structure configuration in either view.

3. To open a new structure in either view, click **Open In View**  in that view.

For either view, the **Open In View** panel is displayed on the right side of your screen.



4. In the **Open In View** pane, click one of the following tabs:

- **Palette**

Allows you to load a structure from the clipboard or to select one from your **Favorites** or **Recent** list.

- **Search**

Allows you to search for a structure.

5. Navigate to the structure you want to open and click **Open** to load the selected structure.

Editing occurrence properties in the context of an assembly

Setting in-context overrides

You can edit the properties of an occurrence in the context of a selected higher level assembly. This in-context override is not defined in the context of the immediate parent but in the context of the next higher level (grandparent level) component.

When a component or sub-assembly is reused in another assembly, some information might have to be overridden to adapt it to fit the context of the new assembly.

Example

Properties such as **Sequence** or **Quantity** can be different for a component when the same component is used in the context of two different assemblies.

When component **C1** is used in the context of the assembly **A1**, the quantity could be **4** and Sequence value could be **10**.

However, it is possible that when the same component **C1** is used in another assembly, namely **A2**, the quantity required could change to **6** instead of **4** and the **Sequence** value could change to **20** from **10**.

In case of such requirements, for the same component **C1**, you can use overrides to define the properties **Quantity** and **Sequence** differently in the context of assembly **A1** and **A2**.

For more information about in-context override, refer to *Structure Manager* in the Teamcenter documentation.

Set in-context overrides

You can set in-context overrides to modify or override certain properties and attributes of elements within a specific structure context without affecting the original item. This helps you add context-specific information, like position and orientation, different mounting positions of an element.

Procedure

- 1. Search for and open the structure.
- 2. Select the grandparent assembly, two-levels above the component, to be defined as the override context.
- 3. Select **Edit Structure** > **In-Context**.
The **Override Context** in the header is updated to reflect the selected assembly. Only the selected assembly remains active. The rest of the structure is made unavailable.
- 4. Click **Edit**.
- 5. Change the properties you want to edit in the context of the selected assembly.
- 6. Save your changes by clicking **Save Edits**. Alternatively, to cancel your edits, click **Cancel Edits**.
The override icon appears against the selected component in the **Overrides** column of the table. On mouse hover, it shows information about overridden properties with the context.

cad drawing1	027022	A	
cad drawing 2	027023	A	
car model	027026	A	
Engine	027027	A	
block	027028	A	
piston	027029	A	
connecting rod	027030	A	
piston ring	027031	A	

Overrides (1):
Context: Engine
Properties: Sequence

The override icons are also shown along with the individual overridden properties.

Tip

To close the **Set In-Context** view, select the same line where you set the in-context override, click **Configuration**, and select **Set In-Context**.

Making mass updates to a structure

Making mass updates

A mass update enables users to update all occurrences of an element, referred to as **Problem Item**, in multiple assemblies simultaneously. This expedites a manual, time-intensive process.

Mass update is used to edit structures by performing any of the following actions:

- Add a new part as a sibling
- Add a substitute
- Remove an occurrence
- Remove a substitute
- Replace a part

You can work with only one **Problem Item** for each mass update action. You can perform a different action on each occurrence of the **Problem Item**. For example, you can choose to add a new part to one occurrence and add a substitute to another occurrence.

Mass updates are always initiated within an Engineering Change Notice or ECN. You can perform a mass update only on the assemblies whose BOM View Revision (BVR) has a release status. The changes applied through the mass update are saved as markups that are attached to the impacted assemblies. You can change or reset the marked-up changes as required through the **Mass Update** panel until the changes are applied.

Once a mass update is initiated for a structure, then no other markups can be made on that structure.

You can apply the marked-up changes to assemblies by initiating the **Review and Apply BOM Markups** workflow process on an ECN for approval. The workflow process will go through the following stages:

Start → Proposed Change → Review ECN → Execute Approve Handler → Finish

The **Execute Approve Handler** step of the workflow applies markups to all assemblies.

Make mass updates to a structure

You can perform mass updates to a structure by using a change notice to propose, manage, and apply changes to multiple impacted assemblies. You can also perform actions like adding, substituting, or replacing items, and then you can submit these markups through a workflow for approval and implementation.

Procedure

1. From **Launcher**, click **Changes** .
2. Search for the change notice you want to work with, select the change notice, and click **Open** .
3. Click the **Affected Items** tab. In the **Problem Items** section, click **Add to** .
4. Click the **Mass Update** tab.

The **Mass Update** tab is not visible when the change notice is in the **Cancelled** or **Closed** state.

Even if multiple problem items are associated with the same change notice or the same problem item is associated with multiple active change notices, you can propose mass update actions for the impacted items of all problem items through the **Mass Update** tab. These actions are applied together as part of a single workflow.

The problem item that you added from the **Affected Items** tab is shown in the **Problem Item** list under the **Impacted Assemblies** section.

All parent assemblies of the selected problem item are listed in a table under the **Impacted Assemblies** section.

5. In the list of impacted assemblies, select the row in which you want to make the update. Only assemblies with the status **Released** can be updated.
6. Choose **More Commands ***** > **Edit**  > **Start Edit**  from the **Page level** toolbar.
7. Click in the **Action** column and select the required action from the list. Only the actions available for the selected row are displayed. The selected action for the row is highlighted.
8. For **Add**, **Substitute**, or **Replace** actions, you can add the required item to the **Proposed Item** column using **Search** or **Palette**.
9. Click **Save Edits**  after updating the required rows.
10. To delete a saved markup, select the row, and select **None** from the **Action** column. The markup is deleted and the assembly is removed from the list of affected items.
11. Set the engineering change notice as Active Change.
12. (Optional) Click the **Affected Items** tab to view the list of items that are marked up for update.
13. Select the affected item that you want to verify from the list and click **Open**.
14. To verify the markup, choose **More Commands ***** > **Markup** .
15. Click the **Mass Update** tab. In the **Mass Update** tab, you can revert the markups before they are applied using the workflow.

If the administrator has configured it for you, the **Mass Update** tab shows the revisions of the parent assemblies for the problem item, considering the global revision rule. Otherwise, the **Mass Update** tab displays all revisions of the parent assemblies for the problem item.
16. To initiate applying the markup, click **Submit to Workflow** .

The **Submit to Workflow** panel with a list of workflow templates is displayed.
17. Select the **Review and Apply BOM Markups** workflow template and click **Submit**.
18. Assign the reviewers for the workflow.
19. Once the workflow is approved, the changes are applied. To verify the changes, go to **Affected Items**, select the row, and click **Show Markup** .

Update multiple structures simultaneously in a change notice

Update multiple impacted structures simultaneously for the problem items that are a part of a change notice by adding siblings, child items, and occurrence substitutes simultaneously as a part of the solution items.

When updating the structure, ensure that the selected item is not in the **Released** state.

When updating multiple structures simultaneously, you can perform the following tasks:

- Add siblings, child items, and occurrence substitutes.
- Revise the structures.
- Replace and remove the items from the structures.

All updates that are made as a part of the change notice are indicated as redline text.

Restrictions and limitations

You can update multiple structures simultaneously if your Teamcenter setup includes BOM Structure Management license.

Procedure

1. In an active change notice, perform the impact analysis for the problem items and then select the impacted items.
The impacted items are the structures or immediate parent assemblies for the problem items. If a problem item is a part of multiple structures, you can select all these structures as impacted items.
2. Click the **Change Content** tab to view all the problem items and their immediate parent assemblies, which are identified as impacted items.

Tip

To navigate to or locate specific items on the **Change Content** tab, use filters and sorting in columns.

After you apply a filter to a column, all items that satisfy the filter criteria are displayed on the **Change Content** tab. For example, suppose that you are searching for an item with ID as 12. In such a case, in the **ID** column, select the **Contains** operator and type **12**. The **Change Content** tab displays all items that have ID as 12. After a filter is applied, the **Change Content** tab considers the redlining and shows the modified content. If an item is removed on the **Change Content** tab as a part of the change, and its content satisfy the filter criteria, then that item is still shown on the **Change Content** tab. For example, suppose that the 8cm nut is removed and you entered the filter specifying the 8cm size. In such a case, the **Change Content** tab shows all 8cm nuts including the one that you removed.

You cannot filter the visual indicator columns like . After applying column filters to locate the required content, if you click another tab in the change and then return to the **Change Content** tab, the column filtering is removed. You must again apply the column filters.

3. Perform the following actions as appropriate to select and show the required items on the **Change Content** tab.

If you want to...	Perform this actions...
Select all items including the parent and child items.	Click Select  > All  .
Select all top parents assemblies.	Click Select  > All top parents .
Select all occurrences of the selected item.	Click Select  > All matching elements .
View all child items for the selected parent node apart from the ones that are a part of the change notice and also view the rest of the siblings and sub-assemblies. After viewing the non-impacted child items, you can update their properties and include them in the change. You might see all child items by default if your administrator has set it for you. Furthermore, if all child items are shown, the parent are always collapsed. You must manually expand them.	Click Show  > All Children  .
View only the child items that are a part of the change notice and do not want to view the siblings and sub-assemblies.	Click Show  > Children in Change  .

4. To revise one or more items, select them and click **Revise** .

If you select only one item for revision, the **Revise** panel appears. Review the information on the **Revise** panel and click **Revise**.

If you select multiple items, a confirmation message appears. Click **Revise** on the confirmation message.

The selected items are revised and are shown with the redlines.

5. If you want to edit the description, quantity, and any other properties of the selected one or more child items, click **Edit** .

After you are finished editing the properties, click **Save Edits** .

6. If you want to add a child item to one or more items:

- a. Select the items under which you want to add the child item and click **Add**  > **Child**.
- b. In the **Add Child** panel, search an existing child item or create a new one, and then click **Add**. The child is added to the selected items.

7. If you want to add a sibling item to one or more items:

- a. Select the items for which you want to add the sibling item and click **Add**  > **Sibling**.

You cannot add sibling to the parent item in the structure.

- b. In the **Add Child** panel, search for an existing child item or create a new one, and then click **Add**.
The child item is added to the selected items.
8. If you want to add occurrence substitutes:
 - a. Select one or more items and click **Add**  **> Substitute** .
 - b. In the **Add Substitute** panel, search for an item or create a new item and click **Add**.
The occurrence of the selected item is added as a substitute. The **Substitutes** and **Primary Component** columns show the information of substitutes and primary items. The **Substitutes** section in the **Overview** tab for the primary item also show the list of the substitutes items. You can further set substitutes as primary components.

After the substitutes are added, if you want to hide them from being shown in the structure, right-click a structure and select **Hide Substitute Components** .
9. If you want to remove some of the items from the structure, select the items and click **Edit Structure**  **> Remove** .

The selected items are removed from the structures and redlines appear for them.

10. If you want to replace the selected items by another item:
 - a. Select one or more items and click **Edit Structure**  **> Replace** .
 - b. In the **Replace** panel, search for an existing item or create a new one, and then click **Replace**.
All selected items are replaced with the chosen item.
11. If required, select the required revision rules and apply date and unit effectivity on the **Change Content** tab.

What to do next

After the impacted structures are updated and the solution items are provided, you can view the change summary and then send the change for downstream process.

Edit a structure in markup mode

You can make changes to a structure in *markup mode*. In this mode, the proposed changes to a structure are highlighted with a different color or a strikethrough, until they are committed to the database. You use mark up changes for reviews and analysis without actually modifying the structure. Any user can mark up a structure regardless of the release status or write access.

You can either reject the changes or apply them to the structure.

Procedure

1. Search for the structure that you want to edit and then click **Open** .
2. To edit the structure in the markup mode, select an element and choose **More Commands *** > Markup**  to switch to this mode. You do not need write access to mark up the structure.
3. Edit the structure as required and then save the edits. You can make the following types of structure edits in markup mode:
 - Add a child or a sibling
 - Remove a structure element
 - Replace a structure element
 - Edit occurrence properties
 - Add or remove substitutes

4. To display or hide the markups for a structure, turn the markup mode on or off.

When you are in the markup mode, you can view a summary of the markup changes on the **Markup** tab. This tab displays a list of markup changes for the current level (all markups for an assembly) or for the current element.

5. To reject an individual markup change listed on the **Markup** tab, click **Cancel Markup** .

You can add your markups in a change context as well. In this case, the markups are related to that change, and you can see them only when you are in the change context. You can create different sets of markups for different changes.

Apply or cancel markups on a structure

You can apply or reject markups from the **Markup** tab or using a workflow.

Apply or cancel markups from the Markup tab

An authorized user can apply the markup changes to a structure through the **Markup** tab.

Procedure

1. Find the structure where you want to apply the markups.
2. If the marked-up assembly is released, revise the assembly.
The markups are carried forward to the new revision.
3. If the structure is not already opened in two panes, choose **Layout Choice > Left Right** .
4. Click the **Markup** tab, select the marked up assembly, and then click **Apply Markup** .

Note

The **Apply Markup**  icon is not visible when the marked up part is selected. It is visible only when the parent assembly is selected.

5. To reject a markup, select the markup change to be rejected, and then click **Cancel Markup** .

Apply markups using a workflow

For the markups made within a change context, you can apply the markup changes to a structure by initiating a workflow process. Submitting content to a workflow sends it through a series of required tasks, such as approvals (signoffs) and reviews.

Procedure

1. To initiate applying the markup, select the change notice containing the markups and then click **Submit to Workflow** .
The **Submit to Workflow** panel is displayed along with a list of workflow templates.
2. From the **Workflow Template** list, select the **Review and Apply BOM Markups** template and then click **Submit**.
3. Assign the reviewers for the workflow.
4. After the workflow is approved, a new revision is created, and the changes are applied.
You can verify the updates in the new structure.

Review active or closed changes for a structure

You can track changes (added, modified, replaced, revised, or deleted) to the assemblies using a change request or a change notice. When the ECN or ECR is an Active change, redlines are shown in the structure. To view the closed changes, use the **Show Redlines** command.

Prerequisites

Your administrator can configure the various methods to effectively track and display relevant changes to an assembly.

Procedure

1. Search for and open the structure. If the structure has changes, they are indicated by redlining. Expand the assembly to view the details of the change.

Element	ID	Revision
Engine	035611	B A
Piston	035612	A
<i>Enhanced Valve</i> Valve	<i>035617</i> 035613	A
Connecting Rod	035614	A
<i>Connecting Rod Assembly</i>	<i>035621</i>	<i>A</i>

- Deleted parts are indicated by a red strikethrough.
- The added parts display in green and are italicized.
- For replaced or revised parts, the old and the new values are shown side by side.

After the structure is revised, the redlines are shown for all changes including the edits that you make for the part attributes.

- (Optional) To disable highlighting the changes in the structure, go to **More Commands *** > View**  icon and turn off **Show Redlines** from the **Page level** toolbar.
- Only active changes are shown by default. To view the closed changes, click **More Commands *** > View**  > **Show Redlines** from the **Page level** toolbar.

These changes are also shown in the packed mode.

You can return a redline change back to its original impacted version by reverting the BOM modifications.

Understanding precise and imprecise assemblies

The precise and imprecise assemblies help you better handle item revisions, configuration updates, and their respective use cases; thus effectively managing product structures and their components.

Imprecise assemblies are dynamic structures of items. An imprecise assembly has links (occurrences) to items of its components.

Note

Videos are not included in PDF. To access videos, use HTML.

Imprecise assemblies allow you to see the product structure (bill of materials) configured with the relevant item revisions, according to the revision rule you choose for the window. For example, you may want to configure the product structure with revisions that are released for production or, alternatively, with the latest working revision of components. All users look at the same underlying product structure, but view different configurations in different circumstances according to the revision rule to suit particular needs.

Teamcenter automatically reconfigures an imprecise structure when new parts are released or created, or other changes are made. Consequently, it is not necessary to make a copy of the structure and manually update it if you make changes.

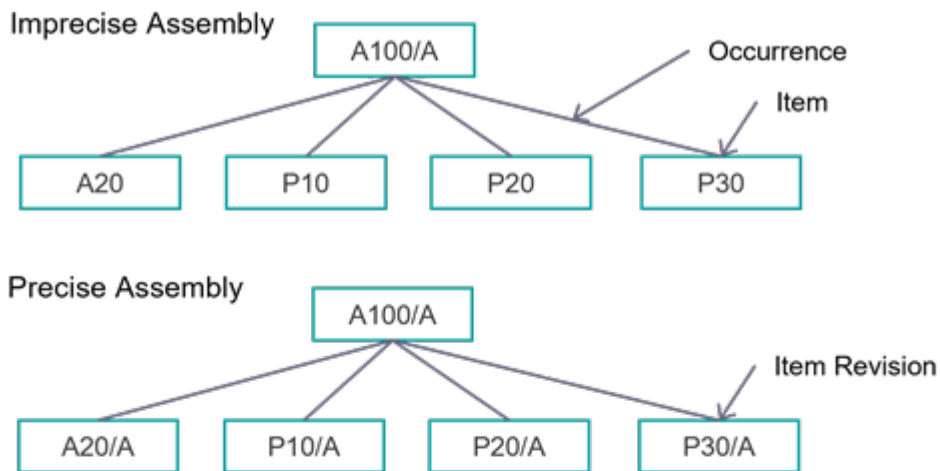
Precise assemblies are fixed structures of specific item revisions. A precise assembly has links (occurrences) to item revisions of its components. When you modify a component to a new revision, you must manually update the assembly by removing the old revision of the component and adding the new revision.

You configure the precise references with a revision rule containing a **precise** rule entry. However, you can dynamically configure a precise assembly (treated it as imprecise) by applying a revision rule that does not contain the precise rule entry, or has the entry at a lower precedence. Precise assemblies are useful in situations where you must control the configuration very carefully, for example, in aerospace environments.

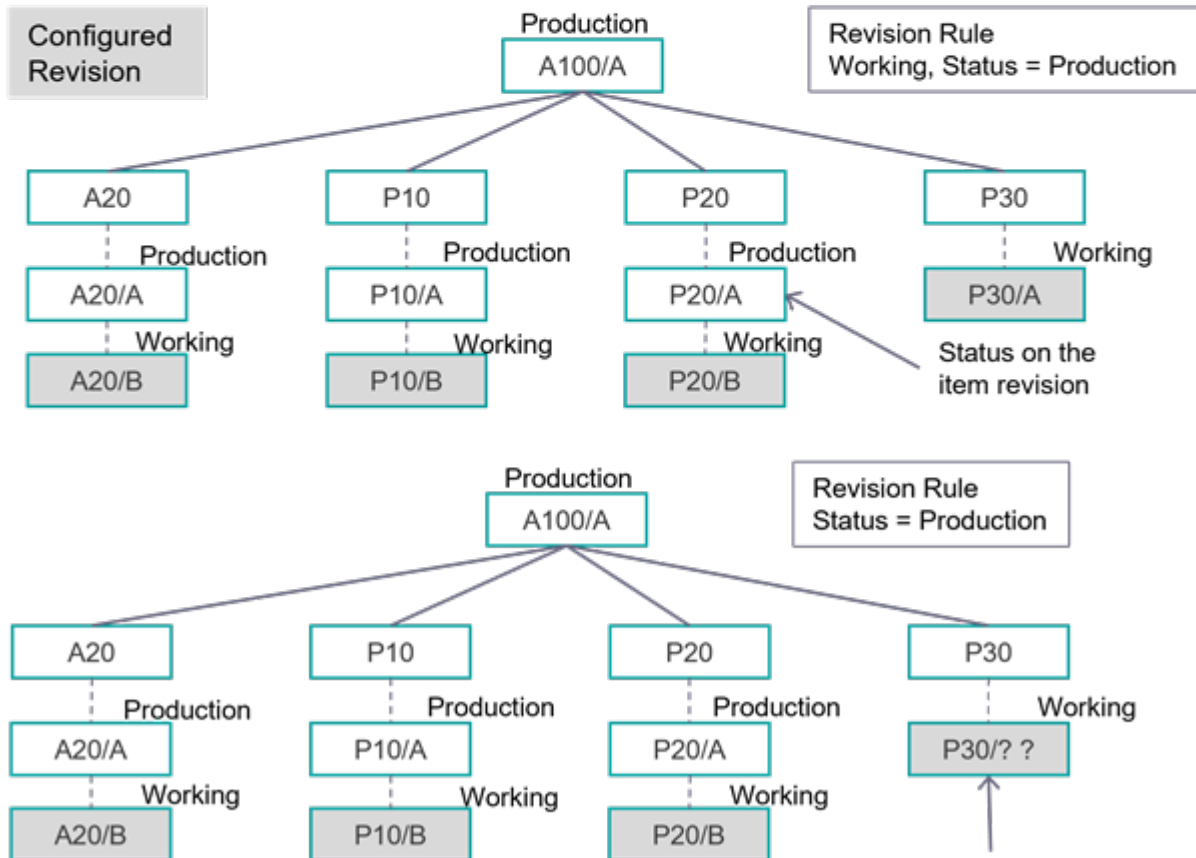
When the parent assemblies are released and can no longer be modified, any change may result in significant *revision ripple*. If you incorporate new revisions of piece parts, every parent assembly to the top of the structure must also be revised. However, you can group changes into batches to minimize this effect.

Item revision configuration

Item Revision Configuration



Example of Imprecise Assembly Configuration



Note - There is no revision that satisfies the rule - Structure Manager shows this as "? ?" for the revision (and if an assembly it cannot be further expanded)

You can change a BOM view revision from precise to imprecise if you have write access to the BOM view revision. This operation causes all the occurrences of that BOM view revision (for example, all of the components for that assembly, everywhere it occurs) to be changed. When you toggle an assembly from imprecise to precise, the precise occurrences reference the item revision that was configured. When you change precise BOM view revisions to imprecise, the Structure Manager window's revision rule is applied to determine the revision of the item for each component in the BOM view revision that should display.

9. View and visualize structures

View a structure in other applications

You can open and view a structure in other applications, such as NX and CATIA. If the structure exists within a partition, you can open the partition itself in some other application, so that one or more structures within that partition are opened.

Application	Action
Teamcenter Lifecycle Visualization	Select Open  > Open in Visualization  .
Rich client	Select Open  > Open in Rich Client  .
Designcenter NX	Select Open  > Open in Designcenter NX  to open the selected element. Select Open  > Open Product in Designcenter NX  to open the entire product structure.
Solid Edge	Select Open  > Open in Solid Edge  .
Illustrator	Select Open  > Open in Illustrator  .
CATIA	Select Open  > Open in CATIA  .
Supplyframe	Select Open  > Open in Supplyframe  .

Viewing structures in the split view

You can use the split view to display the two structures or configurations side-by-side, independently modify them, compare them, and perform drag and drop operations for moving or copying occurrences. You can also filter and retain the context upon exit in the split view.

You can split the structure view into two views using the **Split Context**  command. Initially, identical structures are loaded in both panels. Both panels provide a toolbar with commands relevant for the structure being displayed.

You can also open two different structures side-by-side in the split view by selecting two root nodes and then using the **Open**  command.

In the split view, you can modify the structures independently. Once either structure is updated to a different configuration, then the **Compare** functionality is enabled. You can also use the split view to

move or copy an occurrence from one structure to another using the drag-and-drop functionality. If the two views have the same structure with the same applied configuration, then the occurrence is moved. Otherwise the occurrence is copied.

If the structures are indexed with Smart Discovery, **filter the structures** by clicking **Filter**  in the **Split View**. You can filter the entire structure and apply the filtering criteria to view the required content of the structure.

When you exit the **Split View**, you can continue working on the structure that you worked on the last. For example, in the **Split View**, suppose that the Headlight Assembly is available on the left view and the Front Bumper assembly is available on the right view. Then you changed the configuration of the Front Bumper assembly before exiting the **Split View**. In such a situation, the Front Bumper assembly is retained and you can continue working on it. When you click **Back**  in the **Split View**, you can continue working on the structure in the left view. After clicking **Back**  in the **Split View**, the structure in the left view is opened with its original configuration, that is, the configuration that was applied before splitting the view. If you applied the filter in the **Split View** and then clicked **Back** , the filter criteria is retained on the structure.

You can view connections, for example, welds, and item elements, for example, General Design Elements and ports, in the tree structure and the 3D viewer.

View where an item is used

A where-used analysis in Teamcenter helps you identify all parent assemblies and products in which an item is used by navigating up the structure. Starting from a selected component or assembly as the context, this analysis traces usage throughout all levels of the structure.

You can use a where-used analysis to:

- Assess the impact of engineering changes to the product structure.
- Check if changes in one assembly affect other assemblies.
- Determine which products are affected by item changes.
- View items in the context of their products through traversal paths.

Restrictions and limitations

Your Teamcenter setup must contain Smart Discovery Indexing to access the **TOP LEVEL** section and view top-level assemblies or products.

Procedure

1. Search for and select the item.
2. Click the **Where Used** tab.
3. Expand the tree in the **Used in Structures** section to move up in the structure.

The **Used in Structures** section lists the immediate parents of an item.

The table initially displays parent assemblies and any assemblies where the selected item is used as a substitute. The information displayed is based on the revision rule currently selected in the **Revision Rule** list, which defaults to **Global**. In this case, this refers to the revision that is selected in the global revision rule list.

To view all parent items where the selected child item is configured, select **All** from the **Revision Rule** list. Click **Export to Excel**  to export the information displayed in the table. The number of structure levels to be exported to Excel is set by your system administrator.

Note

If an occurrence of an item is used as a substitute in an assembly, those assemblies are not listed in the **Used in Structures** section.

If the selected item is a substitute, the  column in the **Used in Structures** section of the **Where Used** tab displays  for its parent item.

4. Expand the **Top Level** section if it is not already expanded.

Smart Discovery Indexing is used to find the top-level assemblies or products. Therefore, if your system administrator has not installed Smart Discovery, the **Top Level** section is not displayed. You only see indexed products in the results list.

The table initially displays the top-level products in which the selected item is used. This display is based on the revision rule currently selected in the **Revision Rule** list, which defaults to **Global** (meaning, the revision rule that is selected in the global revision rule list). This is useful for *impact analysis* when you want to see which products are affected if the item is changed. It is also useful for determining the *traversal path* when you want to see the item in the context of one of its products.

To view the top-level products where the selected child item is configured, select **All** from the **Revision Rule** list. Click **Export to Excel**  to export the information displayed in the table.

5. Expand the **Contexts** section if it is not already expanded.

The **Context** section shows various configurations for the selected component. You can select the configuration you want to open.

6. Expand the **References** section if it is not already expanded.

The **References** section lists the parts and documents that reference the selected component.

Visualize structures

You visualize structures to perform visual inspection of the structure elements, and their location and orientation in the structure.

You can view connections (for example, welds) and item elements (for example, General Design Elements and ports) in the tree structure and in the 3D tab.

If the structure is indexed for massive model visualization (MMV), you can view the 3D model of it.

Procedure

1. Open a structure and click **Details**  > **3D** .
2. Select a structure element.
The selected element is highlighted in the **3D** tab.
3. Select an element in the **3D** tab.
The corresponding structure element is selected. However, multi-selecting in the **3D** tab does not select the corresponding elements.

Visualize structure elements located within partitions

You can visualize structure elements that are located within partitions. You can select a partition scheme to view specific partitions and their associated elements highlighted in the 3D viewer. Teamcenter dynamically updates the 3D view when you assign elements to partitions or unassign elements from partitions.

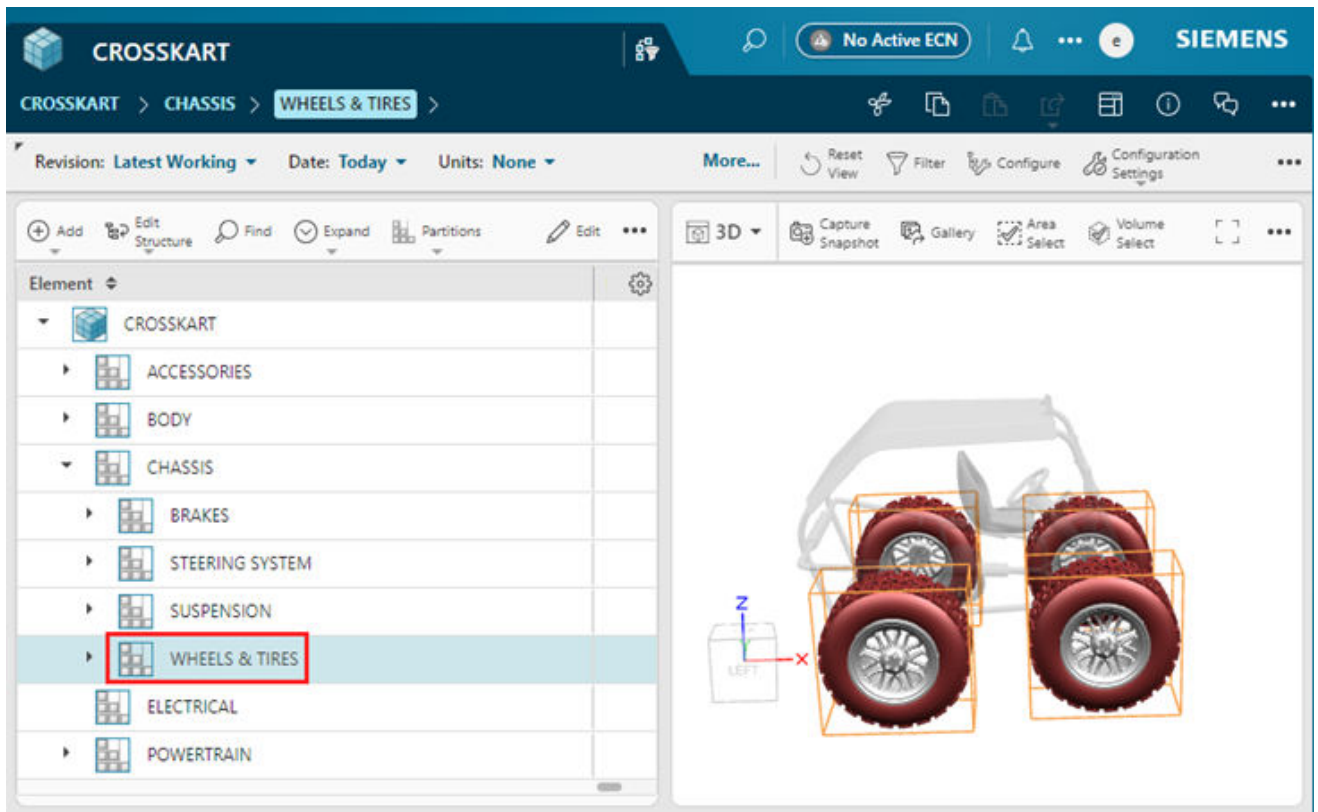
Procedure

1. Open the structure that you want to visualize. By default, the structure opens in the partition scheme that is set as the primary scheme.

Note

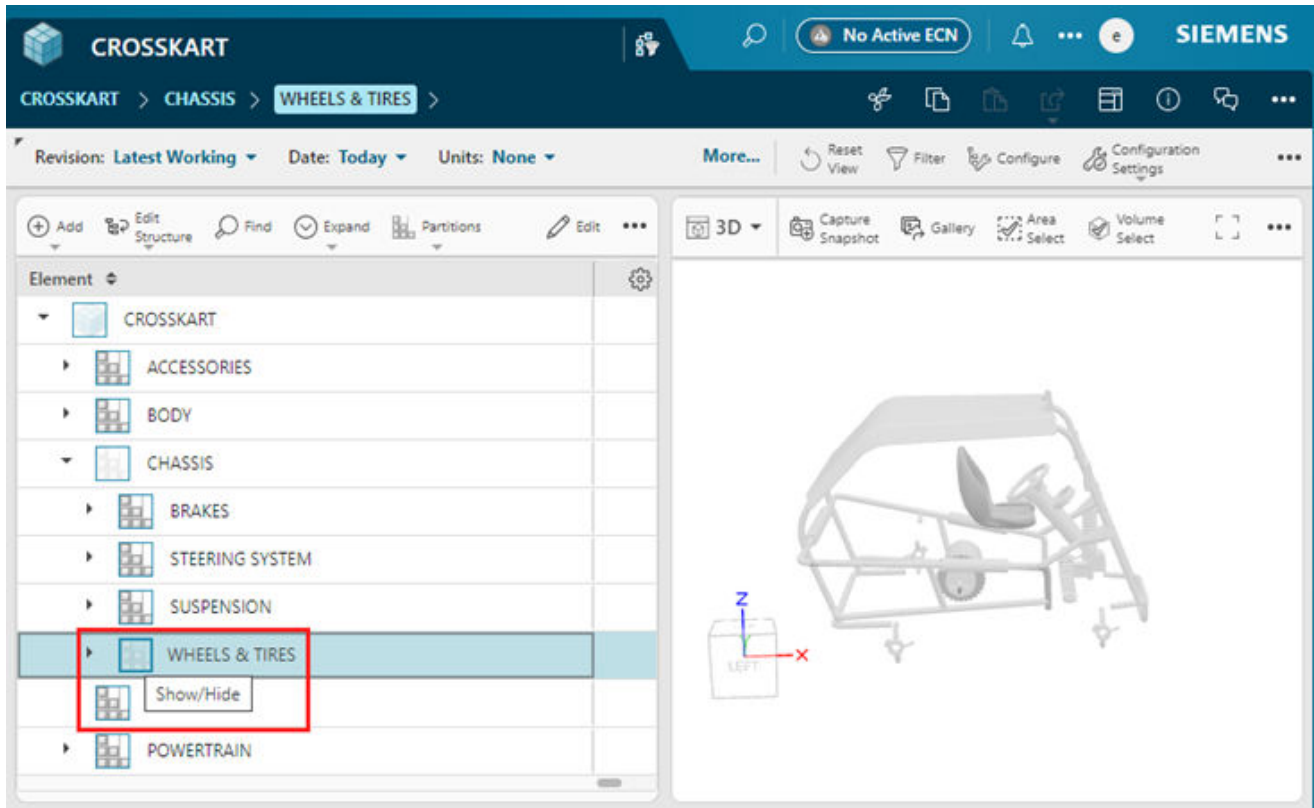
Ensure that a partition scheme exists before you open the structure that you want to visualize. For information on partition schemes and partitions, see *Structure Partitions — Usage*.

2. Select the required scheme from **Partition Scheme** that contains the required partitions.
3. To visualize the structure elements located within a partition, select the partition, and go to the 3D viewer.
All the elements within the selected partition are highlighted with their bounding boxes.



If the selected partition has in turn a child partition within it, all the structure elements within the child partition are also highlighted in the 3D viewer.

You can choose to not visualize the elements of a specific partition by clicking **Show/Hide**. For example, the icon is displayed next to **WHEELS & TIRES**, as shown in the following image. To visualize the elements again, click **Show/Hide** again.



When you set variant criteria on partitions, the structure is configured to either show or hide the partitions and the elements within. Accordingly, the elements are displayed in the 3D viewer as well. In several cases, the configured structure and the 3D viewer might not be synchronized.

- **Scenario 1**

For a structure in a *workset*, a *Functional* scheme is applied and contains a variant criterion, VC1. In this case, the structure elements with a different variant criterion, VC2, are not included in the structure. However, the 3D viewer shows these elements. When you switch to a different partition scheme, the same scenario applies. In such cases, for structures in a workset, you can synchronize the partitioned structure and the 3D viewer by clicking **Replay**.

- **Scenario 2**

A component resides in two different partitions. It is included in the structure configured for one partition, whereas it is not included in the structure configured for the other. When such a mismatch occurs, the component is displayed both in the partitioned structure and in the 3D viewer.

- **Scenario 3**

In a partition scheme, structure elements at different hierarchical levels reside in two different partitions. When you set variant criteria on a partition to configure the structure such that the partition is not included in one case, the structure and the 3D viewer show a mismatch. Siemens recommends configuring the system to restrict the addition of structure elements in partitions in such a manner.

In the **Partition Scheme Assignment** view, the **3D** view for the **Unassigned** pane displays only the unassigned elements. When you drag one or more elements from the **Unassigned** pane to the **Assigned** pane, Teamcenter updates the elements in the **3D** view for the **Unassigned** pane accordingly. Similarly, the **3D** view for the **Assigned** pane gets updated. Thus, in both panes, the elements that you see in the tree structure match the ones you see in the **3D** view.

Working with end-item assemblies

The end-item assemblies enhance the system performance and help you focus by collapsing sub-assemblies. When you search for child components within an end-item assembly, the parent end-item is highlighted.

To improve the system performance and hide information that is not relevant to the user's current task, one or more assemblies in a product structure can be set as *end-item assemblies* for that structure. When you expand a structure containing an end-item assembly, the nodes beyond the end-item assembly are not expanded.

Although you cannot set any occurrence as an end-item, you can view a structure containing end-items. When a structure containing an end-item is opened, the end-item is displayed as a single line in the assembly tree.

The **End Item Assembly State** column identifies the end-items in an assembly.

Element	End Item Assembly State
▼ CROSSKART_DC	
▶ ELECTRIC SYS CK	
▶ EXTERIOR CK	
▶ BRAKE & SUSP CK	
▶ STEERING SYS	☒
▶ TIRE & WHEEL CK	

The 3D viewer displays the full assembly structure regardless of **End Item Assembly State**. To view the full structure of an end-item assembly in the tree, open the end-item as its own root node.

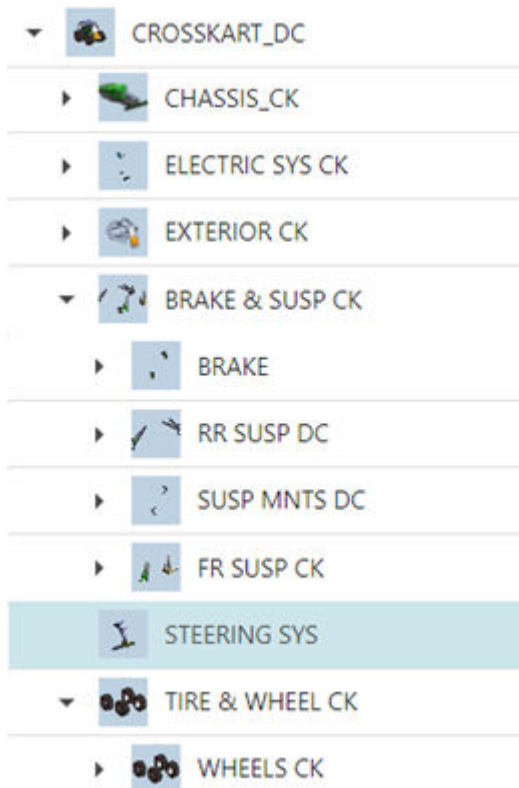
What happens when you search for a child component of an end-item?

To access the components of any subassembly, you expand the relevant subassembly.



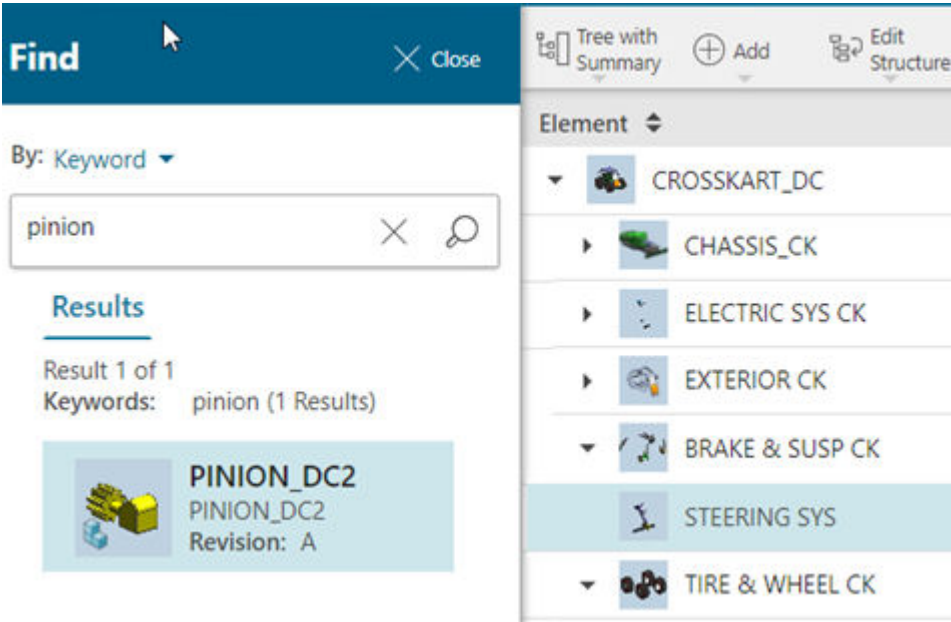
As the assemblies marked as end-items cannot be expanded within the structure, their child components are not visible when the structure is expanded.

For example, if the **steering system** assembly is marked as an end-item in a vehicle structure, its components, namely, **steering mech**, **pinion**, and **rack** are not visible in the structure.



However, when you perform an in-context search for a child component of an end-item, the search results return the child components. When you select the child component of the end-item in the search results, the parent end-item subassembly is selected in the structure.

For example, when you select the **PINION_DC2** child component from the search results, the parent **STEERING SYS** subassembly, which is an end-item, is selected in the structure.



Pack or unpack structure elements

You can group similar elements in an assembly by *packing* them together in a single level of the assembly. This results in a shorter and simpler structure display. The packed elements are represented using a single row in the structure with a numeric value appended to this row to indicate the number of packed items.

You can individually pack or unpack the selected elements in the structure that meet the packing criteria. You can also pack or unpack all elements in a selected structure simultaneously with a single click.

When you select a packed node in the tree, all the part instances packed in that node are highlighted in the 3D viewer.

To pack elements	To unpack elements
To pack the selected elements in the structure that meet the packing criteria: 1. Select one of the elements. 2. Click the Expand ☑ icon. 3. Select Pack.	To unpack the elements in a structure: 1. Select the packed element. 2. Click the Expand ☑ icon. 3. Select Unpack.
To pack all the elements in the structure: 1. Click the Expand ☑ icon. 2. Select Pack All.	To unpack all the elements in the structure: 1. Click the Expand ☑ icon. 2. Select Unpack All.

Note

While finding elements within a structure or filtering a structure (Smart Discovery-indexed), the state of the structure remains the same, regardless of whether the structure elements in it are packed or unpacked manually or by default. However, the search results on the **Find** panel are always displayed as unpacked.

You can specify the reference designator values in the International Organization for Standardization (ISO) and in the American Society of Mechanical Engineers (ASME) formats. In the ASME format, the reference designator value is in the capital letter followed by a number and suffix letter. For example, **S1A**. In an engineering change notice (ECN), if you modify the reference designator value or quantity for one or more elements in the packed and unpacked mode of a structure, these modifications are highlighted in redlines.

View attribute details for a structure occurrence

The columns show the attributes for each structure occurrence. Some columns show the attribute value, while others show an icon. The icon indicates that the structure occurrence has an attribute.

For example, if the **Substitutes**  icon is shown against an occurrence, it means that substitutes are defined for that occurrence.

The following attributes are represented using icons in the table columns.

Icon	Attribute shown on mouse hover
 Element Effectivity	Shows the effectivity information, including end items or dates applied to the occurrence.
 Overrides	Shows information about overridden properties and contexts in which the overrides are applied.
 Variant Condition	Shows the variant condition details.
 Substitutes	Shows the substitute information such as ID, Name, and Revision.
 Global Alternates	Shows the global alternate information such as ID, Name, and Revision.

The **Overview** tab displays the occurrence properties and the other tabs display additional information about the occurrence.

Procedure

1. To add or remove a column, click  to the right of the column headings to display the **Arrange** panel and select or undo the selection of the column.
2. To view any attribute details, hover your mouse over the icon.

Control the display of configured structures

You can control the display of configured structures by showing or hiding occurrences based on effectivity, variants, filtering, and suppression. You can also manage the visibility of connections, joints, and ports within the structure.

When creating variable content or analyzing configured structures, it is important to understand which occurrences will be configured and the ones that will be excluded. When a structure is loaded, only objects that meet the applied effectivity and variant criteria are displayed. However, you can choose to hide or view the occurrences that are not configured as well.

Note

You can do this only when you select an individual structure within the workset or any elements within the same structure. You cannot select multiple structures within the workset or multiple elements across different structures within the workset to hide or view such occurrences.

To perform this step	Do this
To show or hide the occurrences that are excluded by the currently applied effectivity	Click Configuration Settings  > Show Excluded by Effectivity .
If you have Partitions for Structure deployed in your Teamcenter setup, and you have applied variability on partitions, to show or hide occurrences and partitions that are excluded by the variant configuration	Click Configuration Settings  > Show Excluded by Variants .
To view all the structure elements including the ones that are not displayed due to the applied filters	Click Configuration Settings  > Show Excluded by Filtering .
Occurrences in a structure can be hidden by setting the suppress property to True. When a structure is loaded, all occurrences including the suppressed occurrences are not displayed by default. To show or hide the suppressed occurrences	Click Configuration Settings  > Show Suppressed .
Hide all secondary components of the structure.	Click Configuration Settings  > Show Secondary Components . Note In stead of hiding all secondary component of the structure, if you want to hide the secondary components

To perform this step	Do this
	only within an assembly, right-click the assembly and click Hide Secondary Components  .
To show and hide the connections and joints from the structure. For example, welds. The 3D tab still show the 3D view of the connections and joints even if you hide them. This button is available only when the product structure is opened without a session, workset, configuration baseline, saved working context or without a partition scheme.	Click Show Connection Points  > Connections and Joints  .
To show and hide the ports from the structure. The 3D tab still show the 3D view of the ports even if you hide them. This button is available only when the product structure is opened without a session, workset, configuration baseline, saved working context or without a partition scheme.	Click Show Connection Points  > Ports  .

Using structure view types

The *view type* is an attribute of a BOM view revision that indicates its purpose, for example, design or manufacturing. It allows you to distinguish one BOM view revision from another in the same item revision. A structure is loaded based on its view type (for example, view, design, manufacturing, or shipping).

Filtering with Smart Discovery-indexed structures does not support multiple view types. If view types are applied, these are ignored for filtering in Smart Discovery.

Only administrators can add a new view type. You can only configure a structure with an existing view type.

The system administrator defines a set of view types for each site using Business Modeler IDE. Any number of view types can be defined, but most sites only require a single view type.

For more information about adding a new view type, see the *Add a View Type* section of *Configure your business data model in BMIDE* in the Teamcenter documentation.

For more information about setting a default view type, see the *Understanding BOM view revisions, view types, and multiple views* section of *Structure Manager* in the Teamcenter documentation.

Configure and view structure based on view types

You can configure the structure by selecting one of the available view types, but you cannot create a new view type. A new structure is always created with your default view type. You cannot specify the view type when you create a new structure.

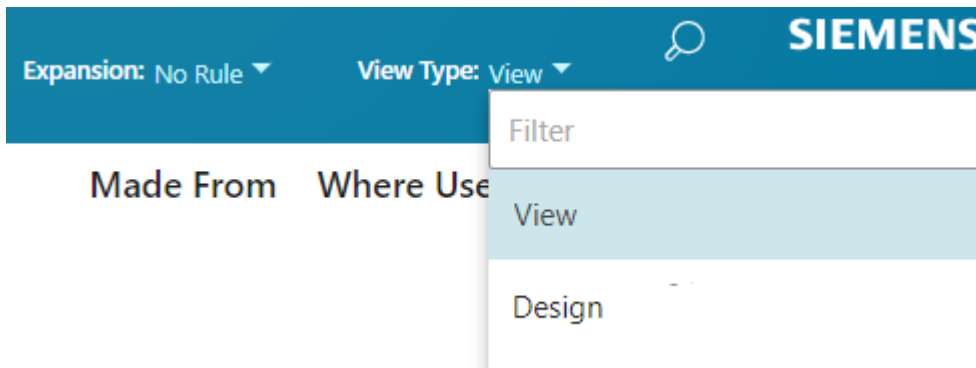
Procedure

1. Search for and open the required structure.

Note

When you open a structure with multiple view types for the first time, it opens in your default view type. To change the view type, you can select a different view type from the list. When you open the same structure again, it opens with the last saved view type in the background working context. To switch to the default view type, you can perform a reset action.

2. To configure the structure with a view type, select it from the **View Type** list in the header area. The **View Type** information is displayed only if multiple view types are defined for the user or site.



Teamcenter refreshes and displays the content based on the selected view type.

3. Edit the structure as required.

Any additions to the structure (for example adding a child or a sibling part) are associated with the selected view type. However, if a new child part is added to a leaf node (a part that does not have any child parts), this new structure branch is created with the user's default view type irrespective of the selected view type.

Viewing structure arrangements

You can view the structure *arrangements* in Teamcenter, which are created in NX to define alternative positions, orientations, and suppression states for components within an assembly, such as open or closed positions for a car door or shock absorber.

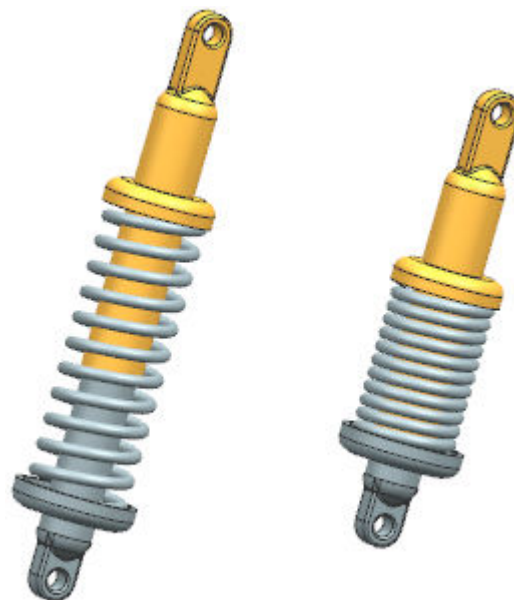
In NX, you can define *arrangements* to specify alternative positions for one or more components in the assembly and store those alternatives with the assembly. Alternate arrangements are created by moving or suppressing components. For example, a car door assembly may have different arrangements when the door is open or closed, and the position of certain components may be overridden accordingly. Similarly, you can suppress components in a particular arrangement, for example, to hide the hinges.

If arrangements are defined for an assembly in NX, you can set and view the arrangements in Teamcenter.

An assembly arrangement determines:

- The position and orientation of the immediate child components.
- The variable component positioning of any subcomponents.
- The used assembly arrangement for each immediate child component.
- The arrangement-specific suppression of components and subassemblies.
- The arrangement-specific assembly constraints.

The following example displays two arrangements of a shock absorber assembly. The arrangement on the left displays the shock absorber in the open position, while the arrangement on the right displays the shock absorber in the closed position.



Set the structure arrangement

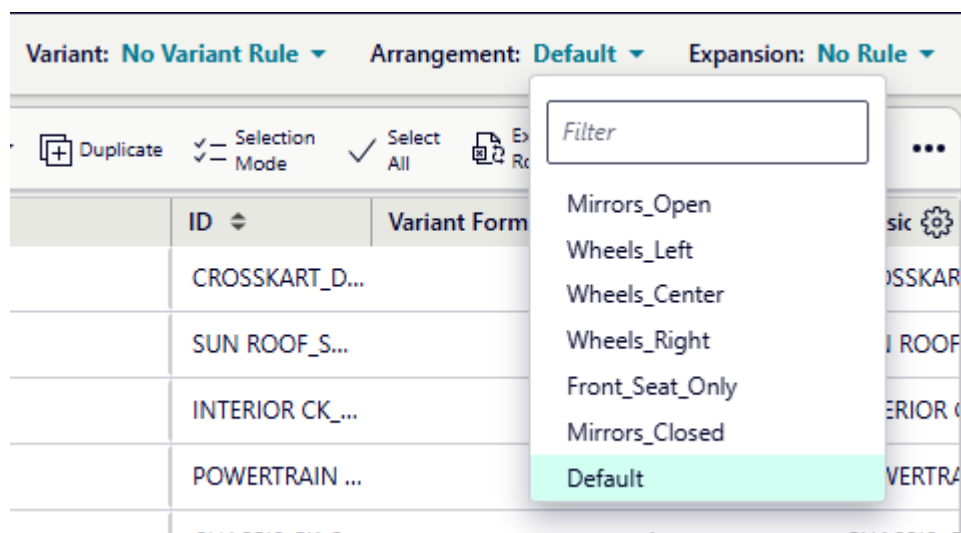
If an arrangement is defined for an assembly in NX, you can view it in Teamcenter. You can set the structure arrangement to view and visualize the structure.

Procedure

Set the structure arrangement in one of the following ways:

- a. Using the header **Arrangement** list:
 - i. The header information for an assembly shows **Arrangement** only if the arrangement is defined for the assembly. When you open an assembly for the first time, it opens with the **Active** arrangement.

To set an arrangement, select it from the **Arrangement** list in the header area. This arrangement persists across user sessions unless you change it.



- ii. To load the **Active** arrangement for the assembly, click **Reset** .
- b. Using the Configure panel:
 - i. Search for and open the product to be configured.
 - ii. Click **Configure**  to display the Configure panel.
 - iii. In the **Arrangements** section, click the list and select one of the arrangements.
 - iv. Click **Configure** to apply the arrangement and close the Configure panel.

View snapshots of a structure

A snapshot captures the current state of the item revisions used in a structure. Snapshots are stored in a snapshot folder. In Active Workspace, you cannot create a snapshot, but you can view any snapshots that were created in the rich client.

You can view the snapshots that you created in the rich client under the **Newstuff** folder in Active Workspace. Snapshots are stored in a snapshot folder .

As a snapshot stores the state of the structure when the snapshot was created, it might not show the latest information. For example, consider that you had three versions of *Engine*, namely A, B, and C initially. While creating the snapshot, you had only revision A configured. Thus, when viewing the snapshot structure, you see only revision A of *Engine*. You do not see the revisions B or C. Also, in the structure, you see the revision rule is automatically created. The name of the revision rule is in the format **Snapshot: <name of the snapshot folder>**.

You can update a snapshot to use a different revision by replacing the item revision with a different revision. After you update the snapshot, you can see a new item revision in the snapshot. In case you do not see one, contact your system administrator.

Attach JT file for product previews

You can attach the JT file, that contains the structure's outline, to the its structure in Teamcenter so that the engineering users can preview the product outline and avoid the performance issues of loading the entire product structure in the Viewer.

Prerequisites

- You must have the write access to the top node of the product structure.
- You have obtained the JT file that contains the product outline. The JT file containing the product outline is typically generated from the CAD software (such as NX). Contact your CAD engineer if you need assistance generating this file.

Procedure

1. In Active Workspace, open the structure for which you want to attach the JT file.
2. In the **Content** tab, select the top node of the product structure, and click the **Attachments** tab.
3. In the **Files** section, click **Add** .
4. In the Add panel, click **Select File** and choose a JT file.
5. In the **Type** list, select **Direct Model**.
6. In the **Relation** list, select **Product Preview**.

7. Click **Add**.

Results

The JT file containing product's outline is attached to its structure. Engineering users can preview the outline by opening the product structure in the Viewer by avoiding the performance issues of loading the entire, large product structure.

View the Supplyframe risk index for a commercial part

If your Teamcenter environment is integrated with the Supplyframe application, you can view the risk index of the vendor part.

Prerequisites

Ensure that the risk index information is already sent from the Supplyframe application to Teamcenter.

Procedure

1. Search for the vendor part in a structure.
2. In the structure table, scroll to view the **Risk Index** column.

If the **Risk Index** column is not available in the structure table, click **Table Settings**  > **Arrange**  to arrange and view the column.

3. (Optional) Click the **Vendor Parts** tab to view the vendor parts for the selected part.

If multiple vendor parts are available for the selected part, the risk index is shown for the one that is marked as **Preferred**.

[View the Supplyframe risk index for a commercial part](#)

10. Create and maintain solution variants

About solution variants

The solution variants in Teamcenter are 100% configured product variants generated from a single variable structure (150% BOM). The solution variants are linked to the source structure. Teamcenter helps you ensure their uniqueness and also facilitates updates.

For effective product definition and management, a range of product variants are managed as a single variable structure (150% BOM) by using a configurator instead of managing them as discrete product variants. You can configure this variable structure to generate a 100% variant by applying a valid variant configuration to create a *solution variant*. The solution variant has a unique item ID, and it is linked to the source structure.

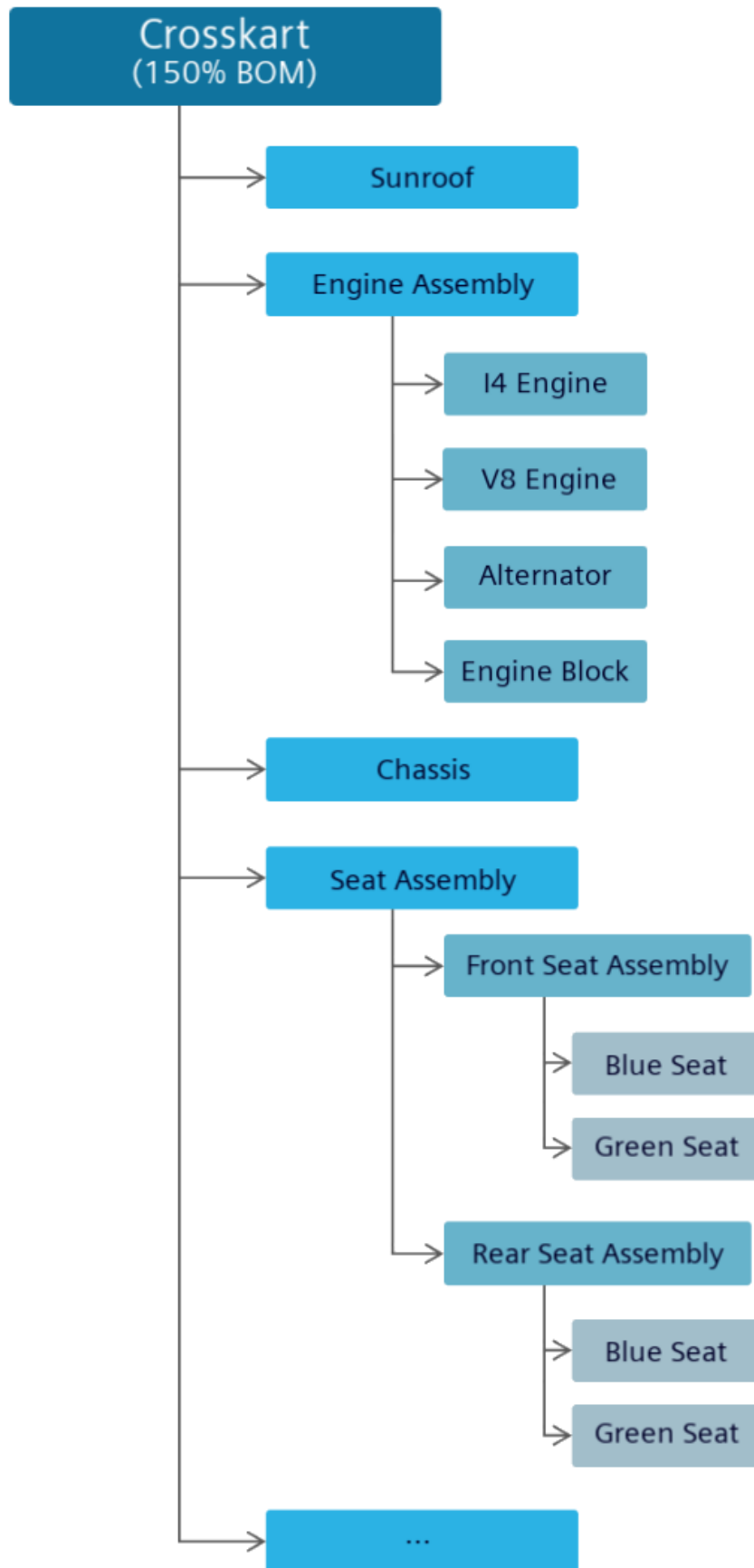
Note

Videos are not included in PDF. To access videos, use HTML.

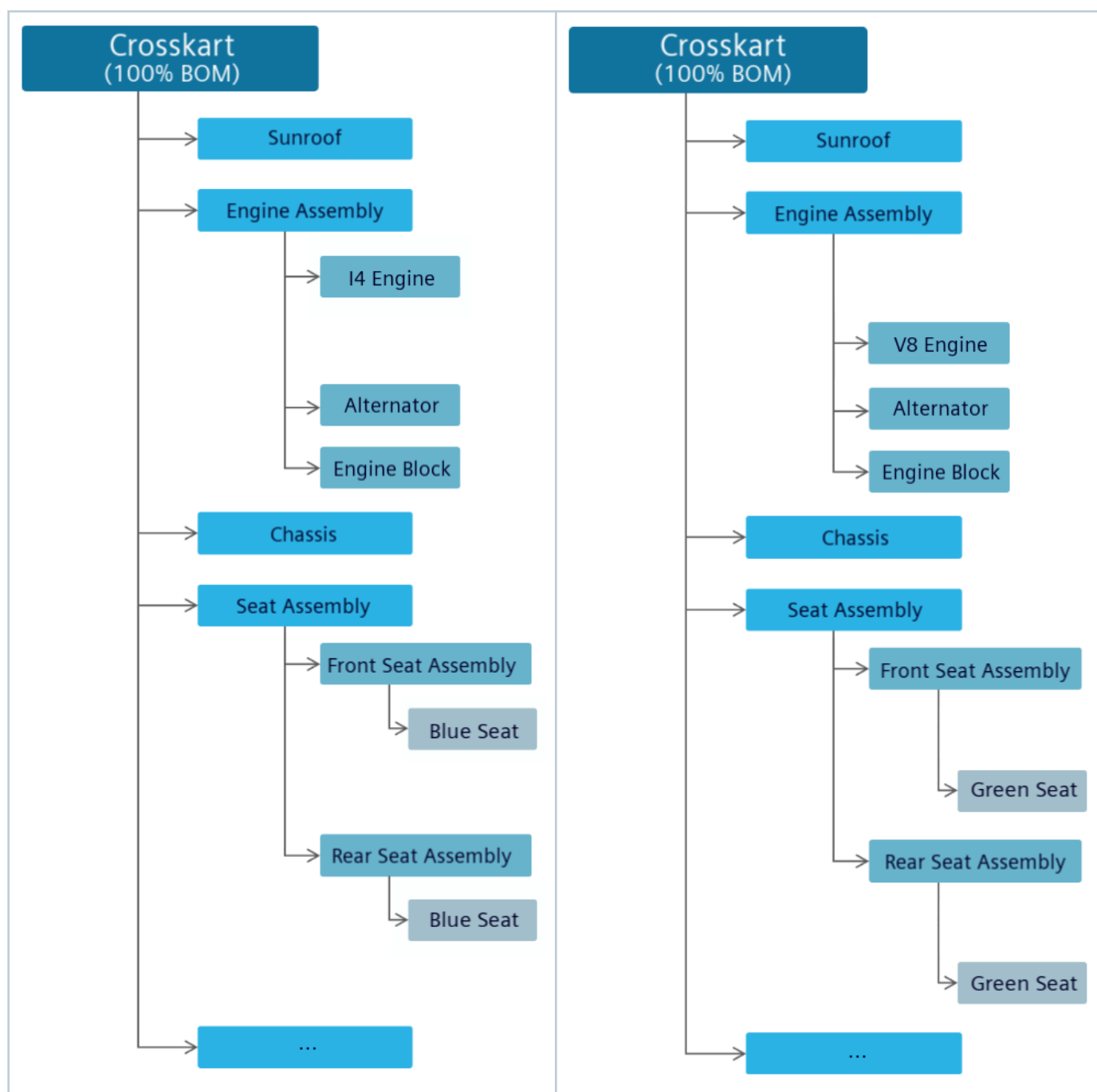
Multiple solution variants can be created for each valid variant configuration of the 150% variable structure. Teamcenter ensures the uniqueness of each solution variant by comparing the content with a list of existing solution variants before creating a new one. As the solution variants are linked to the source variable structure, the solution variants can be updated to easily incorporate the changes made to the source structure.

Example

Consider a product, *Crosskart*, that has two variants, *Base* and *Deluxe*. Both these variants are managed within a single variable structure.

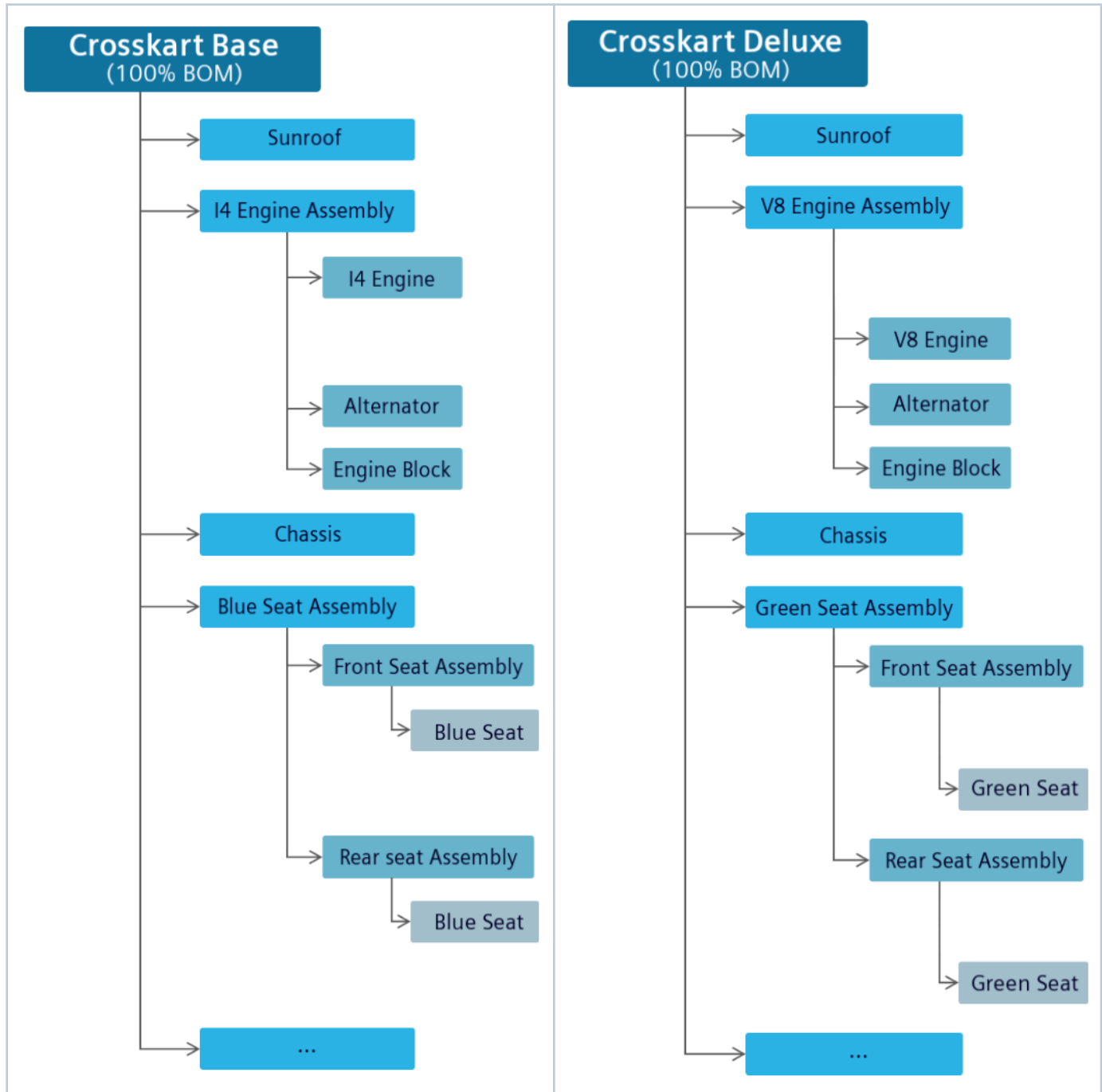


To derive the base and deluxe variants, you apply a valid configuration. This is based on the variant rules already set in Product Configurator for *Crosskart*. As per these rules, *I4 Engine* and *Blue Seat* must be used in the base variant, while *V8 Engine* and *Green Seat* must be used in the deluxe variant.



In these distinct base and deluxe variants, the crosskart, engine assembly, and seat assembly are not uniquely identified even though they contain different parts. Therefore, to identify and represent them statically, you create their solution variants.

When you select *Crosskart*, *Engine Assembly*, and *Seat Assembly* to create their solution variants for the base variant, three separate solution variants are created. Each solution variant has a unique identifier. They are identified as *Crosskart Base*, *I4 Engine Assembly*, and *Blue Seat Assembly*. Similarly, three solution variants can be created for the deluxe variant, namely, *Crosskart Deluxe*, *V8 Engine Assembly*, and *Green Seat Assembly*.



The source structure, *Crosskart*, is maintained as a separate entity and is linked to each solution variant. It can continue to evolve. If you update the source structure, you must update its solution variants. When you update the source structure to correspondingly update its solution variants, solution variants that were

previously created are reused. A new solution variant is created for a new part only if you choose to create a new solution variant for that part.

As solution variants are not just created at the product level but can also be created at the assembly level, this allows the reuse of variable parts. For example, consider that you want to reuse the engine assembly in another product, *Buggy*. According to the variant rules defined in the configurator, an I4 engine is used in the standard variant of *Buggy*. When you create a solution variant of the engine assembly for the standard variant, instead of creating a new solution variant, the one (*I4 Engine Assembly*) that was already created for *Crosskart Base* is reused. This way Teamcenter ensures that each solution variant is unique by comparing it with a list of existing solution variants before creating a new one. These solution variants, generated at the product or assembly level, can be used in downstream processes.

Understand the types of solution variant categories

Explore the various solution variant categories in Teamcenter—Reuse, Managed, and Unmanaged—and understand their distinct synchronization and modification behaviors relative to a source product structure.

You can create the following types of solution variants for a product in Teamcenter:

Reuse

Create a reuse solution variant to link it to the source product structure and synchronize its updates. This means that changes made to the source product structure are propagated to the reuse solution variant. You cannot make direct structural changes (such as adding, removing, or revising child items) to the solution variant independently.

Managed

Create a managed type of solution variant when you want to manually review the updates made to the product structure and then update the solution variant. In other words, if the source product structure changes, you can compare the changes with the solution variant structure and merge the selected changes manually. You can make direct structural changes (such as adding, removing, or revising child items) to the solution variant independently.

Unmanaged

Create an unmanaged solution variant to link it to the source product structure without synchronizing its updates. This means that changes made to the source product structure are not propagated to the unmanaged solution variant. You can make direct structural changes (such as adding, removing, or revising child items) to the solution variant independently.

Create a Reuse solution variant

A Reuse type of solution variant helps you synchronize the contents of the solution variant with the contents of the source product structure, ensuring all updates are propagated.

Create a reuse solution variant to link it to the source product structure and synchronize its updates. This means that the changes made to the source product structure are propagated to the reuse solution variant.

You cannot make direct structural changes independently, such as adding, removing, or revising child items.

Procedure

1. Open the structure that has variability .
2. If not already set, select the top line of the structure and set a variability scope.
 - a. In the **Details**  section, click **Edit** .
 - b. In **Variability Scope**, click **Add**  to search for and add a configurator context.
 - c. Click **Save** .

Additionally, you can set a variability scope to each of the variable structures at the lower levels for which solution variants are not yet created. These variability scopes must be subsets of the variability scope that you applied to the topmost line of the structure.

3. To create the Reuse type of solution variant for the entire product, select the topmost line of the structure, and then set the **Solution Variant Category** as **Reuse**.

When you set the **Solution Variant Category** as **Reuse** for the topmost line of the product structure, you must also set the **Solution Variant Category** as **Reuse** for the child assemblies.

If the **Solution Variant Category** column is not available by default, you must first add it.

4. Enable or disable **Show Excluded by Effectivity** and **Show Suppressed** from **Configure** , as required.
5. Click the **Solution Variants** tab.

Note

If you do not see the **Solution Variants** tab, contact your system administrator.

The **Solution Variants** section displays a list of the existing solution variants.

6. In the **Solution Variants** section, click **Create** .

The **Create Solution Variant** panel appears and displays a list of the variant rules that are available from the associated variability scope. The **Valid for Solution Variant** column indicates whether a solution variant can be created for a given variant rule. **Yes**  indicates that all conditions are met, while **No**  indicates at least one condition is invalid. Ensure that all variant rule conditions are met so that a solution variant can be created.

By default, the following columns in the **Variants** tab act as conditions for a valid variant rule. Ensure these default conditions are met. Contact your administrator for configuring the custom validation criteria.

Columns in the Variants tab	Required values
Snapshot Rule Date Policy	Date
Snapshot Compiled Date	<Valid date>
Valid	True
Complete	True

The **Create Solution Variant** panel also displays other details for the solution variant, including its ID.

7. (Optional) Double-click the **Revision ID** column for a variant rule and select another version, if available.
If the selected version of the variant rule is valid, the **Valid for Solution Variant** column indicates **True** and you can use it for creating the solution variant.
8. Select one or more valid variant rules for creating the solution variant.
9. If required, click the **Run in Background** check box.
10. Click **Create**.
The Reuse type of solution variant is created and the following is carried forward: the absolute occurrence properties—**Position Designator**, **ID in Context**, and **Usage Address**—along with the absolute occurrence data properties for a part within the solution variant.

Create a new solution variant instead of updating an existing one for a variant

You can disable updating an existing solution variant of the type *Reuse* by specifying a status for the solution variant and create a new solution variant.

When a structure is revised multiple times and a solution variant is revised alongside, the changes to the structure might make the solution variant incompatible with the structure. In such cases, you must mark the existing solution variant as obsolete and create a new one for use instead.

Procedure

1. In the **Solution Variants** section on the **Solution Variants** tab, select the solution variant to be marked as obsolete.
2. Specify a valid release status for the solution variant item revision. You can use a workflow, a custom flow, or a business logic configured by your administrator to set the release status on the solution variant item revision.

After you set the release status, the solution variant is obsoleted. Teamcenter changes the solution variant category to **Unmanaged**, restricting any further updates to the solution variant.

You can create a new solution variant for the same configuration to replace the obsoleted solution variant.

3. In the **Solution Variants** section on the **Solution Variants** tab, select the variant for which you want to create a new solution variant.
4. Click **Create** .
5. In the message dialog box, you can choose to create the solution variant in the background or foreground by selecting or clearing the **Run in Background** check box, respectively. Click **Yes** to create the solution variant.
If you choose to create the solution variant in the foreground, the solution variant is created and displayed in the **Solution Variant** column.

If you chose to create it in the background, go to **Alerts**  to open the solution variant.

Results

For the solution variants that are obsoleted at a lower level in the hierarchy, Teamcenter automatically changes the solution variant category to **Unmanaged** and creates a replacement solution variant during the update process.

Update a Reuse solution variant

If the source product structure is modified, you can update the solution variant that is of the reuse type so that all the changes from the source product structure are propagated to the solution variant. You cannot make structural changes, such as adding, removing, or revising child items, directly to the reuse type of solution variant.

Procedure

1. (Optional) **Set an active change notice** that has the appropriate effectivity.
2. Open the structure that has variability .

Warning

Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must remain the same while creating and updating the solution variant.

3. Open the source product structure and go to the **Solution Variants** tab.
If the source structure is modified after the solution variant was created, the **Source Assembly Modified?** column displays **Yes** and indicates that the reuse type of the solution variant also needs to be updated.
4. Select the reuse type of the solution variant that needs to be updated and click **Update** .
5. In the Update Solution Variant panel, perform the following tasks:
 - a. If required, in the **Variant Rule** list, select a different available variant rule.
 - b. In the **Properties** section, review the properties.

- c. If required, in the **Revision Rule** list, select a different revision rule.

Update Solution Variant

 Close

Variant Rule
SV1033-Reuse SVR Internal 1 External 1/001;Reus... ▼

▼ Properties
ID:
SV1033-Reuse SVR Internal 1 External 1
Name:
Reuse SVR Internal 1 External 1
Description:
SVR for Reuse scenario with configuration 1 of Internal families and configuration 1 of External families
Valid:
True
Complete:
True
Snapshot Compiled Date:
14-Aug-2025 02:42
Release Status:
TCM Released
Valid for Solution Variant:
True

Latest Working (Modified) ▼

☐ Run in Background

Update

- d. Click **Update**.

The solution variant that is impacted due to the changes in the source structure is updated. If a solution variant is already released, it is first revised and then updated. If the effectivity of the active change notice completely overlaps with the effectivity of the change notice under which the source structure was updated, the impacted solution variant usage is revised and its **ECN In** is the same as the active change notice.

When you update a solution variant in the context of an active change, the updates are reflected in the change summary.

Depending on whether the automatic propagation of effectivity is set, the following happens:

- **Automatic propagation of effectivity is set**

In this case, if release effectivity is set for the active change, the solution variant occurrences inherit the release effectivity as their element effectivity. Else, the solution variant occurrences inherit the effectivity set for the source structure.

- **Automatic propagation of effectivity is not set**

In this case, the solution variant occurrences inherit the effectivity set for the source structure.

After the solution variant is updated, the absolute occurrence properties for a part within the solution variant are carried forward. The absolute occurrence properties, such as **Position Designator**, **ID in Context**, and **Usage Address**, are also transferred if your administrator has configured them to be carried forward.

Consider that a solution variant with the **Solution Variant Category** set as **Reuse** is reused for more than one variant rule, and either the variant rules or the source structure is modified in such a way that source structure will not result in the same configuration for those variant rules. In such a case, when you update the solution variant, the system splits the existing solution variant to create new solution variants as needed. This ensures an efficient reuse of the existing solution variant.

Example

The SVR(White) and SVR(Grey) variant rules share the same solution variant. The solution variant contains beige seat covers and headrests. Later, *white thread* is added to SVR(White) for the seat cover stitches. In such a situation, a new solution variant is created for SVR(White), and it contains beige seat covers, headrests, and white thread. The SVR(Grey) will still result in the existing solution variant that contained beige seat covers and headrests.

6. To verify if a solution variant is updated correctly, open it from the **Solution Variants** column.
7. If you have set an active change notice, release the active change.
 - a. Click **Changes** , select the change, and click **Release Change**.

Create a Managed solution variant

A Managed type of solution variant helps you manually review and merge updates from the source product structure, and allows independent modifications to be made to the solution variant.

Create a managed type of solution variant when you want to manually review the updates made to the product structure, and then update the solution variant. In other words, if the source product structure changes, you can compare the changes with the solution variant structure and merge the selected changes manually. You can make direct structural changes, such as adding, removing, or revising child items, to the solution variant independently.

Procedure

1. Open the structure that has variability .
2. If not already set, select the top line of the structure and set a variability scope:
 - a. In the **Details**  section, click **Edit** .
 - b. In **Variability Scope**, click **Add**  to search for and add a configurator context.
 - c. Click **Save** .

Additionally, you can set a variability scope to each variable structure at the lower levels for which solution variants are not yet created. These variability scopes must be subsets of the variability scope that you applied to the top line of the structure.

3. For creating the solution variant for the entire product, select the top line of the structure and then set the **Solution Variant Category** as **Managed**. For child assemblies, you can set the **Solution Variant Category** as **Reuse**, **Managed**, or **Unmanaged**.

If the **Solution Variant Category** column is not available by default, you must first add it.

4. Enable or disable **Show Excluded by Effectivity** and **Show Suppressed** from **Configure** , as required.
5. Click the **Solution Variants** tab.

If you do not see the **Solution Variants** tab, contact your system administrator.

6. In the **Solution Variants** section, click **Create** .

The **Create Solution Variants** panel appears and displays the variant rules that are available from the associated variability scope. The **Valid for Solution Variant** column indicates whether a solution variant can be created for a given variant rule. **Yes**  indicates that all conditions are met, while **No**  indicates that at least one condition is invalid. Ensure that all variant rule conditions are met so that a solution variant can be created.

By default, the following columns in the **Variants** tab act as conditions for a valid variant rule. Ensure these default conditions are met. Contact your administrator to configure the custom validation criteria.

Columns in the Variants tab	Required values
Snapshot Rule Date Policy	Date
Snapshot Compiled Date	<Valid date>

Columns in the Variants tab	Required values
Valid	True
Complete	True

The **Create Solution Variant** panel also displays other details for the solution variant, including its ID.

- (Optional) Double-click the **Revision ID** column for a variant rule and select another version if available.
If the selected version of the variant rule is valid, the **Valid for Solution Variant** column indicates **True** and you can use it for creating the solution variant.
- Select one or more valid variant rules for creating the solution variant, and then click **Create**.
- If required, click the **Run in Background** check box.
- Click **Create**.
The Managed type of solution variant is created and the absolute occurrence properties—**Position Designator**, **ID in Context**, and **Usage Address**—along with the absolute occurrence data properties for a part within the solution variant, are carried forward.

Update a Managed solution variant

If the source product structure is modified, you can review the changes and merge them manually with the managed solution variant. You can also make structural changes directly to the managed type of solution variant, such as adding, removing, or revising child items.

Procedure

- (Optional) **Set an active change notice** that has the appropriate effectivity.
- Open the structure that has variability .

Warning

Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must remain the same while creating and updating the solution variant.

- (Optional) If you want to make structural changes directly to the managed solution variant, in the **Solution Variants** tab, click **Open**  to open a managed solution variant. Then, add, remove, or revise its child items.
- Open the source product structure and go to the **Solution Variants** tab.

If the source structure is modified after the solution variant was created, the **Source Assembly Modified?** column displays **Yes** and indicates that the managed solution variant also needs to be updated.

Overview Weight and Balance Cost Parameters Changes Finishes Partners Classification Baselines <u>Solution Variants</u> >					
▼ Solution Variants					
			⊕ Create ↻ Update ✂ ✓ Select ▼		
Solution Variant ↕	Variant Rule ↕	Valid for Solution Variant ↕	Category ↕	Source Assembly Modified? ↕	Creation Date ⚙
Test_SV1037/A;1-C...	SV1033-Classic Mod...	True	Managed	Yes	16-Oct-2025

5. Select the managed type of the solution variant that needs to be updated and click **Update**
6. In the Update Solution Variant panel, perform the following tasks.
 - a. If required, in the **Variant Rule** list, select a different, available variant rule.
 - b. In the **Properties** section, review the properties.
 - c. If required, in the **Revision Rule** list, select a different revision rule.
 - d. In the **Compare and Merge** section, compare the changes in the source structure and the solution variant structure.

The first column displays the items that are added, revised, or removed in the source structure. The second column displays the items that are added, revised, or removed in the selected solution variant structure. The blank cell in the column indicates that the respective structure does not contain the item.

The following example shows that the **Test_034517/A** item is added to the source structure and the **Test_034519-Bumper** item is directly added to the managed solution variant.

Update Solution Variant

Close

▼ Variant Rule

SV1033-Classic Model/001;Classic Model

► Properties

▼ Revision Rule

Latest Working (Modified)

▼ Compare and Merge

✓ Select All

➤ Merge

↶ Undo

	Test_SV1001/A;1-Car wit...	Test_SV1037/A;1-Car wit...	
☐	Test_034517/A;1-Basic w...		
☐		Test_034519-Bumper	

Update

e. Select the required rows in the **Compare and Merge** section and click **Merge** ➤.

Click **Undo** ↶ to remove the last action.

The following example shows that the **Test_034517/A** item is to be added to the managed solution variant.

▼ Compare and Merge

✓ Select All

➤ Merge

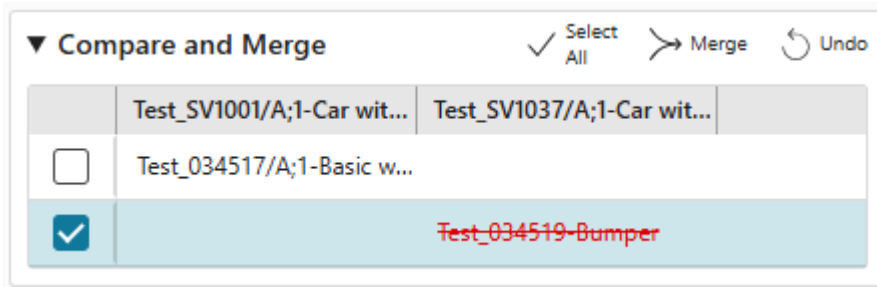
↶ Undo

	Test_SV1001/A;1-Car wit...	Test_SV1037/A;1-Car wit...	
☒	Test_034517/A;1-Basic w...	Test_034517/A;1-Basic w...	
☐		Test_034519-Bumper	

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The following example shows that the **Test_034519-Bumper** item is to be removed from the managed solution variant.



- f. Click **Update**.
The managed type of solution variant is updated.
After the solution variant is updated, the absolute occurrence properties for a part within the solution variant are carried forward. The absolute occurrence properties, that is, **Position Designator**, **ID in Context**, and **Usage Address**, are also transferred if your administrator has configured them to be carried forward.
7. To verify if a solution variant is updated correctly, open it from the **Solution Variants** column.
8. If you have set an active change notice, release the active change.
 - a. Click **Changes** , select the change, and click **Release Change**.

Create an Unmanaged solution variant

An Unmanaged type of solution variant helps you independently modify its structure content without synchronizing it with the source product.

Create an Unmanaged solution variant to link it to the source product structure without synchronizing its updates. This means that the changes made to the source product structure are not propagated to the Unmanaged solution variant. You can make direct structural changes, such as adding, removing, or revising child items, to the solution variant independently.

Procedure

1. Open the structure that has variability .
2. If not already set, select the top line of the structure and set a variability scope:
 - a. In the **Details**  section, click **Edit** .
 - b. In **Variability Scope**, click **Add**  to search for and add a configurator context.
 - c. Click **Save** .

Additionally, you can set a variability scope to each variable structure at the lower levels for which solution variants are not yet created. These variability scopes must be subsets of the variability scope that you applied to the topmost line of the structure.

- To create the solution variant for the entire product, select the top line of the structure and then set **Solution Variant Category** as **Unmanaged**. For child assemblies, you can set the **Solution Variant Category** as **Reuse**, **Managed**, or **Unmanaged**.

If the **Solution Variant Category** column is not available by default, you must first add it.

- Click **Configure**  > **Show Excluded by Effectivity** and **Configure**  > **Show Suppressed**, as required.
- Click the **Solution Variants** tab.

If you do not see the **Solution Variants** tab, contact your system administrator.

- In the **Solution Variants** section, click **Create** .

The **Create Solution Variant** panel appears and displays a list of the variant rules that are available from the associated variability scope. The **Valid for Solution Variant** column indicates whether a solution variant can be created for a given variant rule. **Yes**  indicates that all conditions are met, while **No**  indicates that at least one condition is invalid. Ensure that all variant rule conditions are met so that a solution variant can be created.

By default, the following columns in the **Variants** tab act as conditions for a valid variant rule. Ensure these default conditions are met. Contact your administrator for configuring the custom validation criteria.

Columns in the Variants tab	Required values
Snapshot Rule Date Policy	Date
Snapshot Compiled Date	<Valid date>
Valid	True
Complete	True

The **Create Solution Variant** panel also displays other details for the solution variant, including its ID.

- (Optional) Double-click the **Revision ID** column for a variant rule, and select another version if available.
If the selected version of the variant rule is valid, the **Valid for Solution Variant** column indicates **true** and you can use it for creating the solution variant.
- Select one or more valid variant rules for creating the solution variant and click **Create**.
- If required, click the **Run in Background** check box.
- Click **Create**.
The Unmanaged type of solution variant is created and the absolute occurrence properties—**Position Designator**, **ID in Context**, and **Usage Address**—along with the absolute occurrence data properties for a part within the solution variant are carried forward.

Update an Unmanaged solution variant

You must make direct structural changes, such as adding, removing, or revising child items, to the Unmanaged solution variant independently. The changes made to the source product structure are not propagated to the Unmanaged solution variant.

Procedure

1. (Optional) **Set an active change notice** that has the appropriate effectivity.
2. Open the structure that has variability .

Warning

Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must remain the same while creating and updating the solution variant.

3. In the **Solution Variants** tab, click  to open an Unmanaged solution variant, and then add, remove, or revise its child items.
The solution variant that is impacted due to the changes in the source structure is updated. If a solution variant is already released, it is first revised and then updated. If the effectivity of the active change notice completely overlaps with the effectivity of the change notice under which the source structure was updated, the impacted solution variant usage is revised and its **ECN In** is the same as the active change notice.

When you update a solution variant in the context of an active change, the updates are reflected in the change summary.

Depending on whether the automatic propagation of effectivity is set, the following happens:

- **Automatic propagation of effectivity is set**

In this case, if release effectivity is set for the active change, the solution variant occurrences inherit the release effectivity as their element effectivity. Else, the solution variant occurrences inherit the effectivity set for the source structure.

- **Automatic propagation of effectivity is not set**

In this case, the solution variant occurrences inherit the effectivity set for the source structure.

After the solution variant is updated, the absolute occurrence properties for a part within the solution variant are carried forward. The absolute occurrence properties, that is, the **Position Designator**, the **ID in Context**, and the **Usage Address**, are also transferred if your administrator has configured them to be carried forward.

4. If you have set an active change notice, release the active change.

To do so, click **Changes** , select the change, and click **Release Change**.

Update solution variants for variability changes

When solution variants with the **Solution Variant Category** as **Reuse** and **Managed** are created for variant rules and the variant criteria is modified, you can propagate the new variability to the solution variants. This operation updates the entire hierarchy of the solution variants as applicable.

If the parent structure contains sub-assemblies that are associated with their respective configurator contexts, the variant rules that are associated with these configurator contexts must also be updated based on the modified variant rule of the parent structure.

Example

Consider the Front Door structure, which includes multiple solution variants for mirror assemblies. Based on business requirements, you modify the variant rule for the VX variant and the Front Door structure (150% BOM) by adding a power-folding mirror with a turn signal feature. Following this modification, you assess the impacted solution variants and update them accordingly. The VX mirror assembly variant is updated to include the power-folding mirror with the turn signal feature.

Procedure

1. (Optional) **Set an active change notice** that has the appropriate effectivity.
2. Ensure that the variant rule for the configurator context that is associated with an assembly in the structure is already updated.

When updating the variant rule, ensure that it is in the expanded state and validated and saved.

3. Ensure that the variant rules for the configurator contexts that are associated with the parent structure (150% BOM) and its subassemblies are also already updated.
4. Open the structure that has the impacted solution variants.

Warning

Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must be the same while creating and updating the solution variant.

Because the variant rule is modified after the solution variants are created, the **Variant Rule Modified** column in the **Solution Variants** tab shows **Yes** for the solution variants. This indicates that the solution variants need to be updated to include the latest changes made to the variant rules.

5. Select the solution variant and click **Update** .
The selected solution variant and the parent structure are updated.

Create solution variants for a product through a workflow

You can create solution variants for a product structure using a workflow, and you use the variant rules for creating the solution variants.

Procedure

1. Open the source structure.

Warning

Do not change the state of the **Show Excluded by Effectivity** and **Show Suppressed** toggles. The state of these toggles must remain unchanged while creating and updating the solution variant.

2. Set the **Solution Variant Category** for the source product structure and its sub-assemblies.
You can select **Reuse**, **Managed**, or **Unmanaged**.
3. Click **More Commands** *** > **Manage**  > **Submit to Workflow** .
4. In the **Submit to Workflow** panel, select **Create Solution Variants** in the **Template** list.
5. In the **Targets** section, ensure that the source structure is already selected.
6. (Optional) In the **Targets** section, click **Add** to add an engineering change.
7. In the **References** section, click **Add** to add variant rules.
If a variant rule contains multiple revisions, you can choose a revision for which the solution variant is to be created.
8. Click **Submit**.

What to do next

To verify if the solution variants are created correctly, go to the **Solution Variants** tab and open the required solution variants to view them.

Update the solution variants of a product through a workflow

When a product structure changes, you update its Reuse type of solution variants using the workflow process. When you update a solution variant, you can manage revisions and new elements, and the specific occurrence properties that must be updated in the solution variants.

When a structure is updated, you must also update the corresponding solution variants. Consider the following scenarios before starting the workflow.

Procedure

1. Open the updated source structure.

Warning

Do not change the state of the **Show Excluded by Effectivity** and **Show Suppressed** toggles. The state of these toggles must remain unchanged while creating and updating the solution variant.

2. Click **More Commands** *** > **Manage**  > **Submit to Workflow** .
3. In the **Submit to Workflow** panel, select one of the following templates in the **Template** list:
 - **Update Solution Variants**
Select this template to update all the solution variants of the structure that are impacted due to the structure change.
 - **Update Solution Variants 2**
Select this template, and then specify one or more solution variants in the target to update them.

Alternatively, specify a solution variant in the target and a different revision of the variant rule in the reference. The solution variant is updated with the revised variant rule. In this case, specify only one solution variant in the target.

If you also specify an engineering change with an effectivity in the target, the solution variants with the effectivity mentioned in the engineering change are updated.

The update is applied to all the solution variants that are impacted due to the structure change. If a solution variant is already released, it is revised and then updated. If the effectivity of the active change notice completely overlaps with the effectivity of the change notice under which the source structure was updated, the impacted solution variant usage is revised and its **ECN In** is the same as the active change notice.

For a newly added element in the structure, Teamcenter checks if the **Solution Variant Category** of the element is marked as **Reuse**. If yes, Teamcenter checks if a solution variant is already created for the element and reuses this solution variant in the updated solution variant structure. If a solution variant is not available, Teamcenter creates a new solution variant for the element and reuses this new variant in the updated structure.

Note

Currently, only changes to **Quantity**, **All Notes** (occurrence notes), **Absolute Transform Matrix**, and **Occurrence Effectivity** in a structure are updated in the corresponding solution variants.

4. Click **Submit**.

What to do next

To verify if the solution variants are updated correctly, go to the **Solution Variants** tab and open the required solution variants to see the updates.

After the solution variant is updated, if your administrator has set the preference with the required values, the absolute occurrence properties for a part within the solution variant are updated. These properties include **Position Designator**, **ID in Context**, and **Usage Address**.

View solution variants

You can view a list of the solution variants that are available for a product structure in the **Solution Variants** tab. You can also open a solution variant and view its content.

Procedure

1. Open the structure for which you created solution variants.
2. Go to the **Solution Variants** tab.

The **Solution Variants** section lists the solution variants that were created for the selected structure.

Solution variants for a configurable product, assembly, or component are displayed in the table in the **Solution Variants** section, depending on the fully qualified match of the Saved Variant Rule (SVR) recipe of the top-level product or assembly. The solution variants that do not match the top-level product SVRs are displayed with an empty value in the **Variant Rule** column.

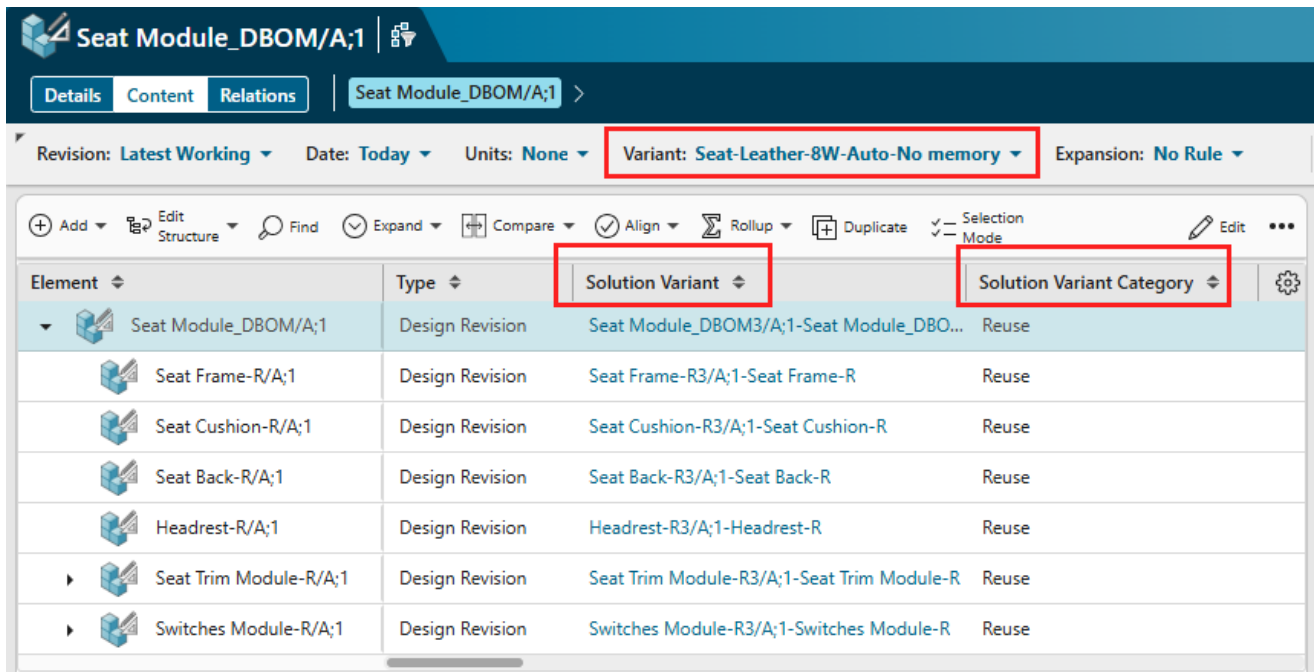
3. To view the content of the solution variant, in the **Solution Variants** tab, select a solution variant and click **Open** . The selected solution variant opens and you can view its content.

View Reused solution variants

You can view Reused solution variants directly in the structure. This information is helpful for manufacturing engineers and process planners, ensuring accurate variant information is available for MBOM and BOP.

Procedure

1. Open the structure for which you already created the reused type of solution variants.
2. In the **Variant** list located in the header, select a variant rule.
The **Solution Variant** column displays the reused type of solution variant that is associated with the selected variant rule. The following screenshot displays an example of the Reused solution variants displayed in the **Solution Variant** column.



Calculate BOM properties dynamically using the variant configuration

You can set the BOM properties based on Product Configurator feature selections so that their values are calculated dynamically by applying the variant configuration. You can then create a solution variant using these BOM properties. The values that are calculated dynamically persist while creating a solution variant.

By setting the Product Configurator feature on BOM properties, you ensure that the product structure is automatically enriched with the variant configuration-specific details that are critical for downstream processes, without manual intervention. This ensures accuracy, reduces errors, and streamlines workflows across the product lifecycle.

Example

When configuring a product like an industrial pump, selecting a specific motor type, such as *Explosion-Proof*, might require unique assembly instructions or special fasteners. By mapping the *Motor_Type* feature to the motor's BOM line properties, you can automatically populate these critical details. This ensures that manufacturing receives precise, configuration-specific guidance for every pump variant, reducing errors and ensuring compliance.

Prerequisites

The administrator has configured the occurrence property that can be used for mapping the BOM property with the configurator feature.

Procedure

1. Open a product structure that has the variability scope associated with it.
2. Understand which the occurrence property to use for mapping the BOM line property with the configurator feature.
3. Arrange the columns of the work area so that the occurrence property column is visible.
4. Create the mapping for the BOM property for which you want to set the properties:

- a. Click **Edit**  to edit the occurrence property column.

For example, **AIE Occurrence Name**.

- b. In the occurrence property column, set the value for the BOM line property.

The BOM properties have display names or system names, and the system names are in the `bl_XXX` format. You must use the system names for the property. For example, if you choose the **AIE Occurrence Note** column to store the mapping and you want to set the value of the Motor Type family on the BOM line property, then enter the mapping in the `<bl_occurrence_note=[Teamcenter]Motor_Type>` format.

Tip

The mapping is in the `<bl_property_name=[Namespace]Feature>` format. If the mapping is not defined in this format, the mapping is not set.

- c. Click **Save Edits**  to save the mapping information.

5. Click **Variant Configuration**, select the feature, and click **Apply Configuration** .

Results

The BOM properties are set based on Product Configurator feature selections so that their values are calculated dynamically using the variant configuration. These values do not persisted on the original product structure.

What to do next

After setting the BOM properties to calculate their values dynamically by applying the variant configuration, you can [create a solution variant](#).

11. Specify alternates and substitutes for parts

About global alternates and substitutes

Understand the difference between global alternates and substitutes in Teamcenter, including their definitions, examples, and key characteristics, to effectively manage interchangeable parts within product structures.

Parts are the different components that make up a product. In certain cases, one part can be interchanged with another part. There are two types of interchangeable parts, namely, global alternates (also known as alternates) and substitute components (also known as substitutes).

Alternates

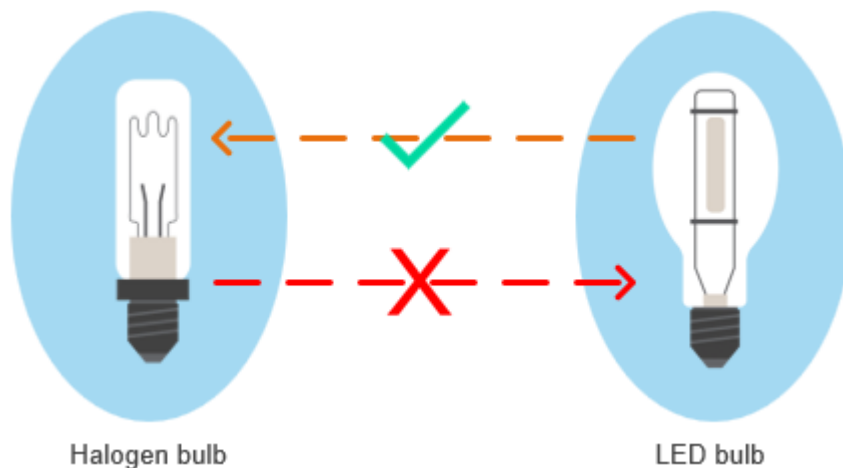
An alternate is a part that can be interchanged with another part in all circumstances, irrespective of where the other part is used in a product structure. Alternates are not specific to a variant or a product. You use alternates to specify that a part is interchangeable in any assembly.

Example

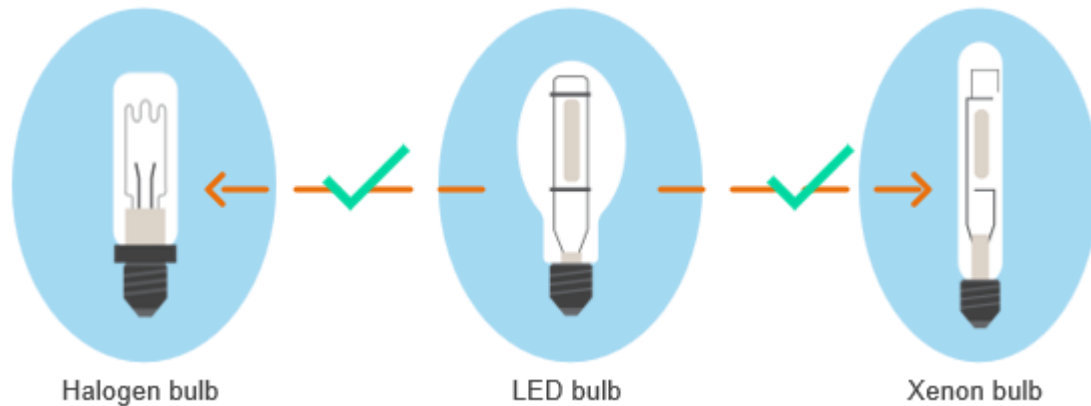
Consider a Halogen bulb that is used in several assemblies across products. You can specify an LED bulb as an alternate for the Halogen bulb. This means that in all assemblies and products, in place of the Halogen bulb, the LED bulb can be used.

Note the following important points about alternates:

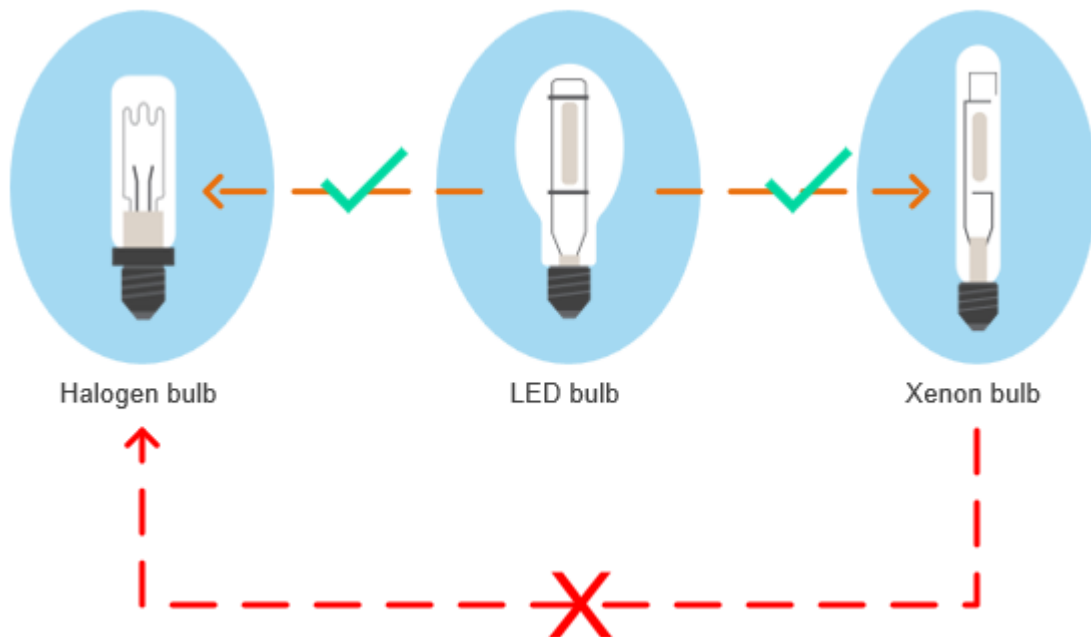
- If you reuse a part in some other assembly, the alternates of the part are carried over.
- Parts and their alternates are related in one way only. For example, the LED bulb is an alternate of the Halogen bulb. However, the Halogen bulb is not an alternate of the LED bulb.



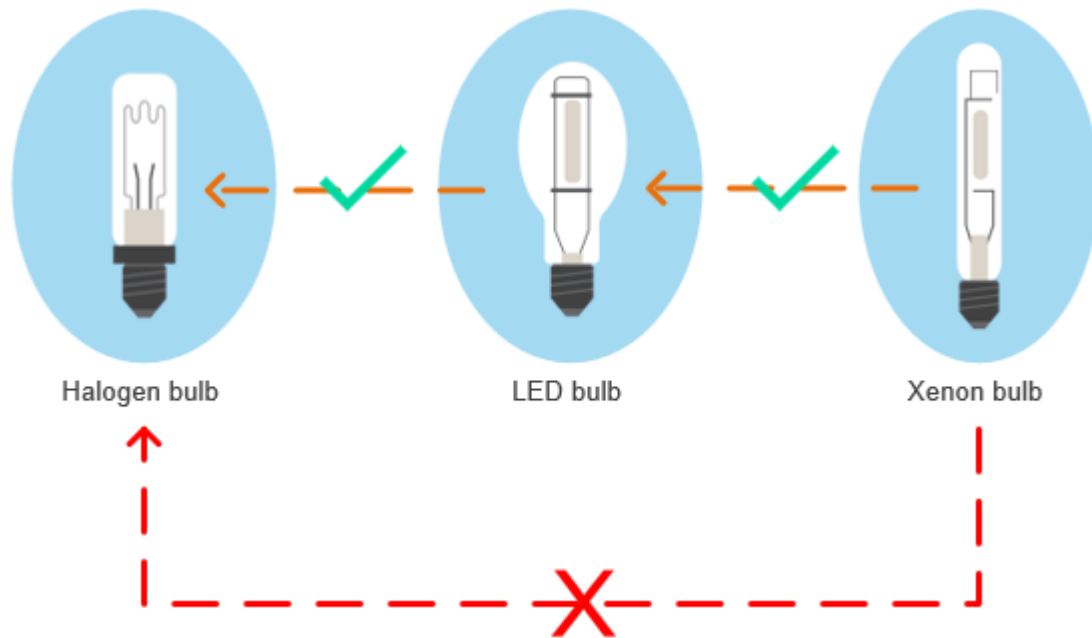
- One part can be an alternate of multiple other parts. For example, the LED bulb is an alternate of the Halogen bulb and of the Xenon bulb.



- Alternate relationships are not shared. For example, the Xenon bulb is not an alternate of the Halogen bulb, even though the LED bulb is an alternate of both bulbs.



- Alternate relationships are not chained. For example, the LED bulb is an alternate of the Halogen bulb. And the Xenon bulb is an alternate of the LED bulb. However, the Xenon bulb is not an alternate of the Halogen bulb.



Substitutes

Substitutes are parts that can be interchanged only with a particular occurrence of a component within a single assembly. You define a substitute for a single occurrence in an assembly and not for an item. You use substitutes to specify that a part is interchangeable only in a specific assembly.

Example

Consider that Crosskart has a lighting assembly that uses the Halogen bulb, by default. You set the Xenon bulb as a substitute for the Halogen bulb within the lighting assembly. A substitute can be used in place of another part but only in a specific assembly. So, you cannot use the Xenon bulb in place of the Halogen bulb in another assembly.

Note the following important points about substitutes:

- You define a substitute for a single occurrence in an assembly and not for an item.
- You can add multiple substitutes for an occurrence of a part.
- You can replace a part in an assembly with a substitute.
- All substitutes of one occurrence share the same occurrence attributes. The occurrence attributes can be a find number, quantity, and notes.

Set global alternates for a part

An *alternate part* is a part that can be used in place of another part in any product. You can set and manage global alternates in Teamcenter, including adding new alternates, selecting from existing ones, and removing alternates for a part, to define interchangeable components across all product structures.

Procedure

1. Select the part for which you want to set a global alternate.
2. If you want to set alternates in the markup mode, choose **More Commands *** > Markup**  to switch to the markup mode.

Caution
If you are working in the markup mode, you must **apply or cancel the markups** to complete the updates.

3. Click the **Overview** tab.
The alternates available for a part are listed under the **Global Alternates** section.

Add an alternate

Procedure

1. In the **Global Alternates** section, click **Add Alternate** .
2. In the **Add Alternate** panel, click one of the following tabs:

Tab	Description
New	On this tab, you can add a new occurrence.
Palette	On this tab, you can paste an occurrence from the clipboard or select one from your Favorites or Recent list.
Search	On this tab, you can search for an occurrence to add.

3. In the **Add Alternate** panel, select the part that you want to add as a global alternate. You can select multiple parts.
4. Click **Add**.
The selected parts are added to the **Global Alternates** list.

Remove an alternate

Procedure

1. From the **Global Alternates** list, select the alternate that you want to remove. You can select multiple alternates.
2. Click **Remove** .

The selected parts are removed from the **Global Alternates** list.

Set substitutes for a part in a structure

A *substitute part* is a part that can be used in place of another part in a specific product configuration. In other words, substitutes are specific to an assembly. You can set, remove, and replace parts with substitutes in a structure.

In a structure, you can add a substitute for a part, remove a substitute, and replace an element with a substitute.

You can set a substitute for only those parts whose **Assembly Indicator** is not set as **Generic Part** and **Configurable Assembly**.

Procedure

1. Select the part for which you want to set a substitute.
2. If you want to set substitutes in the markup mode, choose **More Commands**  > **Markup**  to switch to the markup mode.

Caution

If you are working in the markup mode, you must **apply or cancel the markups** to complete the updates.

3. Click the **Overview** tab.

The substitutes available for a part are listed under the **Substitutes** section.

Add a substitute

Procedure

1. In the **Substitutes** section, click **Add Substitute** .
2. In the **Add Substitute** panel, click one of the following tabs:

Tab	Description
New	On this tab, you can add a new occurrence.
Palette	On this tab, you can paste an occurrence from the clipboard or select one from your Favorites or Recent list.
Search	On this tab, you can search for an occurrence to add.

3. In the **Add Substitute** panel, select the part that you want to add as a substitute. You can select multiple parts.
4. Click **Add**.

The selected parts are added to the **Substitutes** list.

Remove a substitute

Procedure

1. From the **Substitutes** list, select the substitute that you want to remove. You can select multiple substitutes.
2. Click **Remove** .

The selected parts are removed from the **Substitutes** list.

Replace an element with a substitute

Procedure

1. From the **Substitutes** list, select the substitute that you want to use in place of an element.
2. Click **Use** .

In the structure, the element is replaced with the selected part. The element that is replaced in the structure is added to the **Substitutes** list.

Add and manage occurrence substitutes for a part in a structure

A *substitute part* is a part that can be used in place of another part in a specific product configuration. In a structure, you can add a substitute for a part, remove a substitute, and set a substitute as primary item. If you add and manage occurrence substitutes in an active change notice, the changes are indicated by redlining.

Restrictions and limitations

You can add and manage occurrence substitutes for a part in a structure if your Teamcenter setup includes the BOM Structure Management license.

Procedure

1. Select the part for which you want to add a substitute.
2. Click **Add**  > **Substitute** .

Note

You can use the **Substitutes** section in the **Overview** tab to add a substitute only when at least one substitute is already added for the item.

3. In the **Add Substitute** panel, click one of the following tabs:

Tab	Description
New	On this tab, you can add a new occurrence.
Palette	On this tab, you can paste an occurrence from the clipboard or select one from your Favorites or Recent list.
Search	On this tab, you can search for an occurrence to add.

4. In the **Add Substitute** panel, select the part that you want to add as a substitute. You can select multiple parts.
5. Click **Add**.
The selected occurrence of the parts are added to the structure. The **Substitute Indicator** column displays  for the primary part and  for the substitutes. The **Substitutes column** shows the substitute name for the primary item. The occurrence attributes of the primary part, including find number, are also set for the substitutes. The **Substitutes** section of the **Overview** tab for the selected primary part displays the information for the substitutes.

Click **Configuration Settings**  > **Show Substitute Components** to show or hide the substitutes in the structure.

If you copy or cut a primary part to the same structure or to a different structure, all occurrences of the substitutes along with the attribute information are also copied. If you insert a level on the primary part, all substitutes are also moved along with the primary part. If you remove a primary part from the structure, all its substitutes are also removed.

Remove a substitute

If a substitute is no more a valid substitute for any reason, you can remove it.

Procedure

1. Select the substitute that you want to remove and click **Edit Structure**  > **Remove** .
2. Click **Remove** on the confirmation message.

The selected substitute along with its information is removed from the structure.

Set a substitute as a primary part

At times, you might want to mark a substitute as a primary part in the structure. After you mark the substitute as a primary part, the original primary part becomes substitute for it.

Procedure

1. Select a substitute that you want to set as primary part in the structure.
If you select multiple substitutes, the last substitute that you selected is chosen as primary.
2. Right-click and select **Set Primary** .
The selected substitute is set as a primary part and the primary part is set as a substitute in the structure.

Add and manage a substitute group for parts in a structure

When two or more parts in a structure have the same set of substitutes, you can define these substitutes by using a substitute group. For example, a structure contains different caps for the 12V, USB-A, and USB-C charging ports in the car and you want to define similar substitute items for them. In such a situation, you can select these caps as the primary items in the structure, create a substitute group, and then define the substitute parts. If you add and manage occurrence-based substitute groups in an active change notice, the changes are indicated by redlining.

A substitute group can contain multiple sets of substitutes. After a substitute group is created, you can add or remove primary parts, substitutes, and substitute sets to or from it. In addition, you can map a primary part to a substitute part in a substitute group.

Restrictions and limitations

You can add and manage a substitute group for parts in a structure only if your Teamcenter setup includes the BOM Structure Management license.

Procedure

1. Select two or more parts in the structure for which you want to add a substitute group.
2. Click **Add**  > **Substitute Group** .
3. In the **Add Substitute Group** panel, ensure that the **Occurrence Substitute Group** type is selected, and then perform the following tasks:
 - a. In the **Properties** section, specify a name and description for the substitute group.
 - b. In the **Primary Elements** section, ensure that the required parts are listed.
 - c. In the **Substitutes** section, specify a name for **Substitute Set ID** and then click **Add**  to add substitute elements.

Tip

Using the **Add Substitute Elements** panel, you can now **add one or more substitute elements**. All the substitutes that you add become a part of the substitute set.

After you click **Add** ⊕,

For example, in the first substitute set, you can add three caps as substitute parts for the 12V, USB-A, and USB-C charging ports.

- d. To add another substitute set, in the **Substitutes** section, click **Add** ⊕.

You can add multiple substitute sets.

4. Click **Create**.
All the substitutes that you selected are added to the structure. The **Substitute Indicator** column displays  for the primary parts and  for the substitutes. The **Substitutes column** shows the substitute names for the primary parts. The occurrence attributes of the primary part, including find number, are set for the substitutes. The **Substitutes** section of the **Overview** tab for the primary part displays the primary parts, the substitute sets, and the substitutes.
5. To remove a substitute, substitute set, or a primary part, select the item in the **Substitutes** section, and click **Remove** ⊖.

Add a primary part to the substitute group

You can add another part from the structure as a primary part to the existing substitute group. For example, suppose that the USB-B port is added to the newer model of the car. In such a situation, you might want to add the cap for the USB-B port as a primary part to an existing substitute group.

Procedure

1. In the **Substitutes** section, select the **Elements** node, and click **Add Elements** ⊕.
The **Add Elements** panel appears and displays the remaining parts from the structure.
2. Select a part and click **Add**.
The selected part is added as a primary part to the substitute group and the information about the substitutes is also added for it.

Add a substitute set to the substitute group

You might want to add another set of items as substitutes for the primary parts. In such a situation, add a substitute set to an existing substitute group.

Procedure

1. In the **Substitutes** section, select the substitute group, and click **Add Substitute** .
A substitute set is added.
2. Select the newly added substitute set and click **Add Substitute** .
3. In the **Add Substitute Elements** panel, [search for an existing substitute or add a new substitute](#) and click **Add**.
The substitute is added to the selected substitute set.

Map a substitute to a primary part

You can map a substitute to a primary part in the group. For example, map the newly added cap as a substitute for the primary cap for USB-C port.

Procedure

In the **Substitutes** section, select a primary part and a substitute, and then click **Map** .

In the **Substitutes** section, the **Mapping** column displays the name of the primary part.

12. Manage structure effectivity

About structure effectivity

A product structure goes through many changes during the evolution of its product definition. Using structure effectivity, you can capture how the product structure has evolved over a period. You can use element and release effectivity to capture the evolution of product structures over time, manage product variability, and configure different revisions or elements based on date or unit ranges.

An effectivity can be of the following types:

Element effectivity	It denotes from which date or for which unit (and end item) an element is effective.
Release effectivity	It denotes from which date or for which unit (and end item) an element revision is effective. It is necessary to have a release status associated with an element to author release effectivity on it.

Release effectivity helps in configuring different revisions of the same element, wherein with the help of element effectivity you can decide if to show the element itself or not inside the structure.

For example, in an engine cooling block, revision A has 5 cooling fins and in revision B it is changed to 6, now to decide which revision to use, release effectivity is used. Now suppose we decide to use another engine block itself which is liquid-cooled, to decide between the air-cooled or liquid-cooled engine block element effectivity is used.

An effectivity can be expressed as a date range or a range of units or both. When you edit the effectivity range for one occurrence, the change is applied to all occurrences.

If an element does not have an associated effectivity object, it is assumed to be always effective. It is not constrained by any effectivity.

Effectivity can be used for the following purposes:

You can use **group effectivity** to combine multiple end items and range of units for each end item or multiple date ranges to configure multiple occurrences at the same time which have different effectivities on them.

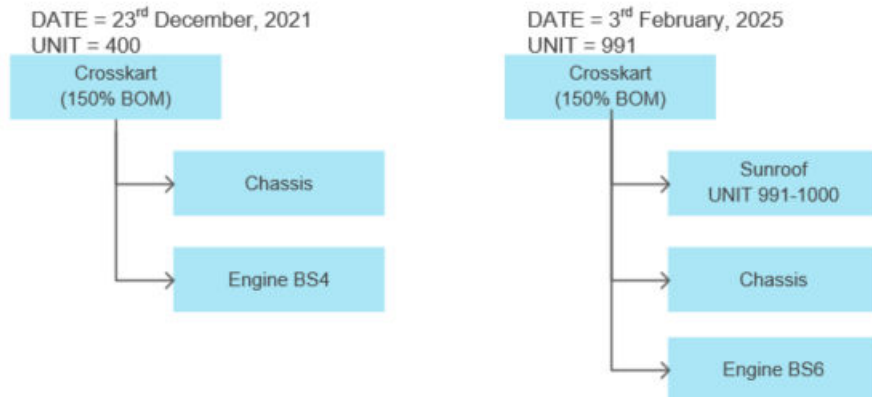
Note

Videos are not included in PDF. To access videos, use HTML.

- To reflect changes to the structure over time when new parts replace old ones.

For example, due to an engineering change, from 1st January 2025 onwards, Engine BS6 must be used instead of Engine BS4. For this, you specify date effectivity on both the engines.

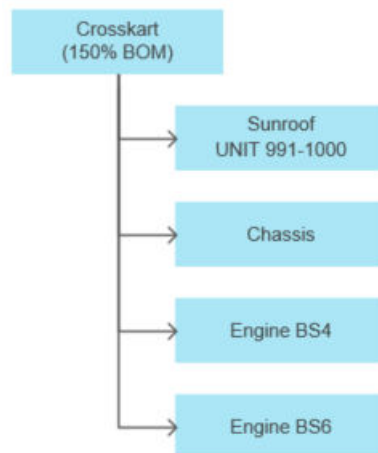
When you configure Crosskart on 23rd December 2021, Engine BS4 is used. However, when you configure Crosskart on 3rd February 2025, Engine BS6 is used.



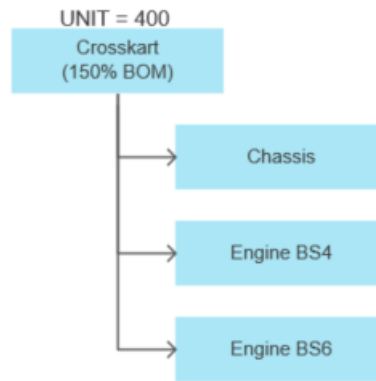
- To state the content of a unit or range of units as a means of managing the variability of the product.

For example, consider that your company manufactures a product, Crosskart. As per the company's business requirement, a total of 1000 units of Crosskart must be manufactured.

However, 10 customers want Crosskart with Sunroof. Due to such special customer requests, there is a slight deviation from the standard configuration of the product. It is inconvenient to maintain unique identifiers for several such small deviations. But with occurrence effectivity, it is easier to accommodate such deviations as the existing structure can be modified to accommodate the changes.



If you specify a unit between 1 and 990, or a range of units between 1 and 990, Sunroof is not included in the configured structure. If you specify a value between 991 and 1000, Sunroof is included.



Add or modify an element effectivity

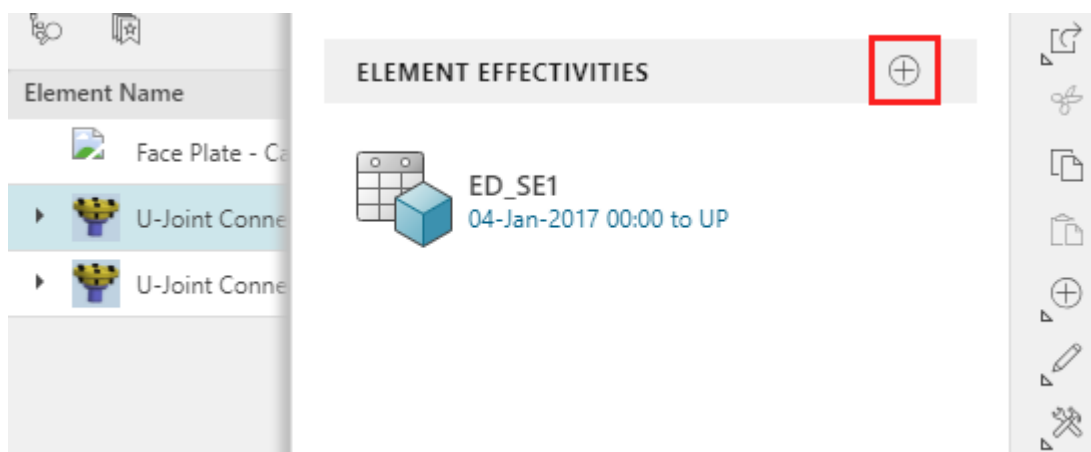
You can add and modify element effectivities using date ranges or unit numbers. You can also create new shared effectivities, and understand how effectivity changes can impact the element visibility.

You can **create** an element effectivity using a range of dates, a unit number or a range of units. You can also **update** an existing element effectivity.

Add an element effectivity

Procedure

1. Select one or more elements and click **Add** ⊕ > **Element Effectivity**.
2. In the **Element Effectivity** panel, click **Add Effectivities** ⊕.



3. You can add multiple effectivities to the selected element.

Note

Add a new effectivity or locate existing effectivities from the **Search** tab.

To author a new date effectivity:

- a. In the **New** tab, select **Date**.
- b. (Optional) Select the **Share** check box to create a shared effectivity.
The **Name** field is displayed only for shared effectivities, so that users can search for the effectivity by name.
- c. Select the **Start** date from the calendar.
- d. Select the **End** date from the calendar. If applicable, you can select **UP (all future dates)** or **SO (stock out)**.
- e. (Optional) Click **Replace** $\leftrightarrow$ to add a new **End Item**, or search for an existing **End Item**.
- f. (Optional) Select the **Protect** check box if you do not want the effectivity to be edited.
- g. Click **Add** to create the effectivity.

The element effectivity is applied to the selected element. If the applied **Revision Rule** or **Date** effectivity is not applicable to this element, then it is excluded from the structure. Select the **Show Excluded By Effectivity** toggle to show this element.

Note

You cannot author multiple date ranges inside a single date effectivity on a single occurrence.

OR

To author a new unit effectivity:

- a. In the **New** tab, select **Unit**.
- b. (Optional) Select the **Share** check box to create a shared effectivity.
The **Name** field is displayed only for shared effectivities, so that users can search for the effectivity by name.
- c. Specify the desired unit or a range of units in the **Unit** field.
- d. Click **Replace** $\leftrightarrow$ to add a new **End Item**, or search for an existing **End Item**.
- e. (Optional) Select the **Protect** check box if you do not want the effectivity to be edited.
- f. Click **Add** to create the effectivity.

The element effectivity is applied to the selected element. If the applied **Revision Rule** or **Unit** effectivity is not applicable to this element, then it is excluded from the structure. Select the **Show Excluded By Effectivity** toggle to show this element.

Note

You cannot edit multiple date ranges inside a single date effectivity on a single occurrence.

Edit or remove an element effectivity

- To update an element effectivity, select it and click **Edit** .
- To remove an element effectivity, select it and click **Remove** .

After the element effectivity is updated for an element, if the applied **Revision Rule** or *effectivity criteria* is not applicable to this element, then it is excluded from the structure. Select the **Show Excluded By Effectivity** toggle to show this element.

Add or modify release effectivity

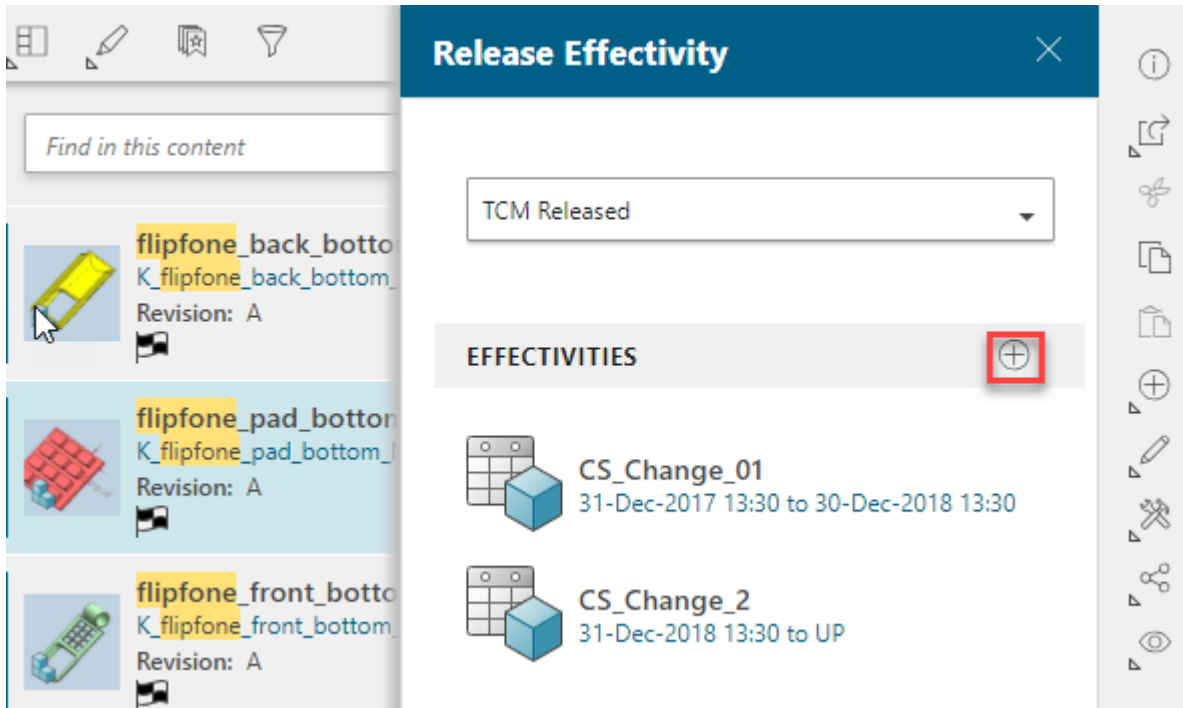
You can add and modify release effectivities using date ranges or unit numbers for elements with an associated release status. You can create new shared effectivities and understand how these changes impact the element configuration.

You can **create** a release effectivity using a range of dates, a unit number or a range of units. You can also **update** existing release effectivity.

Add release effectivity

Procedure

1. Select the element (having some release status associated with it) from the structure and click **Add**  **> Release Effectivity**.
2. You can also search the element (having some release status associated with it) and click **Manage**  **> Release Effectivity**.
3. In the **Release Effectivity** panel, select the appropriate release status.
4. In the **Release Effectivity** panel, click **Add Effectivities** .



5. Add a new effectivity or from the **Search** tab locate existing effectivities. You can add multiple effectivities to the selected element.

To specify the dates for release effectivity:

- a. In the **New** tab, select **Date**.
- b. Select the **Share** check box to create a shared effectivity. Specify the effectivity **Name**.
The **Name** field is displayed only for shared effectivities. You can search for the effectivity by name.
- c. Select the **Start** date from the calendar.
- d. Select the **End** date from the calendar. If applicable, you can select **UP (all future dates)** or **SO (stock out)**.
- e. (Optional) Click the **Replace** $\leftrightarrow$ icon to add a new **End Item** or search for an existing **End Item**.
- f. Select the **Protect** check box if you do not want the effectivity to be edited
- g. Click **Add** to create the effectivity.

After release effectivity is added, the occurrence is configured based on the current revision rules.

OR

To specify units for release effectivity:

- a. In the **New** tab, select **Unit**.
- b. Select the **Share** check box to create a shared effectivity. Specify the effectivity **Name**.

The **Name** field is displayed only for shared effectivities. You can search for the effectivity by name.

- c. Specify the desired unit or a range of units in the **Unit** field.
- d. Click the **Replace**  icon to add a new **End Item** or search for an existing **End Item**.
- e. Select the **Protect** check box if you do not want the effectivity to be edited
- f. Click **Add** to create the effectivity.

After the release effectivity is added, the element is configured based on the current revision rules.

Edit or remove the release effectivity

To update the newly added release effectivity, select it and click **Edit** .

To remove the release effectivity, select it and click **Remove** .

After release effectivity is updated, the part revision is configured based on the current revision rules.

Configure structure using group effectivity

You can configure product structures using group effectivity, which combines multiple end items, unit ranges, or date ranges to simultaneously manage the effectivity of various occurrences. You can also add, modify, or remove these group effectivities.

A group effectivity is a combination of multiple end items and range of units for each end item or multiple date ranges. When you want to configure multiple occurrences at the same time which have different effectivities on them, you can configure using group effectivity.

You can configure the structure by adding group effectivities if your administrator has configured it for you at your site.

Group effectivity is used to configure product structure occurrences of an assembly by:

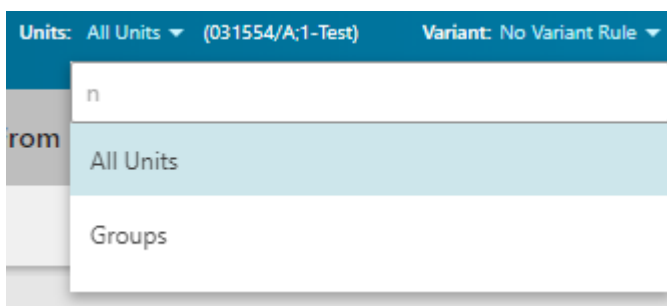
- Specifying multiple end items
- Specifying the unit effectivity ranges for each of those end items
- Specifying multiple date ranges

Configure a structure using a group of unit effectivities and end items

Procedure

1. Search and open the structure that you want to configure with group effectivity.

2. Select the structure and click **Units > Groups**.



3. In the **Group Effectivity** pane, click the **Add Group Effectivity** $\oplus$ icon.
4. Click **New**.

The screenshot shows the 'Group Effectivity' pane with a close button (X). Below the title bar is a section 'ADD GROUP EFFECTIVITY' with tabs 'New' (selected) and 'Search'. Under 'New', there is a 'Name:' field with the text 'Effectivity group u002'. Below this is a table with two columns: 'Units' and 'End Item'. The 'Units' column has a dropdown menu showing '1-8'. The 'End Item' column has a button with a plus sign ($\oplus$). At the bottom of the pane is a large blue 'Add' button.

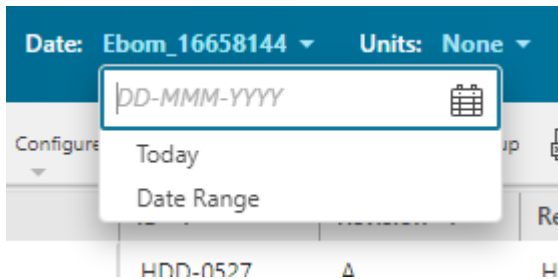
5. Specify the effectivity **Name**.
6. Specify the desired unit or a range of units in the **Unit** field.
7. In the **End Item** field, click **Add** $\oplus$ to add a new **End Item** or search for an existing **End Item**.
Once the unit and end item information is entered, a new row is added to the table.
8. Add more units and unique end items as required.
9. Click **Add**.

A group effectivity is created and applied to the currently displayed structure. As the effectivity criteria set in the header, and in the revision rule are in sync, you may lose any earlier **configuration done using the revision rule**.

Configure a structure using a range of date effectivities containing an end item

Procedure

1. Search for and open the structure that you want to configure with date range effectivity.
2. Select the structure and click **Date > Date Range**.



3. In the **Date Range Effectivity** panel, click **New**.
4. In the **Name** box, specify a name for the date range effectivity.
5. In the **Start** box, specify a start date for the date range effectivity.
6. In the **End** box, choose one of the following values:
 - **Date**
 - **UP - All Future Dates**
 - **SO - Stock Out**
7. In the **End Item** field, click **Add** ⊕.
8. Select an end item from the **Palette** tab or search for and select an end item from the **Search** tab.
9. Click **Add**.
10. Click **Apply**.

Edit or remove group effectivity of units and end items

Procedure

1. Select the structure which is configured with group effectivity and click **Units > Groups**.
2. In the **Group Effectivity** pane, select the effectivity that you want to edit or remove.

The **Remove Group Effectivity** ✕ and **Edit** ✎ icons are displayed.

- a. To remove the group effectivity, click the **Remove Group Effectivity** ✕ icon to remove the effectivity.
 - b. To modify the group effectivity, click **Edit** ✎ and then change the properties you want to edit.
3. Click **Save**.

Edit or remove range of dates

Procedure

1. Select the structure which is configured with group effectivity and click **Date > Date Range**.
2. In the **Date Range Effectivity** pane, select the date range from the **Search** tab.
3. To edit the date range, click **Edit**  and modify the dates and click **Apply**.
4. To remove the applied date range, select **Today** from the header.

13. Configure structures

About configuring structures

Understand the various methods for configuring and viewing specific product structure configurations, including using selection, proximity, effectivities, closure rules, revision rules, variant rules, and custom configurations.

To view a specific configuration of the structures, you can configure a structure using:

- [Selection](#)
- [Proximity](#)
- [Effectivities](#)
- [A closure rule for expansion](#)
- [Revision rules](#)
- [Variant rules](#)
- [A custom configuration](#)

Configure a structure by selection

You can configure a structure based on the Product Configurator-authored variants used in a structure element. This method of configuring a structure is called *configure by selection*. Based on the variability of the selected elements, other elements are configured in or out of the structure, which enables you to validate all possible and valid variant combinations according to your product configurator-authored variant definitions.

Restrictions and limitations

You can configure a structure by selection only if the structure is indexed by using Smart Discovery Indexing and your Teamcenter setup has the Context Management User license.

Procedure

1. Search for a product structure on which variability is set by using Product Configurator variants. You can also search for the workset that contains the structure.
2. Open the structure or workset.
3. Locate a structure element on which variability is set. You cannot configure the structure by selecting its root element or the workset. Additionally, you can configure only one structure at a time within a workset.

4. Click **Configuration Settings**  > **Configure by selection** .

Configure a structure by proximity

You can configure a structure based on the legacy classic variants used in a structure element. This method of configuring a structure is called *configure by proximity*. Based on the variability of the selected elements, other elements are configured in or out of the structure.

Restrictions and limitations

You can configure a structure by proximity only if your Teamcenter setup has the Context Management User license.

Procedure

1. Search for a structure on which variability is set by using legacy variants.
2. Click **Open**  to open the structure.
3. Click **Filter** .
4. Select a structure element from the work area. You cannot configure the structure by selecting its root element.
5. In the **Filter** panel, click **Proximity**.
6. Enter the proximity in **Distance** and click **Add**.

Configure a structure by effectivity

You can configure product structures using either date effectivity (specifying a single date, a date range, or an end item) or unit effectivity (entering a specific unit or using groups). You can understand how these configurations impact the displayed structure.

You can configure a structure by **date** effectivity and **unit** effectivity.

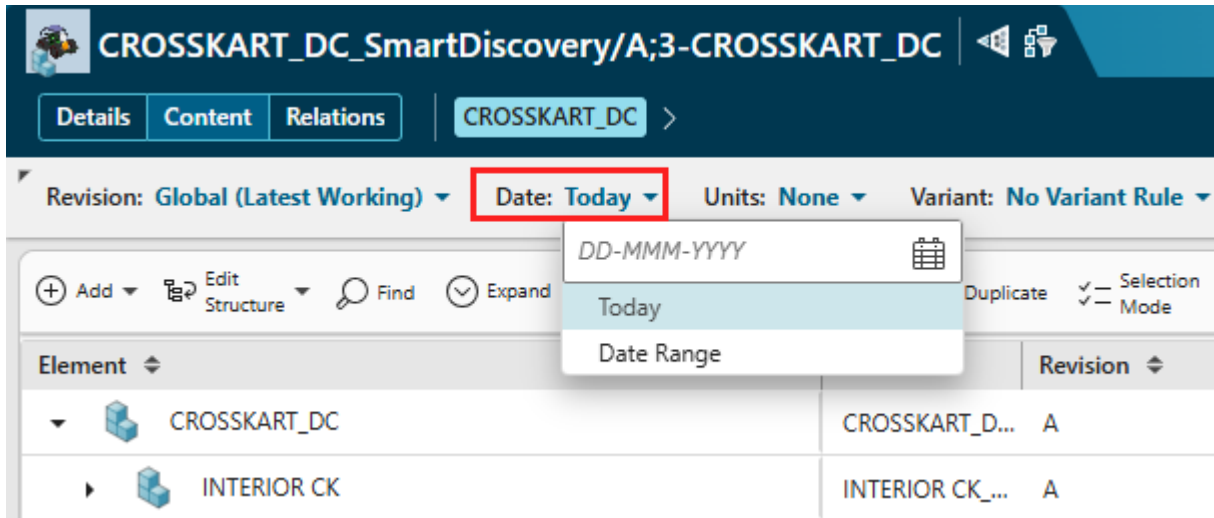
Configure a structure by date effectivity

Procedure

1. Search for and open the product to be configured.
2. Perform one of the following actions:
 - Select **Date** in the header.

- Click **Configure**  to display the **Configure** panel.

In the **Date** section, you can select **Today** or a specific date from the calendar. You can also add a range of dates.



To add a new range of dates:

- Select **Date Range**.
- In the **Date Range Effectivity** panel, you can add a new range of dates or locate the existing date effectivities from the **Search** tab.
- To author a new range of dates inside the **New** tab:
 - Give a suitable name to the new date range.
 - Select the **Start** date from the calendar.
 - Select the **End** date from the calendar. If applicable, you can select **UP** (all future dates) or **SO** (stock out).
 - Click **Apply**.

(Optional) To add an end item in the Configure panel:

- In the **End Item** section of the **Configure** panel, click  to add the end item. By default, the top-level element is displayed in this section.
- Search for an end item and select it or choose an end item from the **Palette** tab.

The end item that you selected is displayed in the **End Item** section on the Configure panel.

- If you are using the Configure panel, click **Configure** to apply the date effectivity.

Note

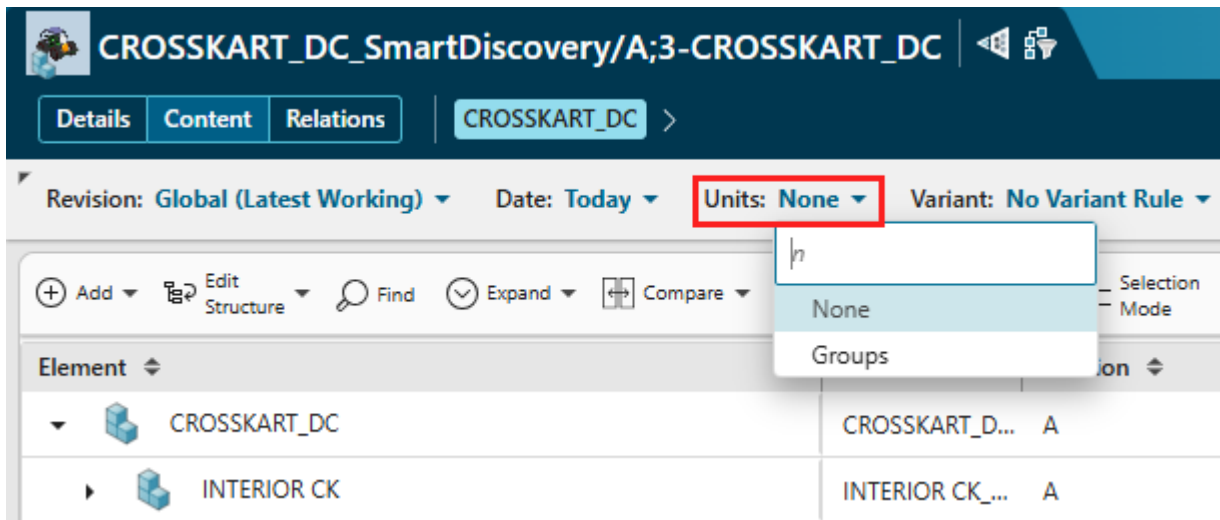
In the Configure panel, if you select two configurations that are of the same type or nature but with different values, the latest configuration is applied. For example, suppose that you first selected a date range from January 1, 2025 through June 30, 2025 for applying the date effectivity, and then selected a revision rule that already contained an effectivity date from February 1, 2025. In such

In a case, the structure is configured using the revision rule that contained the effectivity date from February 1, 2025 because the revision rule with the effectivity date is the latest configuration that is applied to the structure.

Configure a structure by unit effectivity

Procedure

1. Search for and open the product to be configured.
2. Perform one of the following actions:
 - Select **Unit** in the header.
 - Click **Configure**  to display the **Configure** panel.



You can enter a specific integer or select the following options in unit effectivity:

- None** To display all the elements without effectivity.
- Groups** To [author new group effectivity](#) and apply it.

3. If you are using the Configure panel, click **Configure** to apply the unit effectivity.

Configure a structure with a closure rule for expansion

A closure rule holds subsidiary rules that define the objects of interest in a given structure. The rules determine if an object is included in a given structure based on their types, classes, and the relationship between them.

Typically, a Teamcenter administrator creates and adds closure rules.

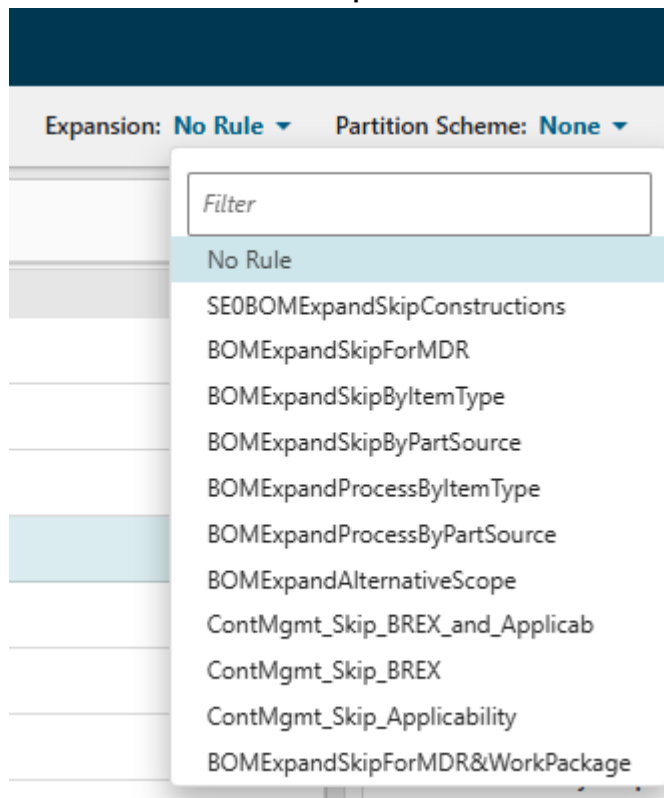
For more information about closure rules, see *Managing closure rules in PLM XML/TC XML Export Import Administration* in the Teamcenter documentation.

In Teamcenter:

- You can configure a non-indexed structure with a closure rule to apply an expansion or filtering logic to the structure.
- You cannot create or edit closure rules.
- You can use only static closure rules.

Procedure

1. Search for and open the structure (only non-indexed structures) to be configured.
2. Perform one of the following actions:
 - Select the rule from the **Expansion** list in the header.



- Use the Configure panel:
 - a. Click **Configure**  to display the Configure panel.
 - b. In the Configure panel, select a rule from the **Expansion** list and click **Configure**.

Results

Teamcenter refreshes and displays the configured content based on the applied closure rule.

Configure a structure with revision rules

Understanding revision rules

You can use revision rules to select specific revisions of parts and assemblies in a product structure by defining criteria such as working revisions, release status, effectivity (date or unit), and override folders. You can understand how these revision rule entries are evaluated to configure the structure.

You can create and apply revision rules that select the appropriate revision of parts and assemblies in a product structure. A revision rule:

- Selects the working revisions and (optionally) specifies the owning user or group.
- Selects revisions by status (according to status precedence) or the latest revision with any based on the using release date.
- Optionally specifies the effectivity against which the revisions are configured. Effectivity may be specified by date or by unit number.
- Selects revisions in a specified override folder.
- Selects the latest revisions according to the revision ID in the following order: alphanumeric, numeric, or creation date. This selection does not depend on whether the revisions are in the working or released state.

You define each of these criteria with a revision rule entry. A revision rule may contain any number of rule entries, each of which attempts to select a revision according to the specified criteria, for example, the status that the revision should have or the user or group that owns the revision.

Teamcenter evaluates rule entries in the order of precedence until a revision is successfully configured. You can include some entries more than once to define the order of precedence. You can modify the order of the rule entries to change the precedence Teamcenter uses when evaluating the revision rule. Certain rule entries can also be grouped so that they are evaluated with equal precedence.

Viewing and updating a revision rule

You can view and update revision rule clauses using Date, End item, Latest, Override, Precise, Status, Unit Number, and Working and then configure product structures.

You can configure a structure with a revision rule. You can also view or update revision rule clauses in the context of a structure.

The following revision rule clauses are supported for viewing and updating in Active Workspace:

Revision	Description
Date	Loads item revision with a specific date.
End item	Configures a structure by effectivity with respect to a product, system, or module. For example, you can configure the structure of unit number 110 in product X400 , where X400 is the end item.
Latest	Loads item revisions regardless of whether they are released or not.
Override	Overrides all item revisions available in an override folder. The revision rule is not evaluated for these item revisions.
Precise	Loads item revisions in a precise product structure.
Status	Loads item revisions that are released with a specific status.
Unit Number	Loads item revisions with a unit number as specified by the user. It is used in combination with the status rule entry with unit number effectivity. Typically, a unit number is a property of the end product or a major module of a product. As Teamcenter may manage many units, you typically qualify a unit number entry with an end item entry.
Working	Loads the working item revision that does not have a release status.

The **Branch** clause is not supported in Active Workspace, while the **Nested Effectivity** clause is read only.

Updating a revision rule

You can view and modify the revision rule clauses. When you modify the clauses of an existing revision rule, a new, temporary instance of that rule is created. This modified instance of the revision rule is session-specific and accessible only to the user who modified it.

You can create multiple modified versions of a revision rule within a single user session. However, these temporary instances are not saved permanently and will be lost upon logging off. The original revision rule remains unaffected by these modifications.

If the original revision rule includes a nested effectivity clause, this clause will be visible and usable within the modified temporary revision rule, but it cannot be updated.

Configure structures with a revision rule

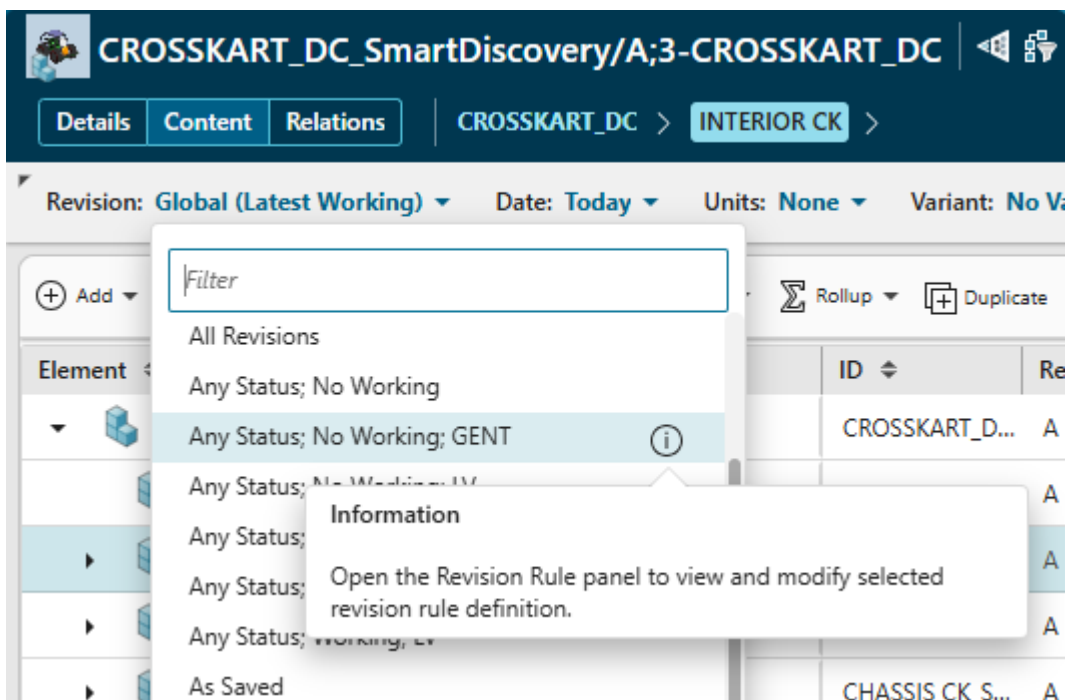
You can configure a structure by using an existing revision rule and by using the **Configuration** panel.

Configure a structure by using an existing revision rule

Procedure

1. Search for and open the product to be configured.
2. To apply a revision rule, select the rule from the **Revision** list in the header.

Depending on how your administrator has configured it, you might see a limited list of revision rules. Click **Show All** to view all revision rules.



Teamcenter refreshes and displays the configured content based on the new configuration.

Configure a structure by using the Configure panel

Procedure

1. Search for and open the product to be configured.
2. Click **Configure**  to display the **Configure** panel.
3. Select the rule from the list in the **Revision** section.

After a revision rule is selected, the Configure panel displays the unit, date, and end item information for the clause, if they exist. You can override any clause information if required.

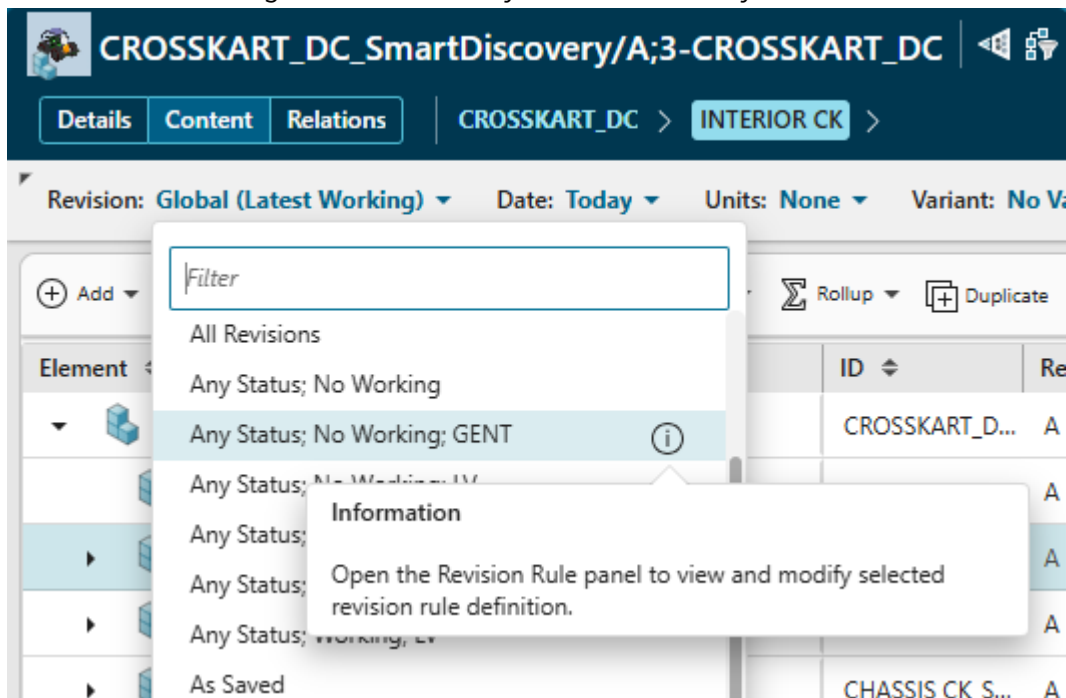
4. Click **Configure** to apply the revision rule and close the Configure panel. Teamcenter refreshes and displays the configured content based on the new configuration. Because the effectivity criteria set in the revision rule and in the header are in sync, you may lose any earlier configuration done using the [effectivity group configuration](#).

Configure a structure with a modified revision rule

You can modify an existing revision rule to configure a structure as per your business requirement.

Procedure

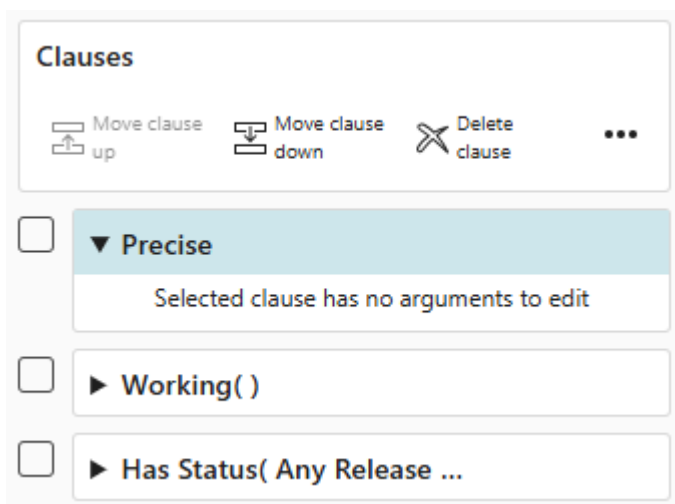
1. Search for and open the product to be configured.
2. To modify a revision rule, perform one of the following actions:
 - In the **Revision** list, go to the rule that you want to modify and click ⓘ **Information**.



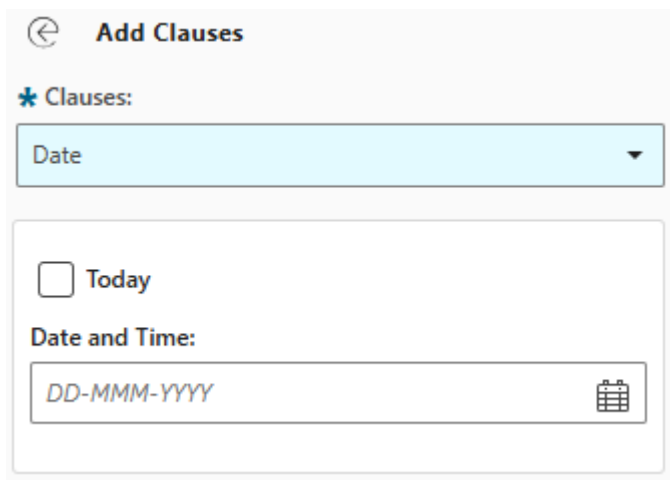
- Click **Configure**  to display the **Configure** panel and then click ⓘ **Information** for a revision that you want to modify.

The revision rule details are displayed with all the clauses.

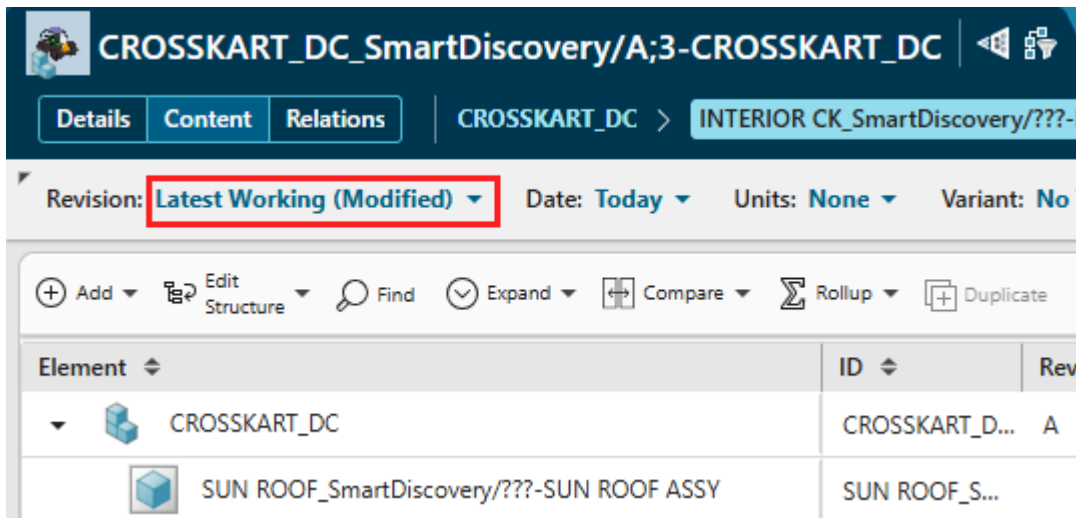
3. To modify the rule, you can add a new clause, delete a clause, or change the order of precedence by moving a clause up or down.



4. To edit a clause, select the clause.
The editable attributes for the clause are displayed. For example, if the **Date** clause is selected, the date-related fields are displayed.



5. To apply the modified revision rule, click **Modify and Configure**.
Teamcenter refreshes and displays the content configured by the modified revision rule. The **Revision** in the header area shows the modified revision rule.



Results

The modified rule is available to you only if you modified the rule. Alternatively, you can use the Configure panel to modify the revision rule clauses and update the **Date** and **Unit** values.

As the effectivity criteria set in the revision rule and in the header are synchronized, you might lose any configuration done earlier using the [effectivity group configuration](#).

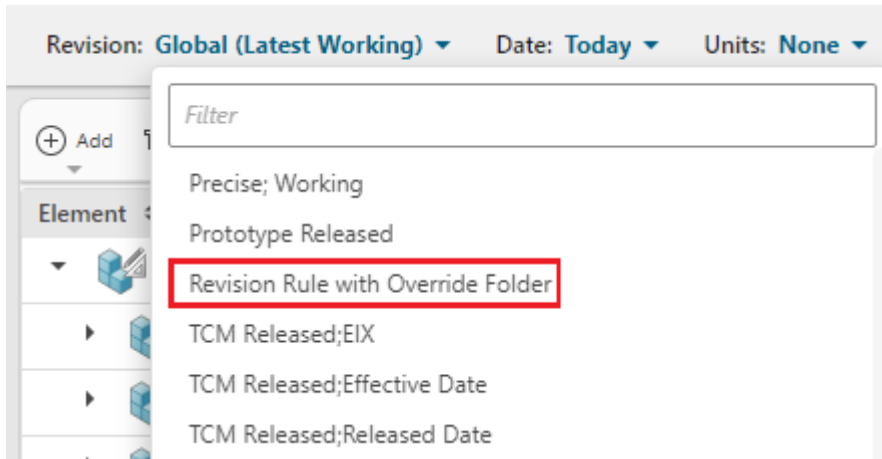
Configure a structure with a revision rule that contains an override clause

You can configure a product structure by using a revision rule with an override clause for various tasks, such as marking the structure as precise, checking temporary changes, performing what-if analyses, and analyzing changes to the structures based on business requirements.

Procedure

1. Identify the folder that is used as an override folder in the revision rule.
If you cannot, contact your administrator.
2. Search for and copy the required items to the folder.
For example, copy item revisions.
3. Search for and open the product to be configured.
4. To apply the revision rule with an override clause, select the rule from the **Revision** list.

The following screenshot shows an example of a revision rule that the administrator has created with an override clause.



Teamcenter refreshes and displays the configured content using the selected revision rule.

Configure a structure with variant rules

Configuring structures with variant rules

You can configure product structures using variant rules, which allow you to define options, their allowed values, and variant conditions to create a single generic product structure that can be tailored for different product variants. You can understand how to manage default options, derived defaults, and variant rule checks.

Manufacturers often want to develop a range of products based on the same generic platform, offering their customers choice, but at the least engineering cost. One approach is to create a single generic product structure that can be configured for each variant of the product offered.

Using Teamcenter, you can define options and the corresponding allowed values and attach them to an item, typically the top-level item in the structure. For example, you can define a Gearbox option with the allowed values of Manual and Automatic. You then attach a logical expression, referred to as a variant condition, to any occurrences of the components that are configurable, for example, the automatic and manual gearboxes. The expression refers to the defined options and can be as complex as necessary.

You choose the desired option values for a configuration and set them in a variant rule. Teamcenter evaluates the variant conditions on the occurrences in the structure against the set option values, and components are configured accordingly. Unconfigured components can be hidden.

A designer can preset the option values in the variant rule. The preset value may be a default option or a derived default option.

- **Default option**

A default value is a value that you preset for an option. For example, the option **aerial** may have a default value set to **standard**.

- **Derived default option**

A derived default is a value that is set to a value that depends on a condition. For example, the option **radio** may have a value **stereo** if **car type = GLX**.

For more information about default and derived default options, see the *Working with option defaults in Structure Management on Rich Client — Usage*.

Designers can define combinations of option values that are not allowed using the variant rule checks. A variant rule check consists of a condition (for example, **engine = 1200 AND gearbox = automatic**) and an error message (for example, **Incompatible engine and gearbox**). An error message containing the condition and message is displayed if the rule check fails when you configure a structure with the variant rule.

You can view existing variant rules and variant configuration data. Additionally, you can change the configuration data and save the updated rule as a new variant rule. You can apply any existing or updated variant rule configuration to your structure.

To view or update saved variant rules, ensure that the **PSEVariantsMode** preference is set to **legacy**.

Configure a structure with a variant rule

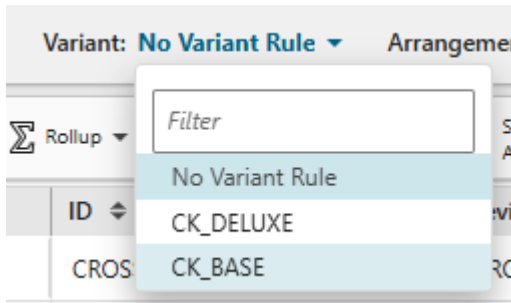
In Teamcenter, you can configure a structure with a variant rule. You can view the existing variant rules and variant configuration data. You can also change the configuration data and save the updated rule as a new variant rule. If you have **Partitions for Structure** deployed in your Teamcenter setup, you can set variability on partitions so that you see only those partitions that are relevant to your configuration.

Example

In a structure where no variant rule or no partition scheme is specified, you choose the variant *Sedan* and then choose the partition scheme *Physical*. Now, you see only the partitions that are relevant to this configuration. Specifically, with the variability applied on the partitions, the partitions *Truck* and *Hatchback* are not displayed in the configured structure.

Procedure

1. Search for and open the product to be configured.
2. To apply a variant rule, select the rule from the **Variant** list in the header.

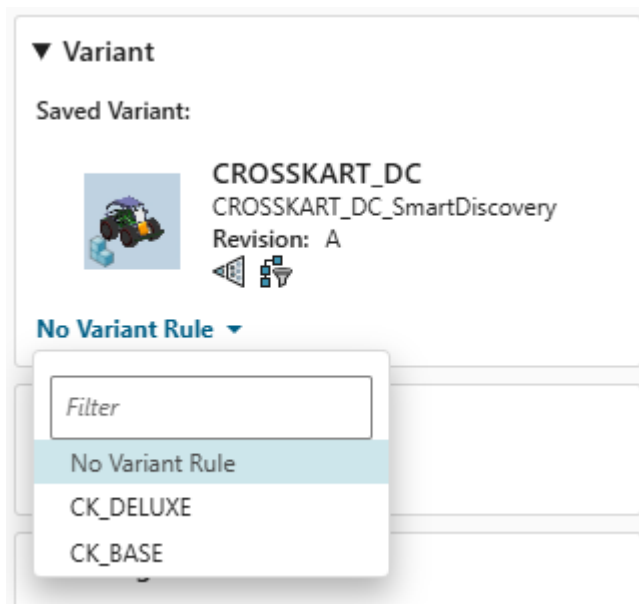


Teamcenter refreshes and displays the configured content based on the new configuration.

Configure a structure with a variant rule by using the Configure panel

Procedure

1. Click **Configure**  to display the **Configure** panel.
2. To apply a variant rule, select the rule from the list in the **Variant** section in the **Configure** panel.



3. Click **Configure** to select the variant and close the Configure panel.
Teamcenter refreshes and displays the configured content based on the new configuration.

Configure a sub-assembly by using the end item's configurator context

Consider that a structure has a configurator context set to it, you open a sub-assembly from that structure, and the sub-assembly does not have its own configurator context set to it. In such a case, the sub-assembly does not carry the configurator context of the parent assembly. However, you can set the end item to the sub-assembly and then configure the sub-assembly by using the end item's configurator context.

Procedure

1. Search for and open the sub-assembly to be configured.
2. Click **Configure** .
3. In the **Configure** panel, select the parent structure as the end item.
The configurator context of the parent structure is selected as the configurator context for the sub-assembly.

What to do next

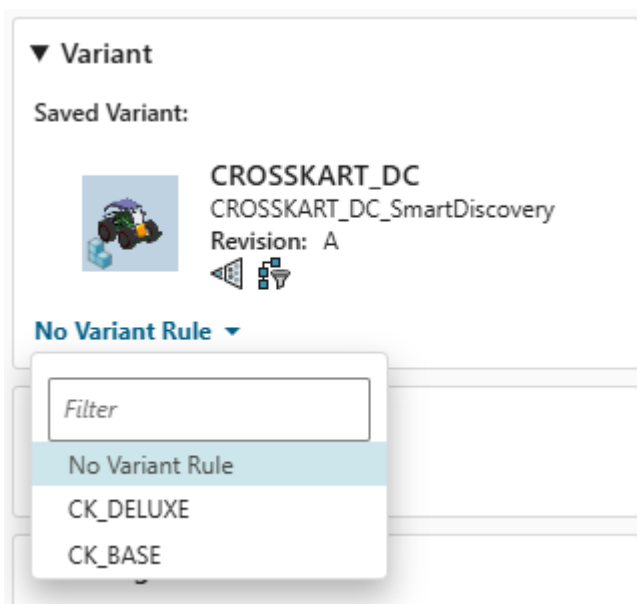
Click the **Variant Configuration** tab and the **Variant Conditions** tab to configure the variants for the sub-assembly.

View variant rule details

You can view the details of an applied variant rule in the Configure panel, including the default values and the derived default values.

Procedure

1. Search for and open the structure to be configured.
2. Click **Configure**  to display the **Configure** panel.
3. In the **Variant** section, select the rule you want to view.



4. To view the details of the applied variant rule, select **Custom** from the variant rule list.
The **Configure** panel shows the options and values defined in the variant rule.

For the options in the variant rule that do not have a defined value, the default values and derived default values are shown.

Configure ✕ Close

← **Custom Configuration**

 Save  Save As

Aerial:

Std ▼

Car Type:

LS ▼

Engine:

1600 ▼

Gearbox:

Manual ▼

Radio:

Stereo ▼

color:

White ▼

Modify

5. Change a default value and click **Modify** to view the derived default values for the selected default value.

Note

Unless the **Modify** button is clicked, the derived default value is not updated, even after the default value is changed.

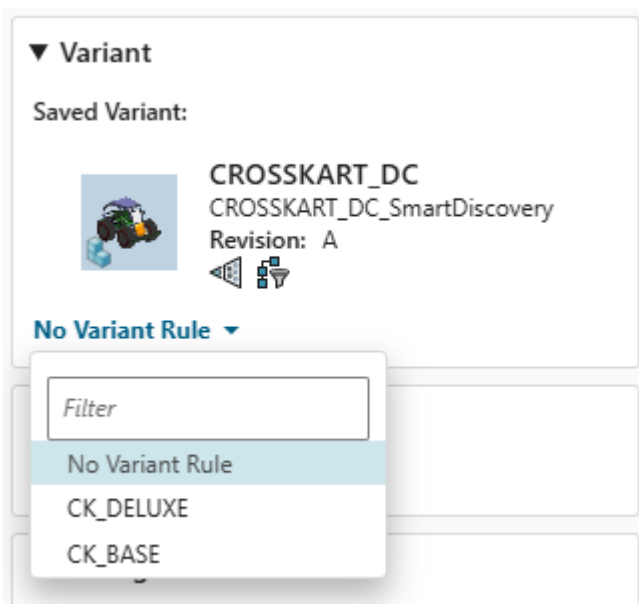
6. Click **Configure** to apply the configuration and close the Configure panel.
The rule checks for the variant rule only after the **Configure** button is clicked.

Update a variant rule

Only users with the requisite access privileges can modify a variant rule.

Procedure

1. Click **Configure**  to display the **Configure** panel.
2. Select the variant rule you want to modify from the list in the **Variant** section in the **Configure** panel.



3. To update the variant rule, select **Custom** from the list.
The **Configure** panel shows the options and values defined in the variant rule.

For the options in the variant rule that do not have a defined value, the default values and derived default values are shown.

Configure ✕ Close

← **Custom Configuration**

Save Save As

Aerial:

Std

Car Type:

LS

Engine:

1600

Gearbox:

Manual

Radio:

Stereo

color:

White

Modify

4. Make the changes to the option values as required and click the **Save** icon.
The variant rule is saved with the updates. You can use it to configure structures.

Save a modified variant rule as new

You can save a modified variant rule as a new rule. When you save the variant rule as a new rule, you can attach this new rule to your current structure or configurator context for future use in configuring structures.

Procedure

1. After [updating a variant rule](#), to save it as a new rule, click the **Save As** icon.
2. Enter a **Name** and **Description** and click the **Save** button.
3. To attach this variant rule to the current structure or the configurator context, select the appropriate option.

Results

A new variant rule is created that you can use to configure structures.

Associate a variability scope to a structure

When you associate a variability scope to a structure, you apply the configurator context information to the structure. The configurator context, which contains information such as inclusion and exclusion rules, is used for creating product variants.

Procedure

1. Search for the required structure element such as item, part, or product design, and open it.
Alternatively, from the item revision, part revision, or design revision, navigate to the respective item, part, or product design using the **Relations** tab.
2. In the **Overview** tab, click **Edit**  > **Start Edit**.
3. In the **Variability Scope** section, click **Add** .
4. Select the required configurator context using **Search** or **Palette** and click **Add**.
5. Click **Edit**  > **Save Edits**.

Create variant conditions for a part

You can create and update variant conditions for a part through the **Variant Conditions** tab and also inline within a structure.

Create and update variant conditions using the Variant Conditions tab

Procedure

1. Search for the product structure that contains the part for which you want to add a variant condition.
2. [Associate a configurator context with the product structure](#), if this is not already done.
3. Select the product design and click **Open** .
4. Select the relevant part and click the **Variant Conditions** tab.
5. Click **Show All Families**  to view all options.
6. To expand each option family, click **Show Children** .
7. Click **Edit** .
8. Click a cell in the grid next to a variant option to set the variant condition for the part:
 - Click once to display a check mark ✓ to include the variant option as a variant condition.
 - Click twice to display a circle backslash  to exclude the variant option as a variant condition.
 - Click three times to display a blank cell to indicate that the variant option is not used as a variant condition.
9. Click **Save Edits**  to update the variant conditions.

Multiple variant conditions are connected using a logical **AND** operation to create the final variant condition.

Create and update variant conditions inline

Procedure

1. Search for the product structure that contains the part for which you want to add a variant condition.
2. [Associate a configurator context with the product structure](#), if this is not already done.
3. Select the product design and click **Open** .
4. Select the relevant part and hover over the **Variant Formula** column and click .
5. Click **Edit**.
6. Type the syntax for the variant condition for the selected part.

Tip

When you are authoring the variant condition, click **Intellisense**  to receive the syntax suggestions to author the variant condition.

Multiple variant conditions are connected using a logical **AND** operation to create the final variant condition.

7. If you want to clear the typed variant condition, click **Clear Text** ✕.
8. If you want to validate the syntax and the variant condition, click **Validate Syntax** ✓.
9. Click **Save**.

Configure a structure using baselines

Overview of capturing the structure configuration using baselines

Baseline is a persistent record of all or part of a product's definition data that can be accessed at any time during or after a product's life. Baselines help you capture and preserve the state of the structure and its sub-assemblies at a particular checkpoint in the product development.

For instance, if you are working on an aircraft structure and have nearly completed the engine assembly while still progressing on the stabilizer and wings assemblies, you can create a configuration baseline for the engine assembly. After the stabilizer assembly is finished, you can establish another baseline for it.

You can create a precise baseline and a configuration baseline. When you create a baseline, you can create a precise baseline and capture the 150% structure of the BOM. After creating a precise baseline, when you open it, you must configure it again to view the exact structure that you want to view. For example, when you open a precise baseline, you must select the revision rule to view the exact configuration of the structure. After a precise baseline is created, you cannot further modify it.

When you create a configuration baseline, you can capture the exact same configuration of the structure. If your Teamcenter environment includes the required license and when you create a configuration baseline, you can choose the required type of the configuration baseline that you want to create.

Using baselines, you can perform the following tasks:

- **Create precise baselines** and capture the 150% structure of the BOM.
- **Create configuration baselines** for the complete structure or for an individual sub-assembly.
- Considering multiple milestones involved in the product development, generate multiple baselines to track different development milestones for the same sub-assembly.

For example, you might first create an ad hoc design review baseline and later a system design review baseline for the engine assembly.

- When multiple baselines are created for the same sub-assembly, **compare configuration baselines** to identify changes between the milestones.
- And finally, **view the entire structure using the baseline configurations**.

Create a precise baseline

You can create a baseline of the item revision that represents a structure at any stage. You create precise baselines to preserve the state of a structure at a particular checkpoint. A baseline revision is created for all the elements in the structure, irrespective of the element selected. After a precise baseline is created, you cannot further modify it.

After creating a precise baseline, when you open it, you must configure it again to view the exact structure that you want to view. For example, when you open a precise baseline, you must select the revision rule and variant to view the exact configuration of the structure.

Procedure

1. Locate the structure that you want to create a baseline for and click **Open** .
2. Click **New**  > **Baseline** .
3. In the Baseline panel, click the **Precise** check box.
4. (Optional) In the **Baseline Workflow** list, modify the default baseline workflow.
By default, the **TC Default Baseline Process** workflow is selected.
5. (Optional) In the **Description** box, enter a description for the precise baseline.
6. Do one of the following:

To create a revision baseline	Do this
Asynchronously (in the background)	Select the Run in Background check box.
Synchronously (in the foreground)	Clear the Run in Background check box.

When creating a precise baseline, the **Run in Background** check box is selected by default. If you clear the **Run in Background** check box, Teamcenter remembers your choice. The next time you log on, the check box is not selected.

Note
If you want to create the revision baseline asynchronously, you must have the AsyncService translator installed, configured, and running for the baseline feature to operate properly.

7. If you want to open the baseline immediately after it is created, click the **Open on Create** check box.
8. Click **Create**.
The precise baseline is created and is available in the **Revision History** table of the **History** tab.
9. If you selected the **Run in Background** check box, a notification is displayed in the **Alerts** panel. Click **Alerts**  and click the baselined structure in the **Target Object** section.

Create a configuration baseline

You can create a configuration baseline for the entire structure. The configuration baseline helps you preserve the state of the structure at a particular checkpoint and visualize it later. You can create a configuration baseline for a structure even if it is released.

Restrictions and limitations

You can create different types of configuration baselines only if your Teamcenter setup has the required license.

Procedure

1. Locate the structure that you want to create a baseline for and click **Open** .
2. Click **New**  > **Baseline** .
3. In the Baseline panel, perform the following tasks:

- a. Clear the **Precise** check box if it is already selected.
- b. In the **Configuration Baseline Type** list, select a value.

If you are creating the configuration baseline for performing an ad hoc design review, select **Ad-hoc Design Review**.

If you are creating the configuration baseline for performing the design review of the entire selected assembly system, select **System Design Review**.

If you are almost done with the system design and want to get the structure reviewed before finalizing it, select **Preliminary Design Review**.

If you are creating the configuration baseline for evaluating whether the structure is complete, meeting all requirements, and is ready to proceed the formal release, select **Critical Design Review**.

If the selected structure is ready and no more changes are expected to it, select **Formal Release Baseline**.

Note

Use the values in the **Configuration Baseline Type** list considering your product development stage. No Teamcenter action is performed on the configuration baseline.

- c. In the **Name** and **Description** boxes, enter a name and description for the configuration baseline.
- d. In the **Action on Working Revisions** list, select a value.

If you want to capture the baseline revisions of the working revisions and apply the workflow, select **Capture and Apply Status**.

If you want to freeze the baseline revisions of the working revisions and apply the workflow, select **Freeze and Apply Status**.

If you want to capture the working revisions as is without applying any workflow, select **Capture Configuration As Is**.

- e. In the **Baseline Workflow** list, select a workflow to apply to the configuration baseline.
4. If you want to create the configuration baseline in the background, select **Run in Background** checkbox.
5. Click **Create**.

Results

The configuration baseline is created for the selected structure. If the structure contains substitutes and remote checked out objects, the configuration baseline is not supported for them. Click the **Baselines** tab to [view the configuration baselines](#) that are already created for the selected structure. If your Teamcenter environment is set up with smart baseline and you close the configuration baseline, the new revisions are created for the associated items whose work in progress revisions are modified since their last baseline was created.

View a configuration baseline

You can view configuration baselines for structures or subassemblies and further configure them by using effectivity and variant configurations.

Procedure

1. Select the structure or the subassembly for which the configuration baseline is created and click **Baselines** tab.
The **Baselines** tab displays all the configuration baselines created for the selected structure.
2. Select a configuration baseline and click **Open** .
The configuration baseline opens. You can identify it with the **Configuration Baseline**  icon.
The configuration baseline displays the same configuration of the structure that it was created in, including the revision rules, partitions, and effectivities. You can also use the Filters panel to view the required content of the configuration baseline if your Teamcenter setup has the Context Management User license.

View the content of the configuration baseline in other applications

You can open and view the content of the configuration baseline in other applications, such as NX and CATIA.

Application	Action
Teamcenter Lifecycle Visualization	Select Open > Open in Visualization .
NX	Select Open > Open in NX .
CATIA	Select Open > Open in CATIA .

Compare configuration baselines

You can compare two configuration baselines within the same structure to view their differences. These configuration baselines can have different types, states, and release statuses.

Procedure

1. Select the structure for which the configuration baselines are created and click the **Baselines** tab. The **Baselines** tab displays all configuration baselines created for the selected structure.
2. Select two or more configuration baselines and click **Compare Structures**.
The **Compare Structures** view appears and displays the comparison between the selected configuration baselines. The **Compare** panel displays the elements that are unique to both the configuration baselines with color bars under the Results. This comparison is performed with the default comparison options.
3. (Optional) Change the comparison level:
 - a. Specify the comparison level by selecting one of the following from **Options**.

Comparison option	Description
All Levels	Compares the configuration baselines at every level in the two structures. If this option is selected for very large structures, performance may be adversely affected.
Components Only	Compares only the elements, or components that are at the lowest level.
Current Level	Compares only the elements that are the immediate children of the selections in the two structures. This is the default level of comparison.
Linked Assemblies or Components	Compares the two configuration baselines by using an accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the elements in one structure have equivalent

Comparison option	Description
	elements in the other structure. You can define your own properties for equivalence.
All Levels Below Selected	Compares the configuration baselines below the levels that you selected.

- b. Specify what you want to display in the comparison results by selecting one or more of the following options from **Options > Display**.

Comparison option	Description
Matched	Displays the elements that are a match in both of the configuration baselines.
Different	Displays the elements that differ between the two configuration baselines.
Multiple Matches	Displays the different numbers of occurrences of the same element in the comparison as a difference.
Unique in Left View	Displays the elements that are only listed in the configuration baseline that is displayed on the left.
Unique in Right View	Displays the elements that are only listed in the configuration baseline that is displayed on the right.

- c. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two configuration baselines might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

- d. (Optional) Select the **Generate Report** check box to generate the comparison report.
- e. Click **Compare**.

The comparison results are listed under **Results**. Clicking an element in the list highlights that element in the configuration baselines.

Comparison results are retained for a specific period. If a comparison result is older than the retention period or if a more recent result exists, the old result is automatically deleted by the system.

If you selected the **Generate Report** check box, the comparison report is generated in Excel. If you are the owner of the configuration baselines, the report is attached to it. You can click the **Attachments** tab of the target structure to download the report. If you are not the owner of the configuration baselines, the report is saved in the **Newstuff** folder. The file name for the comparison report is in the format *Compare_Report_<Source_ID>_AND_<Target_ID>_YYYYMMDD_HH_MM_SS*, where *Source_ID* is the ID of the first configuration baseline that you want to compare and *Target_ID* is the ID of

the configuration baseline with which you want to compare the first configuration baseline. You can also click **Alerts**  to understand the location of the comparison report if the configuration baselines are compared in the background.

Tip

If you closed the **Compare** panel and want to perform any further comparison, click **Compare** on the task bar to view the **Compare** panel again.

When you are comparing the two configuration baselines, you can configure them by unit effectivity, date effectivity, and by variants.

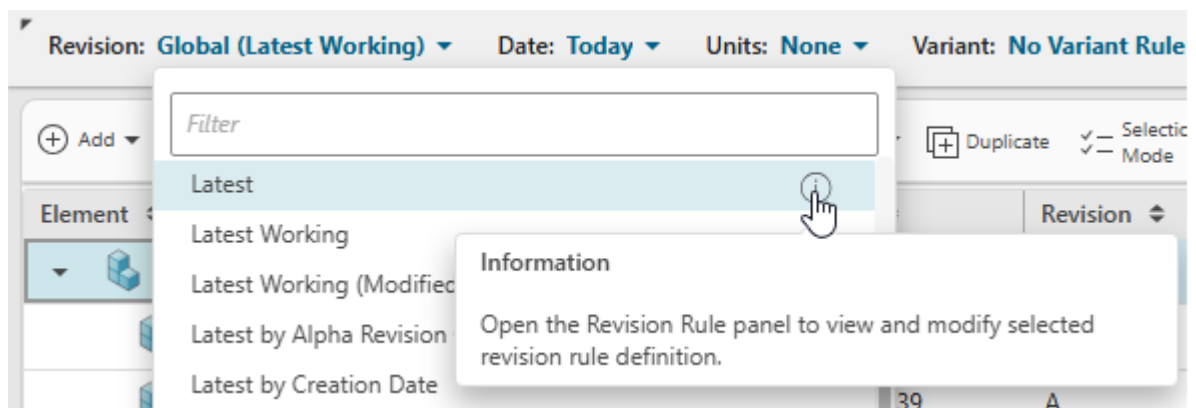
4. Click **Exit Compare** to close the comparison.

View a structure using a configuration baseline

You can modify the revision rule to configure a structure with a configuration baseline and then view it. You can also use effectivity and variants for further analysis.

Procedure

1. Search for and open the structure.
2. **Modify a revision rule** to include the configuration baseline in the structure.
 - a. In the revision rule list, go to the rule that you want to modify and click **Information**.



The revision rule details are displayed with all the clauses.

- b. On the **Clauses** section of the **Revision Rule** panel, click **More ***** > **Add clause**  to add the clause to the configuration baseline.
- c. Perform one of the following tasks:
 - If configuration baselines are created for multiple subassemblies and you want to configure the structure for all of the subassemblies that satisfy the clause you define, in the **Clauses** list, select **Baseline Type** and then select the **Configuration Baseline Type**, and a date.

- To analyze the structure for a specific configuration baseline, in the **Clauses** list, select **Baseline Object** and then add the required configuration baseline.
 - d. Click **Add** to modify the selected revision rule.
 - e. If required, add a new clause, delete a clause, or change the order of precedence by moving a clause up or down.
 - f. Click **Modify and Configure**.
The revision rule is modified and configured to display the content of the configuration baseline.
3. If required, use effectivity and variants to further configure the structure.

Using smart baselines

The standard baseline process may be time-consuming and use significant disk space as it may create many new revisions and (consequently) copies of all the associated data and CAD designs. To avoid this, you can create a *smart* baseline, reusing a baseline revision whose work in progress revision is unmodified since its last baseline was created. You can only create smart baselines of assemblies.

If you try to create a smart baseline of an assembly that has not changed since the previous baseline, a new baseline is not created. No informational message is displayed in Structure Manager to indicate this.

Note

If you make changes to an occurrence property (for example, quantity) the last modified date of the parent BVR changes and Teamcenter therefore considers the parent item revision as a candidate for a new baseline. However, the item revision of the BOM line on which changes were made does not qualify for the new baseline.

Perform a solve using a modular configuration

Why use a modular configuration?

Learn about the benefits of using modular configurations to minimize the engineering and production costs. The configurator administrator or a designated user creates *configurator modules*. The BOM engineers can use these modules to configure products, such as a generic door assembly for refrigerators and freezers, by setting parameters and variant conditions.

The configurator modules are self-contained, plug-compatible units that can be *reused*.

Manufactured goods are often designed and assembled from configurator modules. For example, consider a company that produces a range of refrigerators and freezers in different sizes and colors. The door assemblies are developed in a department that designs a modular door suitable for use in any refrigerator or freezer. They design a generic door assembly that has all possible components for any use—a sheet steel outer door and two internal covers, one for a freezer and one for a refrigerator.

You can then configure the door assembly for a particular use in a refrigerator or freezer by setting various parameters or variant conditions that describe it, for example, **door width**, **door height**, **application** (refrigerator or freezer), and **color** (white or stainless steel). This intelligent door assembly is called a *configurator module*.

The door configurator module is completely self-contained and can be reused in any product by setting its modular configuration. The modular configuration settings only control the features and components contained in the door configurator module itself and make no reference to features in higher- or lower-level configurator modules.

Solve the modular configuration

As a BOM engineer, you can perform a solve using the configurator module.

Procedure

1. Search and open the modular configuration.
2. Click **Details > Variant Configuration**.
3. Choose one of the following modes:

- In the **Guided** mode, only valid features are displayed.

This mode allows you to navigate only valid selections and configurations. If no configuration is applied, a model opens in guided mode, and the system guides you to select a valid group of features in which to configure a product. Every time a selection is made, the system reevaluates the rules and refreshes the list of features.

Guided mode supports both complete and partial configurations.

- In the **Manual** mode, all available features, including variants that are not valid in the current context, are displayed.

In this mode, you can look at the entire variability, choose a variety of features, and assess if those selections are valid based on the configurator rules. You can validate your selections and save your configuration. If there are any conflicting constraints, the system reports violations as error messages and specifies the conflicting selections.

While radio buttons are displayed for making selections in **Guided** mode, check boxes are displayed for **Manual** mode.

4. To **filter configurator data by revision rule, effectivity, and rule date**, and rule date, click  to view the **Settings** panel and change them as appropriate.
5. Set the validation mode.
 - a. Click  to view the **Settings** panel.
 - b. In the **Filter Criteria** section, choose the product hierarchy depth.

The default value is **1** and the maximum is **5**.

Note

By default, five levels of structures are supported. However, this example provides procedures to configure only three levels of structures.

- c. In the **VALIDATION MODE** section, [choose a mode from the drop-down list](#).
 - **Order** (default profile)

This mode is intended for a user who wants to quickly arrive at a 100% BOM. You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.
 - **Overlay**

This mode does not expect you to specify the complete configuration. It is intended for developing a new product or making changes to an existing product.
6. To configure the modular configuration, make the appropriate selections in the **Variant Configuration** view.
7. Select a product model with which you would like to work.
8. Select each group from the list in the left panel and make the necessary feature selections in the work area.
9. (Optional) To find the required selections, in the **Variant Configuration** view, choose **More Commands > Next Required** or **Previous Required** as appropriate.
10. (Optional) To find user and system selections, in the **Variant Configuration** view, click **Full Screen**, choose **Selection Summary**, and select **All** from the **Selection Type** list.

When the product hierarchy depth is greater than **1**, the **Selection Summary** panel shows an additional column, **Module**. This column displays where the selection has been made, that is, the module, family, feature, and type of selection.

The **Selection Summary** panel sometimes displays the same feature in two rows. This happens when either the user or the system (based on constraints) has made a selection on a feature that exists both in the parent module as well as the child module. You can see the name of the parent and the child module for that repeated feature in the **Module** column for that feature.

Selection Summary

★ Selection Type:

All

	Family	Feature	Selection Type	Module
✓	PlatformModelFamily	SPA2	Default	ModConfID-Automotive...
✓	FSCModelFamily	LXI	Default	FscConfigDataModule
✗	AuxiliaryFamily	AuxiliaryFamily	Default	ModConfID-Automotive...
✗	AuxiliaryFamily	AuxiliaryFamily	Default	FscConfigDataModule
✓	CentralLockingSystem	RemoteControl	Default	ModConfID-Automotive...
✓	CentralLockingSystem	RemoteControl	Default	FscConfigDataModule
✓	CoolantInLitre	1.1	Default	ModConfID-Automotive...
✓	CoolantInLitre	1.1	Default	FscConfigDataModule

11. Make all the required selections.

You can make selections at the root level and child level modules (configurator context levels).

In the **Guided** mode, the validation happens for each of your selections, and its configuration completeness is displayed automatically.

If you are in the **Manual** mode, when you click **Expand** or **Validate**, the current selections in the UI are validated or expanded, respectively.

With modular configuration, along with configuration completeness, a module completeness check is also done.

The configuration completeness is standard for all product structures, irrespective of the modular configuration.

The screenshot shows the Variant Configuration interface. At the top, the status is 'Valid, but Incomplete Configuration'. Below this, the 'Module Hierarchy' is displayed on the left, and the 'Products' list is on the right. A yellow callout box labeled 'Configuration validation' points to the status bar. Another yellow callout box labeled 'Modular validation' points to the 'Module Hierarchy' list. The 'Products' list includes items like 'Products', 'ATDDCheckData', 'AutomotiveVehicleGroup', 'EngineOptions', and 'EntertainmentFamilies'. The 'Module Hierarchy' list includes items like 'Partially Complete', 'Incomplete', 'Invalid', and 'Completed'. The 'Configuration validation' message is highlighted in a yellow box.

The completeness indicator at the top of the view displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

- **Valid and Complete** (applicable for both guided mode and manual mode)

Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

When a configuration is valid and complete, you could take this configuration input and execute an order entry in the order system. When a configuration is valid and complete, you can take this configuration input and perform a cost or a weight rollup, compute the center of gravity, or perform a digital analysis of that product.

On the contrary, if you do not see this completeness indicator, you should not typically perform the above action. It means that you have less content or more content.

When a configuration is valid and complete, it assigns a value to every family. This is irrespective of whether the family is mandatory or discretionary.

- **Valid and Incomplete** (applicable for both guided mode and manual mode)

In guided mode, you can select **Next Required** or **Previous Required** to navigate to the next or the previous family or the feature that requires a value to be assigned to complete the configuration.

In the case of modular configuration, **Next Required** or **Previous Required** takes you to the next module after reaching the last incomplete family of the last group.

- **Invalid** (applicable for manual mode only)

It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

The modular validation indicates the following states:

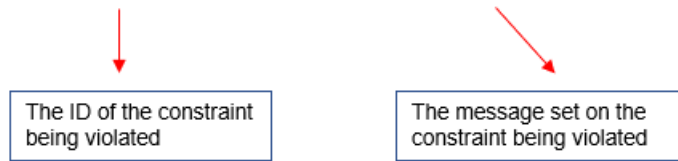
State	Description
Partially Complete	The selections for the parent are complete, but the selections for some of the children are not.
Incomplete	Some required families do not have complete selections.
Invalid	Some violations have occurred due to some constraints.
Completed	All the required selections are complete for the module, and they do not have any conflicts with constraints.

12. In the **Manual** mode, when you click on **Expand** or **Validate**, you might enter into an invalid state because of some constraint being violated. In such cases the system reports violations.

The following are some examples of violation messages in the **Manual** mode:

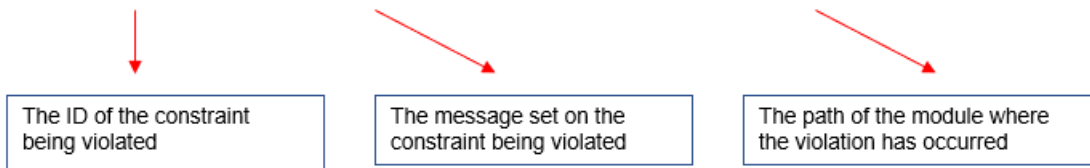
Example of violation message without modular configuration:

CExcRule1/001: Model LXI excludes Seat Heater:



Example of violation message with modular configuration:

ExcRule1/001: Model LXI excludes Seat Heater: CarContext > SeatModule



Note

Violation indicators are available at the child level also.

13. Record your selections by clicking the **Save** button.

If you have not loaded a variant rule or a variant condition, the system prompts you to save the selections as a new variant rule or variant condition. In such cases, specify the name in the **Save** dialog box and select one of the following options:

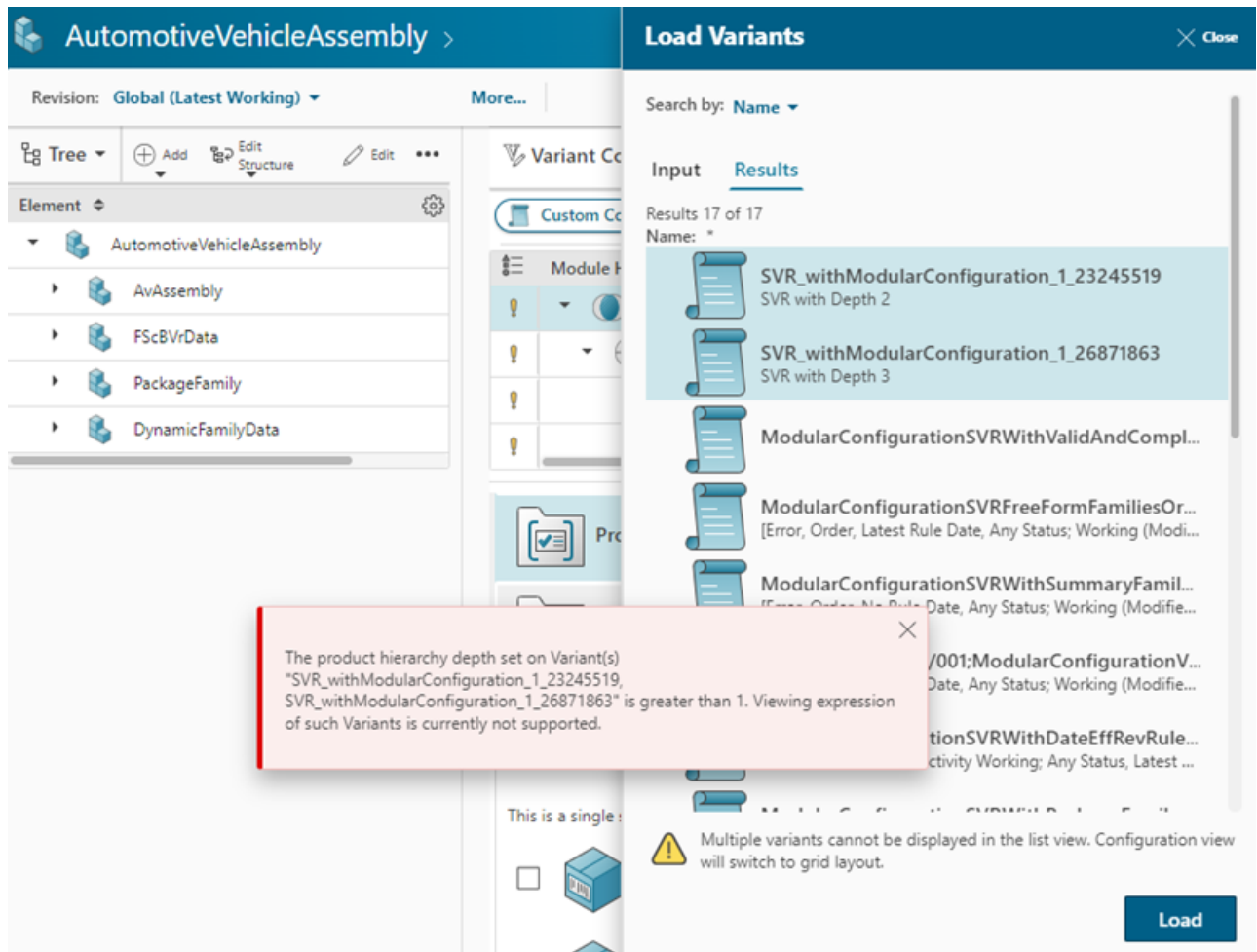
- **Current Structure Revision** to save the variant rule or the variant condition to the current structure.
- **Configurator Context** to save the variant rule or the variant condition to the configurator context. This saved variant rule or variant condition is available in all structures to which the configurator context is attached.

14. To load a variant rule or variant condition, choose **More Commands > Load Variants**.

Currently, when you load multiple variants using the **Load Variant** dialog, the system switches to the variant configuration's **Grid Mode**.

With a modular configuration:

- If the selected variants have a product hierarchy depth greater than **1**, the system does not switch to **Grid Mode** and an error message is displayed. This is currently not supported.
- Any variant with a depth greater than **1** can be viewed only in the variant configuration's **List** view.
- If the selected variants have a mix of depth greater than **1** and depth as **1**, the system switches to the variant configuration's **Grid Mode**. It then displays only the variants with a depth of **1**. The rest of the variants with a depth greater than **1** are skipped with a warning message.



15. Save selections after loading a variant rule or variant condition.
 - Click the **Save** button to save the changes to the existing variant rule or variant condition; OR
 - Click the **Save As** button to save the changes to a new variant rule or variant condition.
16. To apply the selected variant configuration to the current content and to refresh the content in the product, choose **More Commands** > **Apply Configuration**.

Solve the modular configuration using intents and conditional module

You can use intents to scope the modular configuration before performing a solve. Teamcenter configures out modules and their features based on intents specified by the configurator administrator.

You can refine the modular configuration by using variant conditions on modules. For example, when configuring a marketing-specific solution, you can filter out modules not intended for marketing by authoring a variant condition on the module and setting the appropriate intent.

Prerequisites

This topic assumes that the configurator administrator has created a conditional module with **FuelType** family and **Petrol** and **Electric** as its features. Additionally, a variant condition has been created with **FuelType=Electric** on the configurator module and the **Manufacturing** intent selected for the **FuelType** family.

+ Add - Delete Expand Effectivity Selection Mode Select All Refresh Export To Excel				
Object	Type	Parent Context	Child Context	Variant Condition
IntentsCondMod_Car-Car	Configurator Context			
EngineModule	Configurator Module	IntentsCondMod_Car-Car	IntentsCondMod_Engine-Engine	
ElectricEngineModule	Configurator Module	IntentsCondMod_Engine-Engine	IntentsCondMod_ElectricEngine-ElectricEngine	FuelType = Electric
BatteryModule	Configurator Module	IntentsCondMod_ElectricEngine-E...	IntentsCondMod_Battery-Battery	ReplacementBattery = true
PetrolEngineModule	Configurator Module	IntentsCondMod_Engine-Engine	IntentsCondMod_PetrolEngine-PetrolEngine	FuelType = Petrol
SeatModule	Configurator Module	IntentsCondMod_Car-Car	IntentsCondMod_Seat-Seat	
SeatEcoModule	Configurator Module	IntentsCondMod_Seat-Seat	IntentsCondMod_SeatEco-SeatEco	
SeatPremiumModule	Configurator Module	IntentsCondMod_Seat-Seat	IntentsCondMod_SeatPremium-SeatPremium	
ArmrestModule	Configurator Module	IntentsCondMod_SeatPremium-S...	IntentsCondMod_Armrest-Armrest	

+ Add - Remove Expand Reuse Advanced Effectivity Selection Mode Select All Refresh Export To Excel							
Object	Type	Name	Feature Data Type	Optional	Free-form	Multi-select	Intents
IntentsCondMod_Engine-Engine	Configurator Context	Engine					
ICM_EngineDetails	Group	ICM_EngineDetails					
FuelType	Family	FuelType	String	False	False	False	Manufacturing
Unassigned Families							

Procedure

1. In the search box, type the item revision name, open the item revision, and switch from **Details** > **Variant Configuration**.

Revision: Global (Latest Working) ▾ Date: Today ▾ Units: None ▾ Variant: No Variant Rule ▾ Expansion: No Rule ▾				Reset View
+ Add ▾ Edit Structure ▾ Find ▾ Expand ▾ Compare ▾ Rollup ▾ Duplicate ▾ Selection Mode ▾ Select All ▾ Excel Roun... ▾				
Element ▾	ID ▾	Revision ▾	Revision Name ▾	
IntentsCondMod_BOM_Car/A;1	IntentsCondMod_BOM_Car	A	IntentsCondMod_BOM_Car	
000365/A;1-EngineAssembly	000365	A	EngineAssembly	
000368/A;1-PetrolEngineAssembly	000368	A	PetrolEngineAssembly	
000369/A;1-ElectricEngineAssembly	000369	A	ElectricEngineAssembly	
000383/A;1-IL-FuelType-Petrol	000383	A	IL-FuelType-Petrol	
000384/A;1-IL-FuelType-Electric	000384	A	IL-FuelType-Electric	
000367/A;1-SeatAssembly	000367	A	SeatAssembly	
000372/A;1-EconomySeatAssembly	000372	A	EconomySeatAssembly	
000373/A;1-PremiumSeatAssembly	000373	A	PremiumSeatAssembly	
000406/A;1-IL-ComfortOptions-Massage	000406	A	IL-ComfortOptions-Massage	
000407/A;1-IL-ComfortOptions-Recliner	000407	A	IL-ComfortOptions-Recliner	
000408/A;1-IL-SeatModel-S1	000408	A	IL-SeatModel-S1	
000409/A;1-IL-SeatModel-S2	000409	A	IL-SeatModel-S2	
000410/A;1-IL-SeatType-Captain	000410	A	IL-SeatType-Captain	
000411/A;1-IL-SeatType-Folding	000411	A	IL-SeatType-Folding	
000376/A;1-IL-Model-M1	000376	A	IL-Model-M1	
000377/A;1-IL-Model-M2	000377	A	IL-Model-M2	
000378/A;1-IL-DriveSide-LHD	000378	A	IL-DriveSide-LHD	
000379/A;1-IL-DriveSide-RHD	000379	A	IL-DriveSide-RHD	
000380/A;1-IL-Transmission-Auto	000380	A	IL-Transmission-Auto	
000381/A;1-IL-Transmission-Manual	000381	A	IL-Transmission-Manual	

2. Choose one of the following modes:

- In the **Guided** mode, only valid features are displayed.

This mode allows you to navigate only valid selections and configurations. If no configuration is applied, a model opens in guided mode, and the system guides you to select a valid group of features in which to configure a product. Every time a selection is made, the system reevaluates the rules and refreshes the list of features.

Guided mode supports both complete and partial configurations.

- In the **Manual** mode, all available features, including variants that are not valid in the current context, are displayed.

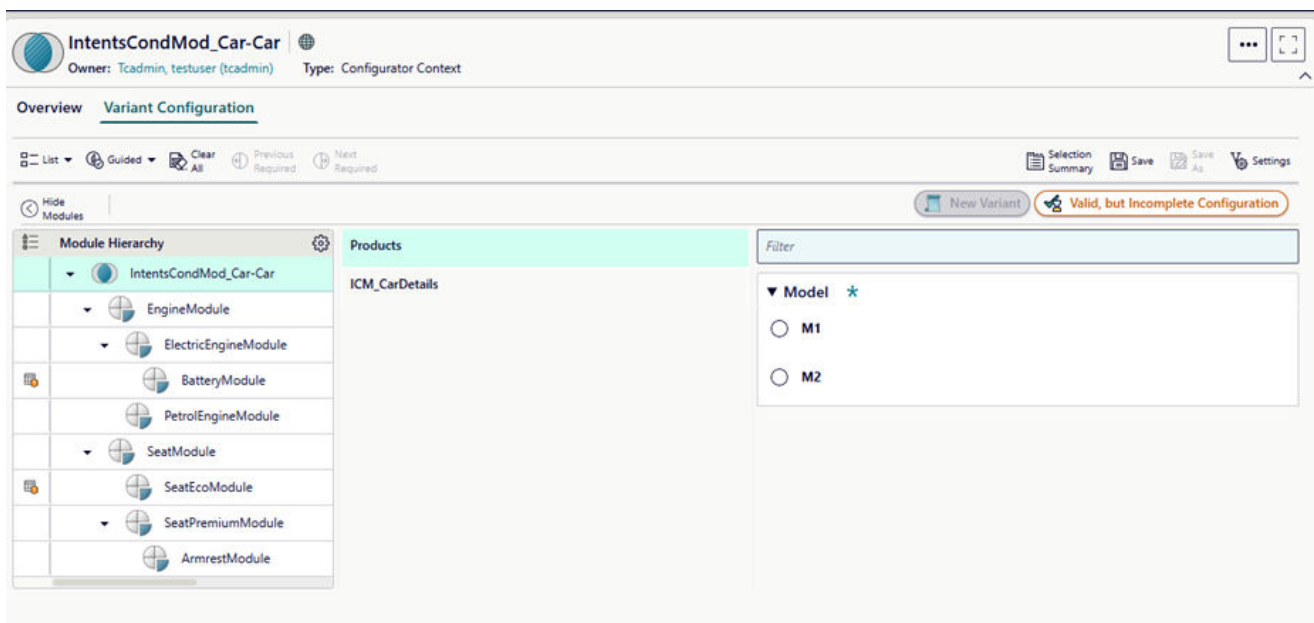
In this mode, you can look at the entire variability, choose a variety of features, and assess if those selections are valid based on the configurator rules. You can validate your selections and save your configuration. If there are any conflicting constraints, the system reports violations as error messages and specifies the conflicting selections.

While radio buttons are displayed for making selections in **Guided** mode, check boxes are displayed for **Manual** mode.

3. Apply filter criteria and validation mode.

- a. Click **Settings**  to view the **Profile Settings** panel.
- b. In the **Filter Criteria** section, choose the product hierarchy depth.
The default value is **1** and the maximum is **5**.
- c. Change the snapshot revision rule, effectivity, and rule date policy as appropriate.
For more information, see [filter configurator data by revision rule, effectivity, rule date, and intent](#).
- d. In the **VALIDATION MODE** section, [choose a mode from the drop-down list](#).
 - **Order** (default profile)
This mode is intended for a user who wants to quickly arrive at a 100% BOM. You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.
 - **Overlay**
This mode does not expect you to specify the complete configuration. It is intended for developing a new product or making changes to an existing product.

4. In the **Variant Configuration** view, observe the module hierarchy.



Note that two modules, **BatteryModule** and **SeatEcoModule** are configured out due to some constraints or rules.

5. Select a product model.
6. Select each group from the list in the left panel and make the necessary feature selections in the work area.
7. (Optional) To find the required selections, choose **More Commands > Next Required** or **Previous Required** as appropriate.
8. (Optional) To find user and system selections, click **Full Screen**, choose **Selection Summary**, and select **All** from the **Selection Type** list.

When the product hierarchy depth is greater than **1**, the **Selection Summary** panel shows an additional column, **Module**. This column displays where the selection has been made, that is, the module, family, feature, and type of selection.

9. Make all the required selections.

You can make selections at the root level and child level modules (configurator context levels).

In the **Guided** mode, the validation happens for each of your selections, and its configuration completeness is displayed automatically.

If you are in the **Manual** mode, when you click **Expand** or **Validate**, the current selections in the UI are validated or expanded, respectively.

With modular configuration, along with configuration completeness, a module completeness check is also done.

The configuration completeness is standard for all product structures, irrespective of the modular configuration.

The completeness indicator at the top of the view displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

- **Valid and Complete** (applicable for both guided mode and manual mode)

Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

When a configuration is valid and complete, you could take this configuration input and execute an order entry in the order system. When a configuration is valid and complete, you can take this configuration input and perform a cost or a weight rollup, compute the center of gravity, or perform a digital analysis of that product.

On the contrary, if you do not see this completeness indicator, you should not typically perform the above action. It means that you have less content or more content.

When a configuration is valid and complete, it assigns a value to every family. This is irrespective of whether the family is mandatory or discretionary.

- **Valid and Incomplete** (applicable for both guided mode and manual mode)

In guided mode, you can select **Next Required** or **Previous Required** to navigate to the next or the previous family or the feature that requires a value to be assigned to complete the configuration.

In the case of modular configuration, **Next Required** or **Previous Required** takes you to the next module after reaching the last incomplete family of the last group.

- **Invalid** (applicable for manual mode only)

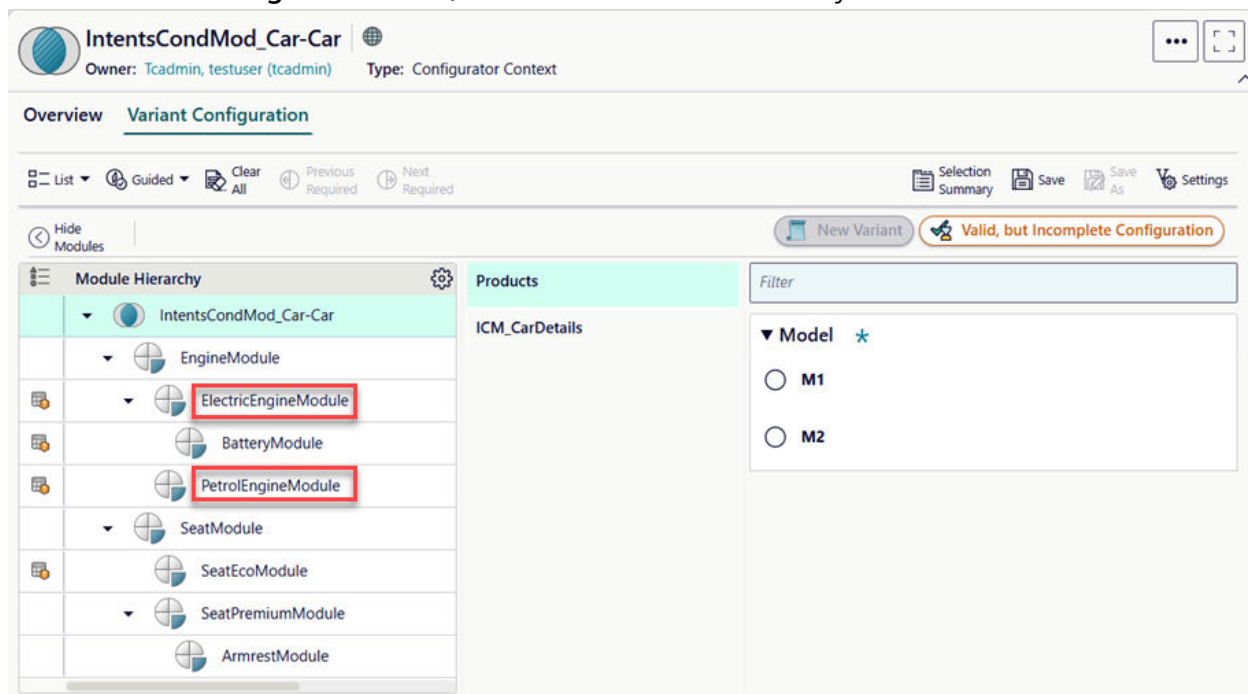
It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

10. Click **Clear All** to clear all your selections.
11. Select an intent to scope down the structure.

Note

Modules that are already configured out due to variant conditions remain configured out when intents are applied.

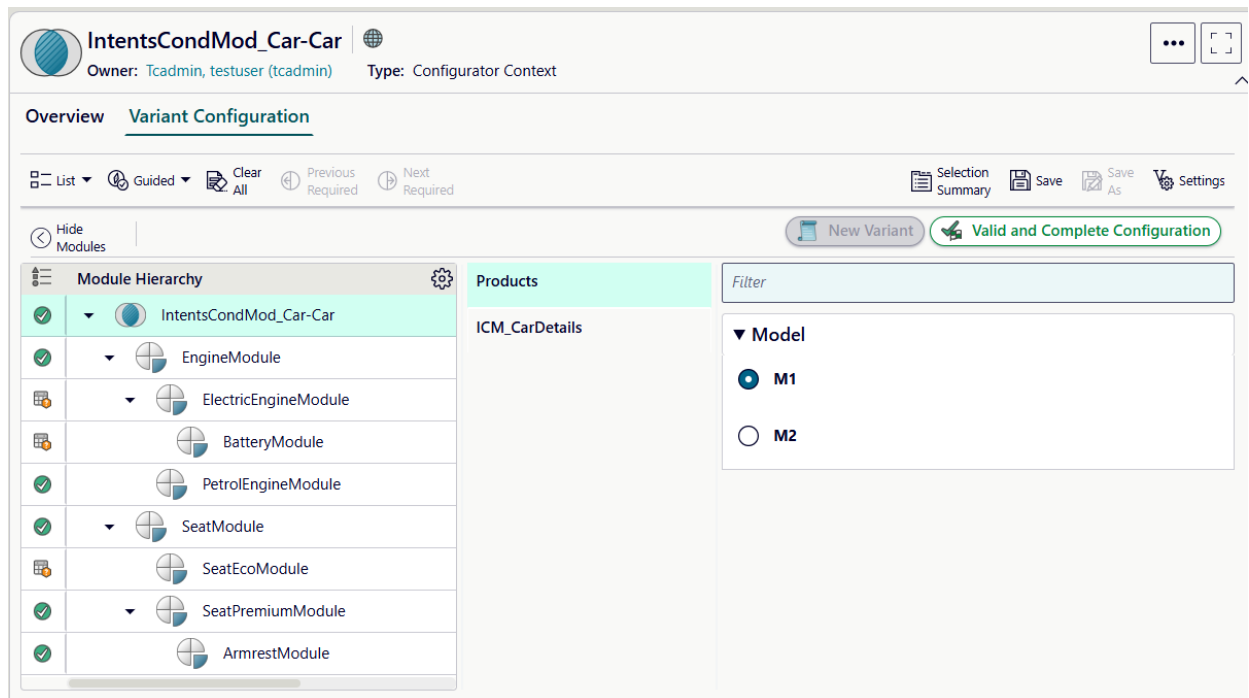
- Click **Settings**  and select **Marketing** from **Snapshot Intents**.
- In the **Variant Configuration** view, observe the module hierarchy.



Note that two additional modules, **ElectricEngineModule** and **PetrolEngineModule** are configured out. This is because you have selected the **Marketing** snapshot intent to scope the configuration before performing a solve.

- Select a product model such as **M1**.

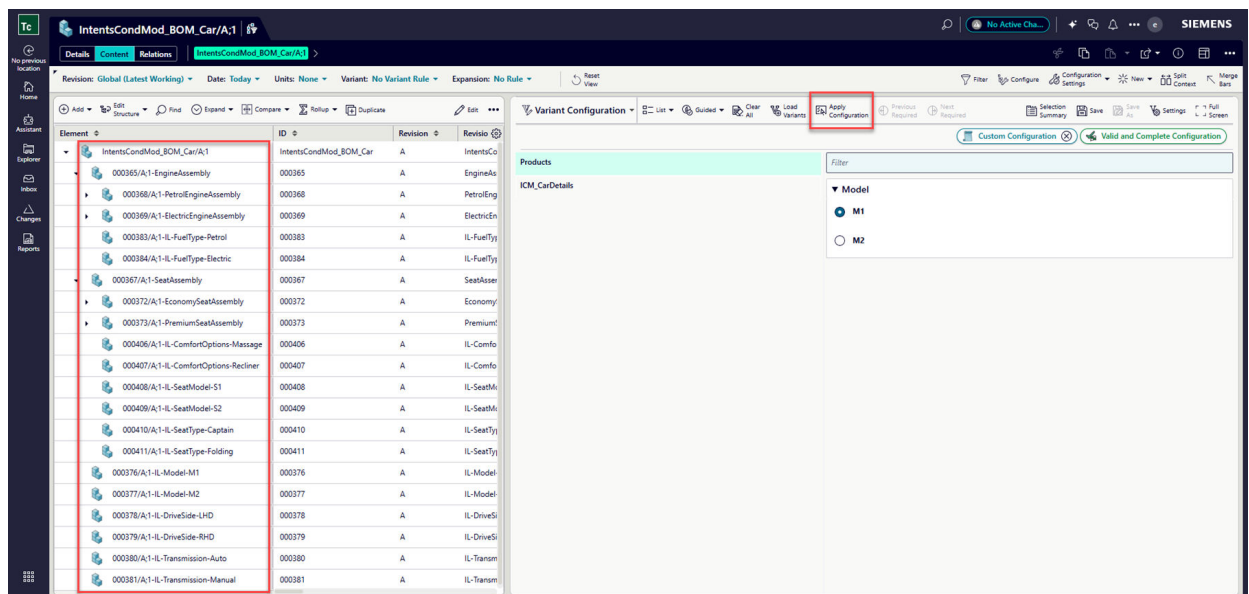
The system makes all the required selections based on constraints and the configuration becomes valid and complete.



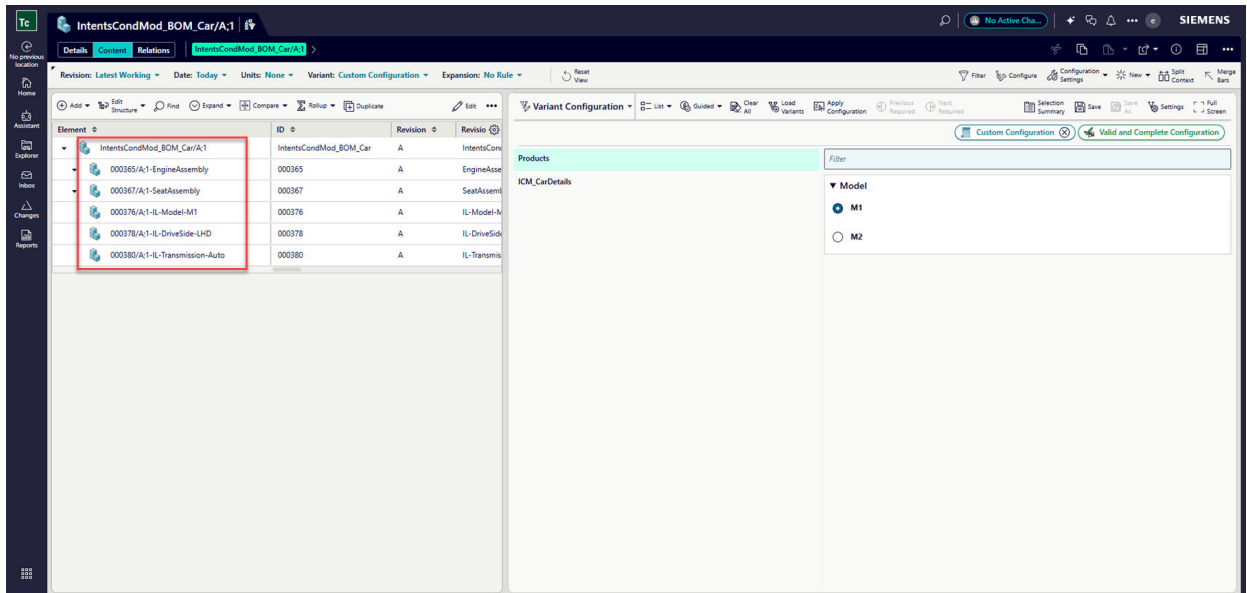
- d. Click **Apply Configuration**.

The BOM lines are configured according to the applied configuration.

Before applying the configuration:



After applying the configuration:



Perform a solve using a custom configuration

Different modes to analyze configurations

You can use **Order** and **Overlay** modes for configuring and analyzing variant configurations. The **Order** mode helps you achieve a 100% Bill of Material (BOM) for shippable products by automatically completing configurations. The **Overlay** mode is used during product development to show compatible features and content without requiring a complete configuration.

You can use the **Order** mode or the **Overlay** mode to configure and analyze the variant configuration.

- **Order** mode

This mode is intended when the product is ready to be shipped. It is for a user who wants to quickly arrive at a 100% BOM. The 100% BOM is an *as sold* product configuration. An example of this is the configuration of a car that is going to be built and shipped to the dealer.

You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.

In this mode, the system considers all the information that it can get to autocomplete the configuration. You can select a certain model and select the country of sale, for example, UK or India, and the system automatically selects the right-hand drive. The system scans the entire knowledge base of the configurator in order to predict what the final 100% product looks like. You can select another color or a different engine. The configurator attempts to consider your inputs and complete the configuration as much as possible as per the default rules.

In cases where your inputs and the default rules are insufficient, the configurator does not achieve the original goal of completing a 100% configuration. When some user options are possibly left open, it achieves only a 90% configuration.

- **Overlay** mode

This mode is intended when the product is in the development stage. It does not expect you to specify the complete configuration. You can, but it is not necessary. It configures all the features and the product content that are compatible with your inputs. For example, a new fuel filter, that you plan to introduce might have collision or interference problems within the existing space. However, at this stage, your focus is in determining all the environments in which the fuel filter can be assembled. The filter can be assembled in the 1.5 liter, 2.0 liter, and 3.0 liter gasoline engine options with or without an automatic transmission. However, the filter cannot be used in a diesel engine. Therefore, in this mode, when you select a fuel filter, you see a large overlay with various types of gasoline engines, but not a diesel engine. You can view only the options that are compatible with your selection.

Therefore, both modes converge for complete configurations, even if they might begin with very different starting positions, based on more inputs provided by the user. While the **Order** mode might automatically complete the empty input configurations to come up with a complete, default product, the **Overlay** mode leaves the same input configuration empty. When the input expression reaches a complete product configuration, the system behavior of both modes is identical.

Set the validation mode

The **Order** mode or the **Overlay** mode is used to configure and analyze the variant configuration. This can set the validation model for the current session.

Procedure

1. Find and open the product to be configured and choose **Details > Variant Configuration**.
2. Click  to view the **Settings** panel.
3. In the **VALIDATION MODE** section of the **Settings** panel, choose a mode from the drop-down list.

- **Order** (default profile)

This mode is intended for a user who wants to quickly arrive at a 100% BOM. You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.

- **Overlay**

This mode does not expect you to specify the complete configuration. It is intended for developing a new product or making changes to an existing product.

For more information about these modes, see [Different modes to analyze configurations](#).

The validation severity, expansion severity, and selection behavior settings are updated based on the selected mode. These settings are defined by your administrator and are therefore, unavailable and read-only.

4. Click **Apply**.
All user selections are retained, but the system selections and violations are cleared.

Tip

Once you change tabs or otherwise start a new session, the validation mode settings will return to the defaults. A **saved variant** stores this setting information, however, so loading the appropriate SVR is a quick way to return to your desired settings.

The individual settings that make up each mode definition are defined by your administrator and are read-only. If you change from one mode to the other, the settings change to reflect the selected mode.

When applied, all user selections are retained, but the system selections and violations are cleared.

Choosing the validation mode sets the severity for reporting and expansion in order to control how the solver behaves and how violations are reported. When a mode is selected, the definition of that mode is shown, including its validation severity, expansion severity, and its selection behavior. These settings determine how strictly you want to evaluate the constraints in the system.

The image displays two side-by-side screenshots of the 'Settings' dialog box, specifically the 'Validation Mode' section. Both screenshots show the same settings for Validation Severity, Expansion Severity, Selection Behavior, Content Configuration, and Expansion Behavior, with checkboxes for Error, Warning, Information, Default, System Enforced Defaults, Allow Multiple Selections for Single Select Families in Manual mode, Apply Constraints, Explicit Configuration, and Allow Validation Rules to Expand.

Left Screenshot (Order Profile):

- Validation Mode:** Profile: Order
- Validation Severity:**
 - ☒ Error
 - ☐ Warning
 - ☐ Information
- Expansion Severity:**
 - ☒ Error
 - ☒ Warning
 - ☒ Information
 - ☒ Default
 - ☒ System Enforced Defaults
- Selection Behavior:**
 - ☐ Allow Multiple Selections for Single Select Families in Manual mode
- Content Configuration:**
 - ☐ Apply Constraints
 - ☒ Explicit Configuration
- Expansion Behavior:**
 - ☒ Allow Validation Rules to Expand

Right Screenshot (Overlay Profile):

- Validation Mode:** Profile: Overlay
- Validation Severity:**
 - ☒ Error
 - ☐ Warning
 - ☐ Information
- Expansion Severity:**
 - ☒ Error
 - ☐ Warning
 - ☐ Information
 - ☐ Default
 - ☐ System Enforced Defaults
- Selection Behavior:**
 - ☒ Allow Multiple Selections for Single Select Families in Manual mode
- Content Configuration:**
 - ☒ Apply Constraints
 - ☐ Explicit Configuration
- Expansion Behavior:**
 - ☒ Allow Validation Rules to Expand

- The **Validation Severity** applies when the configuration is validated and is always higher than the expansion severity.
- The **Expansion Severity** is relevant when an order is expanded by clicking **Apply System Selections**, and the user wants to communicate the severity level to the solver.
- (Not applicable for **Order** mode by default.) The **Selection Behavior** consists of a checkbox which determines whether multiple selections for a single select family will be considered valid or not (in manual mode, where single select families are represented with check boxes for each feature).

Note

The **Selection Behavior** setting is ignored in guided mode.

- (Not applicable for **Order** mode by default.) The **Apply Constraints** check box, if selected, determines if all constraints should be considered while applying the variant expression on the content. If deselected, only user selections are applied.
- (Not applicable for **Overlay** mode by default.) The **Explicit Configuration** check box, if selected, returns only BOM lines where the features have been explicitly selected in **Order** mode. This leads to a configuration that is 100% or less. Conversely, if the **Overlay** mode is selected, the **Explicit Configuration** check box is not selected by default. This leads to a configuration that is more than 100%, for example, a 120% BOM. This happens unless a complete configuration is specified that results in a 100% BOM.

For more information, see *Example: Solve behavior using explicit configuration*.

- The **Allow Validation Rules to Expand** check box, if selected, determines if validation rules should participate during criteria expansion. If deselected, only violations, if present, are reported. The system selection, based on validation rules, does not occur in the expanded criteria.

By default, this check box is selected.

Example

When the validation severity is set to **Error** (as it is for both **Order** and **Overlay** modes), the configuration is considered invalid only when the **Error** severity rules are in conflict. When there are conflicts between **Warning** and **Information** constraints, the configuration is considered valid.

Filter configurator data by revision rule, effectivity, rule date, and intent

You can filter configurator data by applying revision rules, effectivity (date or unit ranges), rule dates (like Latest or Date), and intents (such as Manufacturing or Marketing) to view specific configurations and manage product variations effectively.

You can filter configurator data by [revision rule](#), [effectivity](#), [rule date](#), and intent.

Procedure

1. Find and open the product to be configured and choose **Details > Variant Configuration**.

Note

If your configuration analyst has not associated a configurator context with the product, **Variant Configuration** is not displayed.

2. Click **Settings**  to view the **Settings** panel.
3. Filter configurator data by revision rule.

Content changes can be managed by revising product data. Changes that may necessitate a new revision include addition or removal of components, relationships to other products or assemblies, and changes to properties. You can then apply a revision rule to view the correct revisions of the product data.

Similarly, you can apply revision rules to configure the families and features for a particular configuration. The revision rule (as well as effectivity and rule date) in the **Settings** panel of **Variant Configuration** *only* applies to the configurator families and features. For example, a feature may have a specific rule date, and the **Settings** panel controls whether that feature is available for selection in a configuration.

Note

The revision rule and effectivity for configuration is separate and independent of the revision rule and effectivity for product data.

Rule date is unique to configuration.

You cannot view multiple configurations at the same time; if you want to see another configuration, you must apply a different revision rule.

Revision rules are predefined by your system administrator, and you select a rule from the list of available rules.

To override the default revision rule for the current session:

- a. In the **FILTER CRITERIA** section of the **Settings** panel, click the current revision and select a revision rule from the resulting **Revision Rule** list.

Note

An invalid revision rule is a rule that contains a clause that is not valid for a particular type of object. While both valid and invalid revision rules appear in this list, the system will not let you apply an invalid revision rule.

- b. Click **Apply**.
The system refreshes the displayed content to show only content configured in by the selected revision rule.

4. Filter configurator data by effectivity.

You can configure configurator data with effectivity. When you apply the effectivity, the features are configured according to the unit range or the date range you specify. The system then shows the features that are in effect for the specified date, unit number or range.

You can edit the **Cfg0DefaultEffectivity** user preference to set the default effectivity. While editing this preference, you can specify date ranges or units.

Note

This preference is not supported in rich client.

Examples:

- *Date Range:* <YYYY-MM-dd..YYYY-MM-dd> (Example: **2023-11-08..2023-11-30** or **2023-11-30..UP**)
- *Today:* The present date. (Example: If **Today** is **2023-11-08**, it is displayed in the user interface as **2023-11-08..2023-11-09**)

For more information about setting this preference, see the description of this preference.

For information about retrieving a list of preferences, see *Where can I get a list of preferences?* in *Teamcenter Administration*.

Only the valid features are displayed in the structure views when you set the default effectivity.

Note

Your administrator configures the effectivity scheme used by your business by setting the **PCA_effectivity_shown_columns** preference. This preference can be set to show **Effectivity Date**, **Unit**, or **Both**.

You must ensure that the values specified for both **Cfg0DefaultEffectivity** and **PCA_effectivity_shown_columns** preferences are in sync. For example, if you set the default value as date range, then the effectivity shown column preference should also be set to date. Otherwise, the system displays a warning and the default value is set.

If your site supports date and unit effectivity, both selections are displayed in the **Settings** panel.

To override the default or change the existing dates:

- In the **FILTER CRITERIA** section of the **Settings** panel, click **All Dates** or the displayed date under **Effectivity**.
- Use the calendar to select the desired start date.

To set a date range, use the calendar to select the desired end date.

Using the pull-down menu, you can select **UP** (any date after the one specified) or **SO** (stock out), rather than a specific date.

- Click **Set**.
- To reconfigure the content to match the new date, click **Apply**.

5. Filter configurator data by rule date.

BOM engineers who develop new products typically prefer to use the **Latest** rule date. That is because the configurator engineer is continuously improving configurator rules or adding new ones, and, therefore, the BOM engineer wants to validate the product against the **Latest** rule date.

Order engineers need to validate the variant rules or variant criteria against the rules that were valid at the time of the order. If the order is already in production, they might not want to apply the **Latest** rule date. In such cases, **Date** that is set on the order is used to validate variant rules or variant criteria.

The **Cfg0DefaultRuleDateForContent** user preference specifies the default rule date mode value for the configurator views. By default, it is set to **Latest**.

For information about retrieving a list of preferences, see *Where can I get a list of preferences?* in *Teamcenter Administration*.

- a. Select a rule date as appropriate and click **Set**.

Rule date	Description
Date	Sets the rule date selected from the widget.
Latest	Sets the rule date as the current time on the clock.
System Default	Sets the rule date as the system default.
No Rule Date	Sets the rule date as null.

- b. To apply the settings, click **Apply**.

6. Filter configurator data by intent.

You can filter out configurator data by specific intent while authoring variant conditions. For example, if you are authoring a bill of materials (BOM), you can focus on features with technical intent. Conversely, if you are looking at a bill of process (BOP), you can focus on features with marketing intent. The configurator administrator defines a business purpose or intent, such as manufacturing, marketing, or technical, to categorize families and constraints.

- a. In the **Snapshot Intent** drop-down list, select the intents you want to filter the configurator data.

If configured, you can select the default values such as Manufacturing, Marketing, or Technical or select any custom intents defined by your administrator. The intents selected by default in the Snapshot Intents list may vary according to the configuration set by your administrator. The displayed content refreshes to reflect the filtering for the selected intent. Filtering is applied to all configurator contexts associated with the product.

- b. Click **Apply**.

Note

If a family is used in multiple configurator contexts, it is recommended that you apply the same intents to that family across all contexts. For example, if the **Fuel Type** family has **Manufacturing** and **Technical** intents in the **Car** context, ensure that it also has **Manufacturing** and **Technical** intents in the **Engine** context. This consistency helps maintain predictable behavior and simplifies management across your product structures.

- c. Make your selections to create a valid and complete configuration and click **Save**.

When you filter configurator data by applying **Snapshot Intents**, and then save the valid and complete SVR, the system records the specific intent that was applied when the valid and complete SVR was created and saved. This allows you to create multiple SVRs for the same product configuration, with each considered complete but also under different intents. For example, one SVR might be complete for a Technical intent, while another is complete for a Marketing intent. The variant rule text for each SVR reflects only the families and features that participated in the configuration under its specific snapshot intent.

The **Snapshot Intent** column in the **Variants** tab displays the intent that was used during configuration. If the column is hidden, click **Table Settings > Arrange**, and add **Snapshot Intents** from the **Available Columns** to the **Displayed Columns**, and then click **Save and Arrange**.

Filter configurator objects using intents

You can dynamically filter families and constraints based on their assigned intents and allocation intents. This filtering helps you focus on specific subsets of variability relevant to your current task or domain.

Prerequisites

Your administrator has enabled filtering by intents.

Procedure

1. Open the configurator context and click the **Models, Features, or Constraints** tab.
2. In the **Intents** list, select the intent that you want to specify use for filtering.

You can select multiple values as required. The objects in the view are refreshed based on the selected intents.

Results

- For dynamic families, the standalone features are filtered as specified.
- If a family has an allocation intent assigned, the system uses this allocation intent for filtering. For example, if a family has an allocation intent of **Marketing** and a standard intent of **Manufacturing**, and you select **Marketing** in the breadcrumb, the family is filtered in. If you select **Manufacturing**, the family is filtered out.
- If a family does not have an allocation intent, the system uses its standard intent for filtering.
- If a family has no specified intents, neither standard nor allocation, it is always filtered in, which means that it is displayed regardless of the intents selected in the breadcrumb.

Configure variants in guided mode

You can configure variants using the guided mode. The guided mode simplifies the configuration process by displaying only valid features and helps you create customized variant configurations. It also supports both complete and partial configurations while providing real-time feedback on configuration validity and completeness.

Procedure

1. Find and open the product to be configured and choose **Details > Variant Configuration**.
The system displays a list of available product lines, product models, and groups. If the product has a variant configuration defined, the selections in the configuration are shown with the variability in the context.

Note

Your administrator can deploy the system with the **Cfg0PrimaryBusinessRelevantAttribute** global constant that allows you to use either **Name** or **ID** as the primary business-relevant attribute. When the system is configured with the global constant that points to **Name** as a primary business property, all Product Configurator views display **Name** instead of **ID** in the corresponding areas.

2. Select the **Guided** mode. This is the default mode.

In this mode, only valid features are displayed.

Guided mode allows you to navigate only valid selections and configurations. If no configuration is applied, a model opens in guided mode, and the system guides you to select a valid group of features in which to configure a product. Every time a selection is made, the system reevaluates the rules and refreshes the list of features.

Guided mode supports both complete and partial configurations.

While radio buttons are displayed for making selections in **Guided** mode, check boxes are displayed for **Manual** mode.

Note

If you have an invalid configuration, the system does not allow you to switch from manual to guided mode.

3. To begin creating a customized variant configuration, do one of the following:
 - Select the product line containing the product model with which you would like to work.
The model families in the selected product line are selected by the system.
 - Select a product model with which you would like to work.
The system automatically selects the product line containing the selected product model.
When you make some selection changes and revert them, the system continues to display the **Save** icon.

Note

Product lines are only listed if they are defined in the configurator model.

4. After you have selected the appropriate product model in the work area, select each group from the list in the left panel and make the necessary feature selections in the work area.
 - Use the **Filter** box to help locate the appropriate feature. The text filter is not case-sensitive and is based on the family and feature properties shown (**Name** and **Description**, for example).

Example

You type **mfv** into the **Filter** box.

- The filtered list shows the families containing **mfv** in the family name, along with all the features in that family.
- The filtered list shows the features containing **mfv** in the feature name, along with its parent family.

- An  option button displays when a single-select feature is selected. Click the feature again to deselect.
- In multiselect and Boolean families, you can select as follows:
 -  indicates include.
 -  indicates unselected.
 -  indicates exclude.

To exclude a feature, hover over it and select the **Exclude** icon.

5. Select one feature in a single-select family or one or more features in a multiselect family. The completeness indicator at the top displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

- **Valid and Complete** (applicable for both guided mode and manual mode)

Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

When a configuration is valid and complete, you could take this configuration input and perform a cost or a weight rollup, compute the center of gravity, or perform a digital analysis of that product.

On the contrary, if you do not see this completeness indicator, you should not typically perform any of the above actions. It means that you have less content or more content.

When a configuration is valid and complete, it assigns a value to every family. This is irrespective of whether the family is mandatory or discretionary.

- **Valid and Incomplete** (applicable for both guided mode and manual mode)

In guided mode, you can select the **Next Required** or the **Previous Required** option to navigate to the next or the previous family or feature that requires a value to be assigned to complete the configuration.

- **Invalid** (applicable for manual mode only)

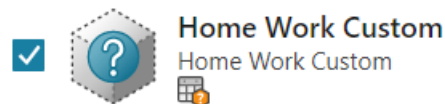
It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

It may also refresh features displayed for other families as it applies the relevant rules by hiding invalid features. If none of the features in a family is still valid, the family itself is hidden.

Default features are selected by the system if they do not already have a value and designated with the  indicator. Constraints are applied, validating the expression and adding any additional expression terms.

6. Select a family or feature that has a specified revision rule, effectivity, or rule date. Apply a different revision rule, effectivity, or rule date. This becomes an unconfigured family or feature.

Such families and features are displayed with a different icon and have a question mark and an indicator that shows it as *configured out*. In such cases, the system switches to the manual mode. To proceed in guided mode, you can clear your selection of the unconfigured family or feature and select another family or feature. The system switches to the guided mode and you can continue to configure the structure.



- If a family is unconfigured and you select a feature within it, then both the family and feature are shown as unconfigured.
- If you deselect an unconfigured feature or family and navigate to a different group, both the family and feature are not displayed by the system. Similarly, if you deselect an unconfigured feature or family, save it, and reopen it, neither the family nor the feature are displayed by the system.

You can optionally save the content with an unconfigured family or feature as an SVR.

7. An optional feature family contains product configurations where users are not required to select a feature for the family. For example, you may have a family with features for optional equipment.

If optional feature families are defined in a product model, an additional option is displayed in the family section header. Click  if you want *none* of the optional family features to appear in the structure.

When Teamcenter configures out all features from an optional family based on constraints, the family itself is hidden from the **Variant Configuration** view. This does not impact the configuration. If you select **Selection Summary**, and **All** from the **Selection Type**, the optional family that is hidden from the GUI is displayed as a **System Negative** type. When you click the optional family that is hidden from the GUI, the system displays a message as shown below.

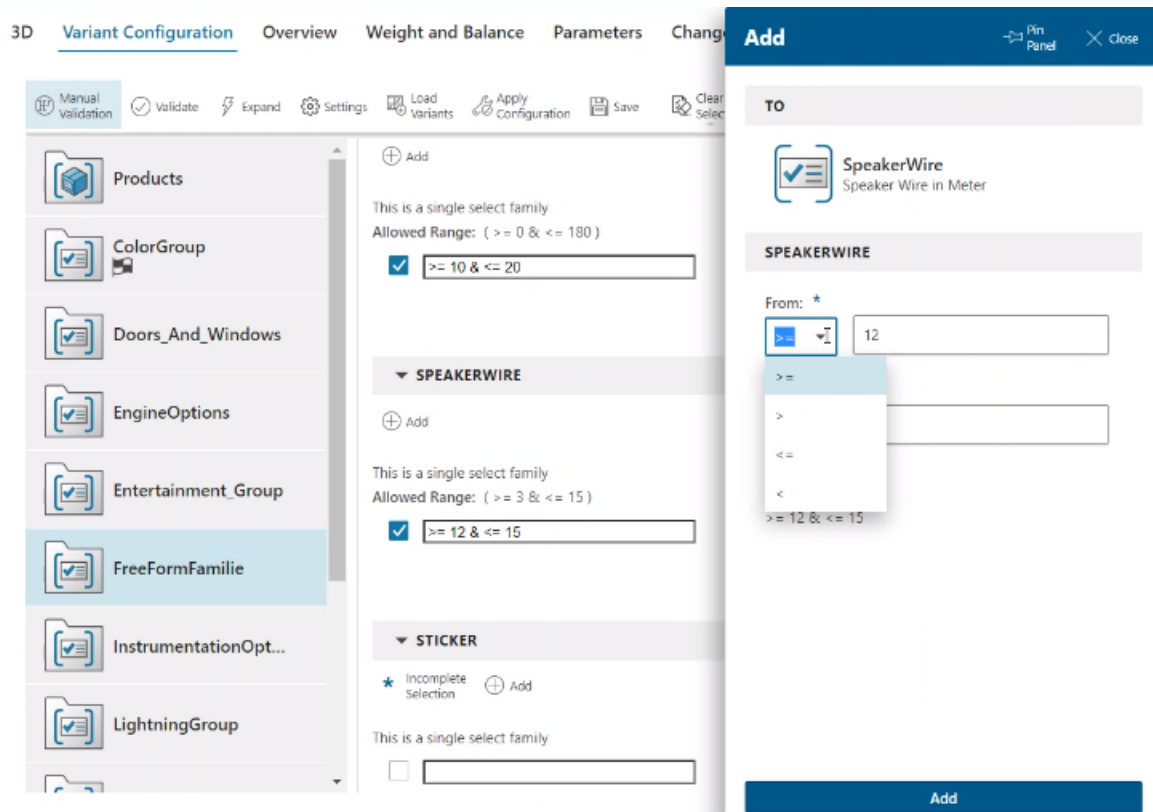
The screenshot shows a configuration interface with a top bar containing buttons for 'Selection Summary', 'Save', 'Save As', and 'Settings'. Below the top bar, there is a 'Filter' input field and a warning message: 'Warning - The selected family is hidden as all its features are configured out. Switch to the Manual Mode to view it.' The main area displays a list of features with their selection status. The 'Selection Type' is set to 'All'. The '11 features selected' section shows a table of features and their selection status.

Selection Type	Family	Feature	Group	
☒	User Positive	Models	LXI	Products
☒	Default Negative	DymFamA	None	DynamicGrp
☒	Default Negative	TransmissionType	None	EngineOptions
☒	Default Negative	Auxiliary	Auxiliary	EngineOptions
☒	System Negative	Engine	None	EngineOptions
☒	Default Negative	EnginePower	None	EngineOptions
☒	Default Negative	InputSupported	Bluetooth	EngineOptions
☒	Default Negative	InputSupported	USB	EngineOptions
☒	Default Negative	DriveSystem	None	FreeFormGrp
☒	Default Negative	MaxSpeedSupported	None	FreeFormGrp
☒	Default Negative	DymFamB	None	DynamicGrp

- Typically, features are predefined for each family, but free-form families allow you to enter a value for a feature at any time. If a family is specified as free-form, you can enter numbers, strings, or dates, depending on the data type defined.

Select an assembly that contains free-form families in the **Variant Conditions** tab, and click the **Add Feature** option. You can use operators such as $\geq$, $>$, $\leq$, and $<$. If you select $\leq$ or $<$ in the **From** field, the **To** field is disabled.

If you have a **SpeakerWire** free form family in your configuration context, then you can enter the free form value as a discreet value or a range in the text box or use the **Add** panel to author the range. According to the range or the discreet value you specified, the configuration context displays only the correct BOM lines. For combining the values, you can use the AND (&) operator. The OR (|) operator is not supported.



When you apply the configuration to the structure, the system filters the structure to show only the items that meet the condition for the feature.

9. As you make selections for the custom configuration, in the **Selection Summary** , filter your selections using one of the following options from the **Selection Type** list.

Selection Type	Description
User	Displays the features selected by the user.
All	Displays all features that are selected. These include user and system selections.
Incomplete Families	Displays the families from which you have not made any selections yet.

Selection Summary

Save

Save As

Settings

ModularConfigurationValidAndCompl...Valid, but Incomplete Configuration

* Selection Type:

All

28 features selected

	Selection Type	Family	Feature	Group
	Default Negative	InputDevicesSupported	CDPlayerDevice	EntertainmentGroupFam...
	System Positive	ColorTypes	Metallic	EntertainmentGroupFam...
	System Negative	ColorTypes	NonMetallic	EntertainmentGroupFam...
	User Positive	ExteriorColors	Orange	EntertainmentGroupFam...

In the selection summary table, you can:

- Identify the type of selection by viewing the icon in the first column. For example, user positive selections are indicated by . To know more about the icons and the type of selection they indicate, hover over the icon in the column header.
- View information about the **Family**, **Feature**, and **Group** of the selected features.
- Sort the columns in ascending or descending order as required. To do this, click the header cell of the specific column and select the required option.
- Filter the column contents. To do this, click the header cell of the specific column, select the option from the **Contains** list, enter the filter criteria in the search box and click **Filter**.

The selection summary options help you cross-probe to narrow down and select specific features to complete your configuration.

10. At any time, click **Clear All** to clear all user and system selections as well as any reported violations across all groups.
This provides a blank configuration from which to start.
11. (Optional) **Save the custom configuration**.
12. Click **Apply Configuration** to apply the selected variant configuration to the current content and to refresh the content in the product.

Configure variants in manual mode

You can configure variants using the manual mode. The manual mode help you view all the available features, including potentially invalid ones, and manually select and validate configurations. You can

identify and resolve conflicting constraints, understand system indicators for violations, and apply configurations to customize product structures.

You can view all the product lines and product models in a configurator context. Product lines are sometimes referred to as *series* and provide a high-level organization of product models.

Procedure

1. Find and open the product to be configured and choose **Details > Variant Configuration**. The system displays a list of available product lines, product models, and groups. If the product has a variant configuration defined, the selections in the configuration show with the variability in the context.

Note

Your administrator can deploy the system with the **Cfg0PrimaryBusinessRelevantAttribute** global constant that allows you to use either **Name** or **ID** as the primary business-relevant attribute. When the system is configured with the global constant that points to **Name** as a primary business property, all Product Configurator views display **Name** instead of **ID** in the corresponding areas.

2. Switch to **Manual** mode by selecting **Guided > Manual**. You are now in manual mode.
 - You see all available features, including features that are not valid in the current context.
 - Summary models are listed below the models in a particular product group and act as a dependency chain to other places in the model. For example, one particular car model requires gasoline, while another requires diesel. Choosing one of these fuel categories triggers changes in other parts of the model, such as the engine itself, the need for spark plugs, the need for a fuel receptacle, and so on.
 - You can look at entire variability, choose variety of features, and assess if those selections are valid based on the configurator rules. You can validate your selections and save your configuration. If there are any conflicting constraints, the system reports violations as error messages and specifies the conflicting selections.
 - Single select groups are designated as such to show the intent of the groups. However, when you select a feature, other features in the same family remain displayed, allowing you to select additional features.
 - Configurator rules and defaults are *not* automatically applied. You must manually apply configurations by clicking .
 - Automatic selection, filtering, and indications of invalid selections are *not* automatically provided. You must manually validate  and apply system selections .

Note

If you have an invalid configuration, the system does not allow you to switch from manual to guided mode.

3. Select the appropriate model and the summary model.

When you make some selection changes and revert them, the system continues to display the **Save** icon.

4. Select the desired group to see the features.

You can only view one group at a time.

- Use the **Filter** box to help locate the appropriate feature. The text filter is not case-sensitive and is based on both the family and the feature names.

Example

You type **mfv** into the **Filter** box.

- The filtered list shows the families containing **mfv** in the family name, along with all the features in that family.
- The filtered list shows the features containing **mfv** in the feature name, along with its parent family.

- Because manual mode allows multiple selections for multiselect and Boolean families, you can select as follows:
 - ☒ indicates include.
 - ☐ indicates unselected.
 - ☐ indicates exclude.

To exclude a feature, hover over it and select the **Exclude** icon.

5. Select the appropriate features for this custom variant configuration.

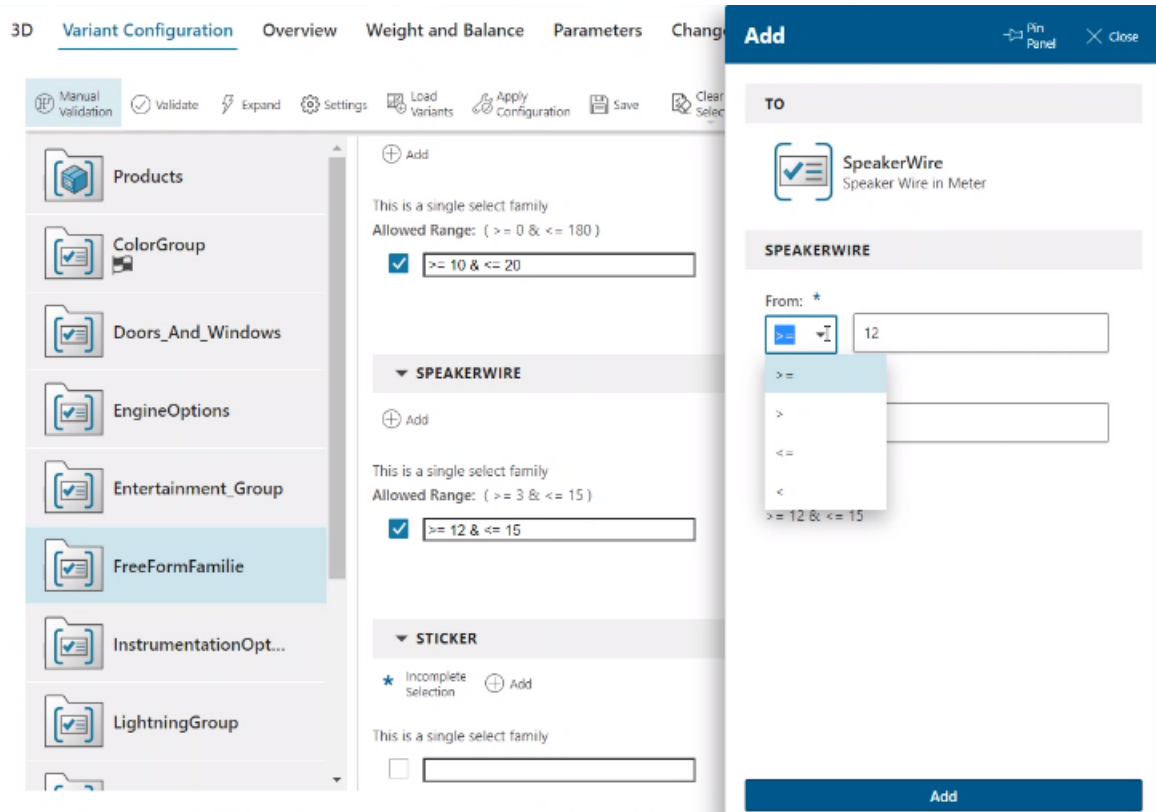
6. An optional feature family contains product configurations where users are not required to select a feature for the family. For example, you may have a family with features for optional equipment.

If optional feature families are defined in a product model, two options display in the family section header. Click  to allow *any* of the optional family features to appear in the structure. Click  if you want *none* of the optional family features to appear in the structure.

7. Typically, features are predefined for each family, but free-form families allow you to enter a value for a feature at any time. If a family is specified as free-form, you can enter numbers, strings, or dates, depending on the data type defined.

Select an assembly that contains free-form families in the **Variant Conditions** tab, and click the **Add Feature** option. You can use operators such as **>=**, **>**, **<=**, and **<**. If you select **<=** or **<** in the **From** field, the **To** field is disabled.

If you have a **SpeakerWire** free form family in your configuration context, then you can enter the free form value as a discreet value or a range in the text box or use the **Add** panel to author the range. According to the range or the discreet value you specified, the configuration context displays only the correct BOM lines. For combining the values, you can use the AND (**&**) operator. The OR (**|**) operator is not supported.



When you apply the configuration to the structure, the system filters the structure to show only the items that meet the condition for the feature.

8. At any time, click **Validate** to validate the configuration.
Each violation is identified using the  error indicator.

Note

The type of violation assigned to a feature or feature group is based on the severity assigned by your configuration analyst. You cannot change the severity level in the current Active Workspace release. The severity is currently set to *error* by default.

Hover over the indicator icon next to a feature to view the reason for the violation.

- Next to a feature to view the reason for the violation.
- Next to a group to view the features within that group that have a violation.

The system identifies violations in the configuration using the following indicators:

-  error
-  warning
-  information

Note

The type of violation assigned to a feature or feature group is based on the severity assigned by your configuration analyst. You cannot change the severity level in the current web client release. The severity is currently set to *error* by default.

Hover over the indicator icon:

- Next to a feature to view the severity and reason for the violation.
- Next to a group to view the features within that group that have a violation.

9. As you make selections for the custom configuration, in the **Selection Summary** , filter your selections using one of the following options from the **Selection Type** list.

Selection Type	Description
User	Displays the features selected by the user.
All	Displays all features that are selected. These include user and system selections.
Incomplete Families	Displays the families from which you have not made any selections yet.

 Selection Summary
  Save
  Save As
  Settings

 ModularConfigurationValidAndCompl...
  Valid, but Incomplete Configuration

* Selection Type:

All

28 features selected 

	Selection Type 	Family 	Feature 	Group 
	Default Negative	InputDevicesSupported	CDPlayerDevice	EntertainmentGroupFam...
	System Positive	ColorTypes	Metallic	EntertainmentGroupFam...
	System Negative	ColorTypes	NonMetallic	EntertainmentGroupFam...
	User Positive	ExteriorColors	Orange	EntertainmentGroupFam...

In the selection summary table, you can:

- Identify the type of selection by viewing the icon in the first column. For example, user positive selections are indicated by . To know more about the icons and the type of selection they indicate, hover over the  icon in the column header.
- View information about the **Family**, **Feature**, and **Group** of the selected features.

- Sort the columns in ascending or descending order as required. To do this, click the header cell of the specific column and select the required option.
- Filter the column contents. To do this, click the header cell of the specific column, select the option from the **Contains** list, enter the filter criteria in the search box and click **Filter**.

The selection summary options help you cross-probe to narrow down and select specific features to complete your configuration.

Note

You can export the contents of the selection summary if required. To do this, in the **Variant Configuration** tab, in the **Selection Summary**  view, click **Export to Excel** .

10. At any time, click **Expand** to apply system selections.

The system makes selections based on configurator rules or constraints and displays the  indicator to designate the system selections.

The completeness indicator at the top displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

- **Valid and Complete** (applicable for both guided mode and manual mode)

Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

When a configuration is valid and complete, you could take this configuration input and perform a cost or a weight rollout, compute the center of gravity, or perform a digital analysis of that product.

On the contrary, if you do not see this completeness indicator, you should not typically perform any of the above actions. It means that you have less content or more content.

When a configuration is valid and complete, it assigns a value to every family. This is irrespective of whether the family is mandatory or discretionary.

- **Valid and Incomplete** (applicable for both guided mode and manual mode)

In guided mode, you can select the **Next Required** or the **Previous Required** option to navigate to the next or the previous family or feature that requires a value to be assigned to complete the configuration.

- **Invalid** (applicable for manual mode only)

It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

11. At any time, click **Clear Selections** > **Clear System Selections** to clear any system selections that may have been previously calculated, while preserving your user selections.

Alternatively, click **Clear Selections** > **Clear All** to clear all user and system selections, as well as any reported violations across all groups.

12. (Optional) **Save the custom configuration**.

13. Click **Apply Configuration** to apply the selected variant configuration to the current content.

Note

If you save the current context (for example, the subset definition or workset), the variant configuration displays in the list of saved variants.

Configure by saved variant or variant criteria

You can configure a specific product variant by applying the predefined saved variants or variant criteria. You can load these stored configurations that consist of families and features (for example, color = red).

Variants allow you to select and apply feature families (for example, color) and allow features of those families (for example, red and blue) to configure a part or assembly.

To configure a **particular variant of a product**, apply the appropriate variant (a group of families and features such as **color = red, material = cotton**) or variant criteria (a revisable version of a saved variant). These items are predefined by the configuration analyst and are stored in Teamcenter for users to retrieve later.

To configure content by a saved variant or variant criteria (VC):

Procedure

1. Find and open the product to be configured and choose **Details > Variant Configuration**.

Note

If your configuration analyst has not associated a configurator context with the product, **Variant Configuration** is not displayed.

2. To open the SVR or VC, click **Load Variants** .
3. Search for the appropriate configuration by typing the name (or part of the name) of the SVR or VC to be loaded.

The available SVRs and VCs is obtained from the configurator context associated with the product or from the product itself. Only SVRs and VCs directly associated with the product can be loaded. A variant that is created for a product is not available for other products, even if the products share a configurator context.

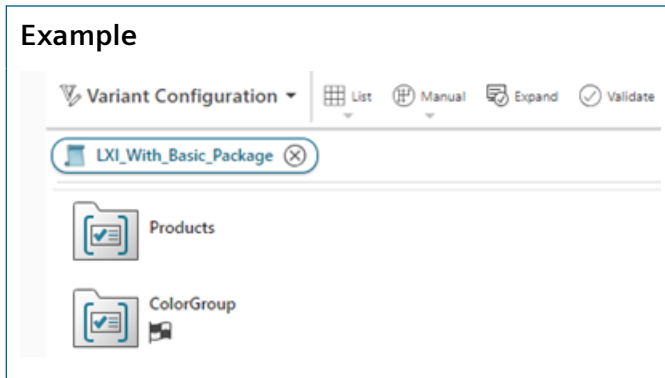
The SVRs and VCs matching the search criteria are listed.

4. Select an SVR or VC and click **Load**. The system loads the selected configuration.

Note

In this example, a single variant configuration is applied. Multiple variants can be applied to 4G *structures* from the **Variant** section of the **Configuration** panel. Access the **Configuration** panel by clicking **Configure** .

- If no variant is applied to the structure, the model opens in guided mode.
 - If a configuration is applied, the model displays in manual mode.
5. Remove the variant rule or variant condition applied to the structure.



Solve behavior for default rules

Default rules anticipate common settings that align with most users' preferences. When conflicts arise, Teamcenter resolves them by determining which default rules to suppress based on their sequence number and affiliation with the input configuration.

Default rules, created by the configuration specialist, enhance usability by assisting users in forming a complete product configuration. These rules anticipate common settings that align with the preferences of most users.

Default rules that are in conflict with the user input or constraint rules with a higher severity are automatically skipped when expanding the configuration. Teamcenter automatically resolves all conflicts that involve default rules by determining the optimum set of default rules to be suppressed based on their affiliation with the input configuration and their ranking based on sequence number. Default rules with a higher sequence number are evaluated later and therefore have a lower ranking. Those with the same sequence number have the same ranking. The system infers the rank of default rules based on site preferences set by the administrator.

Prerequisites

This topic assumes that the administrator has set the required site preferences for default rules as described in *Setting preferences for default rules in Product Configurator on Rich Client — Usage*.

Procedure

1. Create a **Car_2024** configurator context with a positive or negative bias.
2. Create a model family and product models as follows:

029321-Car_2024						
Overview Models Features Constraints Variants						
Revision: Any Status; Working (Modified) Effectivity: All Dates Owner: Engineer,Ed (ed) More... 029321-Car_2024						
Add Delete Remove Effectivity Add Selection Mode Select All Refresh						
Object	Type	Name	Description	Optional	Multi-select	
029321-Car_2024	Configurator Context	Car_2024				
Car_2024_Model_Family	Model Family	Car_2024_Model_Family		False	False	
LXI	Product Model	LXI				
VXI	Product Model	VXI				
ZXI	Product Model	ZXI				

3. Create a group, family, and features as follows:

Object	Type	Name	Feature Da...	Optional	Free-form	Multi-select
029637-Car_2024	Configurator Context	Car_2024				
Engine group	Group	Engine group				
Diesel engine family	Family	Diesel engine family	String	False	False	False
R3 diesel engine	Feature	R3 diesel engine				
R4 diesel engine	Feature	R4 diesel engine				
R5 diesel engine	Feature	R5 diesel engine				
Gasoline engine family	Family	Gasoline engine family	String	False	False	False
V4 gasoline engine	Feature	V4 gasoline engine				
V6 gasoline engine	Feature	V6 gasoline engine				
V8 gasoline engine	Feature	V8 gasoline engine				
Unassigned Families						

4. Verify the value of the **Cfg0DefaultRuleSortSequence** site preference. The default value of this preference is **cfg0Sequence:ASC**. This means that only sequence numbers are considered for default rules and the lower sequence takes priority.

Override this preference by setting the value as **Cfg0DefaultRule:object_name:ASC**.

5. Create availability rules as follows in the **Grid Editor**.

Availability rules are required only if you create a configurator context with a negative bias.

Variability Content	1964/001 ⓘ Availability Rule	1973/001 ⓘ Availability Rule	1974/001 ⓘ Availability Rule
▼ Subject			
▼ Diesel engine family			
R3 diesel engine			✓
▼ Gasoline engine family			
V4 gasoline engine		✓	
V6 gasoline engine	✓		
▼ Condition			
▼ Car_2024_Model_Family			
LXI	✓	✓	✓

6. Create default rules as follows in the **Grid Editor**.

Variability Content	1976/001 ⓘ Default Rule	1977/001 ⓘ Default Rule
▼ Subject		
▼ Gasoline engine family		
V4 gasoline engine	✓	
V6 gasoline engine		✓
▼ Condition		
▼ Car_2024_Model_Family		
LXI	✓	✓

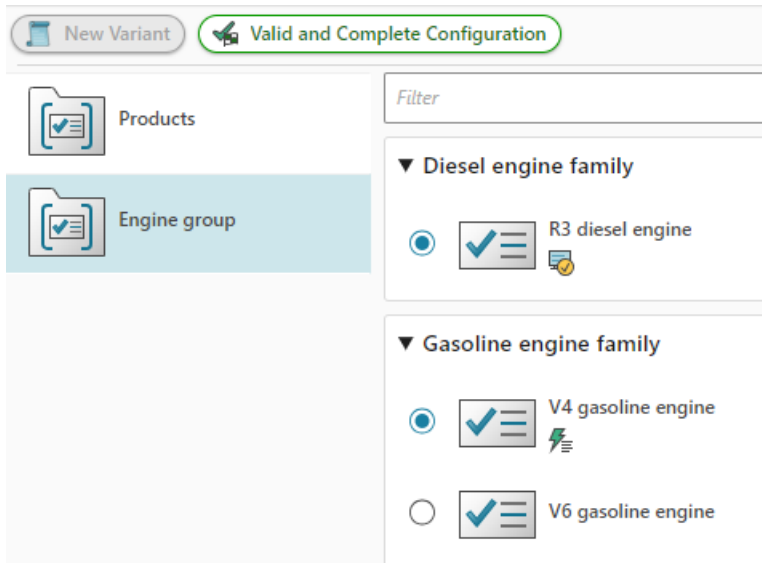
7. Click **Settings > Arrange**. In **Available** columns, search for **object_name**, select the **Name** column, and move it to **Displayed Columns**. Ensure that this is next to the **Sequence Number**. Click **Save and Arrange**.

object_name is the internal name for the **Name** column.

8. Edit the default rules to specify the **V4 gasoline engine** and **V6 gasoline engine** names as **V4_gasoline** and **V6_gasoline**, respectively.

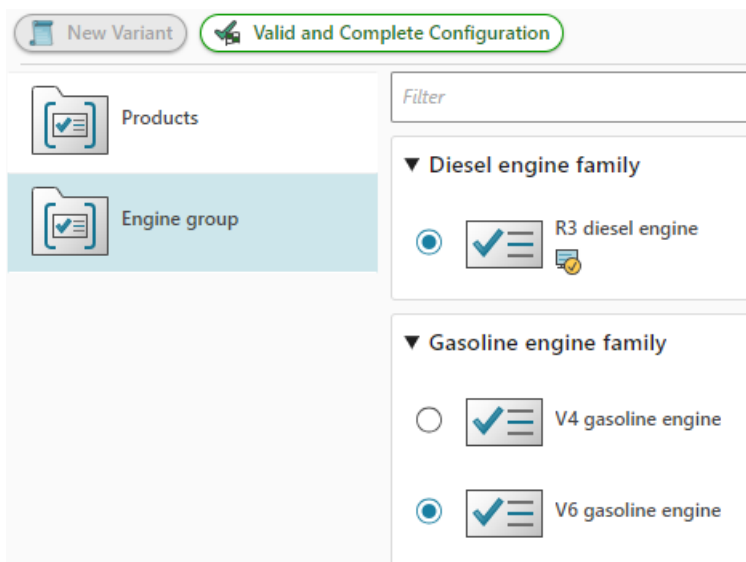
Severity	Name	Subject	Condition
Error	1964	'Gasoline engine family' = 'V6 gasoline engine'	Car_2024_Model_Family = LXI
Error	1973	'Gasoline engine family' = 'V4 gasoline engine'	Car_2024_Model_Family = LXI
Error	1974	'Diesel engine family' = 'R3 diesel engine'	Car_2024_Model_Family = LXI
	V4_gasoline	'Gasoline engine family' = 'V4 gasoline engine'	Car_2024_Model_Family = LXI
	V6_gasoline	'Gasoline engine family' = 'V6 gasoline engine'	Car_2024_Model_Family = LXI

9. Open the **Variants** tab and click **Variant Configuration**. It shows a valid and complete configuration.
10. Navigate to the **Engine** group.



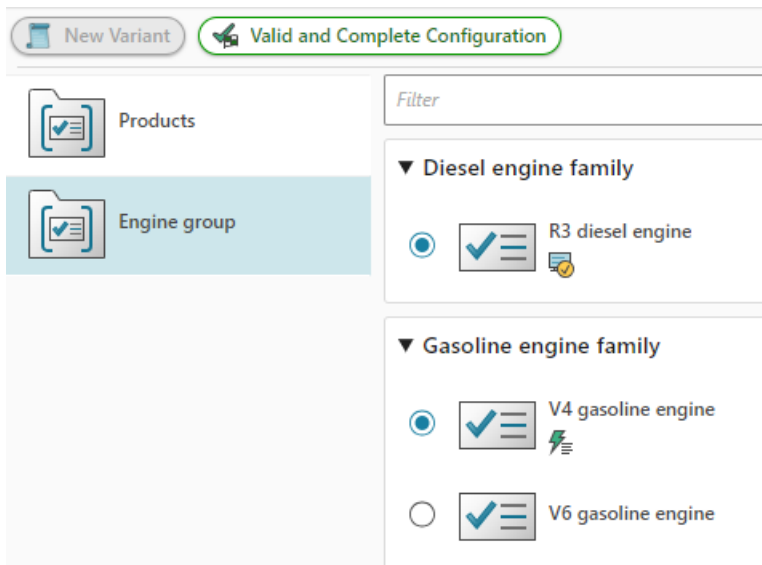
The **R3 diesel engine** is a system selection because of an availability rule. The **V4 gasoline engine** is the default selection. This is because the default rule name was **V4_gasoline** for **V4 gasoline engine** and **V6_gasoline** for **V6 gasoline engine**. The alphabetical order of the default rule name takes priority for the default rules since you have overridden the **Cfg0DefaultRuleSortSequence** site preference and set the value to **Cfg0DefaultRule:object_name:ASC**.

11. Edit the names of the default rules to specify **y_gasoline** and **x_gasoline** for **V4 gasoline engine** and **V6 gasoline engine**, respectively.
12. In the **Variant Configuration** view, navigate to the Engine group.



The **V6 gasoline engine** is the default selection now. This is because the default rule name was **y_gasoline** for **V4 gasoline engine** and **x_gasoline** for **V6 gasoline engine**. The alphabetical order of the default rule name takes priority for the default rules since you have overridden the **Cfg0DefaultRuleSortSequence** site preference and set the value to **Cfg0DefaultRule:object_name:ASC**.

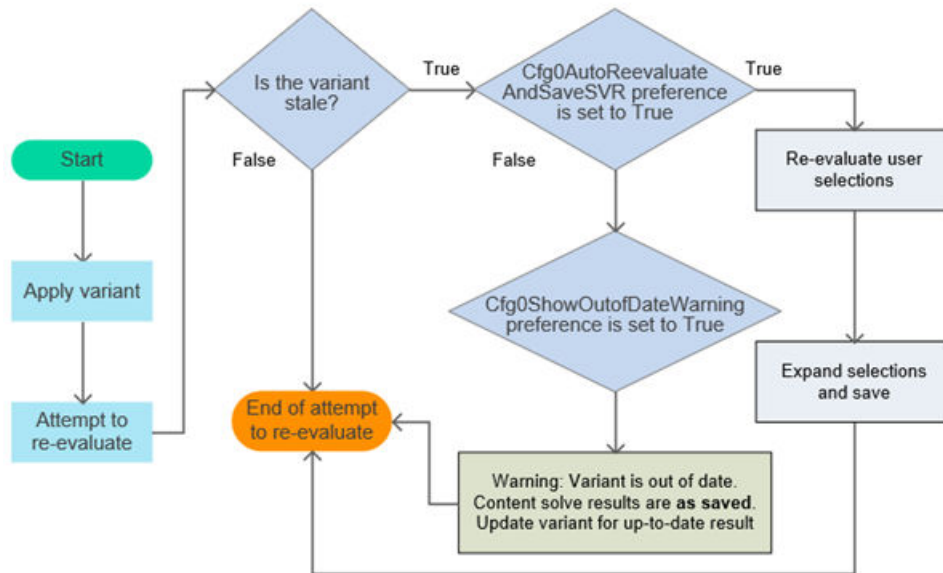
13. Override the **Cfg0DefaultRuleSortSequence** site preference and set the value as follows:
Cfg0DefaultRule:object_name:ASC
Cfg0DefaultRule:cfg0Sequence:ASC
14. Edit the names of both the default rules and specify **gasoline** as the value.
15. Edit the **Sequence Number** property of the default rule and specify **10** as the value for **V6 gasoline engine**. The sequence number of **V4 gasoline engine** remains unchanged as **0**.
16. Save your changes.
17. In the **Variant Configuration** view, navigate to the **Engine** group.



The **V4 gasoline engine** is selected. This is because the names (**gasoline**) are the same for both the default rules. However, the sequence number was **0** for **V4 gasoline engine** and **10** for **V6 gasoline engine**. The system, based on the preference value, first considers the names of the default rules and since they are the same, it considers the sequence number next to prioritize the rules. In this case, the lower sequence number is prioritized.

Apply a valid and complete SVR for a faster BOM solve

You must create a valid and complete SVR to improve system performance while performing a BOM solve. You can do so by making the desired selections and by using the **Expand** command in the **Variant Configuration** view. Then, you can save it as an SVR.



The previously created SVRs become outdated if the configurator data such as family, features, and rules are modified or newly created. If a stale SVR is applied on a structure, then the system displays a warning saying that the applied SVR is stale and should be refreshed. To view this warning, you must set the **Cfg0ShowOutOfDateVariantWarning** user preference to **True**. By default, it is set to **False**.

Manual revalidation can be done by executing the **Expand** command on an SVR after opening it in the **Variant Configuration** view. If you want the system to revalidate stale SVRs, set the **Cfg0IsAutoReEvaluateAndSaveVariant** user preference to **True**. By default, it is set to **False**.

Guidelines for selecting search methods

The example in this topic provides search guidelines while configuring product structures.

While configuring a product structure, you can search for features using the **Feature** or **Expression** option in the **Search by** list. Understanding when to use each method is crucial for efficient variant configuration.

To illustrate the various scenarios for using these search methods, consider the following families in a vehicle configuration:

Engine Type Family	Transmission Type Family	Drivetrain Type family
Petrol Engine	Manual Transmission	FWD Drivetrain
Diesel Engine	Automatic Transmission	RWD Drivetrain
Hybrid Engine	CVT Transmission	AWD Drivetrain
Electric Engine	-	-

For searching using basic feature combinations that span across different families, such as Petrol Engine AND Manual Transmission, selecting either **Feature** or **Expression** will generate identical results. Similarly, when searching within a single feature family, such as all Petrol Engine configurations, you can choose either search method based on your preference, as both approaches will provide the same comprehensive results.

When you use expression-based filtering, the system performs solving operations in the background. As a result, searching by expression is slower compared to using the standard search by feature option.

The following table suggests the recommended guidelines for searching using the **Expression** or **Feature** option.

Search method	When to use	Example	Special considerations
Search by Feature	When features are connected by OR across different families.	Petrol Engine OR Manual Transmission	Recommended for inclusive searches.
	When dealing with potentially conflicting criteria.	Electric Engine with Petrol Engine AND Manual Transmission	Provides more predictable results.
	When searching across multiple families with OR conditions.	Petrol Engine OR Manual Transmission OR AWD Drivetrain	More user-friendly for multiple selections.
Search by Expression	When using nested conditions or multiple logical operators.	Petrol Engine AND (Manual Transmission	Provides more precise control when you

Search method	When to use	Example	Special considerations
		OR Automatic Transmission)	specify the search criteria.
	When excluding specific features or combinations.	Petrol Engine AND Not Automatic Transmission	Required for NOT conditions.
	When using specific combinations of multiple features.	(Petrol Engine AND Manual Transmission) OR (Electric Engine AND Automatic Transmission)	Allows precise logical grouping of the search criteria.

Filter variability data by intent while authoring variant conditions

You can filter out configurator objects by specific intent while authoring variant conditions. For example, if you are authoring a bill of materials (BOM), you can focus on objects with technical intent. Conversely, if you are looking at a bill of process (BOP), you can focus on objects with marketing intent.

The configurator administrator defines a business purpose or intent such as manufacturing, marketing, or technical to categorize families and constraints. For more information, see [Categorize and filter families and constraints by business purpose or intent](#).

Procedure

1. Find and open the product or item revision to be configured.
2. Click the **Models, Features, or Constraints** tab.
3. To filter the variability data by intents, in the breadcrumb, select the required option from the **Intents** drop-down list. For example, you can filter using the **Technical, Manufacturing, or Marketing** intent.

When you apply intent filters for a Bill of Materials (BOMs) with its standard variant conditions, the **Variant Configuration** view displays the specific intent filters that are applied.

Any BOM line falling outside of the applied intent filter is treated as if it is configured out, and this status influences how other constraints are evaluated. For example, if a family A has an intent X that is outside of the current intent filter, then Family A will be configured out, and subsequent constraint evaluations will proceed based on this exclusion.

Note

You can filter using intents if your administrator has included this ability in your configuration.

Author variant formula in the grid

You can author variant formula by selecting features from the grid. This is the recommended method.

You can also **author variant formula by typing the formula in the text editor**. This is recommended only for complex formula or combinations that cannot be selected on the grid.

Procedure

1. Find and open the product for which you want to author variant formula.
2. Select the BOM lines for which you want to apply variant conditions and click the **Variant Conditions** tab.
The BOM lines you selected are displayed as columns at the top. You can hover over the column header to view the complete name of the BOM line.
3. (Optional) Enable **Variant Formula** column in your work area, if it is not available.
This column displays the variant formula for each BOM line, and you can edit this in the grid.
 - a. In your work area, click **Settings** .
 - b. Click **Arrange** and search for the **Variant Formula** column.
 - c. Select the **Variant Formula** column and click **Move up**  to bring it to the top of the **Arrange** panel.
4. Click **Start Edit**  on the top of the grid
5. Make the required selections in the grid.
6. Click **Save Edits** .

Note

You can export the contents of the Variant Conditions grid view to Microsoft Excel if required. To do this, click **Export to Excel** . The **Export to Excel** panel appears if you select **Expand All**. The icons for the user positive, user negative, and configured out objects, and the corresponding text is displayed in the panel. You modify the default names if required, and click **Export**. When you specify names for the icons, other than the default ones, you can include spaces in the names. However, the text box cannot be blank, you must specify a name or retain the default values.

If there are no icons, a message which indicates that the contents are exported to Microsoft Excel is displayed.

Author variant formula in the editor

You can author variant formula by typing the formula in the text editor. This is recommended only for complex formula or combinations that cannot be selected on the grid.

You can also **author variant formula by selecting features from the grid**. This is the recommended method.

In the **Variant Formula Editor**, you can author complex variant conditions that cannot be authored in the grid. This view is used by expert users who want to write the expression directly in the editor. When you save such a variant condition, it is not available in the grid.

Procedure

1. Find and open the product for which you want to author variant formula.
2. Select the BOM lines for which you want to apply variant conditions and click the **Variant Conditions** tab.

The BOM lines you selected are displayed as columns at the top. You can hover over the column header to view the complete name of the BOM line.

3. (Optional) Enable **Variant Formula** column in your work area, if it is not available.

This column displays the variant formula for each BOM line, and you can edit this in the grid.

- a. In your work area, click **Settings** .
- b. Click **Arrange** and search for the **Variant Formula** column.
- c. Select the **Variant Formula** column and click **Move up**  to bring it to the top of the **Arrange** panel.
4. Select one of the BOM lines, click on the top of the column, and click **Show Variant Formula Editor**.
Even if you select multiple BOM lines in the **Variant Conditions** view, you can edit only one BOM line at a time.
5. To edit the variant condition, in the **VARIANT FORMULA EDITOR**, click **Start Edit** .
6. Click inside the box to start creating your formula.

The **Intellisense** autocompletion feature is turned on by default. When it is turned on, you can get suggestions by pressing either Ctrl + space bar or space bar and then make a selection. After making the selection, press the space bar again to get the appropriate logical operators. Continue to create a complete formula and click **Validate Syntax** to validate the syntax of the formula you have created manually.

Example: `ExteriorColor = NeptuneBlue`

When the autocompletion feature is turned on, the system honors the name or the ID as the primary business attribute depending on how the attribute is set at your site by your administrator. This means that if name is set as the primary attribute, you must type the name in the formula editor and not the ID to complete the formula. Additionally, the administrator at your site can set the visibility of option families and the display of family namespace. By default, option families are visible, and the family namespace is not displayed. In such cases, you can type the option families in the formula editor, but not the family namespace to complete the formula. The system honors the configurations set by the administrator and makes suggestions based on these configurations. Consult the administrator at your site for more information.

Note

If you have families or features with identical names in the configurator context, they create ambiguity when creating an expression in the formula editor. In such cases, you should use the object IDs instead of the names to create the expressions. In case the object IDs are also identical, then contact your administrator to enable the necessary preferences, allowing you to create the expression by prefixing the namespace.

When you type the suggestions, the system performs validations for each suggestion when the feature is turned on. However, when the feature is turned off, the system does not perform any validation.

When the autocompletion feature is turned off, you must author the formula using the internal Teamcenter formula format.

You can also use custom formula in the formula editor. By default, custom formula is not enabled. Consult your administrator to enable it.

- 7. Make the required changes to the variant formula for the selected BOM line.
The formula displayed in the **Variant Formula** column is the display formula whereas it is an internal formula for the editor. The server converts the internal formula to the display formula.

Examples where the logical operators are different:

Variant Formula column	Formula in editor
Engine = V4 OR Engine =V8	(Engine = V4 Engine =V8)
Music Player = Radio AND Music Player = CD Player	(Music Player = Radio & Music Player = CD Player)

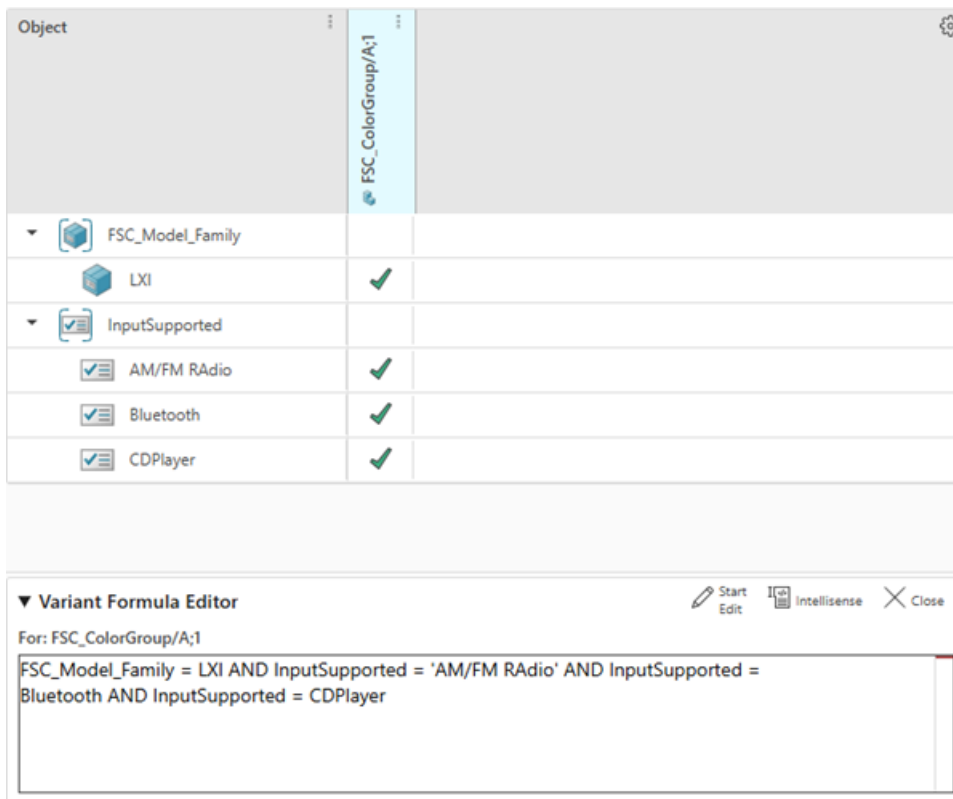
- 8. Click **Save Edits** .
The system validates the expression syntax only after you save the changes.
If you specify a wrong formula in the editor and click **Save Edits**, the system displays a syntax error or invalid error as appropriate.
You cannot save the changes if they are invalid.
- 9. (Optional) To validate the expression against the constraints associated with the configurator context, after saving multiple changes, click **Validate All**  on the top of the grid.

Examples

The following are some examples of the differences when selections are made for single select and multiselect families.

- 1. Open a structure, select a BOM line, and make selections in the **Variant Conditions** tab as follows. Additionally, select the BOM line, click on the top of the column, and click **Show Variant Formula Editor**.
- 2. In a multiselect family, feature selections are connected using **AND**.

Example:



The screenshot shows a software interface for managing variants. At the top, there's a header bar with 'Object' on the left and a gear icon on the right. Below this is a table with three columns. The first column lists features, the second column shows their status (checked or unchecked), and the third column shows a green checkmark. The features listed are FSC_Model_Family, LXI, InputSupported, AM/FM RAdio, Bluetooth, and CDPlayer. Below the table is a 'Variant Formula Editor' window. It has a title bar with 'Start Edit', 'Intellisense', and 'Close' buttons. The main area of the editor contains the following formula:

```
FSC_Model_Family = LXI AND InputSupported = 'AM/FM RAdio' AND InputSupported = Bluetooth AND InputSupported = CDPlayer
```

3. In a single select family, feature selections are connected using **OR**.

Example:

Object

FSC_ColorGroup/A;1

FSC_Model_Family

LXI

Interior_Color

Beige

Black

✓

✓

✓

✓

▼ Variant Formula Editor

Start Edit

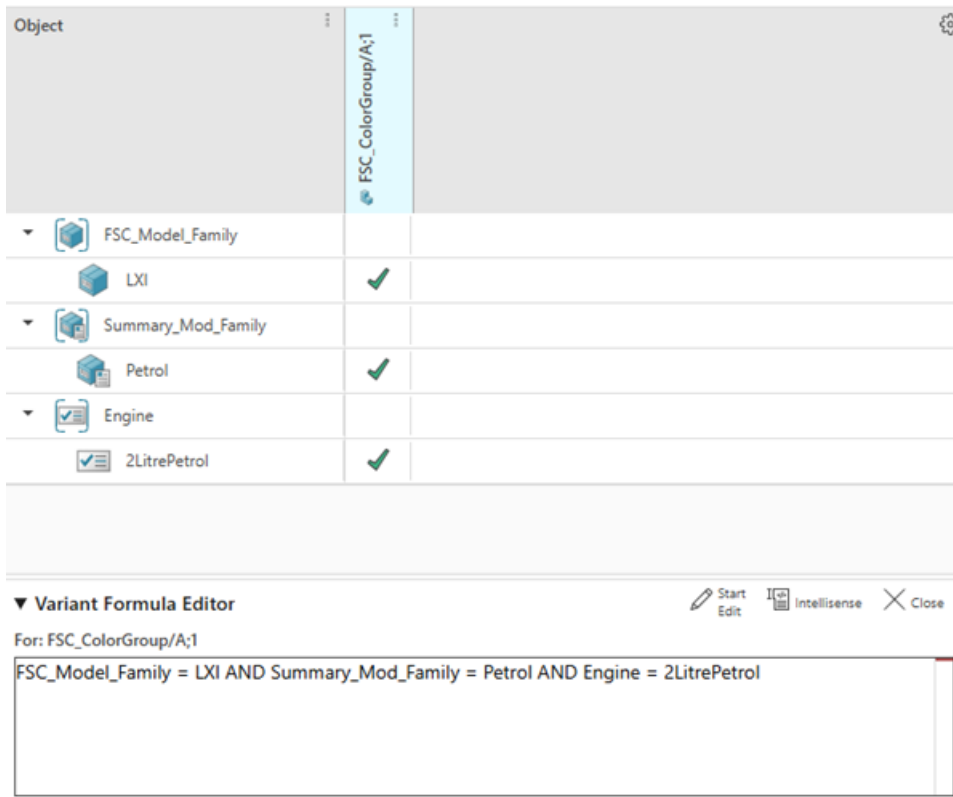
Intellisense

Close

For: FSC_ColorGroup/A;1

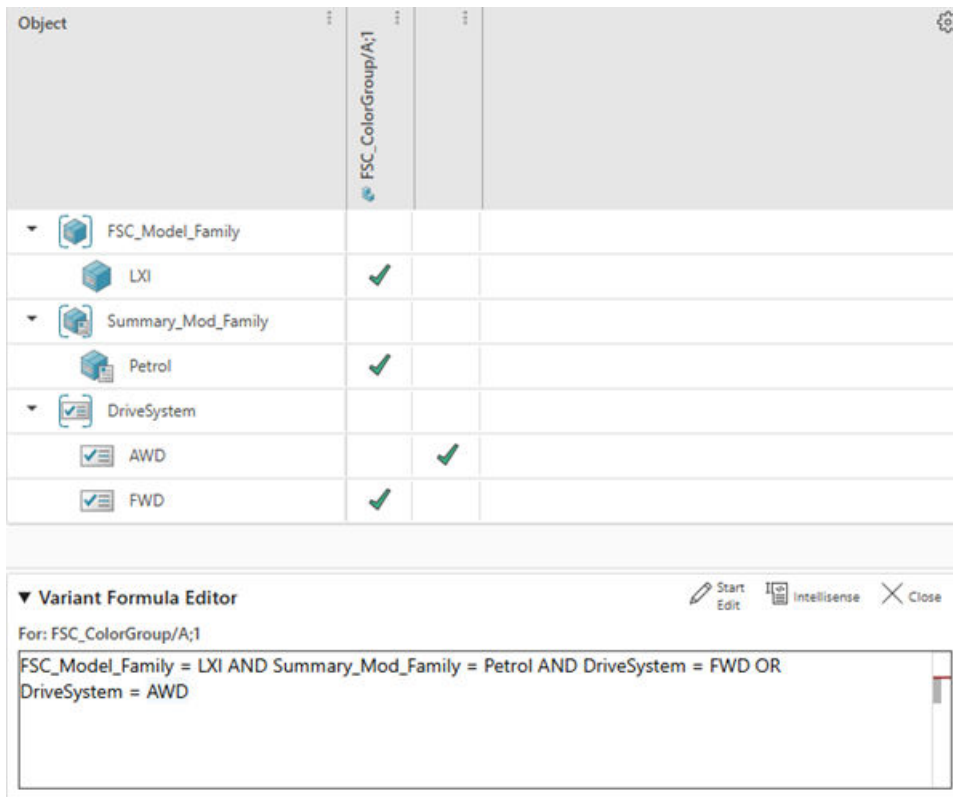
FSC_Model_Family = LXI AND (Interior_Color = Beige OR Interior_Color = Black)

4. Feature selections from different families are connected using **AND** in the same column.
- Example:



- For split columns, the expression generated by feature selections across columns are connected using **OR**.

Example:



Save changes in a configuration

You can save changes in your existing variant or save your custom configuration as a new variant rule. Alternatively, you can save them as a variant criteria. This depends on how the administrator configures your Teamcenter environment.

When an SVR or VC is saved, the user and system selections are saved, along with everything in the **Settings** panel, including the validation severity, expansion severity, and selection behavior settings, as well as the filter criteria information (revision rule, effectivity, and rule date).

In earlier versions, the system only saved the user selections when saving an SVR. If you load an SVR that was saved in a previous version, it loads the saved user selections but keeps the current settings in the **Settings** panel.

To save changes in your configuration,

Procedure

1. To open the **Save** panel, choose **More Commands *** > Save** .
2. Enter a name.
3. (Optional) Type a description.

4. Select one of the following options in **Attach to**:
 - **Current Structure Revision** to save your changes to the structure you have opened. These changes are available only for the specific structure.
 - **Configurator Context** to save your changes to the configurator context. These changes are available to all the structures to which this configurator context is associated.

The default selection for this option is determined by how the administrator configures your Teamcenter environment. By default, it is **Configurator Context**.

5. Click **Save**.

14. Filter structures

About filtering structures

You filter the product structures to work with the specific product definitions. You can filter a product structure by spatial proximity and volume, creation or modification period, properties (like owner or type), and partitions. You can also filter the product structure using Copilot if Teamcenter Artificial Intelligence Services are installed in your Teamcenter environment

You **filter a structure** so that you can work with a specific product definition, which is comparatively smaller in size than the entire structure.

You can filter structures if your Teamcenter setup has the Context Management User license. Additionally, you can filter only those structures that are indexed using Smart Discovery Indexing. Such structures can be identified by the *Indexed* indicator . Hovering over the indicator shows the start time of the last successful index update indicating whether the modified structure is available for filter. You can further visualize the 3D design data of those structures if they are indexed for massive model visualization (MMV) using Smart Discovery Indexing. Such structures can be identified by the *Massive model visualization indexed* indicator .

The different ways to filter a product structure are as follows:

- **Filter by spatial proximity and volume**

Filter a structure by locating parts that are at a certain proximity from a specific part. You can also filter a structure by locating parts that are inside or outside a certain spatial zone. The spatial zone is represented by a bounding box. You can also choose to locate parts that are on the edges of the bounding box.

- **Filter by period**

Filter a structure by selecting a date range. You can select the date range when the structure was created or modified.

- **Filter by properties**

Filter a structure by specifying certain properties such as *owner*, *type*, and *occurrence notes*.

You cannot find an element within a structure by specifying occurrence properties. You can do so by specifying item revision properties. However, you can filter a structure by specifying both occurrence properties and item revision properties.

- **Filter by partitions**

Filter a structure by selecting partitions.

You can filter by partitions only if your Teamcenter setup has the Smart Discovery license.

Filter a structure using Copilot

Apart from using the regular [filters](#), you can also [filter your currently open structure](#) using Copilot if Teamcenter Artificial Intelligence Services are installed in your Teamcenter environment. If the structure is indexed using Smart Discovery Indexing, you can filter the structure in Copilot:

- **by spatial proximity and volume**

If the spatial properties exist for the structure, you can use Copilot to filter the structure by locating parts that are at a certain proximity from a specific part. You can also filter a structure by locating parts that are inside or outside a certain spatial zone. The spatial zone is represented by a bounding box. You can also choose to locate parts that are on the edges of the bounding box.

- **by properties**

You can use the following properties only in Copilot to filter the structure: *name*, *last modified*, *owning group*, *owning user*, *quantity*, *sequence number*, *description*, and *item ID*.

Tip

When you filter a structure using Copilot, provide a single instruction at a time.

You cannot filter the structure in Copilot by period and partitions.

Create hierarchy for freeform zones

Create a hierarchy for freeform zones to allow for efficient filtering and searching within your product structure. Engineers can use this hierarchy to identify components that intersect with specific spatial zones, such as maintenance access areas.

Freeform zones are spatial regions within a product structure that help engineers visualize, filter, search, and collaborate. You can obtain the JT files that contain the freeform zones already defined in CAD software, for example, NX, and then create a hierarchy to associate zones with relevant product components and assemblies.

In Teamcenter, a freeform zone hierarchy is created using a secondary component of type **Spatial Zone Container** at the top, **Item** or **Design** for middle nodes, and **Spatial Zone** for the lowest-level (leaf) child items. Each assembly can contain one such freeform zone hierarchy.

Example

A coffee vending machine product structure can have separate freeform zone hierarchies for Brewing Unit Assembly, Water System Assembly, and Waste Management Assembly.

Prerequisites

- JT files containing freeform zones are available.
- You are the administrator of the product structure and your role has the permission to create the freeform zone hierarchy for the structure.
- The Teamcenter environment has the Smart Discovery Indexing license.

Procedure

1. Obtain the JT files that contain the freeform zones already defined in CAD software, for example, NX.
2. Search for a structure, and click **Open** .
3. Expand the structure and select an assembly or sub-assembly for creating the freeform zone hierarchy.
4. Click **Add**  > **Secondary Component**.
5. In the Add Secondary Component panel, perform the following steps:
 - a. In the **Secondary Component Type** list, select **Spatial Zone Container**.
 - b. Enter the name and description for the freeform zone container and click **Add**.
6. Select the newly added spatial zone container, and click **Add**  > **Child** to create the hierarchy.
7. In the Add Child panel, perform the following steps:
 - a. In the **Type** list, perform one of the following actions: .
 - If you want to create a lowest-level node in the hierarchy, select the **Spatial Zone**
 - If you want to create different levels in the hierarchy, select **Item** or **Design** as per your requirement.
 - b. Enter the required details and click **Add**.
Add more child items to create the freeform zone hierarchy, considering your business requirement.
8. Perform the following steps to attach the JT file to each lowest-level child of the hierarchy.
 - a. Select the lowest-level child of the hierarchy and click the **Attachments** tab.
 - b. In the **Files** section, click **Add** .
 - c. In the Add panel, upload the JT file and click **Add**.

Results

The freeform zone hierarchy is available in the **Spatial** section of the **Filter** panel for engineers, allowing them to filter and search the product structure content.

Filter a structure

You filter a structure so that you can work with a specific product definition, which is smaller in size than the entire structure.

Restrictions and limitations

You can filter structures if your Teamcenter setup has the Context Management User license. Additionally, you can filter only those structures that are indexed using Smart Discovery Indexing. Such structures can be identified by the *Indexed* indicator . Hovering over the indicator shows the start time of the last successful index update indicating whether the modified structure is available for filtering. You can further visualize the 3D design data of those structures if they are indexed for massive model visualization (MMV) using Smart Discovery Indexing. Such structures can be identified by the *Massive model visualization indexed* indicator .

Procedure

1. Search for a structure, and click **Open**  to open it.
2. Click **Filter** .
3. In the **Filter** panel, click **Settings** , enable or disable the following options, and click **Apply**:
 - **Auto-update filters**: By default, this option is enabled, and the structure is filtered in real-time as you apply each filter. When not enabled, you select all filters first and then apply them on the structure.
 - **Filter/Append Elements with Children**: By default, this option is enabled, and the children of the structure element selected as a filter are automatically added to the filter criteria. When not enabled, the children are not added to the filter criteria.
4. In the work area, right-click a structure element to choose a filter. The filters that you can choose and how these filters are applied on the structure depends on whether the administrator has configured the advanced search or not.
 - *SCENARIO 1*: The administrator has not configured the advanced search.

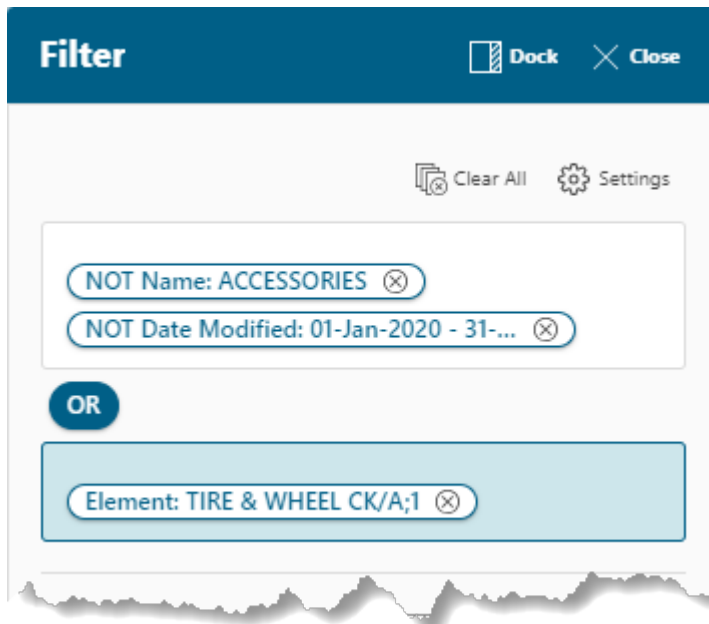
If the administrator has not configured the advanced search options, you can use the following filters:

Filter	Description
Filter Selected Elements with children	Use this filter to narrow down the structure by including the selected elements along with the children. The elements and the children are added to the filter criteria using the AND operator.

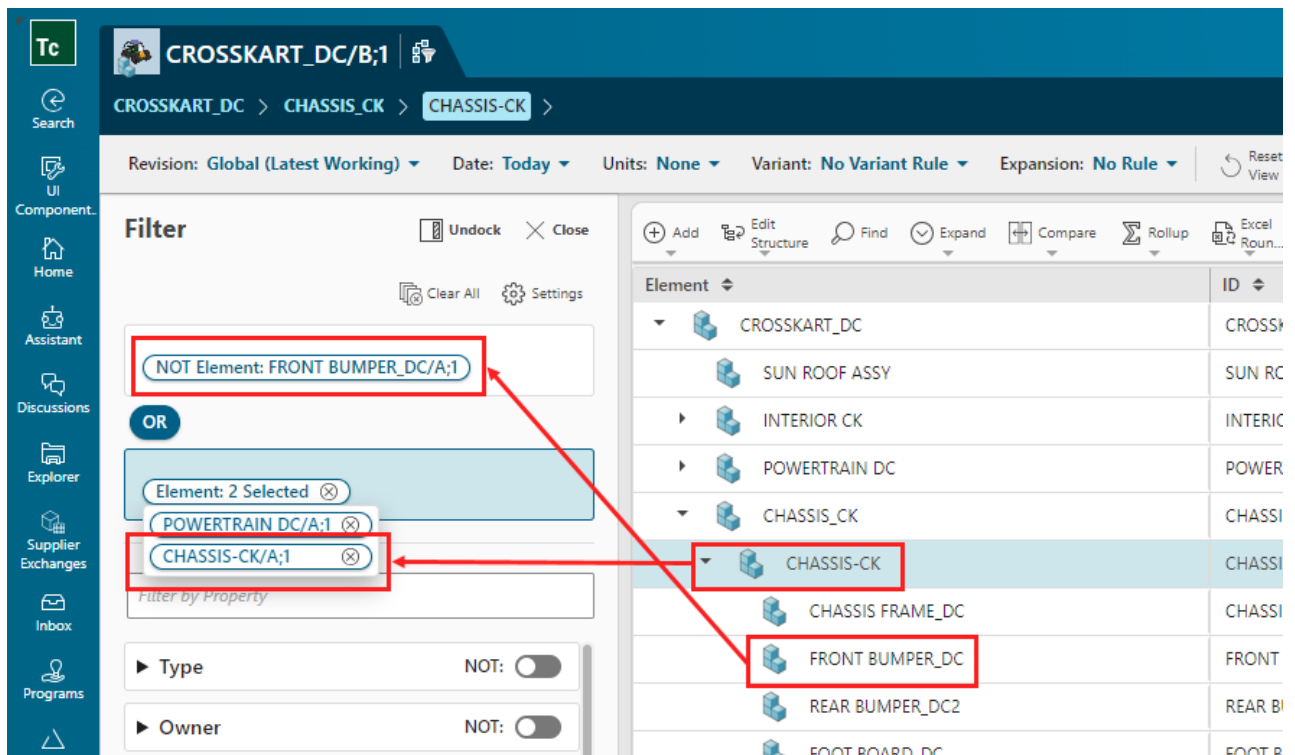
Filter	Description
	This filter option is displayed when the Filter/Append Elements with Children setting is enabled.
Filter Selected Elements	Use this filter to narrow down the structure by including only the selected elements and not their children. The selected elements are added to the filter criteria using the AND operator.
Append Selected Elements with children	Use this filter to broaden the filtered structure by including the selected elements along with the children. The elements and the children are added to the filter criteria using the OR operator. This filter option is displayed when the Filter/Append Elements with Children setting is enabled. Additionally, this option is displayed only when after you first apply another filter.
Append Selected Elements	Use this filter to broaden the filtered structure by including only the selected elements and not their children. The selected elements are added to the filter criteria using the OR operator. This filter option is displayed when the Filter/Append Elements with Children setting is disabled. Additionally, this option is displayed only when after you first apply another filter.
Exclude Selected Elements	Use this filter to narrow down the structure by excluding the selected elements along with the children. The elements and children are added to the filter criteria using the NOT operator. For example, if you want to exclude an assembly and its children parts, select the assembly and click Exclude Selected Elements .

The filters are applied in groups. If the advanced search options are not configured, a maximum of three groups with the **OR** operator can be created and you can use the same filter expression in NX.

The filters applied using the **OR** operator are always placed last in the filter criteria.



Sometimes, when you append selected elements with children, a filter applied using the **OR** operator overrides a filter applied using the **NOT** operator. For example, if you want to exclude **FRONT BUMPER_DC** from the structure but want to append its parent **CHASSIS-CK** with children, **FRONT BUMPER_DC** is automatically included in the structure.



- **SCENARIO 2:** The administrator has configured the advanced search.

If the administrator has configured the advanced search options, you can use the following filters:

Filter	Description
Filter Selected Elements with children	Use this filter to narrow down the structure by including the selected elements along with the children. The elements and the children are added to the filter criteria using the AND operator. This filter option is displayed when the Filter/Append Elements with Children setting is enabled.
Filter Selected Elements	Use this filter to narrow down the structure by including only the selected elements and not their children. The selected elements are added to the filter criteria using the AND operator.
Exclude Selected Elements	Use this filter to narrow down the structure by excluding the selected elements along with the children. The elements and children are added to the filter criteria using the NOT operator. For example, if you want to exclude an assembly and its children parts, select the assembly and click Exclude Selected Elements.

If the advanced search options are configured, the  **OR Group** and  **NOT Group** buttons are available in the **Filter** panel. You can use these buttons to create the filter expression which is a combination of the **OR** and **NOT** groups. You can create multiple **OR** groups in the expression but only one **NOT** group can exist in the filter expression.

When the filter expression is a combination of **OR** groups and a **NOT** group, the filter shows all results from the **OR** groups by excluding the result from the **NOT** group. For example, suppose that you want to filter the structure that was not created between 1 January 2020 through 31 December 2020, contains **CHASSIS_CK**, and does not contain **ACCESSORIES**. In such a case, you can use the filters shown in the following image.

The screenshot shows the Teamcenter software interface. On the left is a vertical navigation bar with icons for Search, UI, Component, Home, Assistant, Discussions, Explorer, Supplier Exchanges, Inbox, Programs, Changes, and Program Deliverables. The main header area displays 'CROSSKART_DC/B;1' and a search bar with 'CROSSKART_DC'. Below the header, a status bar shows 'Revision: Global (Latest Working)', 'Date: Today', 'Units: None', 'Variant: No Variant Rule', and 'Expansion: No'. The central 'Filter' dialog is open, showing an 'OR' group of filters. The first filter is 'NOT Date Modified: 01-Jan-2020 - 31-...' and the second is 'Element: CHASSIS CK/A;1-CHASSIS_CK'. Below these is a 'NOT' group with the filter 'Element: ACCESSORIES/B;1'. At the bottom of the dialog are sections for 'Filter by Property' with 'Type' and 'Owner' filters. On the right, a table lists the structure elements:

Element	ID
CROSSKART_DC	CROSSKART_DC
SUN ROOF ASSY	SUN ROOF
INTERIOR CK	INTERIOR CK
POWERTRAIN DC	POWERTRAIN
CHASSIS_CK	CHASSIS CK
ELECTRIC SYS CK	ELECTRIC SYS
EXTERIOR CK	EXTERIOR CK
BRAKE & SUSP CK	BRAKE & SUSP
STEERING SYS	STEERING SYS
TIRE & WHEEL CK	TIRE & WHEEL
MUS SYSTEM CD	MUS CD

If the administrator has configured a wild card search in the Teamcenter environment, in a text-based filter, you can use operators like **Contains**, **Does not contain**, **Equals**, **Does not equal**, **Begins with**, **Ends with**, and **Like** operators. For example, suppose that you want to search for the parts in the structure that are owned by John. In such a case, use the **Contains** operator and then type **John** in the **Owner** filter. Similarly, if you want to search for all the parts that are not owned by John, enter **John** and then change the operator to **Does not contain** in the **Owner** filter.

Tip

To use a negative operator, use its positive operator first and then change to its negative one. For example, in order to use **Does not contain** operator, first select the **Contains** operator, enter a value in the filter, and then change the operator to **Does not contain**.

In a numeric filter, you can use operators like **Greater than or equals**, **Greater than**, **Less than or equals**, **Less than**, **Equals**, **Does not equal**, **Range inclusive**, and **Range exclusive**.

After you select a value in a filter, the values in the remaining filters are dynamically populated. For example, imagine that the structure contains parts that are created by John who belongs to the Engineering group. In such case, if you select **John** in the **Owner** filter, the values in the **Group ID**

filter are dynamically populated and only show **Engineering**. If advanced filters are configured, the values in the filters are dynamically populated even when you switch between groups.

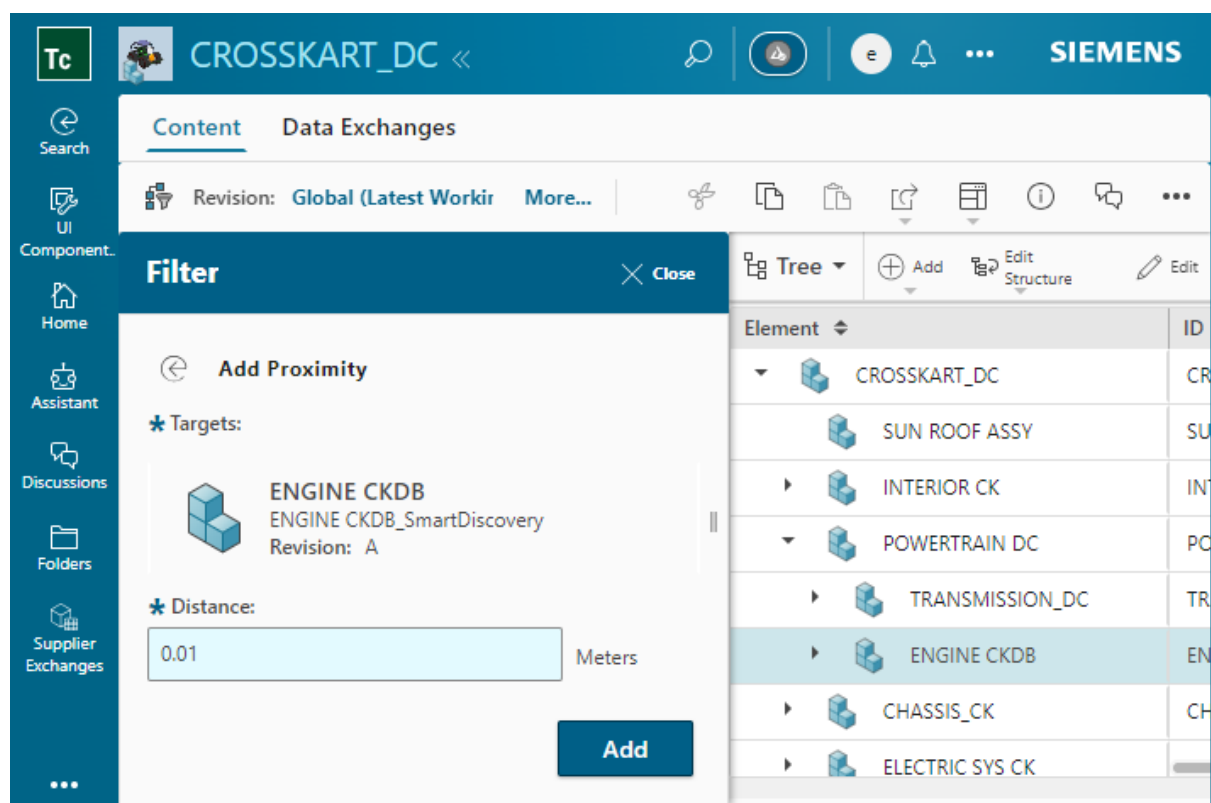
You can use an asterisk (*) to filter the parts in the structure. For example, consider that the structure contains parts owned by different engineers whose names are John, Johan, and Johnnie. In such a case, if you want to search the parts that are owned by all these engineers, use the **Like** operator, and then type **Joh*** in the **Owner** filter.

The advanced search might not provide you with the intended results.

5. In the **Filter** panel, apply additional filters, as required:

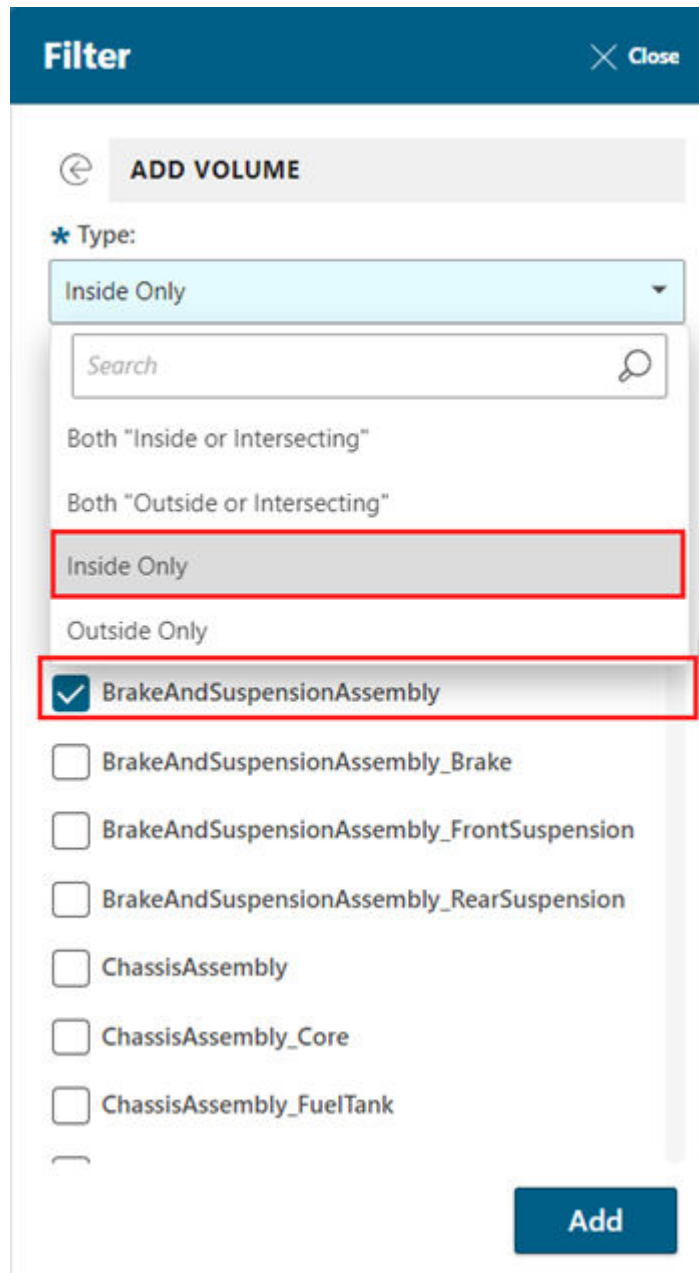
To filter by spatial proximity:

- a. From the work area, select a structure element to locate other structure elements that are at a certain proximity from it.
- b. In the **Filter** panel, select **Proximity** from the **Spatial** section.
- c. Enter the proximity in **Distance** and click **Add**.



To filter by spatial volume:

- a. In the **Filter** panel, select **Volume** from the **Spatial** section.
- b. Select the volume for which you want to perform the search. For example, you can choose to locate parts that are inside the brake and suspension assembly.



- c. Click **Add**.

To filter the structure by spatial freeform zone:

If the structure administrator has set up the freeform zone hierarchy and configured its display in the **Filter** pane, use the following steps:

- a. In the **Filter** panel, select **Freeform Zone** from the **Spatial** section.
- b. In the **Add Freeform Zone** section, select a type from the list.
For example, **Inside or Intersects**.
- c. In the freeform zone hierarchy, search for and select the zones for filtering.

For example, you can choose to locate parts that are inside the brake and suspension assembly.

- d. Click **Add**.

To filter the structure by properties:

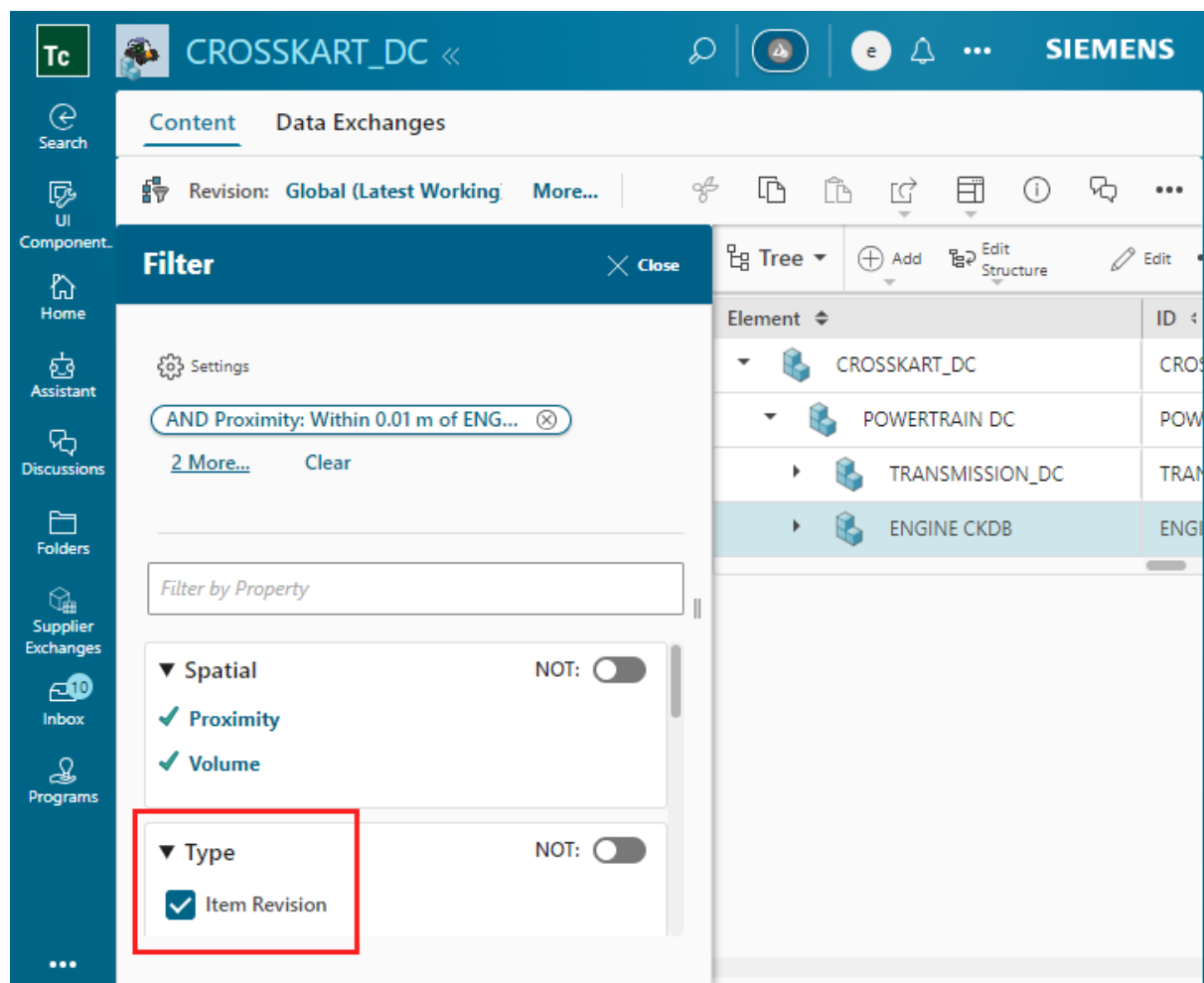
Note

The top element in a structure or BOM is not considered for the **Find** or **Filter** actions. Therefore, the properties of the top element are not available while you perform these actions.

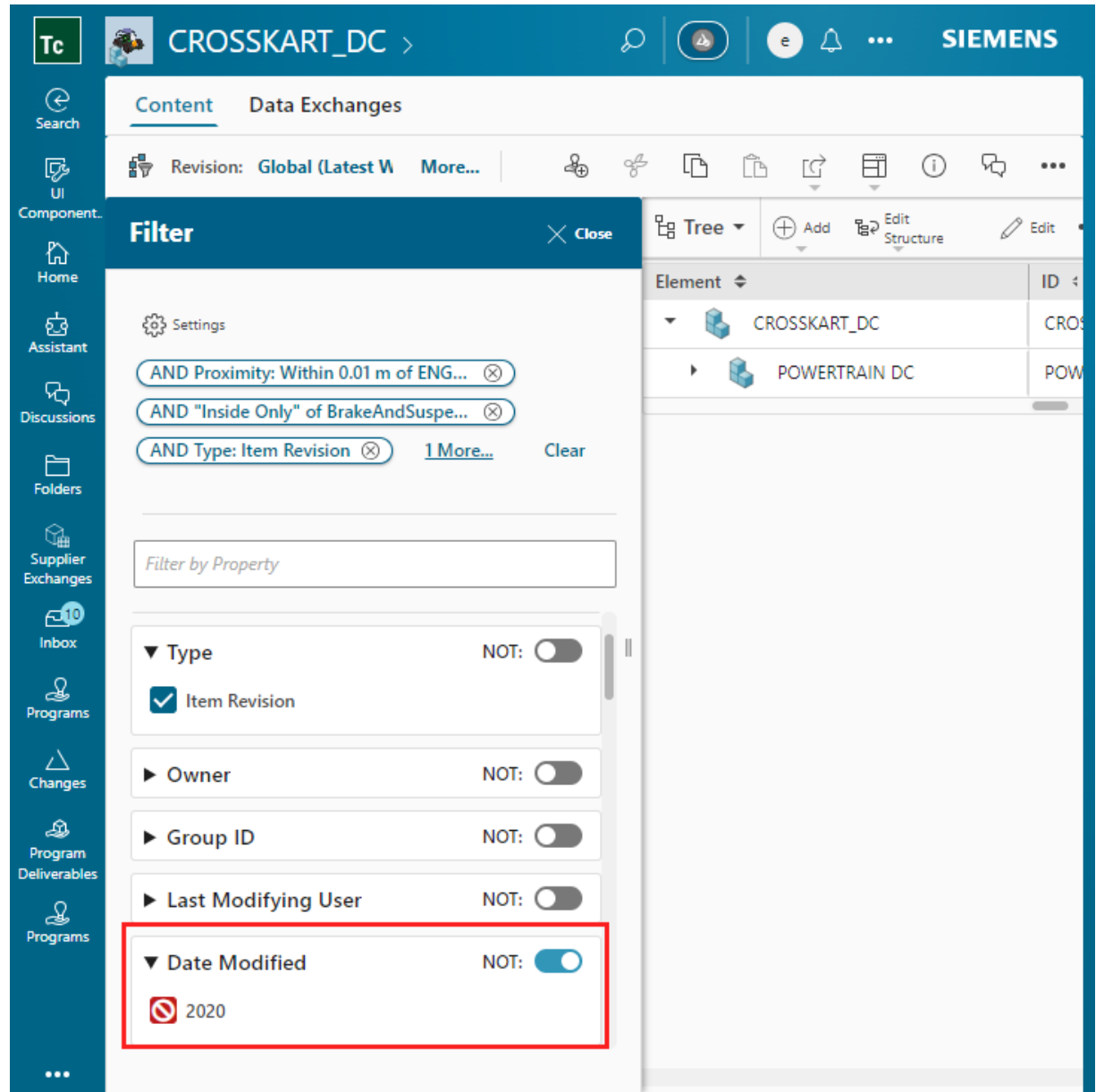
Note

If you do not have the Read access privilege for the structure element selected in the work area, you are not be able to view the value of its indexed occurrence properties in the **Filter** panel.

- a. Select the attributes, such as **ID**, **Occurrence Type**, and **Date Modified**, by which you want to further filter the structure In the **Filter** panel. For example, include all elements that are of the type, **Item Revision**.



To exclude a property as a filter, select the property and enable the **NOT** option. For example, to exclude all elements that were not modified in 2020, select **2020** and enable the **NOT** option in the **Date Modified** section.



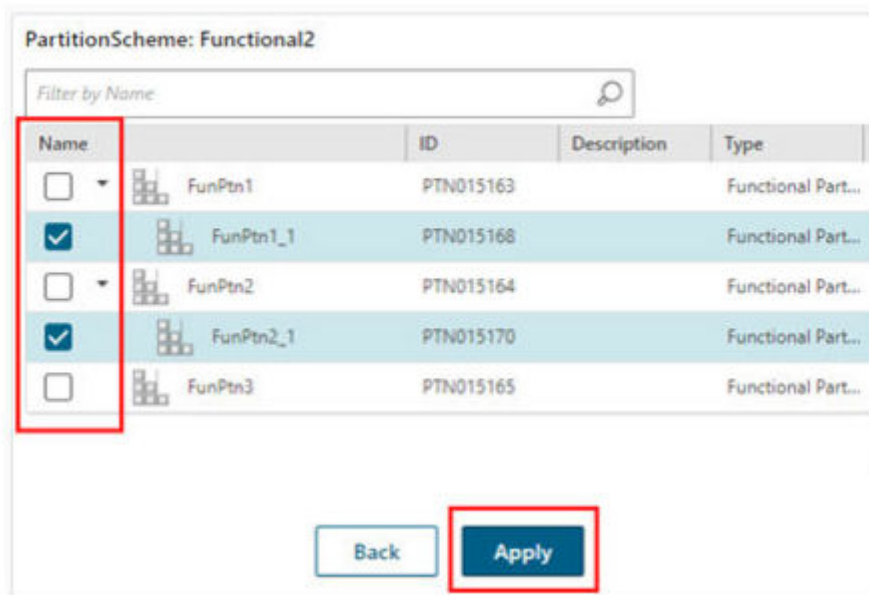
To filter by partitions:

Tip

To quickly filter a structure by a partition using a shortcut menu, right-click the partition and then click **Filter Selected Elements with children**. However, to filter a structure by a partition using the **Filter** panel, follow step **a** and later in this section.

If you use the shortcut menu option, the structure is narrowed and the selections are included with children as filters. The selections are added to the filter criteria using the AND operator. This action is synchronized with the method of filtering by using the **Filter** panel.

- a. In the **Filter** panel, select a partition scheme.
- b. Select the partitions that contain the required structure elements and click **Apply**.



If a selected partition has child partitions, the elements of the child partitions are also considered.

As you select different filters in the **Filter** panel, an expression is built at the top.

To remove any filter from the expression, click ⊗ next to it.

To remove all filters, click **Clear All**.

6. If you disabled auto-update, click **Filter** to filter the structure.

What to do next

To view all the structure elements including the ones that are not displayed due to the applied filters, click **Configure** > **Show Excluded by Filtering**.

You can save the filtered structure in a **workset** or **session** for easy retrieval.

Reset a filtered structure

Reset removes all filters and sets the configuration criteria to its default settings.

Procedure

1. Search for the structure.

2. Click **Open**  to open the structure.
3. Select **Configure**  > **Reset View**  to remove the filters applied to the structure.

15. Save filtered and configured structures within a session

About sessions

Sessions help you save and retrieve specific product structures, along with any applied filters and configurations. You can store either static or dynamic data, enable 3D visualization and snapshots in the sessions. You can share the sessions with other users.

To easily retrieve a structure that you are currently working with, you save it within a [session](#). You can further apply filters to the structure or configure it to derive a smaller structure that is more relevant to you. The session retains the filters and configurations applied to the structure.

When you open a session, you view the structure as it was previously saved retaining its applied filtering and configuration criteria. Depending on how your application administrator has configured the system, the session displays:

- Static data already stored in the session.
- OR
- Dynamic data where the stored data is re-evaluated against the current data and the latest data is loaded.

You can override the value of this preference by creating a new instance of the preference at a higher-precedence location.

You can make [changes](#) to the session by applying new filters and configuration, and can also [save it as a new session](#). Additionally, you can capture a snapshot of the product within the session to include the 3D measurement data applied on the structure. You can also [share a session](#) with other users.

Depending on how structures are indexed, you can perform the following tasks:

Scenarios		Tasks
Structures are indexed using Active Content Structure Indexing.	Structures are filtered.	Create a session. But the session will not retain the filters that are applied. Cannot visualize structures and capture snapshots.
	Structures are not filtered.	Create a session. Visualize large structures and massive structures, and capture snapshots.

Scenarios		Tasks
Structures are indexed using Smart Discovery Indexing.	Structures are filtered.	Create a session. Visualize large structures and capture snapshots.
	Structures are not filtered.	Create a session. Visualize large structures and capture snapshots.

When a structure is indexed by using both types of indexing, select an appropriate revision rule to view the structure. To view the structure as indexed by Active Content Structure Indexing, select a revision rule above the rule separator. To view as indexed by Smart Discovery Indexing, select a revision rule below the rule separator.

Note

When applying the **Global Latest Working** revision rule, the system will reload the entire structure to ensure the components are the latest working versions.

Create a session

You can create a session for a regular structure and also for a structure within a workset with a partition scheme applied. You can save this session and share it with other applications.

Procedure

- 1. Search for and **Open**  the desired structure.
- 2. Filter the structure as per your requirement.
- 3. Click **New**  > **Session** .
- 4. In the **Session** panel, specify the access level for other users:
 - a. Select the **Allow others to view** check box to provide *read* access to other users. Clear the check box to deny read access.
 - b. Select the **Allow others to edit** check box to provide *write* access to other users. Clear the check box to deny write access.

Note

The **Allow others to edit** check box is displayed only when the **Allow others to view** check box is selected.

- 5. Click **Create** to create the session.

Search for a session

You can search for a session based on the properties associated with the top-structure node in that session. To quickly search for a session, you can further filter sessions by their properties such as **Name**, **ID**, and **Type**.

Procedure

1. Search for sessions by performing a global search.

You can search with a keyword or the wildcards characters ***** or **?**.

The **Filters** panel is displayed.

2. In the **Filter** search box under the **Type** list, select the **Session** check box.

The session that you are searching for should be displayed in the panel to the right of the **Filters** panel.

You can further refine your search by applying more filters, such as **ID Contained in**, **Name Contained in**, and **Type Contained in** and many other filters.

Save a session as a new session

You can save an existing session as a new session. thereby preserving the original session while creating an updated version.

Procedure

1. Search for a session.
2. Click **Open**  to open the session.
3. Click **Filter**  or select **Configure**  > **Configuration**  to modify the filters or the configurations applied on the session.
4. Click the session and then select **Session**  > **Save As** .
5. In the **Save As Session** panel, enter the required details and click **Save**.

View and update a session

When you view a session, the session is loaded either with the static or dynamic data depending upon how the administrator has configured it for you. You can restore the sessions after unexpected logouts, replay to see the latest filters and configurations, modify applied settings, and save changes—either by

overwriting the original session (if permitted) or saving it as a new one. You can also reset a session to its last saved state.

Procedure

1. Search for a session.
2. Click **Open**  to open the session. Your site administrator sets whether to load a session with the static data stored in the session or with the latest data.

Sometimes, when a session is opened, you get a message asking you to choose to restore the session. Due to some issues, such as getting logged out while making updates to a session, results in this message. Click **Restore** if you want to autorecover the updates that you earlier made to the session. If you do not want to restore, you can ignore the displayed message, and continue with your updates to the session.

Session information that gets restored includes snapshots, quick measurements, part orientation, queries, sections, and workspace settings such as floor options and material.

3. To view the latest filters and configurations applied on the structure within the session, click **Replay**.

If you opened the structure in another window and edited it, the updates are not reflected in the session when you replay it. You must refresh the browser window to reload the changes in the session.

4. Click **Filter**  or select **Configure**  > **Configuration**  to modify the filters or the configurations applied on the session.
5. Click **Save Session** .

If the session is opened by another user at the same time and the other user made some changes to the session, you can choose to overwrite the session or save the session as a new one. You can overwrite the session only if your administrator has allowed overwriting of the session.

If another user removes some structure elements from the session, you receive a message with the list of the structure elements that are no longer available in the session.

6. To save a session as a new one:
 - Click **Session**  > **Save As** .
 - In the **Save As Session** panel, enter the required details and click **Save**.
7. To undo the updates to the session, click **Session**  > **Reset View** .

On resetting, the session goes back to its previously saved state.

Configure a session

You can configure a session to save specific filtering and configuration criteria for the product structures. You can retain the applied configurations (like variants and effectivity) and filters, and control the display of excluded or suppressed occurrences.

Sessions are used to save the filtering and configuration criteria applied to a structure. When you open a session, you view the structure as it was previously saved retaining its applied filtering and configuration criteria.

When a new session is created for a structure, objects in the structure are displayed based on the configurations applied prior to creating the session. The status of the following **Show...** sets a preference. The value of the preference is not saved when the session is saved.

- Show Excluded by Effectivity
- Show Excluded by Variants
- Show Suppressed

You can use these **Show...** toggles in the **Configuration Settings** menu to control the display of the structure elements. You can apply any filters on the configured structures as required. The configuration is retained with the filters applied.

When the session is saved, the applied configuration (variant, effectivity, filtering, etc.) is also saved. However, the status of the **Show...** commands is not saved.

Note

The system will reload the entire structure to reflect the changes in the configuration settings. This is due to the dependency on the BOM and the need to reprocess the entire structure.

Show or hide occurrences excluded by effectivity

When a structure is loaded, all objects including those which are not effective for the specified revision rule are displayed by default. However, you can choose to display only those occurrences that are effective for a specified date, unit, range of dates, or unit numbers.

To show or hide the occurrences that are excluded by the currently applied effectivity, in the work area toolbar, click **Configuration Settings**  > **Show Excluded By Effectivity**.

Show or hide occurrences excluded by variant

When a structure is loaded, all occurrences including those occurrences which are not configured by the variant are displayed by default. However, a user can choose to hide or show the that are not configured.

To show or hide the occurrences that are excluded by the variant configuration, in the work area toolbar, click **Configuration Settings**  > **Show Excluded By Variants**.

Show suppressed occurrences

Occurrences in a structure can be hidden by setting the suppress property to *True*. When a structure is loaded, all occurrences including the suppressed occurrences are displayed by default.

To show or hide the suppressed occurrences, in the work area toolbar, click **Configuration Settings**  > **Show Suppressed**.

Share a session with other users

You can share a session with other users. As a prerequisite, the system administrator must set the **Has Class(Fnd0AppSession)** Access Manager rule. To share the session, you must be the creator of the session. After you share the session, other users can view or edit the session that is shared by you.

Make a session shareable while saving the session

Procedure

1. Search for and open the structure.
2. Configure the structure as required.
3. Click **More Commands**  > **New**  > **Create Session** .
4. In the **Create Session** pane, specify the access level for other users:
 - a. Select the **Allow others to view** check box to grant *read* access to other users. Clear the check box to deny read access.
 - b. Select the **Allow others to edit** check box to grant *write* access to other users. Clear the check box to deny write access.

Note

The **Allow others to edit** check box is displayed only when **Allow others to view** check box is selected.

5. To create the session, click **Create**.

Initiate a workflow to share a session

When you share a session, you must inform the person you have shared the session with. You can communicate using any method such as emailing the session URL. You can also use a workflow to share sessions and assign tasks to other users.

Procedure

1. Search for and select the session that you want to share.
2. Click **More Commands**  > **Manage**  > **Submit to Workflow** .
The **Submit to Workflow** panel with a list of workflow templates is displayed.
3. Select the **Session Collaboration Workflow** template.

4. Enter a description for the workflow participants, and select the appropriate workflow template.
A default workflow is defined for sharing the session. It is automatically selected as the workflow template.
5. Click **Assignments** to assign users and resource pools to the workflow.

Note

You must have selected the **Allow Others to View** check box while creating a session in order for the assigned user to receive an email about the shared session.

6. Click **Submit**.

Release a session

You can release a session by submitting it to a workflow. When you release a session, you ensure that the proper release maturity is set for your session.

Procedure

1. Search for the session you want to release.
 2. Click **Open**  to open the session.
 3. Click **More Commands** **...** > **Manage**  > **Submit to Workflow** .
- The **Submit to Workflow** panel with a list of workflow templates is displayed.
4. Select the **Session Release Workflow** template.
 5. Enter the required details and click **Submit**.

Find an element within a session

You may want to search for a structure element within a session. Searching for an element is not limited to the filtered structure displayed in the session. The element is searched for across the unfiltered structure.

Procedure

1. Open a session.
2. Click **Find** .
3. Enter the search keyword.
4. (Optional) To narrow your search, select **Find within** to find the element within a certain assembly.

Capture a snapshot of a session

Snapshots capture the 3D data associated with your product. When you load the product in the 3D viewer and take a snapshot, you capture the current 3D view, including camera, visibility, selection, view port (pan, zoom, and rotate modes), orientation, sections, measurements, queries, and markups. You also capture the configuration and filtering criteria.

You can capture a snapshot of the current view of an assembly as an in-session snapshot. In-session snapshots are contained within the session. You cannot share these snapshots independently with other users.

View a session in other applications

You can a session and its associated structures in other applications like Teamcenter lifecycle visualization, NX, or CATIA so that you can leverage saved filters and configurations for 3D visualization, design modifications.

To easily retrieve a structure that you are currently working with, you save it within a *session*. The session also retains any filters and configurations applied to the structure. You can view a saved session and update the structure in applications such as Teamcenter lifecycle visualization, NX, or CATIA.

When you make changes to a session, and then open the unsaved session in other applications such as Lifecycle Visualization, NX, or CATIA, the changes must be saved or discarded before opening the structure in the other application. You can save or discard the unsaved changes.

Procedure

1. Search for a session that you want to view.
2. Click **Open**  to open the session.
3. To work with the 3D data associated with the structure, open the session in Teamcenter lifecycle visualization by clicking **Open**  > **Open in Visualization**.

Before opening the session in Teamcenter lifecycle visualization, you must first replay and save the session in Active Workspace.
4. To make design modifications to the structure, open the session in CATIA or NX by clicking **Open**  > **Open in CATIA** or **Open in NX**.

16. Save filtered and configured structures within a workset

About worksets

A workset is your personal working context. You can save your product definition in a workset. It is a container that holds one or more structures together. These structures can be the different views of the same structure or two entirely different structures. You can perform all BOM operations for a structure within a workset even when a partition scheme is applied to the structure.

You can create and use worksets only if your Teamcenter setup has the Smart Discovery license.

Note

Videos are not included in PDF. To access videos, use HTML.

Create a workset

A workset is your personal working context in which you can save your product definition. You can create a workset to bring separate structures together. A workset can be created for all structures, including those on which a partition scheme is applied.

Procedure

1. Open the structure that has variability data created using Product Configurator.
You cannot create a workset for a structure that has variability data defined using classic variants.
2. Click **New**  > **Workset** .
3. In the **Workset** panel, do the following:
 - a. In the **Type** list, select either **Workset** or **Workset Custom** depending on whether you want to create a workset or a custom workset.
 - b. In the **ID** box, enter an ID for the new workset.
 - c. In the **Revision** box, enter a revision for the new workset.
 - d. In the **Name** box, enter a name for the new workset.
 - e. (Optional) In the **Description** box, enter a description for the new workset.
 - f. (Optional) Click **Add Project** to assign a project to the new workset.
4. Click **Add**.

Results

A workset is created for your structure.

If your administrator has defined the settings, you can configure the structure in a workset using [arrangements](#).

Update a workset

When you change the filter parameters of a configuration in a workset, the updates you made to the workset and the structures within it are saved automatically. You can also update a structure (within the workset) for which a partition scheme is applied. However, the tree structure does not reflect the change. Therefore, you must perform a recalculation. Subsequently, the partition scheme displays the configured BOM lines under a partition, and this is based on the applied configuration settings and filter criteria.

Procedure

1. Search for a workset using the global search or **Quick Access**.
2. Click **Open**  to open the workset.
The workset is always loaded with static data.
3. To update the workset with the latest applied filters and configurations, select the workset and then click **Replay** . All modified structures within the workset are updated.
OR
To update just a structure within the workset, select the structure and then click **Replay**.
4. Select a product within a workset and then click **Filter**  or select **Configure**  to apply filters or configurations to the workset.
5. Select the workset and click **Replay** to update it.

Save a workset as a new workset

When you change the filtering aspects of a configuration in a workset, the updates you made to the workset and the structures within it are saved automatically. When you save a workset as a new workset, the name of the workset is changed, but its configurations are retained. You can also save a structure (within the workset) for which a partition scheme is applied as a new structure.

Procedure

1. Search for a workset using the global search or **Quick Access**.
2. Click **Open**  to open the workset.

3. Select a product within a workset and then click **Filter**  or select **Configure**  to modify the filters or the configurations applied on the workset.
4. Click the workset and then select **More Commands** *** > **New**  > **Save As** .
5. In the **Save As** panel, enter a name for the new workset.
6. (Optional) In the **Description** box, enter a description for the new workset.
7. (Optional) Click **Add Project**  to assign a project to the new workset.
8. Click **Save**.

Revise a workset

When you change the filter parameters of a configuration in a workset, the updates you made to the workset and the structures within it are saved automatically. When you update or revise a workset, the revision of the workset is changed, but its configurations are retained. You can also revise a structure (within the workset) for which a partition scheme is applied.

Procedure

1. Search for a workset using the global search or **Quick Access**.
2. Click **Open**  to open the workset.
The workset is always loaded with static data.
3. To update the workset with the latest applied filters and configurations, select the workset and then click **Replay** . All modified structures within the workset are updated.

OR

To update just a structure within the workset, select the structure and then click **Replay** .

4. Select a product within a workset and then click **Filter**  or select **Configure**  to modify the filters or the configurations applied on the workset.
5. Click the workset and then select **More Commands** *** > **New**  > **Revise** .
6. On the **Revise** panel, do the following:
 - a. In the **Revision** box, enter a revision for the workset.
 - b. In the **Name** box, enter a name for the workset.
 - c. (Optional) Enter a description for the workset in the **Description** box.
 - d. (Optional) Click **Add Project**  to assign a project to the new workset.
 - e. Click **Revise**.

Find elements within a workset

You can find elements across the different product structures (indexed using Smart Discovery Indexing) saved within a workset.

Procedure

1. Open a workset from the search results.
2. Click **Find** .
3. In the **Find** panel, enter the search keywords and click **Search** .

The search result displays elements that match the search keywords. These elements can belong to the different structures within the workset.

You can also search within a specific structure. However, this structure must be indexed using Smart Discovery Indexing. To search within a structure in the workset:

- Select the structure and click **Find** .
- In the **Find** panel, select the **Find within <Product Name>** check box, enter the search keywords, and click **Search** .

You can also find an element in a single structure (within the workset) for which a partition scheme is applied.

Compare worksets

You can compare two different worksets to view their differences. You cannot open the same workset for comparison.

Procedure

1. Search for and select two different worksets.
2. Click **Open** .
The two worksets are opened in the Split View.
3. Click **Compare**.
4. (Optional) Change the comparison level:
 - a. In the **Options** list, select a comparison level. The following table explains the different options.

Comparison option	Description
All Levels	Compares the elements at every level in the two worksets.
All Levels Below Selected	Compares all levels below the selected level in the two worksets.

Comparison option	Description
Components Only	Compares only the elements that are at the lowest level, that is, components.
Current Level	Compares only the elements that are immediate children of the selections in the two worksets. This is the default level of comparison.
Linked Assemblies or Components	Compares two worksets by using an accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the elements in one structure have equivalent elements in the other structure. You can define your own properties for equivalence.

- b. Specify what you want to display in the comparison results by selecting one or more of the following options from **Options > Display**:

Comparison option	Description
Matched	Displays the elements that are a match in both worksets.
Different	Displays the elements that differ between the two worksets.
Multiple Matches	Displays the different numbers of occurrences of the same element in the comparison as a difference.
Unique in Left View	Displays the elements that are only listed in the workset displayed on the left.
Unique in Right View	Displays the elements that are only listed in the workset displayed on the right.

- c. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two structures might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

- d. If required, select the **Generate Report** check box to generate the comparison report.

- e. Click **Compare**.

The comparison results are listed under **Results**. Clicking an element in the list highlights that element in the workset.

Comparison results are retained for a specific period. If a comparison result is older than the retention period, or if a more recent result exists, the old result is automatically deleted.

If you selected the **Generate Report** check box, the comparison report is generated in Excel. If you have the permissions to edit the target structure, the report is attached

to it. You can click the **Attachments** tab of the target structure to download the report. If you do not have permissions to edit the target structure, the report is saved in the **Newstuff** folder. The file name for the comparison report is in the format *Compare_Report_<Source_ID>_AND_<Target_ID>_YYYYMMDD_HH_MM_SS*, where *Source_ID* is the ID of the first structure that you want to compare and *Target_ID* is the ID of the structure with which you want to compare the first structure. You can also click **Alerts**  to understand the location of the comparison report if the structures are compared in the background.

Tip

If you closed the **Compare** panel and want to perform further comparison, click **Compare** on the task bar to view the **Compare** panel again.

Note

If the content of one of the worksets is modified, compare the worksets again.

5. When you are comparing two worksets, if the content of a workset is modified, click **Replay**  to bring the latest content to the workset.

Create a snapshot of a workset

You can create a snapshot of all worksets, irrespective of whether a partition scheme applied to the structures within the worksets.

Snapshots capture the 3D data associated with your product. When you load the product in the 3D viewer and take a snapshot, you capture the current 3D view, including camera, visibility, selection, view port (pan, zoom, and rotate modes), orientation, sections, measurements, queries, and markups. You also capture the configuration and filtering criteria.

When working with worksets, you can capture snapshots of a workset to preserve the working context and save it for later use.

Working with Product Configurator data in a workset

You can work with the Product Configurator data within a workset. A workset helps you manage multiple structures, enabling you to author variant conditions and load Saved Variant Rules (SVRs) directly from the workset or its included structures, while noting that classic variants are not supported in this context.

A workset is a container that holds two or more structures together. Therefore, to view configuration information, select the relevant structure within the workset.

You can perform Product Configurator-related tasks, such as authoring variant conditions and loading a Saved Variant Rule (SVR), from within the workset and from the structures included in the workset. Classic variants are not supported in a workset.

Modifying the data in a workset

You can modify data within a workset, which initially loads with the static data. The changes like adding or deleting elements are immediately visible. However, the changes like property edits, edits made to variant condition, or authoring occurrence effectivity require replaying the workset or the structure within the workset to fully take effect.

After you open a workset, it is always loaded with static data. You can modify the data in a workset, for example, you can add a structure element, remove an element, and edit the properties of an element. When you do this, you must replay the workset or a structure within the workset for the changes to take effect.

When you insert an element in a structure within a workset, the element is immediately displayed in the tree structure. However, when you reopen the workset or refresh the browser page, the element is not displayed. Now, when you replay the workset or a structure within the workset, the element might be displayed in the tree structure based on the applied filter criteria.

When you delete an element from a structure within a workset, the element is removed from the tree structure.

When you edit the properties of an element or edit a variant condition or author occurrence effectivity, the updates are immediately done. However, to display the changes, you must replay the workset or the structure within the workset. Now, when you reopen the workset or refresh the browser page, the updates are maintained even though the changed property does not satisfy the filtering or configuration criteria.

Export and import structures along with worksets

You can export and import structures along with worksets to and from other Teamcenter sites.

A workset is a context collector for other structures. If your organization has very large structures with a multitude of occurrences, these occurrences can be collated within a workset to do a *what-if* analysis. But when you send the workset to another site by using a Multi-Site environment or a Briefcase, you might not want to export such a large number of occurrences along with the workset. Therefore, while exporting, the **Include entire BOM** option is not enabled by default. Consequently, the TCXML-based Multi-Site functionality works to share workset and related objects but not the product item or item revisions or their content.

Worksets also support site-consolidation activities through the TCXML-based Multi-Site functionality.

To export and import structures along with worksets, you can use:

- Briefcase files
- Multi-Site Collaboration
- PLM XML

View a workset in other applications

You can open a workset in other applications, such as NX or CATIA. You can use Smart Discovery filtering in combination with a workset to update your CATIA drawings. In CATIA, you can open the workset with its individual structures. Also, you can open the selected elements from the workset in CATIA.

To make design modifications to the structure, open the workset in CATIA or NX by clicking **Open**  > **Open in CATIA** or **Open in NX**.

17. Compare structures

Compare the content in structures

You can compare two structures, to view their differences by using rules. You can compare different configurations of a structure as well. For this, you must set the revision rule and variant rule separately for each structure. Your administrator sets up the properties that can be compared.

Procedure

1. Perform the action listed in the following table depending on your task:

Task	Action
To compare two structure elements	Select two structure elements. Click Compare  > Compare Structures  .
To compare two structures	Select two structures from one of the following, and then click More Commands *** > View  > Compare Structures  : • Search results • Change History table on the History tab • Favorites
To compare two revisions of an element on the History tab	In the Revision History section on the History tab, select two revisions of an element and click More Commands *** > View  > Compare Structures  .

In the **Compare** panel, the elements that are unique to both structures are displayed with color bars under **Results**. This comparison is performed with the default comparison options.

2. (Optional) Change the comparison level:
 - a. Specify the comparison level by selecting one of the following from **Options**:

Comparison option	Description
All Levels	Compares the elements at every level in the two structures. If this option is selected for very large structures, performance may be adversely affected.
All Levels Below Selected	Compares all levels below the selected level in the two structures.
Components Only	Compares only the elements that are at the lowest level (components).

Comparison option	Description
Current Level	Compares only the elements that are immediate children of the selections in the two structures. This is the default level of comparison.
Linked Assemblies or Components	Compares two structures by using accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the elements in one structure have equivalent elements in the other structure. You can define your own properties for equivalence.

- b. Specify what you want to display in the comparison results by selecting one or more of the following options from **Options > Display**:

Comparison option	Description
Matched	Displays the elements that are a match in both structures.
Different	Displays the elements that differ across the two structures.
Multiple Matches	Displays the different numbers of occurrences of the same element in the comparison as a difference.
Unique in Left View	Displays the elements that are only listed in the structure displayed on the left.
Unique in Right View	Displays the elements that are only listed in the structure displayed on the right.

- c. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two structures might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

- d. (Optional) Select the **Run in Background** check box to perform the comparison in the background. This is recommended when you compare all levels or only the components in large structures. You can view the results later by selecting the relevant notification from **Alerts**.

Note

The **Run in Background** check box is not displayed when you select the comparison level as **Current Level**.

- e. (Optional) Select the **Generate Report** check box to generate the comparison report.
- f. Click **Compare**.

The comparison results are listed under **Results**. Clicking an element in the list highlights that element in the structure. If your administrator has configured it for you, the parent elements of the structures are also included in the comparison.

Comparison results are retained for a specific period. If a comparison result is older than the retention period or if a more recent result exists, the old result is automatically deleted by the system.

If you selected the **Generate Report** check box, the comparison report is generated in Excel. If you have the permissions to edit the target structure, the report is attached to it. You can click the **Attachments** tab of the target structure to download the report. If you do not have permissions to edit the target structure, the report is saved in the **Newstuff** folder. The file name for the comparison report is in the format *Compare_Report_<Source_ID>_AND_<Target_ID>_YYYYMMDD_HH_MM_SS*, where *Source_ID* is the ID of the first structure that you want to compare and *Target_ID* is the ID of the structure with which you want to compare the first structure. You can also click  to understand the location of the comparison report if the structures are compared in the background.

Tip

In case you closed the **Compare** panel and want to perform any further comparison, click **Compare** on the task bar to view the **Compare** panel.

You can select some other elements either from the same structure or one each from the two structures displayed side-by-side for further comparison.

To reconcile the differences found in the structure, you can drag an element from one structure to another and compare them again.

Compare the content in partitions

You can compare the content that is in two partitions when two different configurations are available for comparison, for example, when the content in each partition has a different revision, date, units, variants, and partition schemes. You can compare the content from a single structure where two different partition schemes are applied and where only one structure has a partition scheme applied. However, you must select the same element on both sides in a split view. You can compare the content in partitions that are present in two different structures as well. The partitions themselves are not listed in the comparison result.

Procedure

1. Open a structure.
2. Click **Split Context** .
The **Split View** displays the structure side by side in two views.
3. Change the structure configuration, as required. You can do this in both views.
4. Click **Compare** on the task bar.

5. In the **Compare** panel, specify the comparison level by selecting one of the following **Options**:

Comparison option	Description
All Levels	Compares the elements at every level in the two structures. If this option is selected for very large structures, performance may be adversely affected.
All Levels Below Selected	Compares all levels below the selected level in the two structures.
Components Only	Compares only the elements that are at the lowest level (components).
Current Level	Compares only the elements that are immediate children of the selections in the two structures. This is the default level of comparison.
Linked Assemblies or Components	Compares two structures by using accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the elements in one structure have equivalent elements in the other structure. You can define your own properties for equivalence.

6. In the **Compare** panel, from **Options > Display**, select one or more of the following options to specify what you want to view in the comparison results:

Comparison option	Description
Matched	Displays the elements that are a match in both views.
Different	Displays the elements that differ across the two views.
Unique in Left View	Displays the elements that are only listed in the view displayed on the left.
Unique in Right View	Displays the elements that are only listed in the view displayed on the right.

7. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two structures might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

8. (Optional) Select the **Run in Background** check box to perform the comparison in the background. This is recommended when you compare all levels or only the components in large structures. You can view the results later by selecting the relevant notification from **Alerts**.

Note

The **Run in Background** check box is not displayed when you select the comparison level as **Current Level**.

9. Click **Compare**.

The comparison results are listed under **Results**. Clicking an element in the list highlights that element in the structure. Clicking an element in the structure highlights that element in the list.

If your system administrator has configured it for you, the top elements of the structures are also included in the comparison result.

Note

If you close and relaunch the **Compare** panel by clicking **Compare** on the task bar, the panel is reset, and it does not display the previous results.

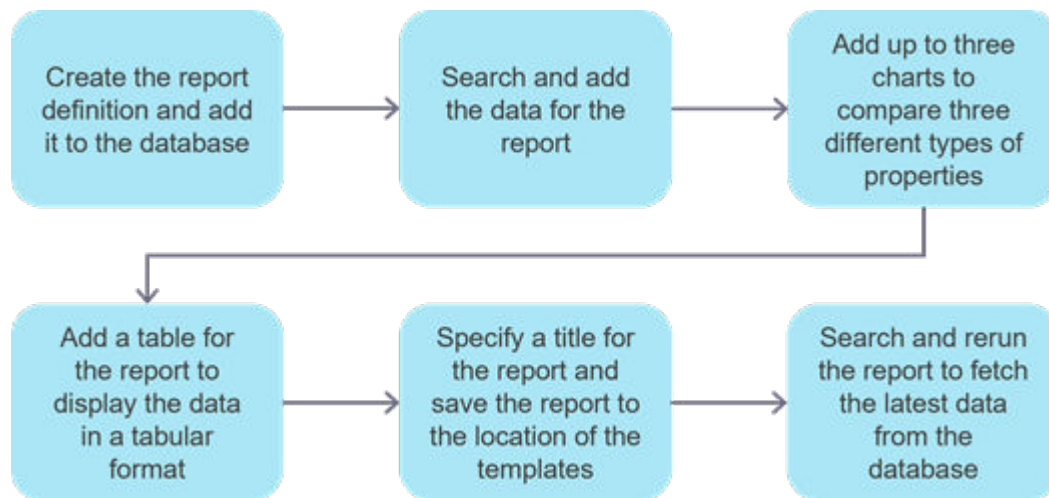
18. Create dashboard reports for structures

Create dashboard reports for structures and related content

You can create dashboard reports for structures and their related content. This includes generating active item reports that display information like BOM lines or attachments for a selected product structure, and customizing these reports with charts and tables to monitor aspects like structure readiness or associated problem reports.

Active item reports are generated in the context of a specific object, for example, reports that show all BOM lines or attachments for a selected product structure (**Item Revision**).

You can create report definitions for active item reports dynamically and add them to the database. The report definition identifies the content you want to include in the report. You can select a layout for the report and add charts, and a table.

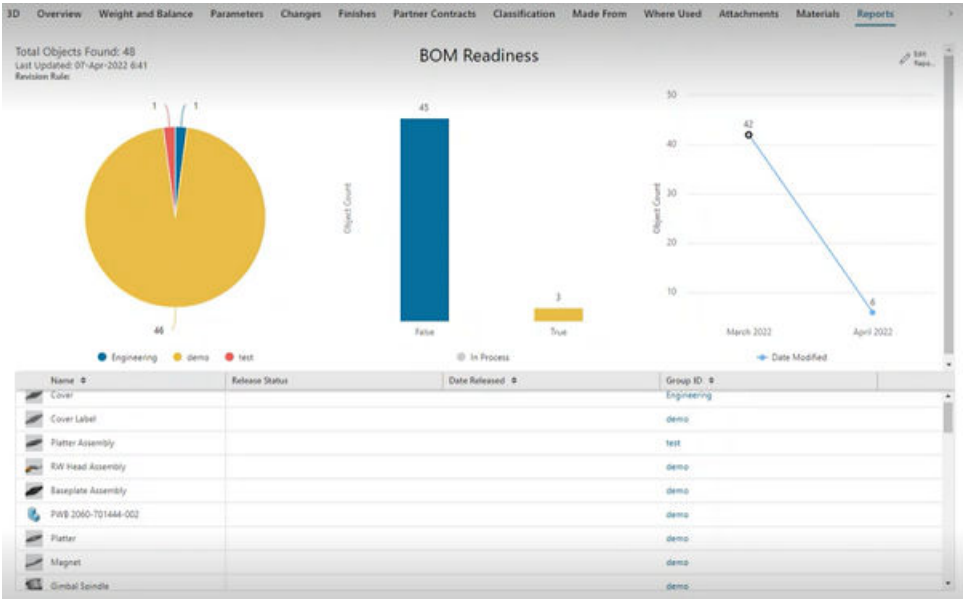


- Create a dashboard report for structure readiness (example)
- Create a dashboard report for problem reports associated with a product structure (example)

Create a dashboard report for structure readiness (example)

You can create a dashboard report for structure readiness to gather data about the components of the selected structure.

The following is an example of how to create a structure readiness report. After creating the report definition, you can search and select a product structure, and generate a report. This report traverses the selected structure to display information about its components, such as, their release status, how many of them are in process, the date released, and the date modified.



Procedure

1. Create the report definition and add the data for the report.
 - a. From **Launcher**, click **Reports**.
If this option is unavailable, search for it. You can pin it from the search results for future access.
 - b. In **My Dashboard** or **Templates**, choose **More Commands** > **New** ✨ > **Create Report**.
 - c. From **Type**, select **Active item report**.
2. Search for the data you want to include in the report.
In the **Search Item** panel, for example, type <hard drive> to search for an item revision such as **Hard Drive Assembly**, and click **Select**.
3. To add the relations, click **Add** and click **Pin Panel** to add multiple relations.
The system displays all the possible relations of the source object.

Column name	Description
Relation (Object)	Displays the relation name and the type of object this particular relation returns.
Relation Type	Displays the relation type such as primary, secondary, reference, or referenced by.
Count	Displays the number of objects that are related.

4. Select the appropriate relation object and click **Add**. Keep adding relations as appropriate.
5. Select a layout for the report and add charts

You must first create the report definition and add the data for the report before you can select a layout and add charts and a table.

- a. Specify a report title.
- b. Specify the number of charts you want to add for this report by selecting the appropriate option.
- c. Select **Bar Chart**, **Pie Chart**, or **Line Chart** for the type of chart you want to create.
- d. To create a layout for the report, click **Edit**.
- e. To add a property for the chart, from the **Chart On** list, select a value, for example, **Owner**.

You can specify a title for the chart type or accept the default title.

- f. To add the second and third charts, repeat step 5.c through step 5.e. You can choose other properties such as **Date modified** and **Release Status** in the **Chart On** panel for the second and the third chart.

To use additional **Chart On** properties, use the **REPORT_AW_ItemReport_Objects_FilterProperties** preference.

You can work with all Teamcenter preferences from within Active Workspace by using the **Preferences** page.

For information about retrieving a list of preferences, see *Where can I get a list of preferences?* in *Teamcenter Administration* on Support Center.

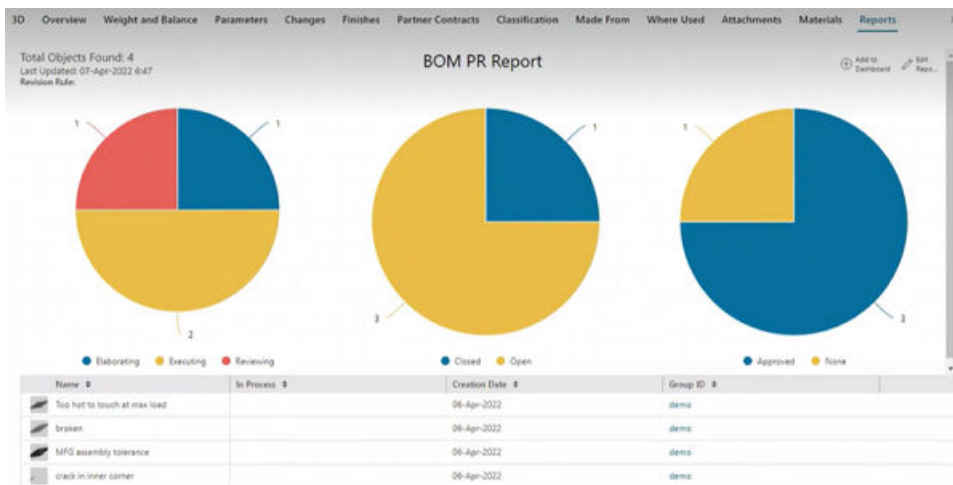
- g. After creating more than one chart, to move the chart, click ... **More Commands** and select the appropriate option.
6. To display the data in a tabular format, add a table for the report.
 - a. In the **TABLE** area, click **Table Settings > Arrange**.
 - b. From **Available Columns**, select column name properties such as **Name** and **ID**, and click **>** to move them to **Displayed Columns**.
Be sure to add the **Type** column.
 - c. (Optional) To rearrange the columns in the table, in the **Column Name** area, select a column name, click the column widget, and move the column up or down.
 - d. To save the column arrangement, click **Arrange**.
 - e. (Optional) To add additional columns to the table, use the **REPORT_AW_ObjectType_Properties** preference.
 7. Save the report as a template.
 - a. To save the report as a template, click **Save as Template**.
 - b. (Mandatory) Specify a name for the report.
 - c. (Optional) Include a description.
 - d. Select the appropriate thumbnail option.
 - e. Click **Save**.

8. Edit the report from **Templates**.
 - a. Navigate to the **Templates** location.
 - b. Search for the report definition you want to edit, select it, and choose **More Commands > Edit**  **> Edit Report**.
 - c. Make the necessary changes and save the report.
9. Generate the report.
 - a. Search for the item revision for which you want to generate the report, for example, **Hard Drive Assembly**.
 - b. Select the item revision and then choose **More Commands > New**  **> Generate Report**.
 - c. Select the report definition you created and click **Generate**.
The report is generated by fetching the latest data from the database.
 - d. To apply a different revision rule, click **Edit Report Criteria** in the **Reports** tab in the secondary work area, and select a revision rule. The report is updated based on your selection.
By default, the system displays the **Latest Working** revision rule.

Create a dashboard report for problem reports associated with a product structure (example)

You can create a dashboard report for problem reports associated with a product structure.

The following is an example of how to create a dashboard report for problem reports associated with a product structure. This report is used to traverse a configured structure and locate problem report objects attached to every node of the structure. After creating the report definition, you can search and select a product structure, and generate a report. The report displays how many problem reports are open, closed, in process, in review, and approved for the selected structure.



Procedure

1. (Optional) Create the sample data for this report.

The following procedures to create problem reports are optional. At your site, you can search for the appropriate object to add as the sample source to validate the related data.

- a. Open a product structure such as **Hard Drive Assembly**.
- b. Select a node level BOM line and choose the **Changes** tab in the secondary work area.
- c. Choose **Create Change**, type **Problem Report**, select it, specify a synopsis (mandatory), and click **Create**.
- d. Select a leaf level BOM line, choose the **Changes** tab in the secondary work area, and create a problem report (see step 1.c).

2. Create the report definition and add the data for the report.

- a. From **Launcher**, click **Reports**.
If this option is unavailable, search for it. You can pin it from the search results for future access.
- b. In **My Dashboard** or **Templates**, choose **More Commands** > **New** ✨ > **Create Report**.
- c. Select **Active item report**.

3. Search for the data you want to include in the report.

- a. In the **Search Item** panel, for example, type *<hard drive>* to search for an item revision such as **Hard Drive Assembly**, select it, and click **Save**.
- b. To add the relations, click **Add** and click **Pin Panel** to add multiple relations.

The system displays all the possible relations of the source object.

Column name	Description
Relation (Object)	Displays the relation name and the type of object this particular relation returns.
Relation Type	Displays the relation type such as primary, secondary, reference, or referenced by.
Count	Displays the number of objects that are related.

- c. Select the appropriate relation object and click **Add**. Continue adding relations as appropriate.

4. Select a layout for the report and add charts

You must first create the report definition and add the data for the report before you can select a layout and add charts and a table.

- a. Specify a report title.
- b. Specify the number of charts you want to add for this report by selecting the appropriate option.
- c. Select **Bar Chart**, **Pie Chart**, or **Line Chart** for the type of chart you want to create.

- d. To create a layout for the report, click **Edit**.
- e. To add a property for the chart, from the **Chart On** list, select a value, for example, **Closure**.
You can specify a title for the chart type or accept the default title.
- f. To add the second and third charts, repeat step 4.c through step 4.e. You can choose other properties such as **Disposition** and **Maturity** in the **Chart On** panel for the second and the third chart.

To use additional **Chart On** properties, use the **REPORT_AW_ItemReport_Objects_FilterProperties** preference.

You can work with all Teamcenter preferences from within Active Workspace by using the **Preferences** page.

For information about retrieving a list of preferences, see *Where can I get a list of preferences?* in *Teamcenter Administration* on Support Center.

- g. After creating more than one chart, to move the chart, click ... **More Commands** and select the appropriate option.
5. To display the data in a tabular format, add a table for the report.
 - a. In the **TABLE** area, click **Table Settings > Arrange**.
 - b. From **Available Columns**, select column name properties such as **Name** and **ID**, and click **>** to move them to **Displayed Columns**.
Be sure to add the **Type** column.
 - c. (Optional) To rearrange the columns in the table, in the **Column Name** area, select a column name, click the column widget, and move the column up or down.
 - d. To save the column arrangement, click **Arrange**.
 - e. (Optional) To add additional columns to the table, use the **REPORT_AW_ObjectType_Properties** preference.
 6. Save the report as a template.
 - a. To save the report as a template, click **Save as Template**.
 - b. (Mandatory) Specify a name for the report.
 - c. (Optional) Include a description.
 - d. Select the appropriate thumbnail option.
 - e. Click **Save**.
 7. Edit the report from **Templates**.
 - a. Navigate to the **Templates** location.
 - b. Search for the report definition you want to edit, select it, and choose **More Commands > Edit**  **> Edit Report**.
 - c. Make the necessary changes and save the report.

8. Generate the report.

- a. Search for the item revision for which you want to generate the report, for example, **Hard Drive Assembly**.
- b. Select the item revision, and then choose **More Commands > New  > Generate Report**.
- c. Select the report definition you created and click **Generate**.

The report is generated by fetching the latest data from the database. It displays the problem reports associated with the structure.

- d. To apply a different revision rule, click **Edit Report Criteria** in the **Reports** tab in the secondary work area, and select a revision rule. The report is updated based on your selection.

By default, the system displays the **Latest Working** revision rule.

Create a dashboard report for problem reports associated with a product structure (example)

19. Work in a context

What is a context?

When you select a product or subassembly, the selected structure with revision rules, variant rules, and effectivity form your *working context*. There are two types of working contexts, a *background working context* and a *saved working context*.

The background working context is your current environment. It is not necessary to save your current background working context before you exit. This is because you always return to your previous background context when you open the same content the next time.

When you perform a find-in-context search, the search results are obtained in the context of the product or subassembly and do not extend to other products or subassemblies.

If you want to preserve your current environment for later retrieval or sharing, you must specifically save the context. Teamcenter preserves your environment for future retrieval in a saved working context. This allows you to return to the product or subassembly at a later time without having to re-establish the same context. You can also share a saved working context with other users.

The following information are captured for a saved working context:

- Structure configuration with context applied, including revision rules, effectivity, and variant rules
- Current search results and criteria

Working in a context

A working context provides a way to set your work aside and then pick up where you left off when you return. Your working context can be retrieved from any device, allowing you to, for example, work on a desktop machine during office hours and then review your work on a tablet in a meeting or on the shop floor later in the day.

You can also save a working context to share it with other users, stakeholders, and collaborators. You can save multiple working contexts, up to limits defined by the system administrator.

As a user, you can:

- Open a saved working context that you saved or was shared with you by other users.
- Open a saved working context object from search results as a referenced item on a change object (for example, a problem report) or any other reference.
- View all the saved working contexts that are associated with a part that you are viewing.
- Open a saved session that you saved or was shared with you by other users.

- Open a saved session object from search results as a referenced item on a change object (for example, a problem report) or any other reference.
- View all the saved sessions that are associated with a part that you are viewing.

Save a working context

You can save your current working context for later recall. For example, you may save a context to create a study or to capture an issue.

You can share a working context with others informally or formally by using change management and issue reporting features.

Your background session is saved automatically during your current session. It is not necessary to manually save before exiting if you only wish to resume where you left off later. However, saving the context with a name allows you to create multiple contexts and recall any of them later.

Procedure

1. To save a working context, click **More Commands ***** > **New**  > **Working Context** .
2. In the **Save Working Context** pane, enter a name and a description.

This action creates a saved context that you can optionally recall or share with others.

- A default name is assigned when you save a context based on the content. You can assign your own name if you prefer to.
- If you are working in a previously saved context and want to save changes to the existing context, click **Save**.
- If you are working in a previously saved context and want to save the changes as a new context and not overwrite the existing context, click **Save As**.
- The properties that are required when you save a context are configured by the system administrator. For example, your administrator can create the working context types using Business Modeler IDE. Subsequently, you can specify the working context type when you save a working context.

Teamcenter deletes background working contexts to free up disk space, based on when they were created last and the permitted maximum number of contexts allowed for each user. For example, if the administrator specifies that the maximum number of days a working context must be retained

for is 30 (days), when a user opens a structure, all working contexts older than 30 days are deleted. Similarly, if the administrator permits each user to retain a maximum of 50 working contexts, when a user opens a structure, the oldest working contexts from the set of that user's contexts above the limit (of 50) are deleted.

Delete a saved working context

Users with appropriate privileges can delete a saved working context. If you have pinned a saved working context to the home page or added it to your favorites, the working contexts are visible either on the home page or in the favorites list. You can also search for the saved working context.

Procedure

1. Search for the saved working context you want to delete.
2. Select the working context.
3. Click **Edit**  > **Delete** and confirm that you want to delete the working context.

Reset the background working context

The background working context is your current environment. It is automatically saved. When you open the same structure again in a different login session, you get the option of returning to the previous background working context.

If you do not wish to continue working with the saved working context, you can reset it. The following message is displayed when you open the content again in the same login session:

Restored "HDD-0527/A;1-Hard Drive Assembly" from "Tue May 10 00:36:29 GMT+530 2016". Select **Reset** to discard automatically saved changes.

- To reset the working context and to discard automatically saved changes, select **Reset**.
- To continue with the same working context, ignore the message. It disappears automatically.

Restore your working context

The background working context is your current environment. You get the option of returning to your previous working context when you login and to open the structure you were previously working on. The following message is displayed:

 Welcome back! Restore from where you left off? (Today, 14:28) **Restore**

- To go back to the previous working context, click **Restore**.
- To continue with the default working context, ignore the message and start working. The message disappears.

If you do not restore your previous working context, the structure is loaded with the following default configuration:

- Default configuration (global revision rule) is applied.
- No variant rules, effectivity, expansion, or filters are restored from the previous session.
- The structure is expanded to one level only.
- The structure opens with the default tab. The previous tab is not restored.

View where an element is used in a context

Select an element and click the **Where Used** tab to view where the element is used in a context.

- The result shows where an element is used in parent assemblies or in end items (products).
- The results show the context if a configuration contains the element, allowing you to open the specific configuration.
- You can find a part using a global full text search to find an element by name or part of a name and then use the *where used* capability to find where it is used. Wildcard characters may be included in the search string.
- Use the **References** section to find what parts and documents reference the object.

Results

When you open a context from your search results, Teamcenter takes you to the occurrence that was the subject of the *where used* query.

Opening an element in context

When you open a context for which the seed part appears multiple times, the navigation pane displays all the occurrences. This allows you to navigate to each occurrence of the element in context.

Note

The same navigation pane allows you to find other parts in context.

Recalling a working context

You can open a working context that you saved or was shared with you. You can find the working context object by using search, from a reference item on a change object (for example, a problem report), or from any other reference.

To open a working context shared by another user, click the URL (link) provided or search for it. The sharing user's complete working context opens for you.

Share a working context with other users

You can share a working context with others. As a prerequisite, the system administrator must enable this by adding the **Has Class(Awb0SavedBookmark)** rule to the Access Manager (AM) rule tree. To share the working context, you must be the owning user of a saved working context. On sharing, other users can view your working context or edit your saved working context.

Procedure

1. Search for the saved working context you want to update.
2. Select the working context.
3. Click **More Commands**  > **New**  > **Working Context** .
4. In the **Save Working Context** pane, specify the access level for other users:
 - a. Select the **Allow others to view** check box to grant read access to other users. Clear the check box to deny read access.
 - b. Select the **Allow others to edit** check box to grant write access to other users. Clear the check box to deny write access.

Note

The **Allow others to edit** check box is displayed only when **Allow others to view** check box is selected.

5. To save changes to the context, click **Save**.

Including multiple structures in a working context

To include multiple structures in the working context, click the **Add**  icon in the context and browse to the required structure. By default, Teamcenter applies the global revision rule but you can change to another configuration rule if necessary. You can configure each structure in a working context independently.

By default, when you open a working context and nothing is selected, the first structure in the working context is the context of the configuration. The parent of the selected structure becomes the context and remains the context even if you drill down into the structure.

You can identify if a working context contains multiple structures by selecting the context in the breadcrumb and verifying if more than one structure is shown in the **Content** pane.

When opening a working context that has multiple structures, each of the included structures is configured as it was when the working context was saved.

Sorting of content in a working context works at the sub-levels and not at the root level. The root levels are always the products in working context.